Proof reductions by squashing (reductions)

Contract, version 0.22.0

This contract replaces a physical receiver, and in trusted-source mode a physical source as well, by finite models a security analysis can run on, and states exactly what each replacement transfers. A squashing relation asks that the adjoint of a channel pull every target POVM element back to the physical one, together with carried operators that move device facts into the squashed model; because the physical measurement channel then factors through the squashing channel, the physical and the squashed protocol have the same output state. The core module has four main results, in the order the contract states them. Security of the squashed model over analysed inputs that contain the squashed image of every physical input transfers to the physical protocol with the same ε, for every criterion (Theorem 3.11), and the attack-set licence places every squashed physical attack in the admissible set the consuming key-rate engine restricts to, whenever each carried operator's pull-back is bounded above or below by, or equal to, a multiple of the identity, a spectral certificate (Theorem 6.7). In trusted-source mode the source replacement, which exchanges the specified signal ensemble for its virtual one, and the squashing reduction compose, in either order, into one exact operational reduction whose image lies in the admissible attack set (Theorem 8.6), and the target a reduction instance emits is exactly admissible and squashes the declared model with no error term (Theorem 11.2). The two submodules, wpfss (flag-state and weight-preserving squashers) and domination (the link that relates a dominated family to the one dominating it, kept for a declared device weight), follow the core as parts of their own and add main results, listed with these four in the front matter. Every construction is a reduction instance, emitted as a reduction chain of typed links that each carry their channel witness, and every condition on declared numbers is decided exactly, every declared and emitted operator entry and declared real scalar lying on a declared lattice of binary64 (double-precision) values; where the emission has to move an instance's datum, exact arithmetic having left that lattice, the emission theorem is about the moved, declared model, which nothing proved relates to the handed one; a moved class is disclosed, and a run that reads it rests on the premise that what it concludes from the emission holds of the declared model, as it does where the device measures that model. Where the declaring instance hands a reference of its measurement beside its snap, the same construction run on the reference, with no normalizer, is emitted beside the artefact as its reference companion (Corollary 12.1). The supplier's device facts, the independence of the measuring party's private setting and the round structure are assumptions the caller carries, while the soundness of the emitted constraints is a conclu-

sion of the chain, emission and licence theorems and not an assumption; the adversary's registers are finite-dimensional, as the round model's are, and the consuming engine carries the security statement to an adversary whose memory has any dimension (Corollary "Security of the protocol and its net key" (meat-core/cor:composed-security)).

Grey text, here and throughout, is non-normative: it explains, and nothing in it is assumed or proved.

The design notes (non-normative) explain why the module is built as it is and draw its processing flowchart. The dependencies between the sections are drawn in the section dependency matrix, and the graph index opens every section's dependency graph. The statements digest holds every definition, assumption, result and requirement without the proofs. How to read a contract explains this layout and its links.

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Main results

The module's most important results, those its overview names as what it establishes, with every assumption each rests on, directly or through the claims it uses. Each links to its statement and proof; the statements digest holds them in full.

Theorem 3.11 Security transfer from a constrained squashed model

If the analysed squashed inputs contain the squashed image of every physical input, a bound ε on any criterion of the output state of the squashed protocol holds for the physical protocol with the same ε. No property of the criterion is used.

Rests on A-MEAS, A-ROUND-SQ, A-SEQ

Theorem 6.7 The attack-set licence

If each operator the squashing carries pulls back to a physical operator $\leq$, $=$ or $\geq$ a multiple of the identity, every squashed physical attack respects the resulting constraint list in either source mode, factor by factor for per-factor bounds. Security against such attacks transfers with the same ε.

Rests on A-CONST, A-MEAS, A-ROUND-SQ, A-SEQ

Theorem 8.6 The two-sided reduction in trusted-source mode

In trusted-source mode the source map and the squashing reduction compose, in either order, into one exact operational reduction into the admissible attack set. Security of the virtual squashed model thus transfers with the same ε to the specified physical protocol, against attacks of its type.

Rests on A-CONST, A-MEAS, A-ROUND-SQ, A-SEQ, A-SRC

Theorem 11.2 Exact emission of a reduction instance

The emitted target is computed exactly, decided exactly admissible before emission, and squashes the declared model exactly; nearness to the handed model bounds nothing. On blocks the record names as resting on A-SNAP, the run's conclusions about the device rest on that premise.

Rests on A-CONST, A-MEAS, A-ROUND-SQ, A-SEQ, A-SNAP

Corollary 12.1 The reference companion of a reduction instance

The same instance, run on the reference in place of the snap, or on its own snapped blocks where none is handed, is a reduction instance of the emitted shape that squashes that model exactly. Its entries are enclosed from the reference, requested ones where their certificates hold there.

Rests on A-CONST, A-MEAS, A-ROUND-SQ, A-SEQ, A-SNAP

Proposition 19.4 Flag-state squashers: existence, chain and encoded weights

With total certified weight below one, the physical POVM squashes to both targets, and the squashers compose exactly through a flag-noise channel. A weight the weight-preserving target encodes holds on every squashed state; under the flag-state target it holds only on reachable ones.

Rests on no assumption

Corollary 21.3 Exact emission of the flat instance

The flat instance's emitted family is exactly admissible and pulls back exactly to the declared model on the channel system, its weights rounded down yet certified. Where the normalizer moved a block below the cutoff, nearness to the unrounded family bounds nothing.

Rests on A-CONST, A-EIG, A-EXACT, A-MEAS, A-MODES, A-NOSEL, A-PASSIVE, A-PORT, A-PROBE, A-RECV, A-ROUND, A-ROUND-SQ, A-SEQ, A-SNAP, A-THRESH, A-TRUST, A-TSM, A-VAC

Theorem 23.2 The noise channel of a domination

If G dominates F sector by sector, G squashes exactly to F with added flag sectors, or to F itself where F has flag sectors, through a noise channel that keeps each sector

with probability one minus its weight. The adjoint returns each sector projector scaled by that probability.
Rests on A-CONST, A-MEAS, A-ROUND-SQ, A-SEQ

Assumptions at a glance

Every assumption this contract states or its claims assume, with its category and the claims of this contract that assume it directly. An assumption of another contract links into that contract's PDF.

Assumption	Category	Title	Stated as	Assumed directly by
A-MEAS	receiver	Each measuring party's device as one POVM per round	Assumption 13.1	Lem. 2.4, Thm. 3.1, Rem. 3.7, Rem. 3.8
A-ROUND-SQ	receiver	Round homogeneity of the physical measurement	Assumption 13.2	Thm. 3.1, Rem. 3.6, Rem. 3.7
A-SNAP	receiver	The device measures the snapped family where the run reads the snap	foundations/ass: snapped-measurement	Thm. 11.2, Cor. 12.1, Req. 17.3
A-SEQ	protocol structure	Sequential round structure	transmitter/ass: seq	Thm. 3.1, Lem. 8.3
A-TSM	protocol structure	Output-state test statistics	Assumption 13.6	Rem. 4.5, Prop. 21.2, Cor. 21.3, Cor. 21.5, Cor. 21.6
A-PROBE	local randomness	The station's private setting drawn from its own declared randomness	Assumption 13.5	Prop. 21.2, Cor. 21.3, Cor. 21.5, Cor. 21.6
A-CONST	supplied artefact	Declared constants of the constraint list	Assumption 13.7	Rem. 3.14, Thm. 6.7, Rem. 6.8, Cor. 11.3, Rem. 11.8, Rem. 19.9

Assumption	Category	Title	Stated as	Assumed directly by
A-RECV	supplied artefact	Supplier artefact whose claims hold of the device	Assumption 13.3	Rem. 6.8, Prop. 20.4, Rem. 20.7, Rem. 20.10, Rem. 20.11, Rem. 20.12, Prop. 21.2, Cor. 21.3, Cor. 21.4, Cor. 21.5, Cor. 21.6

Notation

The symbols of the repository's notation catalogue (spec/notation.md) that this contract uses, in the catalogue's order, with their meaning.

Symbol	Meaning
$\mathcal{H}(A)$	Hilbert space of the system label A
$\mathcal{L}(A)$	Bounded linear operators on $\mathcal{H}(A)$
$\mathcal{L}(A, B)$	Linear maps $\mathcal{H}(A) \to \mathcal{H}(B)$
$\mathrm{Pos}(A)$	Positive semidefinite operators
$S_=(A)$	Normalized states
$S_\leq(A)$	Subnormalized states
$\mathbb{1}$, $\mathbb{1}_C$	Identity operator (contextual / named)
id	Identity channel
$\oplus$	Direct sum of system labels
P_n, P_n^C	Photon-number projector
$M^{(\leq N)}$	Operator block restriction
$\leq N$, $> N$, $= n$	Block indices for `\blk`
$\mathfrak{D}$	Protocol declaration $\big(A, \tau_A;\ Q, \mathbb{S};\ \{M_x\};\ \{F_v\};\ \mathfrak{P};\ a_A, a_B;\ \phi;\ r, \mathsf{h};\ \mathfrak{L}\big)$ — the tuple of meat-core/def:declaration: the party partition $\mathfrak{P}$ is a slot, and the marginal slot τ_A takes a state or the no-marginal value (trusted- vs untrusted-source mode)
$\mathbb{S}$	Sector index set of Q
Q	The declared sector-carrying measured system
$\mathfrak{P}$	The declared party partition: the readout maps π_A, π_B onto the alphabets $\mathcal{O}_\mathsf{A}$, $\mathcal{O}_\mathsf{B}$, with a party label for A and, for each sector s, a declared tensor factorisation of $\mathcal{H}(Q_s)$ with a party label per factor
$\mathcal{X}$	Outcome alphabet of the measurement on A — Alice's under the default partition
$\mathcal{V}$	The measuring parties' declared joint outcome alphabet — Bob's alone under the default partition
$\widehat{C}$, $\widehat{\mathcal{C}}$	The acceptance-statistic register $\widehat{C}$ and its alphabet $\widehat{\mathcal{C}}$

Symbol	Meaning
$\mathfrak{L}$	Declared constraint list $\{(\hat{O}_k, \bowtie_k, \Theta_k)\}$
$\mathcal{M}$	Single-round protocol channel $AQ \to Z\widehat{C}I$
$\mathcal{P}_{\mathbb{S}}$	Sector pinching on Q
C, P	Channel input; probe input
Flg	Flag system — post-squashing only
$\mathcal{Y}$	Bob's coarse-grained detection alphabet, $\lvert\mathcal{Y}\rvert = 6$
$\mathcal{T}_{\text{set}}$	Bob's private setting set, weights q_t (Def. 4.4); in the optical instance the probe settings (register T_{set})
on, off, dc, sq, mc, nc, sc, vac	Mode / outcome abbreviations
$\lambda_{\min}$	Minimum eigenvalue
T	Transpose, X^{T}
$\langle a\vert b\rangle$	Inner product
$\boldsymbol{v}$	Vector (bold)

Terms

The phrases this contract defines, and the terms of other contracts it uses, each with a one-line gloss and the definition that fixes it. In the text, a linked term opens its definition.

Term	Meaning	Defined in
admissible attack sequence	an attack sequence whose every round lies in the attack set of a given constraint list	foundations/def: round-model
admissible datum	A datum that is Hermitian, positive semidefinite and class complete in every data sector; the set D_adm.	Definition 10.1
admitted	Said of a declaration with a non-empty source segment and a constraint list that meets the admission rule: licensed entries only, in trusted-source mode.	Definition 7.2
analysed strategy set	The strategy set of the presentation an exact operational reduction lands in, which the analysis runs over; also called admissible.	Definition 2.3
attack sequence	one channel per round whose output is the per-round system, with Alice's system where the adversary supplies it, and Eve's new memory	foundations/def: round-model
attack set	the channels whose adjoint maps every listed operator into the stated relation with its constant	foundations/def: round-model

Term	Meaning	Defined in
attack strategy	An element of the strategy set of a protocol presentation; a state, a channel or any other object.	Definition 2.3
bare form	The input form of the weight route in which the supplied preserved blocks are those of the conditional POVM before the dark-count channel, so that step 1 acts on them.	Definition 20.3
binary64 numbers	the finite values of the IEEE 754 binary64 format, each a dyadic rational	foundations/def: binary64
block list	one square complex matrix per block of a sector map, of that block's dimension, identified with their direct sum	foundations/def: sector-map
carried operators	Hermitian operators on the squashed system, listed beside the target POVM, whose pull-backs a squashing relation prescribes.	Definition 1.3
certificate	The hypothesis that makes a transported constraint hold: an operator identity or inequality on the pull-back, or an assumption on the input states; not a source-replacement certificate.	Definition 1.8
certified entries	The constraint entries of a reduction instance, each a lattice combination of transported operators whose certificate is discharged at every admissible datum.	Definition 10.1
certified weights	Non-negative numbers, one per outcome, each an operator lower bound on that outcome's POVM element above the cutoff.	Definition 19.1
channel modes	The declared input modes through which the adversary's signal enters.	receiver/def: instrument
channel witness	The squashing channel a link carries, relating its source family to its target together with the carried operators.	Definition 7.1
class complete	Said of a datum whose elements in each class sum exactly to the class weight times the identity, in every data sector.	Definition 10.1

Term	Meaning	Defined in
class weights	The declared probabilities of the settings: positive lattice values that sum to one exactly.	Definition 4.4
classes	The blocks of the joint alphabet, one per setting: every outcome paired with that setting, in the order of the outcome alphabet.	Definition 4.4
class-marginal flag	The flag of a reduction instance recording whether its construction establishes the class-marginal identity of the model at every admissible datum.	Definition 10.1
class-marginal identity	The operator identity that the elements of each class of a joint family sum to the class weight times the identity.	Definition 4.4
complex enclosure	a Gaussian-rational centre and a non-negative rational radius that bounds the distance of a complex number from the centre; a matrix is enclosed entry by entry	receiver/def: complex-enclosure
composable	Said of a receiver segment in which each link's target is the next link's source family.	Definition 7.2
composite	The pair of a chain's source composite and receiver composite.	Definition 7.2
consumer	a module other than a supplier that reads declared data	foundations/def: declared-data
dark-composed family	the click patterns of each setting composed with a dark-count channel, and their sums over the fibres of the declared coarse-graining	receiver/ detector-imperfections/ def:dark-composed
dark-composed form	The input form of the weight route in which the supplied preserved blocks are a declared complete family that the model takes below the cutoff after the dark counts, so that step 1 is not applied to them.	Definition 20.3
dark-count channel	A column-stochastic matrix on click patterns that can add clicks and never removes one.	receiver/ detector-imperfections/ def:darkadm
data sector	A sector of a reduction instance's target whose blocks are the datum's matrices.	Definition 10.1

Term	Meaning	Defined in
datum	A family of square matrices, one per data sector and outcome, with entries in the substrate and zero off each sector's support.	Definition 10.1
declared coarse-graining	A surjection from the raw click patterns onto the declared emitted alphabet, supplied as data by the protocol declaration.	receiver/def: coarse
declared dark-count route	The dark-count channel is a matrix R declared as such.	receiver/ detector-imperfections/ def:darkadm
declared data	the data an instance states and the data a producer emits as its own declaration, with the sets such data define	foundations/def: declared-data
declared datum	The datum the emission normalizer outputs: exactly admissible on the declared lattice, and carried entry by entry in the emitted target.	Definition 11.1
declared descriptor family	The family of source descriptors for which a source link's channel identity is asserted.	transmitter/def: source-link
declared element graph	A finite directed graph whose vertices are catalogue elements, each with an entry type, a parameter tuple and a support.	receiver/def: graph
declared mode space	A finite, totally ordered set of optical mode labels together with the Hilbert space whose orthonormal basis they index.	receiver/def: modes
declared model	The handed model with its datum replaced by the declared datum, as the instance's model map defines it.	Definition 11.1
declared order	a total order on a finite alphabet, fixed together with the alphabet	foundations/def: declared-order
declared source modes	The declared input modes that carry the receiver's own probe state.	receiver/def: instrument
declared tolerance	a binary64 number, at least zero and below one, that an instance states in its run configuration	foundations/def: tolerance
declared value	a binary64 number stated as a datum, which denotes exactly that dyadic rational and approximates nothing	foundations/def: binary64

Term	Meaning	Defined in
default party partition	The assignment of every factor of every target sector to Bob, the party whose system the handed model lives on.	Definition 10.1
domination link	A receiver-stage typed link of kind family-dominating from a finite target family to a family it dominates at the domination weights, whose channel witness is the noise channel with added or shared flag sectors.	Definition 23.1
domination weights	The weights, one per sector of a finite target system, at which the source family of a domination link dominates its target family.	Definition 23.1
eigenvalue slack	The declared number subtracted from each computed block eigenvalue of an exact window.	receiver/def: pmcex
emission displacement	Per data sector, the largest entrywise modulus of the difference between the declared datum and the handed datum.	Definition 11.1
emission normalizer	The deterministic procedure that puts a reduction instance's exactly formed datum back on the declared lattice: it keeps the classes already there, snaps every other class for fidelity without reviving a zero row or column or, failing that, by a flagged last resort, and decides the six emission conditions last.	Definition 11.1
enclosed weight	The per-setting weight formed on enclosed bound data and, at an exact window, on rigorous window bounds, with no computed eigenvalue and no eigenvalue slack.	receiver/def: enclosed-weight
entrywise enclosure	a Gaussian-rational Hermitian centre together with symmetric non-negative rational radii that bound, entry by entry, the modulus of the difference between a matrix and that centre	foundations/def: exact-reference

Term	Meaning	Defined in
evaluated off its snap	said of a block of a snap in a run: no number from which the run computes its key length reads the snapped block, save the points at which those numbers are evaluated and the values other than the output length that the run fixes before its first round	foundations/def: evaluated-off-snap
evaluation model	The arithmetic of this contract: its exact claims evaluated in binary64 with diagnostic residuals and no rigorous enclosure of a floating-point number beyond the eigenvalue-slack assumption, beside exact rational enclosures of the blocks and a rigorous evaluation of the weights.	receiver/def: evaluation-model
exact flag	The flag of a setting, exact or general; it may read exact only where every prefactor is one by construction.	receiver/def: setting-family
exact island	the computations of the engine, its instances' declarations included, by which a consumer of data or the declarer about to declare them decides whether they satisfy the conditions their declaration states, a declarer computes and decides the reference of a measurement family, a consumer's replacement of an emitted scalar bound is decided to lie on the bound's weakening side, a producer constructs the operator data it emits, the acceptance side of the optimisation is constructed and certified, or a lower bound on a linear minimum over the feasible side of the optimisation is certified	foundations/def: exact-island
exact operational reduction	A map of attack strategies between two presentations over the same output registers under which every strategy and its image produce the same full output.	Definition 2.3
exact reference	a reference all of whose radii are zero, so that each block is its Gaussian-rational centre	foundations/def: exact-reference

Term	Meaning	Defined in
exactly admissible	said of a declared operator family whose sector map agrees with the given one, whose blocks are exactly Hermitian and positive semidefinite, and whose sum in the declared order equals the constant times the identity on every block and rounds nowhere; the constant is one where none is named	foundations/def:exact-admissibility
exactly admissible datum	An admissible datum every entry of which satisfies the lattice condition of the number model; the set D_κ.	Definition 10.1
exactly admissible on the declared lattice	A family that is exactly admissible, in the sense of foundations, with the constant the invoking rule names, and whose every entry satisfies the lattice condition of this definition.	Definition 9.1
exact-window depth	The declared integer up to which the weight of a setting uses, block by block, the larger of the analytic bound and a computed block eigenvalue less the eigenvalue slack.	receiver/def:pmcex
factor enclosure	a matrix box of columns together with an exact positive diagonal core, such that the matrix is the product of a matrix L the box contains, the core and the adjoint of L	foundations/def:factor-enclosure
factored loss	The factor t, between the largest declared efficiency and one, by which a normalized station divides every efficiency.	receiver/detector-imperfections/def:normalized-station
flag-state target	The target that keeps the POVM below the cutoff and replaces everything above it by one flag per outcome: the target of the flag-state squasher.	Definition 19.1
Fock space	The direct sum over n of the symmetric n-photon subspaces of a finite-dimensional mode space.	receiver/def:fock
full output	The state on every output register that a protocol produces under one attack strategy, and no marginal of it.	Definition 2.3

Term	Meaning	Defined in
generalized instrument	A finite family of completely positive maps on one input system whose traces add up to the trace; each map is a branch, and its output system may depend on the branch.	Definition 5.2
generated dark-count route	The dark-count channel is the independent product form of the declared per-port probabilities.	receiver/detector-imperfections/def:darkadm
handed datum	The admissible datum a reduction instance is handed, formed exactly from the numbers it was given.	Definition 10.1
handed model	The model map of a reduction instance at its handed datum.	Definition 10.1
handed reference	a reference of a snapped measurement family that its declarer passes, with its centres and radii and unaltered, to every consumer of the snap	foundations/def:evaluated-off-snap
Hermiticity-and-positivity defect	The larger of the spectral norm of X minus its adjoint and the negative part of the smallest eigenvalue of the Hermitian part of X: one number, measured in the spectral norm.	receiver/def:residuals
hypotheses (H)	The finite list of hypotheses a reduction instance makes about the parts of the model its datum does not describe.	Definition 10.1
image	The set of analysed strategies that an exact operational reduction reaches from the physical strategy set.	Definition 2.3
independent product form	The dark-count channel built from per-port dark-count probabilities, each port firing independently.	receiver/detector-imperfections/def:darkadm
indicator sectors	One one-dimensional sector per outcome, on which every element of a target family is one at its own outcome and zero at every other.	Definition 23.1
instance	the party that fixes one protocol's device data, protocol data and run configuration	foundations/def:declared-data

Term	Meaning	Defined in
interface conditions	The conditions (S1) to (S4) a source segment satisfies: descriptor match, index preservation, the source identity, and the slots each link carries.	transmitter/def: source-segment
joint alphabet	The product of the outcome alphabet and the settings, ordered with the outcome varying fastest.	Definition 4.4
joint family	The POVM over the joint alphabet whose element at an outcome and a setting is the class weight times the conditional element of that outcome given that setting.	Definition 4.4
lattice exponent	an integer k with $1 \leq k \leq 53$	foundations/def: binary64
lattice value	at a lattice exponent k, a real number that is an integer multiple of two to the power $-k$ and has modulus at most one; a complex number or a matrix is one when the real and the imaginary part of every entry are	foundations/def: binary64
licence record	The record of the attack-set licence a link carries exactly when its constraint list is non-empty.	Definition 7.1
licensed	Said of a constraint list every entry of which has a real constant as its right-hand side, either untagged in the form its source mode admits or in the party-local form.	Definition 6.5
matrix box	a matrix of Gaussian-rational centres together with non-negative rational radii that bound, entry by entry, the modulus of the difference between a rectangular complex matrix and that centre	foundations/def: factor-enclosure
measured system	The quantum system the measuring parties receive in one round, and no other system; Bob's alone where he alone measures.	Definition 2.1
measurement channel	The channel that measures a POVM on a quantum input and writes the outcome into a classical register.	Definition 1.10

Term	Meaning	Defined in
measuring parties	The parties a declaration names as measuring; their measurements are the factors of the round's measurement channel.	Definition 2.1
model map	The map assigning to every admissible datum a POVM on the instance's system, under its hypotheses (H).	Definition 10.1
module boundary	the passage of declared data from their declarer to a consumer	foundations/def: declared-data
moved	a block of a snap of an operator family read as a measurement that is not kept: the snap is not decided exactly to equal on it the data it is made from	foundations/def: snap
multiclick set	The outcomes in which at least two ports click.	receiver/def: clickalphabet
normalized station	A receiver instrument with the same channel modes and ports whose port Grams are those of the receiver divided by the factored loss t.	receiver/ detector- imperfections/ def:normalized- station
n-round protocol	A channel that in each of n rounds applies one measurement channel to the round's measured system and then processing that touches no measured system, followed by end-of-protocol processing.	Definition 2.1
obligation record	What a theorem-supplied source link carries in place of a decided identity: its kind, the soundness claim, the assumptions the obligation carries, and the family member its descriptors are.	transmitter/def: source-link
outcome alphabet	The finite ordered set of device outcomes that a private setting is joined to.	Definition 4.4
output registers	The tuple of registers on which the full outputs of a protocol presentation live.	Definition 2.3
outside the sector support	Said of the elements of a datum at outcomes outside a data sector's support: exactly zero, and read, rounded or repaired by no step.	Definition 10.1

Term	Meaning	Defined in
party-local form	The form of a constraint entry scoped to one factor of a product measured system: tagged party-local and read as an operator inequality against the marginal actually present.	Definition 6.5
per-setting weight	The high-photon multiclick weight of one setting: its collapse value, its window and tail, or its exact window.	receiver/def:pmc
physical inputs	The input states the physical protocol is analysed over, one set per admissible adversary system.	Definition 3.10
physical strategy set	The strategy set of the presentation an exact operational reduction starts from.	Definition 2.3
post-processed datum	The preserved photon-number blocks of the joint family formed by steps 1 and 2 of the weight route.	Definition 20.3
protocol presentation	A set of attack strategies, a tuple of output registers, and the full output each strategy produces.	Definition 2.3
pull-back	The image of an operator on the squashed system under the adjoint of the squashing channel.	Definition 1.3
pure-loss channel	The channel on the channel system that sends every channel mode through a beamsplitter of transmittance t with vacuum in its other input and discards the reflected light.	receiver/ detector-imperfections/ def:pure-loss
reachable	Said of a squashed state that is the image of some physical input under the n-fold squashing channel.	Definition 3.10
reachable-set statement	The set of reachable squashed states of a link, for its channel witness and the declared family of inputs.	Definition 7.1
receiver composite	The composition of the channel witnesses of a chain's receiver segment.	Definition 7.2
receiver instrument	A declared mode space with channel and probe inputs, disjoint detector ports and a contractive instrument matrix.	receiver/def: instrument
receiver segment	The list of receiver-stage typed links of a reduction chain.	Definition 7.2

Term	Meaning	Defined in
receiver tolerances	The eight declared numbers at which the receiver decides its exact conditions on declared data, bounds its residuals and spends its eigenvalue slack; not the instance's declared tolerance.	receiver/def: tolmap
receiver-stage typed link	A tuple of ten slots relating a source family to a target through a channel witness, with its carried operators, reachable-set statement, transported entries, licence record and status.	Definition 7.1
reduction chain	A pair of a source segment and a receiver segment of typed links.	Definition 7.2
reduction instance	A handed model with a finite datum and hypotheses about the rest, a target built from the datum and scalar values, a squashing relation with a transported set, and certified entries, all established for every admissible datum.	Definition 10.1
reference	a family of Hermitian positive semidefinite block lists complete at the constant on every block, read together with an entrywise enclosure of every block of every element	foundations/def: exact-reference
residual family	on a block with weight q, the family (G − (1−q)F)/q where q is positive, and G itself where q is zero	foundations/def: domination
retained-register sector map	An optional matched pair of sector structures, on Alice's retained system and on the descriptor the certificate is written against.	transmitter/def: source-link
round model	n rounds of a per-round system S, with the attack sequences and attack sets below	foundations/def: round-model
round-homogeneous	one fixed list of objects describes every round, and every round behaves as they describe whatever the other rounds do	foundations/def: round-homogeneity
rounds nowhere	said of an accumulation every one of whose binary64 additions is exact	foundations/def: declared-order

Term	Meaning	Defined in
routed weights	The weights the weight route assigns to the joint outcomes: at a bounded letter, the class weight times that setting's certified weight sharpened by the dark counts, rounded down onto the lattice; zero elsewhere.	Definition 20.3
scalar sector	A sector of a reduction instance's target on which every target element is a scalar value times the identity.	Definition 10.1
scalar value	The number in the unit interval, satisfying the lattice condition, that a scalar sector assigns to an outcome; the values of a sector sum exactly to one.	Definition 10.1
scoped to a factor	Said of a constraint entry whose operator acts on one factor of a product measured system and which carries the party-local tag.	Definition 6.5
sector map	a block kind together with a non-empty, ordered list of block identifiers, each with a dimension	foundations/def: sector-map
sector set	The finite ordered list of a reduction instance's target sectors, data and scalar, each with a declared dimension.	Definition 10.1
security criterion	A family of functions, one per admissible adversary system, of the state of the protocol's output registers together with that system.	Definition 2.2
setting family	A finite set of settings, each with a photon-number block-diagonal POVM whose joint no-click elements have the prefactor form, and an exact flag.	receiver/def: setting-family
settings	The finite ordered set of values of a private per-round setting of the receiver.	Definition 4.4
single-bound case	The weight-preserving target in which only one certified weight is non-zero.	Definition 19.1

Term	Meaning	Defined in
snap	at a lattice exponent k, data of the same shape as the given data, over the same alphabet in the same order, with every entry a lattice value at k, that satisfy exactly the conditions the declaration of the given data states	foundations/def: snap
sound	Said of a family of squashed states, or of a constraint, that contains every reachable state; not the soundness of a transmitter source link.	Definition 3.10
soundness registry	The closed list, emitted with every artefact, of admitted source-link kinds with their soundness claims and carried assumptions.	transmitter/def: source-segment
source composite	The composition of a chain's source links, the identity when the source segment is empty.	Definition 7.2
source family	The system and POVM a receiver-stage link starts from.	Definition 7.1
source link	A link of a reduction chain's source stage that maps one Alice-side ensemble to another with the same labels and weights.	transmitter/def: source-link
source model	A submodule's named definition of the signal ensemble as mathematical objects, possibly infinite dimensional, rather than as matrices.	transmitter/def: source-spec
source segment	The ordered, possibly empty list of source links leading from the virtual ensemble to the specified one.	transmitter/def: source-segment
source specification	The declared description of Alice's side of a protocol: alphabet, retained system with sectors, measurement, signal ensemble and source mode.	transmitter/def: source-spec
source-replacement certificate	An operator on Alice's retained system and the virtual transmitted system whose marginal on her side is her declared marginal and which, measured on her side by her POVM, leaves the weighted virtual members, with retention attested.	transmitter/def: srcert

Term	Meaning	Defined in
squashed protocol	The protocol with the physical system and POVM of each round replaced by the squashed system and the target POVM, all other processing unchanged.	Definition 2.1
squashed sequence	The attack sequence followed, in every round, by the squashing channel on the measured system and the identity on everything else.	Definition 3.2
squashing channel	A channel from the physical system to the squashed system whose adjoint realises a squashing relation.	Definition 1.3
squashing relation	The relation in which the adjoint of a channel takes every target POVM element to the physical one and every carried operator to its prescribed physical operator.	Definition 1.3
stability clause	The requirement that no hypothesis in (H) read an entry of the datum and that parts (I2) to (I4) hold at every admissible datum.	Definition 10.1
stage-ordered	Said of a chain in which every source-stage link precedes every receiver-stage link.	Definition 7.2
strengthening slot	The list of disjoint raw-outcome sets with per-setting high-photon lower-bound weights that the artifact carries.	receiver/def:slot
structural support	of a block of an element: the set of indices whose row or column holds an entry that is not structurally zero	foundations/def:snap
substrate	An ordered field F with $\mathbb{Q} \subseteq F \subseteq \mathbb{R}$, together with its Gaussian extension F + iF, in which every exact decision of this contract is taken.	Definition 9.2
support	The subset of the alphabet on which a data sector's elements may be nonzero; it meets every class.	Definition 10.1
system setting	The setting of this contract: a separable, possibly infinite-dimensional physical system and finite-dimensional, declared target spaces and side registers other than the adversary's.	Definition 1.1

Term	Meaning	Defined in
target family	The target POVM of a reduction instance at a datum, assigned one sector block at a time from the datum and the scalar values.	Definition 10.1
target POVM	The POVM on the squashed system that a squashing relation pulls back to the physical POVM.	Definition 1.3
theorem constructor	A named claim of a submodule that supplies a source link's channel for each member of a declared family of descriptors.	transmitter/def: source-link
total weight	The sum of the certified weights over all outcomes.	Definition 19.1
transmitted system	The system Alice sends into the channel, carrying the specified signal ensemble.	transmitter/def: source-spec
transported constraint entries	Entries of a consumer's constraint list carried across a link, each with the certificate case that discharges it.	transmitter/def: source-link
transported set	The finite set of sector-block-diagonal Hermitian operators on the target, every sector projector among them, that a reduction instance transports with their pull-backs.	Definition 10.1
trusted-source mode	The source mode in which Alice's retained system never reaches the adversary, with a declared marginal and, above dimension one, a certificate.	transmitter/def: source-spec
tuple-chained	Said of a receiver segment in which each link's target-side carried operators are the next link's source-side ones.	Definition 7.2
typed link	A link of a reduction chain, at the source stage or the receiver stage, carrying what is needed to re-check it.	Definition 7.1
uniform weight of an entry	The minimum over the settings of a slot entry's weights; it bounds the entry at every setting.	receiver/def:slot
untrusted-source mode	The source mode with no certificate and no marginal, in which the adversary may have prepared Alice's retained system.	transmitter/def: source-spec

Term	Meaning	Defined in
virtual ensemble	The finite-dimensional ensemble the source segment starts from and the certificate is written against.	transmitter/def: source-segment
virtual transmitted system	The transmitted system of the virtual ensemble, where the source segment starts.	transmitter/def: source-segment
weight route	The three ordered steps that carry a receiver artifact's blocks and certified weights through the dark counts and the private setting into the flat instance.	Definition 20.3
weight-preserving target	The target that keeps the POVM below the cutoff and gives each outcome a flag block carrying its certified weight on every flag and, in addition, one minus the total weight on its own flag: the target of the multi-bound weight-preserving flag-state squasher.	Definition 19.1

Part I
Mathematics

1 Squashing relations and their statistics

A squashing relation replaces the physical measurement on Bob's system by a target measurement on a finite system, through a channel whose adjoint pulls every target element back to the physical one, and a tuple of carried operators travels along with prescribed pull-backs. The relation is equivalent to agreement of all statistics, side system included (Lemma 1.4), and it turns a device fact about a carried operator's pull-back into an operator constraint on every state the squashing channel produces, the fact being the constraint's certificate (Corollary 1.7). The setting comes first, a separable physical system with finite declared targets and side registers (Definition 1.1), and with it the infinite-dimensional bookkeeping behind adjoints and tensor channels (Lemma 1.2); the section ends with the measurement channel, through which the relation becomes a factorisation of channels (Lemma 1.11).

The physical system Bob's receiver acts on has a separable Hilbert space, possibly infinite-dimensional, while every target space and every side register other than the adversary's is finite-dimensional with declared dimensions. The squashing relations and their transport are stated in it.

Bob's physical receiver acts on a system whose Hilbert space may be infinite dimensional; in every optical application it is a Fock space. Every other system of the analysis, apart from the adversary's registers, is finite-dimensional and declared. Separability is no restriction on such a receiver: a Fock space over countably many modes has the countable occupation-number basis. It is used only to type the systems, and for a branch register whose dimension is nowhere declared it is a convention.

Definition 1.1 (The system setting of the reductions)**.** The *system setting* of this contract is the following:

(a) the Hilbert space $\mathcal{H}(B)$ of the system B that Bob's physical receiver acts on is separable and may be infinite-dimensional;

(b) every target space $\mathcal{H}(Q)$ is finite-dimensional, with declared dimensions;

(c) every side register other than the adversary's registers, that is, Alice's retained system and the classical registers, is finite-dimensional, with declared dimensions.

Used by Lem. 1.2, Thm. 8.6, Cor. 10.2, Rem. 11.9, Prop. 21.2, Cor. 21.6 · newly derived · accepted

States separate bounded operators on a separable space; trace-preserving completely positive maps of any output dimension have unital completely positive adjoints; and tensor products of channels into finite targets, or from finite inputs, are channels. Transport arguments rest on this.

Bob's physical receiver acts on a system whose Hilbert space may be infinite dimensional; in every optical application it is a Fock space. Adjoints are therefore set up for a separable input and an output of any dimension; tensor products of channels are set up for separable inputs and finite-dimensional outputs, the case of the squashing channels, and for finite-dimensional inputs and separable outputs, the case of an attack on finite registers and of a source channel on a finite virtual system. The system setting and the finite dimension of the adversary's registers are what make part (ii) available and what make the trace pairing on a side system separate points; the statistics equivalence of Lemma 1.4, Corollaries 1.7 and 3.13 and the licence, Theorem 6.7, use them. Besides its hypotheses and notation, the Setting paragraph carries assertions of the lemma: that the adjoint it names exists, is unique and is completely positive, unital where the map is trace preserving and sub-unital where it is trace non-increasing. Steps 2 to 4 of the proof prove them, beside conclusions (i) to (iii).

Lemma 1.2 (Separation and tensor adjoints in the infinite-dimensional case)**.** *Setting.* B is the system Bob's physical receiver acts on and Q the system it is squashed to. $\mathcal{H}(B)$ is separable and may be infinite-dimensional, while $\mathcal{H}(Q)$ and every side register are finite-dimensional (Definition 1.1, and, for the adversary's registers, the round model of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model), every register of which is finite-dimensional). $\mathcal{T}(B)$ denotes the trace-class operators on $\mathcal{H}(B)$; a *channel* $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ is a completely positive trace-preserving map, and its *adjoint* $\Lambda^\dagger\colon \mathcal{L}(Q) \to \mathcal{L}(B)$ is the unital completely positive map fixed by

$$\operatorname{tr}\left[X\,\Lambda(\rho)\right] = \operatorname{tr}\left[\Lambda^\dagger(X)\,\rho\right] \qquad \text{for all } X \in \mathcal{L}(Q),\ \rho \in \mathcal{T}(B). \tag{1.1}$$

More generally, if $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(B')$, for a Hilbert space $\mathcal{H}(B')$ of any dimension, is only completely positive and trace non-increasing ($\operatorname{tr}\Lambda(\rho) \le \operatorname{tr}\rho$ for $\rho \succeq 0$), then (1.1) for all $X \in \mathcal{L}(B')$ and $\rho \in \mathcal{T}(B)$ fixes a completely positive map $\Lambda^\dagger\colon \mathcal{L}(B') \to \mathcal{L}(B)$, again called its *adjoint*, and $\Lambda^\dagger$ is subunital: $\mathbb{1}_B - \Lambda^\dagger(\mathbb{1}_{B'}) \succeq 0$. Where such a Λ is moreover trace preserving, $\Lambda^\dagger$ is unital, $\Lambda^\dagger(\mathbb{1}_{B'}) = \mathbb{1}_B$, whatever the dimension of $\mathcal{H}(B')$.

Let $\mathcal{H}(B_1), \dots, \mathcal{H}(B_n)$ be separable and $\mathcal{H}(Q_1), \dots, \mathcal{H}(Q_n)$ finite-dimensional.

(i) *(Separation.)* If $X \in \mathcal{L}(B_1)$ satisfies $\operatorname{tr}[X\rho] = 0$ for every $\rho \in S_=(B_1)$, then $X = 0$.

(ii) *(Tensor channels and their adjoints.)* If $\Lambda_i\colon \mathcal{T}(B_i) \to \mathcal{T}(Q_i)$ are channels, then exactly one trace-norm continuous linear map $\Lambda_1 \otimes$

$\cdots \otimes \Lambda_n$ from $\mathcal{T}(B_1 \cdots B_n)$, the trace class of the completed tensor product of the $\mathcal{H}(B_i)$, to $\mathcal{T}(Q_1 \cdots Q_n)$ takes every $\rho_1 \otimes \cdots \otimes \rho_n$ with $\rho_i \in \mathcal{T}(B_i)$ to $\Lambda_1(\rho_1) \otimes \cdots \otimes \Lambda_n(\rho_n)$; it is a channel, and $(\Lambda_1 \otimes \cdots \otimes \Lambda_n)^\dagger = \Lambda_1^\dagger \otimes \cdots \otimes \Lambda_n^\dagger$ on all of $\mathcal{L}(Q_1 \cdots Q_n)$. The same holds with any of the Λ_i replaced by id_{R_i} for a finite-dimensional R_i.

(iii) *(Tensor channels from finite inputs.)* Let $\mathcal{H}(R_1), \ldots, \mathcal{H}(R_n)$ be finite-dimensional, $\mathcal{H}(B_1'), \ldots, \mathcal{H}(B_n')$ separable, of any dimension, and $\Phi_i \colon \mathcal{L}(R_i) \to \mathcal{T}(B_i')$ completely positive and trace preserving. Then exactly one linear map $\Phi_1 \otimes \cdots \otimes \Phi_n$ from $\mathcal{L}(R_1 \cdots R_n)$ to $\mathcal{T}(B_1' \cdots B_n')$, the trace class of the completed tensor product of the $\mathcal{H}(B_i')$, takes every $X_1 \otimes \cdots \otimes X_n$ with $X_i \in \mathcal{L}(R_i)$ to $\Phi_1(X_1) \otimes \cdots \otimes \Phi_n(X_n)$, and it is completely positive and trace preserving. The identity map id_R of a finite-dimensional R is such a Φ_i.

Uses Def. 1.1 · *Used by* Def. 1.3, Lem. 1.4, Lem. 1.11, Prop. 3.4, Cor. 3.13, Prop. 4.1, Prop. 4.8, Lem. 5.3, Thm. 5.13, Lem. 6.6, Thm. 6.7, Rem. 6.10, Lem. 8.1, Lem. 8.3, Thm. 10.4, Lem. 10.6, Thm. 11.5, Lem. 12.3, Prop. 19.4 · newly derived · accepted

Proof. Idea. Separation is tested on rank-one operators. The adjoint is built on rank-one operators by the Riesz representation theorem and extended to the whole trace class because finite-rank operators are dense in trace norm. For (ii) the output is finite-dimensional, so the tensor map is defined from the tensor product of the adjoints on a basis of matrix units; it acts on products as required, and products are dense. For (iii) the input is finite-dimensional, so the tensor map is defined on a basis of matrix units of the input, and its complete positivity is read off its Choi operator, a tensor product of positive operators.

Step 1 (separation, part (i)). Every $\tau \in \mathcal{T}(B_1)$ is a complex linear combination of at most four positive trace-class operators (split τ into Hermitian and anti-Hermitian parts, then each into positive and negative parts), and every non-zero positive trace-class operator is a positive multiple of a state. Hence $\mathrm{tr}[X\tau] = 0$ for every $\tau \in \mathcal{T}(B_1)$. If $X \neq 0$ there are unit vectors $|\psi\rangle, |\varphi\rangle$ with $\langle\varphi| X |\psi\rangle \neq 0$; take $\tau = |\psi\rangle\langle\varphi|$, which is trace class of rank one, and $\mathrm{tr}[X\tau] = \langle\varphi| X |\psi\rangle \neq 0$, a contradiction.

Step 2 (the adjoint exists and is unique). Let $\Lambda \colon \mathcal{T}(B) \to \mathcal{T}(B')$ be completely positive and trace non-increasing; a channel is the case $B' = Q$ with the trace preserved. It is bounded in trace norm. Write $\tau \in \mathcal{T}(B)$ as $\tau = H_1 + iH_2$ with the Hermitian $H_1 = \frac{1}{2}(\tau + \tau^\dagger)$ and $H_2 = \frac{1}{2i}(\tau - \tau^\dagger)$, each of trace norm at most $\|\tau\|_1$ since $\|\tau^\dagger\|_1 = \|\tau\|_1$, and each H_l as $H_l^+ - H_l^-$ with positive trace-class $H_l^\pm$ and $\|H_l^+\|_1 + \|H_l^-\|_1 = \|H_l\|_1$ (its spectral decomposition). A positive operator has trace norm equal to its trace, and Λ is positive (as every completely positive map is) and trace non-increasing,

so

$$\|\Lambda(\tau)\|_1 \le \sum_{l,\pm} \operatorname{tr} \Lambda(H_l^{\pm}) \le \sum_{l,\pm} \operatorname{tr} H_l^{\pm} = \|H_1\|_1 + \|H_2\|_1 \le 2\|\tau\|_1.$$

Now fix $X \in \mathcal{L}(B')$. For $\psi, \chi \in \mathcal{H}(B)$ the rank-one operator $|\psi\rangle\langle\chi|$ has trace norm $\|\psi\|\,\|\chi\|$, so $s(\chi,\psi) \coloneqq \operatorname{tr}[X\,\Lambda(|\psi\rangle\langle\chi|)]$ is linear in ψ, conjugate-linear in χ, and bounded: $|s(\chi,\psi)| \le \|X\|\,\|\Lambda(|\psi\rangle\langle\chi|)\|_1 \le 2\|X\|\,\|\psi\|\,\|\chi\|$. For fixed ψ the map $\chi \mapsto \overline{s(\chi,\psi)}$ is a bounded linear functional on $\mathcal{H}(B)$, so the Riesz representation theorem gives exactly one vector $\eta_\psi \in \mathcal{H}(B)$ with $s(\chi,\psi) = \langle\chi|\eta_\psi\rangle$ for every χ. Set $Y\psi \coloneqq \eta_\psi$: uniqueness makes Y linear, and $\chi = \eta_\psi$ gives $\|Y\psi\| \le 2\|X\|\,\|\psi\|$, so $Y \in \mathcal{L}(B)$. Then $\operatorname{tr}[Y\,|\psi\rangle\langle\chi|] = \langle\chi|\,Y\,|\psi\rangle = \operatorname{tr}[X\,\Lambda(|\psi\rangle\langle\chi|)]$, which is (1.1) for Y at every rank-one ρ, hence by linearity at every finite-rank ρ. Finite-rank operators are trace-norm dense in $\mathcal{T}(B)$ (truncate the singular value decomposition $\rho = \sum_j s_j\,|u_j\rangle\langle w_j|$, whose tail beyond J terms has trace norm at most $\sum_{j>J} s_j \to 0$ since $\sum_j s_j = \|\rho\|_1$), and both sides of (1.1) are trace-norm continuous in ρ, bounded by $2\|X\|\,\|\rho\|_1$ and $\|Y\|\,\|\rho\|_1$; so (1.1) holds for Y at every $\rho \in \mathcal{T}(B)$. Two operators satisfying (1.1) for the same X have a difference that (i), read on $\mathcal{H}(B)$, shows is zero, so $\Lambda^\dagger(X) \coloneqq Y$ defines the one map on $\mathcal{L}(B')$ that satisfies (1.1). It is linear, since $c\Lambda^\dagger(X) + c'\Lambda^\dagger(X')$ satisfies (1.1) for $cX + c'X'$.

Step 3 (the adjoint is completely positive). Let $k \ge 1$, let $\{|a\rangle\}_{a=1}^k$ be the standard basis of $\mathbb{C}^k$, let $Z = \sum_{a,b=1}^k |a\rangle\langle b| \otimes Z_{ab} \succeq 0$ on $\mathbb{C}^k \otimes \mathcal{H}(B')$, so that $(\mathrm{id}_k \otimes \Lambda^\dagger)(Z) = \sum_{a,b} |a\rangle\langle b| \otimes \Lambda^\dagger(Z_{ab})$, and let $\xi = \sum_{a=1}^k |a\rangle \otimes \xi_a \in \mathbb{C}^k \otimes \mathcal{H}(B)$, so that $|\xi\rangle\langle\xi| = \sum_{a,b} |a\rangle\langle b| \otimes |\xi_a\rangle\langle\xi_b|$. By (1.1) at the rank-one operators $|\xi_b\rangle\langle\xi_a|$,

$$\langle\xi|\,(\mathrm{id}_k \otimes \Lambda^\dagger)(Z)\,|\xi\rangle = \sum_{a,b} \langle\xi_a|\,\Lambda^\dagger(Z_{ab})\,|\xi_b\rangle = \sum_{a,b} \operatorname{tr}\left[Z_{ab}\,\Lambda(|\xi_b\rangle\langle\xi_a|)\right] = \operatorname{tr}\left[Z\,(\mathrm{id}_k \otimes \Lambda)(|\xi\rangle\langle\xi|)\right],$$

which is non-negative: $T \coloneqq (\mathrm{id}_k \otimes \Lambda)(|\xi\rangle\langle\xi|)$ is positive by complete positivity of Λ, and for its spectral decomposition $T = \sum_m t_m\,|e_m\rangle\langle e_m|$, with $t_m \ge 0$ and $\{e_m\}$ orthonormal, $\operatorname{tr}[ZT] = \sum_m t_m\,\langle e_m|\,Z\,|e_m\rangle \ge 0$, the trace being taken in an orthonormal basis that extends $\{e_m\}$ by vectors of the kernel of T. As ξ is arbitrary, $(\mathrm{id}_k \otimes \Lambda^\dagger)(Z) \succeq 0$.

Step 4 (sub-unitality, and unitality where the trace is preserved). Sub-unitality is trace non-increase read through (1.1): for every $\psi \in \mathcal{H}(B)$, $\langle\psi|\,\Lambda^\dagger(\mathbb{1}_{B'})\,|\psi\rangle = \operatorname{tr}[\Lambda(|\psi\rangle\langle\psi|)] \le \operatorname{tr}|\psi\rangle\langle\psi| = \|\psi\|^2$, so $\mathbb{1}_B - \Lambda^\dagger(\mathbb{1}_{B'}) \succeq 0$. Where Λ is trace preserving, unitality is trace preservation read the same way: we have $\operatorname{tr}\left[\Lambda^\dagger(\mathbb{1}_{B'})\rho\right] = \operatorname{tr}[\Lambda(\rho)] = \operatorname{tr}\rho$ for every $\rho \in \mathcal{T}(B)$, so $\Lambda^\dagger(\mathbb{1}_{B'}) = \mathbb{1}_B$ by (i), read on $\mathcal{H}(B)$. Nothing in this step reads the dimension of $\mathcal{H}(B')$; a channel is the case $B' = Q$.

Step 5 (the tensor map, part (ii)). Fix orthonormal bases of the finite-dimensional $\mathcal{H}(Q_i)$. The tensor products of their matrix units form a basis of $\mathcal{L}(Q_1 \cdots Q_n)$, so there is exactly one linear map $\Psi\colon \mathcal{L}(Q_1 \cdots Q_n) \to$

$\mathcal{L}(B_1 \cdots B_n)$ with $\Psi(X_1 \otimes \cdots \otimes X_n) = \Lambda_1^\dagger(X_1) \otimes \cdots \otimes \Lambda_n^\dagger(X_n)$ for every product operator, both sides being multilinear in $(X_1, \ldots, X_n)$; it is the map the statement writes $\Lambda_1^\dagger \otimes \cdots \otimes \Lambda_n^\dagger$. For an orthonormal basis $\{e_r\}$ of $\mathcal{H}(Q_1 \cdots Q_n)$ and $\rho \in \mathcal{T}(B_1 \cdots B_n)$ put

$$\Lambda(\rho) \coloneqq \sum_{r,s} \operatorname{tr}\left[\Psi(|e_r\rangle\langle e_s|)\,\rho\right]\ |e_s\rangle\langle e_r|\,.$$

Then $\operatorname{tr}[X\Lambda(\rho)] = \operatorname{tr}[\Psi(X)\rho]$ for all $X \in \mathcal{L}(Q_1 \cdots Q_n)$: the identity holds at every $X = |e_r\rangle\langle e_s|$, and both sides are linear in X. Λ is linear, and bounded in trace norm, $\|\Lambda(\rho)\|_1 \le \sum_{r,s} \|\Psi(|e_r\rangle\langle e_s|)\|\,\|\rho\|_1$.

Step 6 (action on products, and uniqueness). For $\rho = \rho_1 \otimes \cdots \otimes \rho_n$ and $X = X_1 \otimes \cdots \otimes X_n$ the traces below factor, and the pairing of Step 5 with (1.1) for each Λ_i gives

$$\begin{aligned}\operatorname{tr}\left[X\,\Lambda(\rho)\right] &= \operatorname{tr}\left[\Psi(X)\,\rho\right] = \prod_i \operatorname{tr}\left[\Lambda_i^\dagger(X_i)\,\rho_i\right] \\ &= \prod_i \operatorname{tr}\left[X_i\,\Lambda_i(\rho_i)\right] = \operatorname{tr}\left[X\,\big(\Lambda_1(\rho_1) \otimes \cdots \otimes \Lambda_n(\rho_n)\big)\right].\end{aligned}$$

Product operators span $\mathcal{L}(Q_1 \cdots Q_n)$, and an operator σ on this finite-dimensional space with $\operatorname{tr}[X\sigma] = 0$ for every X is zero (take $X = \sigma^\dagger$), so $\Lambda(\rho_1 \otimes \cdots \otimes \rho_n) = \Lambda_1(\rho_1) \otimes \cdots \otimes \Lambda_n(\rho_n)$. Product trace-class operators span a trace-norm dense subspace of $\mathcal{T}(B_1 \cdots B_n)$: finite-rank operators are dense there (Step 2); a rank-one $|\psi\rangle\langle\chi|$ is the trace-norm limit of $|\psi_m\rangle\langle\chi_m|$ for finite sums of product vectors $\psi_m \to \psi$ and $\chi_m \to \chi$, which are dense in the completed tensor product, since $\|\,|\psi\rangle\langle\chi| - |\psi_m\rangle\langle\chi_m|\,\|_1 \le \|\psi - \psi_m\|\,\|\chi\| + \|\psi_m\|\,\|\chi - \chi_m\|$; and each $|\psi_m\rangle\langle\chi_m|$ is a finite sum of products $|\alpha_1\rangle\langle\beta_1| \otimes \cdots \otimes |\alpha_n\rangle\langle\beta_n|$. So Λ is the only trace-norm continuous linear map that acts on products as the statement says, and $\Lambda_1 \otimes \cdots \otimes \Lambda_n \coloneqq \Lambda$.

Step 7 (the tensor map is a channel). Every $\Lambda_i^\dagger$ is unital, so $\Psi(\mathbb{1}) = \mathbb{1}$ and $\operatorname{tr}\Lambda(\rho) = \operatorname{tr}[\Psi(\mathbb{1})\rho] = \operatorname{tr}\rho$. For complete positivity let $k \ge 1$, let $\omega = \sum_{a,b=1}^k |a\rangle\langle b| \otimes \omega_{ab} \succeq 0$ be trace class on $\mathbb{C}^k \otimes \mathcal{H}(B_1 \cdots B_n)$, and let $z = \sum_{a=1}^k |a\rangle \otimes z_a \in \mathbb{C}^k \otimes \mathcal{H}(Q_1 \cdots Q_n)$. With $P \coloneqq (\mathrm{id}_k \otimes \Psi)(|z\rangle\langle z|)$ the pairing of Step 5 gives

$$\langle z|\,(\mathrm{id}_k \otimes \Lambda)(\omega)\,|z\rangle = \sum_{a,b} \langle z_a|\,\Lambda(\omega_{ab})\,|z_b\rangle = \sum_{a,b} \operatorname{tr}\left[\Psi(|z_b\rangle\langle z_a|)\,\omega_{ab}\right] = \operatorname{tr}\left[P\,\omega\right],$$

and $\operatorname{tr}[P\omega] \ge 0$ once $P \succeq 0$, by the spectral decomposition of ω as in Step 3. P is bounded, so $P \succeq 0$ follows from $\langle\xi|\,P\,|\xi\rangle \ge 0$ for every ξ in the dense subspace of finite sums of product vectors $|a\rangle \otimes \beta_1 \otimes \cdots \otimes \beta_n$. Such a ξ lies in $\mathbb{C}^k \otimes S_1 \otimes \cdots \otimes S_n$ for finite-dimensional subspaces $S_i \subseteq \mathcal{H}(B_i)$. Let J_i be the inclusion of S_i into $\mathcal{H}(B_i)$ and $J \coloneqq \mathbb{1} \otimes J_1 \otimes \cdots \otimes J_n$, with $\mathbb{1}$ on $\mathbb{C}^k$. The map $\Psi_i(X) \coloneqq J_i^\dagger \Lambda_i^\dagger(X) J_i$ from $\mathcal{L}(Q_i)$ to $\mathcal{L}(S_i)$ is completely

positive, since $(\mathrm{id}_m \otimes \Psi_i)(Z) = (\mathbb{1} \otimes J_i)^\dagger (\mathrm{id}_m \otimes \Lambda_i^\dagger)(Z)(\mathbb{1} \otimes J_i)$ for every m, and expanding $|z\rangle\langle z|$ in product operators gives $J^\dagger P J = (\mathrm{id}_k \otimes \Psi_1 \otimes \cdots \otimes \Psi_n)(|z\rangle\langle z|)$. Between these finite-dimensional spaces, $\mathrm{id}_k \otimes \Psi_1 \otimes \cdots \otimes \Psi_n$ is the composite, over $i = 1, \ldots, n$, of the maps that apply Ψ_i to the i-th of the last n tensor factors and the identity to the others; after a unitary reordering of the factors each of these is $\mathrm{id}_m \otimes \Psi_i$, for m the finite dimension of the other factors, and so positive. Hence $\langle \xi | P | \xi \rangle = \langle \xi | J^\dagger P J | \xi \rangle \geq 0$, with ξ read as a vector of $\mathbb{C}^k \otimes S_1 \otimes \cdots \otimes S_n$ on the right.

Step 8 (its adjoint). By the pairing of Step 5, Ψ satisfies (1.1) for the channel $\Lambda_1 \otimes \cdots \otimes \Lambda_n$, whose input space is separable and whose output space is finite-dimensional, so by Step 2 it is the one map that does, the adjoint, and it is unital and completely positive (Steps 3 and 4). The id_{R_i} case is the same computation with $\Lambda_i = \mathrm{id}$.

Step 9 (tensor products of trace-class operators). Let $T_i \in \mathcal{T}(B_i')$ have the singular value decompositions $T_i = \sum_k s_{ik} |u_{ik}\rangle\langle w_{ik}|$, with $s_{ik} \geq 0$, $\sum_k s_{ik} = \|T_i\|_1$ and orthonormal families $\{u_{ik}\}_k$, $\{w_{ik}\}_k$. The series $\sum_{k_1,\ldots,k_n} \left(\prod_i s_{ik_i}\right) |u_{1k_1} \otimes \cdots \otimes u_{nk_n}\rangle\langle w_{1k_1}$ of rank-one operators on the completed tensor product, each of trace norm $\prod_i s_{ik_i}$, converges absolutely in trace norm, its sum of norms being $\prod_i \|T_i\|_1$; its sum is the operator $T_1 \otimes \cdots \otimes T_n$, which is therefore trace class, with $\mathrm{tr}(T_1 \otimes \cdots \otimes T_n) = \sum_{k_1,\ldots,k_n} \prod_i s_{ik_i} \langle w_{ik_i} | u_{ik_i} \rangle = \prod_i \mathrm{tr}\, T_i$. Where every $T_i \succeq 0$ its spectral decomposition is such an expansion with $u_{ik} = w_{ik}$, so the partial sums are positive and so is their trace-norm limit.

Step 10 (the tensor map from finite inputs, part (iii)). Fix orthonormal bases of the $\mathcal{H}(R_i)$ and write $E_{ab}^{(i)}$ for the matrix units. Their tensor products $E_{\mathbf{ab}} \coloneqq E_{a_1 b_1}^{(1)} \otimes \cdots \otimes E_{a_n b_n}^{(n)}$ form a basis of $\mathcal{L}(R_1 \cdots R_n)$, so exactly one linear map Φ takes each $E_{\mathbf{ab}}$ to $\Phi_1(E_{a_1 b_1}^{(1)}) \otimes \cdots \otimes \Phi_n(E_{a_n b_n}^{(n)})$, a trace-class operator by Step 9; both sides of the product rule being multilinear in $(X_1, \ldots, X_n)$, Φ takes every product $X_1 \otimes \cdots \otimes X_n$ to $\Phi_1(X_1) \otimes \cdots \otimes \Phi_n(X_n)$, and a linear map is fixed by its values on a basis, which gives uniqueness. It is trace preserving: by Step 9 and trace preservation of each Φ_i, $\mathrm{tr}\, \Phi(E_{\mathbf{ab}}) = \prod_i \mathrm{tr}\, E_{a_i b_i}^{(i)} = \mathrm{tr}\, E_{\mathbf{ab}}$, and both sides are linear. For complete positivity, let $\Omega_i \coloneqq \sum_a |a\rangle \otimes |a\rangle$ on two copies of $\mathcal{H}(R_i)$; complete positivity of Φ_i makes $J_i \coloneqq \sum_{a,b} E_{ab}^{(i)} \otimes \Phi_i(E_{ab}^{(i)}) = (\mathrm{id}_{R_i} \otimes \Phi_i)(|\Omega_i\rangle\langle\Omega_i|)$ positive, and after a unitary reordering of the factors $J \coloneqq \sum_{\mathbf{a},\mathbf{b}} E_{\mathbf{ab}} \otimes \Phi(E_{\mathbf{ab}})$ is $J_1 \otimes \cdots \otimes J_n$, positive by Step 9. Let $k \geq 1$ and let $x = \sum_{\mathbf{a}} |y_{\mathbf{a}}\rangle \otimes |\mathbf{a}\rangle$ be a vector of $\mathbb{C}^k \otimes \mathcal{H}(R_1 \cdots R_n)$, with $y_{\mathbf{a}} \in \mathbb{C}^k$, and put $V \coloneqq \sum_{\mathbf{a}} |y_{\mathbf{a}}\rangle\langle \mathbf{a}|$. Then

$$(\mathrm{id}_k \otimes \Phi)(|x\rangle\langle x|) = \sum_{\mathbf{a},\mathbf{b}} |y_{\mathbf{a}}\rangle\langle y_{\mathbf{b}}| \otimes \Phi(E_{\mathbf{ab}}) = (V \otimes \mathbb{1})\, J\, (V \otimes \mathbb{1})^\dagger \succeq 0.$$

Every positive operator on the finite-dimensional $\mathbb{C}^k \otimes \mathcal{H}(R_1 \cdots R_n)$ is a finite sum of such $|x\rangle\langle x|$, so $\mathrm{id}_k \otimes \Phi$ is positive for every k, and Φ is completely positive. The identity map id_R is linear, completely positive and trace preserving from $\mathcal{L}(R)$ to $\mathcal{T}(R) = \mathcal{L}(R)$. □

A squashing relation asks that the adjoint of a channel pull every target POVM element back to the physical one, and every carried operator back to its prescribed physical partner. The carried operators let one reduction move device facts into the squashed model as well as outcome statistics.

Squashing replaces one *model* of the receiver by another. A model is what the analysis runs on: the system Bob receives and the POVM he measures on it. The physical model is (B, M), the one the security statement is wanted for; the squashed model (Q, F) lives on a system a proof technique can handle. The squashed model must reproduce the physical model's outcome statistics *exactly*, since a protocol reads Bob's device through nothing else (Definition 2.1), and it must be at least as generous to the adversary. Handing Λ to the adversary accomplishes that: the squashing channel is counted as part of her attack rather than of Bob's device, so every attack on the physical model, followed by Λ, is an attack on the squashed model. Both verbs the definition introduces, squashing and transporting, run from the physical side to the target side, in the direction of the symbol $\preceq$ and of Λ; the identities that define them are written with the adjoint $\Lambda^\dagger$, which runs the other way.

Definition 1.3 (Squashing relation)**.** Let $M = \{M_v\}_{v\in\mathcal{V}}$ be a POVM on B and $F = \{F_v\}_{v\in\mathcal{V}}$ a POVM on Q over the same finite alphabet $\mathcal{V}$, and let $\hat{O} = (\hat{O}_k)_{k=1}^m$ and $O = (O_k)_{k=1}^m$ be tuples of Hermitian operators $\hat{O}_k \in \mathcal{L}(Q)$, $O_k \in \mathcal{L}(B)$. For a channel $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$, with adjoint $\Lambda^\dagger$ as in Lemma 1.2, write

$$(M;O) \;\preceq_\Lambda\; (F;\hat{O}) \quad :\Longleftrightarrow \quad \Lambda^\dagger(F_v) = M_v \;\forall v \in \mathcal{V} \text{ and } \Lambda^\dagger(\hat{O}_k) = O_k \;\forall k \in \{1,\dots,m\}, \tag{1.2}$$

and call (1.2) a *squashing relation*: M *squashes to* the *target POVM* F via the *squashing channel* Λ, which *transports* O to $\hat{O}$. The operators $\hat{O}_k$ are the relation's *carried operators*. The empty case $m = 0$ is written $M \preceq_\Lambda F$, and $\preceq$ replaces $\preceq_\Lambda$ when only the existence of a squashing channel is asserted. We call $\Lambda^\dagger(\hat{O}_k)$ the *pull-back* of $\hat{O}_k$.

Uses Lem. 1.2 · *Used by* Lem. 1.4, Rem. 1.5, Rem. 1.6, Rem. 3.15, Prop. 4.1, Lem. 4.3, Prop. 4.6, Prop. 4.8, Lem. 4.9, Lem. 5.1, Prop. 6.1, Def. 10.1, Lem. 10.6, Cor. 12.1, Lem. 12.3, Def. 23.1, Thm. 23.2 · newly derived · accepted

A squashing relation holds exactly when the physical and the squashed model give the same statistics for every POVM element and carried operator, correlations with a finite side system included. Two channels realising one relation therefore have adjoints that agree on the span of those operators.

The two lists in (1.2) contain the same kind of object, an operator on Q whose pull-back is prescribed, and differ in role, not in type: the F_v are what Bob measures and must form a POVM, while the $\hat{O}_k$ are never measured and serve only to carry statements about the physical device into the squashed

model.

Lemma 1.4 (Squashing as statistics equivalence)**.** Let $M, F, O, \hat{O}$ be as in Definition 1.3 and let $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel. The following are equivalent.

(i) $(M;O) \preceq_\Lambda (F;\hat{O})$.

(ii) For every $\rho \in S_=(B)$, every v and every k, $\mathrm{tr}[M_v\rho] = \mathrm{tr}[F_v\Lambda(\rho)]$ and $\mathrm{tr}[O_k\rho] = \mathrm{tr}\Big[\hat{O}_k\Lambda(\rho)\Big]$.

(iii) For every finite-dimensional side system R, every $\rho \in S_=(BR)$, every $X \in \mathcal{L}(R)$, every v and every k, writing $\sigma \coloneqq (\Lambda \otimes \mathrm{id}_R)(\rho)$,

$$\mathrm{tr}\left[(M_v \otimes X)\rho\right] = \mathrm{tr}\left[(F_v \otimes X)\sigma\right] \qquad \text{and} \qquad \mathrm{tr}\left[(O_k \otimes X)\rho\right] = \mathrm{tr}\left[(\hat{O}_k \otimes X)\sigma\right].$$

Moreover, two channels Λ, Λ' that both realize the relation satisfy $\Lambda^\dagger = \Lambda'^\dagger$ on $\mathrm{span}\left(\{F_v\}_{v\in\mathcal{V}} \cup \{\hat{O}_k\}_{k=1}^m\right)$.

Uses Lem. 1.2, Def. 1.3 · *Used by* Cor. 1.7, Lem. 1.11, Rem. 3.9, Rem. 3.14, Rem. 3.15 · newly derived · accepted

Proof. (i)⇒(iii). By the tensor-adjoint identity of Lemma 1.2 (ii), $(\Lambda \otimes \mathrm{id}_R)^\dagger(Y \otimes X) = \Lambda^\dagger(Y) \otimes X$. Taking $Y = F_v$ and $Y = \hat{O}_k$ and inserting (1.2) turns the two right-hand sides of (iii) into the two left-hand sides, by (1.1) for $\Lambda \otimes \mathrm{id}_R$.

(iii)⇒(ii). Take $\mathcal{H}(R) = \mathbb{C}$ and $X = 1$.

(ii)⇒(i). By (1.1), the first identity in (ii) reads $\mathrm{tr}\left[\left(\Lambda^\dagger(F_v) - M_v\right)\rho\right] = 0$ for every $\rho \in S_=(B)$, so $\Lambda^\dagger(F_v) = M_v$ by the separation property, Lemma 1.2 (i). The same argument applied to the second identity gives $\Lambda^\dagger(\hat{O}_k) = O_k$.

For the last claim, $\Lambda^\dagger$ and $\Lambda'^\dagger$ take the same prescribed values on the operators F_v and $\hat{O}_k$, hence agree on their span by linearity. □

The physical POVM stands on the left of the squashing symbol, as in the squashing-model literature, while the measurement-theory literature writes the same relation the other way round. The direction is to be checked before a statement is imported from there.

The side system R in Lemma 1.4 (iii) is not a technicality: in a security proof it holds Alice's retained register and Eve's side information, both of which are present whenever a squashed constraint is evaluated.

Remark 1.5 (Direction of the squashing symbol in the literature). In Definition 1.3 the *physical* POVM stands on the left and the target on the right, in the direction of $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ and of the QKD usage in which Λ maps physical states to squashed states. The identity (1.2) is the one the squashing-model literature writes: it is [1, Eqs. (1)–(2)], which states the statistical equivalence $\mathrm{tr}[\rho_{\mathrm{in}}F_M(i)] = \mathrm{tr}[\Lambda(\rho_{\mathrm{in}})F_Q(i)]$ and observes that it holds for all inputs *if and only if* $F_M(i) = \Lambda^\dagger(F_Q(i))$, with $\Lambda^\dagger$ completely positive and unital but not necessarily trace preserving; and it is [8, Eq. (146)] in measurement-channel form. The

measurement-theory literature writes the same relation the *other way round*: [2, Def. 6.1] defines $P \succ Q$, "P is cleaner than Q", to mean $Q = \mathcal{E}(P)$ for a channel $\mathcal{E}$. The paper's channels acting on effects are its identity-preserving completely positive maps, so $\mathcal{E}$ is what this contract writes $\Lambda^\dagger$ for a trace-preserving Λ, and the left-hand argument P is our *target*. The direction is worth checking before importing a statement from there; this contract imports none.

Uses Def. 1.3 · newly derived · accepted

The pull-back identity of the POVM alone does not confine a squashed analysis to the image of the physical inputs. Operators with a prescribed pull-back carry device facts across, which is why the squashing relation carries a tuple of them from the start.

Remark 1.6 (Carried operators and the image of the squashing channel). The relation of Definition 1.3 carries more than the POVM. Once Λ is handed to the adversary, that is, counted as part of her attack, nothing in the identity $\Lambda^\dagger(F_v) = M_v$ confines the squashed analysis to states a physical input can produce, and optimizing over all of $S_=(Q)$ instead of over the image is where a squashed analysis loses. What recovers the loss is that the identity keeps its meaning for operators that are *not* POVM elements: if $\Lambda^\dagger(\hat{O}) = O$, the expectation of $\hat{O}$ on a squashed state equals that of O on its physical preimage, so a device fact about O transfers to a statement about $\hat{O}$. The relation therefore carries a designated tuple of such operators, its carried operators, from the start.

Uses Def. 1.3 · newly derived · accepted

A carried operator's partial expectation on a squashed state equals its pull-back's on the physical input, as an operator on the side system. An identity, positivity or spectral bound on the pull-back therefore becomes an operator constraint on every state the squashing channel produces.

The following corollary is the recipe every constraint of this contract is built by: pick an operator $\hat{O}$ on Q, compute its pull-back, and read off what the physical device already guarantees about it.

Corollary 1.7 (Pull-back of operator bounds)**.** Let $(M; O) \preceq_\Lambda (F; \hat{O})$, let R be a finite-dimensional side system, let $\rho \in S_=(BR)$ and let $\sigma = (\Lambda \otimes \mathrm{id}_R)(\rho)$. Then for every k,

$$\mathrm{tr}_Q\left[(\hat{O}_k \otimes \mathbb{1}_R)\,\sigma\right] = \mathrm{tr}_B\left[(O_k \otimes \mathbb{1}_R)\,\rho\right] \tag{1.3}$$

as operators on R. Consequently, on every such σ:

(i) *Exact values.* If $O_k = c\,\mathbb{1}_B$ then $\mathrm{tr}_Q[(\hat{O}_k \otimes \mathbb{1}_R)\sigma] = c\,\sigma_R$; in particular $\mathrm{tr}\left[\hat{O}_k \sigma\right] = c$, and at $c = 0$, $\mathrm{tr}\left[(\hat{O}_k \otimes X)\sigma\right] = 0$ for every $X \in \mathcal{L}(R)$.

(ii) *Positivity.* If $O_k \succeq 0$ then $\mathrm{tr}_Q[(\hat{O}_k \otimes \mathbb{1}_R)\sigma] \succeq 0$; in particular $\mathrm{tr}\left[\hat{O}_k \sigma\right] \geq 0$.

(iii) *Spectral bounds.* If $a\,\mathbb{1}_B \preceq O_k \preceq b\,\mathbb{1}_B$ then $a\,\sigma_R \preceq \mathrm{tr}_Q[(\hat{O}_k \otimes \mathbb{1}_R)\sigma] \preceq b\,\sigma_R$; in particular $a \le \mathrm{tr}\left[\hat{O}_k\sigma\right] \le b$.

Uses Lem. 1.4 · *Used by* Def. 1.8, Cor. 3.13, Rem. 3.14, Lem. 5.3, Thm. 5.13, Thm. 6.7, Def. 7.1, Def. 10.1, Thm. 10.4, Cor. 11.3, Req. 16.4, Prop. 19.4, Rem. 19.9, Cor. 24.1 · newly derived · accepted

Proof. Test (1.3) against an arbitrary $X \in \mathcal{L}(R)$. By the defining property of the partial trace and the side-system form of the statistics equivalence, Lemma 1.4 (iii),

$$\mathrm{tr}_R\Big\{X\ \mathrm{tr}_Q\left[(\hat{O}_k\otimes\mathbb{1}_R)\sigma\right]\Big\} = \mathrm{tr}\left[(\hat{O}_k\otimes X)\sigma\right] = \mathrm{tr}\left[(O_k\otimes X)\rho\right] = \mathrm{tr}_R\Big\{X\ \mathrm{tr}_B\left[(O_k\otimes\mathbb{1}_R)\rho\right]\Big\},$$

and since R is finite-dimensional the trace pairing on $\mathcal{L}(R)$ separates points, so the two operators on R coincide.

(i) With $O_k = c\,\mathbb{1}_B$ the right-hand side of (1.3) is $c\,\mathrm{tr}_B\,\rho = c\,\rho_R$, and $\rho_R = \sigma_R$ because Λ is trace preserving and acts only on B. (ii) For $X \succeq 0$ we have $O_k \otimes X \succeq 0$, so $\mathrm{tr}[(O_k \otimes X)\rho] \ge 0$; hence $\mathrm{tr}_R\{X\ \mathrm{tr}_Q[(\hat{O}_k \otimes \mathbb{1}_R)\sigma]\} \ge 0$ for every $X \succeq 0$, which for a Hermitian operator on a finite-dimensional space is positive semidefiniteness. (iii) Apply (ii) to $O_k - a\,\mathbb{1}_B \succeq 0$ and to $b\,\mathbb{1}_B - O_k \succeq 0$, then use (i) on the identity terms; the scalar forms follow by taking traces and using $\mathrm{tr}\,\sigma_R = 1$. □

The certificate of a transported constraint is the hypothesis that makes it hold: for a case of Corollary 1.7 an operator fact about the device and the channel, valid whatever the input, and for the assumed case of Corollary 3.13 an assumption on the input states.

A constraint is a condition on squashed states that an analysis imposes. Corollary 1.7 derives one from an identity, a positivity or a spectral bound on a carried operator's pull-back, and Corollary 3.13, in the section on output-state equality, reads the same derivation over n rounds and also admits a bound assumed to hold on the physical inputs.

Definition 1.8 (Certificates of transported constraints)**.** The *certificate* of a transported constraint is the hypothesis that makes it hold: for a constraint from Corollary 1.7, the hypothesis of the case that yields it, an operator identity in its case (i) and an operator inequality in its cases (ii)–(iii), both facts about the device and the channel, valid whatever the input; and, for an assumed constraint of Corollary 3.13 (ii), an assumption on the input states.

Uses Cor. 1.7 · *Used by* Cor. 3.13, Def. 7.1, Def. 10.1, Lem. 12.3 · newly derived · accepted

The universal squash model establishes a statistical equivalence and cutoff-based dimension reduction controls the effect of a cutoff, neither by exhibiting a squashing channel. Without such a channel the output-state equality behind Theorem 3.11 is not available.

Remark 1.9 (Reductions that are not squashings). Not every dimension reduction used in QKD comes with a squashing channel, and one that does not cannot be fed into the security transfer, Theorem 3.11. The universal squash model of [4, Sec. II] establishes a statistical equivalence between the threshold-detector setup and an ideal squashed one by routing both through classical post-processing of a photon-number-resolving measurement, and the dimension-reduction method of [9, Sec. III] controls the effect of a finite cutoff rather than exhibiting a channel. Neither proceeds through the existence of a Λ with $\Lambda^\dagger(F_v) = M_v$, which is exactly the hypothesis the output-state equality of Theorem 3.1 runs on and the hypothesis [6, Lemma 6] states for a narrower model. The rule that follows for the emitted artefact, a reduction chain of links, is Requirement 14.1: an explicit channel witness for every link. The companion rule for constraint entries, exactly one named certificate each, is Requirement 16.1.

Used by Req. 14.1 · newly derived · accepted

The measurement channel of a POVM takes a quantum input to the classical register holding its outcome. It separates Bob's measurement from the protocol's later processing.

Replacing Bob's physical POVM M by a squashed target F can only overestimate the adversary, for a short reason. First, $M \preceq_\Lambda F$ says exactly that Bob's physical measurement *channel* is the squashing channel followed by the squashed one. Second, Bob's device enters the protocol only through that channel. Third, therefore, the physical protocol and the squashed protocol have the same output state, not merely close ones, which is why the transfer costs no penalty term, no correction factor, and the same ε on both sides. What remains to check is that the squashed analysis runs over a set of inputs containing the images of the physical ones, and that is Theorem 3.11.

Definition 1.10 (Measurement channel)**.** For a POVM $M = \{M_v\}_{v\in\mathcal{V}}$ on B and a classical register Y with alphabet $\mathcal{V}$, the *measurement channel* of M is

$$\mathcal{N}_M \colon \mathcal{T}(B) \to \mathcal{T}(Y), \qquad \mathcal{N}_M(\rho) \coloneqq \sum_{v\in\mathcal{V}} \operatorname{tr}\left[M_v \rho\right] \, |v\rangle\langle v|_Y \,. \tag{1.4}$$

Used by Def. 2.1, Prop. 3.4 · newly derived · accepted

A POVM squashes to a target via a channel exactly when its measurement channel is the target's measurement channel after that channel. This is the channel identity the protocol-level argument inserts round by round.

Lemma 1.11 (Measurement-channel factorization)**.** Let M, F be POVMs on B, Q over $\mathcal{V}$ and let $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel. Then

$$M \preceq_\Lambda F \qquad \Longleftrightarrow \qquad \mathcal{N}_M = \mathcal{N}_F \circ \Lambda. \tag{1.5}$$

Uses Lem. 1.2, Lem. 1.4 · *Used by* Thm. 3.1, Cor. 3.3 · newly derived · accepted

Proof. ($\Rightarrow$) For every $\rho \in \mathcal{T}(B)$, $\mathcal{N}_M(\rho) = \sum_v \operatorname{tr}\left[\Lambda^\dagger(F_v)\rho\right] |v\rangle\langle v| = \sum_v \operatorname{tr}[F_v \Lambda(\rho)]\, |v\rangle\langle v| = \mathcal{N}_F(\Lambda(\rho))$ by (1.1). ($\Leftarrow$) Comparing the coefficient of $|v\rangle\langle v|$ on both sides gives $\operatorname{tr}[M_v\rho] = \operatorname{tr}[F_v\Lambda(\rho)]$ for every state ρ, which is condition (ii) of the statistics equivalence, Lemma 1.4, with $m = 0$. □

2 Protocols, security criteria and exact operational reductions

The protocols of this contract pass each round's measured system through one measurement channel and process only its classical outcome afterwards. A security criterion is any family of functions of the protocol's final output together with the adversary's system, and an exact operational reduction is a map of attack strategies under which every strategy and its image produce the same full output. Such a map transfers every bound on every criterion with the same ε, and reductions restrict, enlarge and compose (Lemma 2.4). The section closes with the reason both reductions of this contract are exact rather than approximate.

An n-round protocol passes each round's measured system, held by the measuring parties, through one measurement channel, the same in every round, and no later processing touches that system. The squashed protocol is the same channel with the target POVM in place of the physical one.

The protocols this contract reduces touch each round's measured system only through a measurement; what follows reads its classical outcome and never the system itself. The definition fixes that structure. The declaration it speaks of is the protocol declaration that defines the consuming engine's protocol (Definition "Protocol declaration" (protocol-compiler/def:declaration)); it names the parties that measure. The structure rests on a round order, every public message of a round sent after every measurement of that round, which is Assumption A-SEQ of the transmitter's contract. This definition places the measurement channel before the round's remaining processing and fixes one POVM and one channel for all rounds; the claims that rest on that structure declare A-SEQ and A-ROUND-SQ (Assumption 13.2), a definition assuming nothing. Assumption 13.1 and Assumption 13.2 leave

the round order out. The embedding by which a declaration names Alice as a measuring party is that of Remark "The two source modes on the program" (meat-core/rem:modes-da1) (b).

Definition 2.1 (Measurement-factorizing n-round protocol)**.** An *n-round protocol* with receiver POVM M is a channel

$$\mathcal{E}_M \;=\; \mathsf{K} \circ \mathsf{G}_n \circ \cdots \circ \mathsf{G}_1, \qquad \mathsf{G}_j \;=\; \bar{\mathsf{G}}_j \circ \big(\mathrm{id} \otimes \mathcal{N}_M\big), \tag{2.1}$$

in which the measurement channel $\mathcal{N}_M$ of Definition 1.10 acts on the round-j *measured system* B_j, the quantum system that the *measuring parties*, the parties the declaration names as measuring, receive in round j, and no other system, and the identity acts on everything else. $\bar{\mathsf{G}}_j$ is the round's remaining processing (Alice's measurement on A_j, the round's announcements appended to the public record, and the retained per-round data), and K is the end-of-protocol processing: sifting, parameter estimation and the accept/abort decision, error correction, verification, privacy amplification and the key maps. The *squashed protocol* $\mathcal{E}_F$ is (2.1) with B_j replaced by Q_j, M by F, and $\bar{\mathsf{G}}_j$, K unchanged. One POVM and one channel serve all n rounds: the same M, and in the squashed protocol the same F and the same squashing channel, stands in every G_j, with no dependence on j. That the measuring devices measure by the same M in every round, with no memory between rounds, is Assumption 13.2; F and the squashing channel are objects of the analysis, chosen once for all rounds. That is what makes the composite of the per-round channels a tensor power, such as $\Lambda^{\otimes n}$ for the squashing channel, rather than a product of n different maps.

One measuring party, or several. M is one POVM on the measured system, over the alphabet of the measuring parties' joint outcome. Where one party measures, which is the one-party case, B_j is that party's system, Bob's in a prepare-and-measure declaration. Where several parties measure, $B_j = \bigotimes_f B_{f,j}$ is the product of their party-scoped parts, each factor $B_{f,j}$ belonging to one of them, and (2.1) places *all* of their measurements at the one step $\mathcal{N}_M$, before the round's remaining processing $\bar{\mathsf{G}}_j$ and so before every public message of the round. The declaration says on which side Alice's measurement sits, and each reading is covered by exactly one clause of the first paragraph: the measurement on A_j listed in $\bar{\mathsf{G}}_j$ is that of a party the declaration does *not* name as measuring (Alice's, in a prepare-and-measure declaration), while where she *is* named as measuring, her system is a factor $B_{f,j}$ of the measured system and her measurement is a factor of $\mathcal{N}_M$, not a step of $\bar{\mathsf{G}}_j$. Either way $\bar{\mathsf{G}}_j$ and K touch no B_j, which is all the theorems of this contract read. A declaration realizes the second reading by an embedding: it declares Alice's system A one-dimensional, $d_A = 1$, folds her measured system into the sectors of Q and her outcome components into v, and names the components she holds in its party partition (Definition "Protocol declaration" (protocol-compiler/def:declaration)). That the physical

protocol has that order, every public message of round j sent after every measurement of round j whichever party sends it and whichever party measured, is Assumption A-SEQ of the transmitter (Assumption "Sequential round structure" (transmitter/ass:seq)). The definition says nothing about which party sends which message or about a message's dependence on earlier messages of its own round, and a quantum operation of round j that depended on a message of round j is outside it, as it is outside A-SEQ.

Uses Def. 1.10 · *Used by* Thm. 3.1, Def. 3.2, Cor. 3.3, Prop. 3.4, Thm. 3.11, Lem. 8.3, Ass. 13.1, Ass. 13.2, meat-core/lem:honest-attacks, meat-core/lem:memory-lift · newly derived · accepted

A security criterion is any family of functions, one per admissible adversary system, of the joint state of the protocol's output registers and that system. Equality of those states therefore transfers every criterion at once, with no proof for its particular formula.

The protocol $\mathcal{E}_M$ of Definition 2.1 on which a criterion is evaluated factors in each round as $\mathsf{G}_j = \bar{\mathsf{G}}_j \circ (\mathrm{id} \otimes \mathcal{N}_M)$, and that factorization rests on Assumption 13.1, together with the round order and the one POVM for all rounds that Definition 2.1 declares: each measuring party's interaction with the quantum system it receives is described by a single POVM, and M is their product on the round's measured system, Bob's POVM where he alone measures. The measurement emits the classical outcome v and nothing else; no post-measurement quantum system is retained or reused, and all later processing reads the measured system only through that outcome.

Definition 2.2 (Security criterion)**.** A *security criterion* is a family $f = (f_E)_E$ of functions, one for every admissible adversary system E, defined on states of the protocol's output registers together with E; an *admissible adversary system* is a finite-dimensional system, the form the round model of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) gives the adversary's registers. Its value on the protocol $\mathcal{E}_M$ at an input ρ, a state of the systems the protocol acts on together with E, is

$$\mathsf{sec}(\mathcal{E}_M, \rho) \coloneqq f_E\big((\mathcal{E}_M \otimes \mathrm{id}_E)(\rho)\big). \tag{2.2}$$

Used by Lem. 2.4, Rem. 2.5, Thm. 3.1, Def. 3.2, Rem. 3.8, Rem. 3.9, Thm. 3.11, Cor. 3.12, Thm. 8.6 · newly derived · accepted

A protocol presentation is a set of attack strategies with the full output each produces; an exact operational reduction maps strategies between two presentations over the same registers so that every strategy and its image give the same output. Bounds carry back with no property of the criterion.

The two reductions this contract performs, replacing a physical receiver by a finite model and replacing a physical source by a virtual one, are the same kind of object, and everything either of them transfers is transferred by that kind of object alone. This definition states the object once and Lemma 2.4

states what it buys; Corollary 3.3 and Theorem 3.11 are the receiver-side instance, Lemma 8.3 the source-side one, and Theorem 8.6 their composition. Nothing in the definition is about squashing, about measurement, or about any particular protocol: it speaks of a set of strategies, of the state each of them produces, and of nothing else.

Definition 2.3 (Exact operational reduction)**.** A *protocol presentation* is a triple $\mathsf{P} = (\mathbb{A}, W, \mathsf{out})$: a set $\mathbb{A}$, each element of which is called an *attack strategy*, a tuple W of *output registers*, and a map $\mathsf{out}\colon \mathbb{A} \to S_{\leq}(W)$ assigning to each strategy the *full output* the protocol produces under it. The full output is the state on *every* register a security criterion may read (Definition 2.2): both parties' key registers, the public transcript with the acceptance and abort data, and the adversary's final register, and no marginal of them.

Let $\mathsf{P} = (\mathbb{A}, W, \mathsf{out})$ and $\mathsf{P}' = (\mathbb{A}', W, \mathsf{out}')$ be presentations *over the same output registers* W. An *exact operational reduction* from P to P' is a map $r\colon \mathbb{A} \to \mathbb{A}'$ with

$$\mathsf{out}(a) \;=\; \mathsf{out}'\big(r(a)\big) \qquad \text{for every } a \in \mathbb{A}. \tag{2.3}$$

We call $\mathbb{A}$ the *physical strategy set* and $\mathbb{A}'$ the *analysed strategy set*, equivalently the *admissible* strategy set, and $r(\mathbb{A}) \subseteq \mathbb{A}'$ the reduction's *image*.

The conditions, in full.

(R1) *(one output tuple)* the two presentations carry literally the same output registers, and which register of P' is which register of P is part of the data rather than inferred. Where a model protocol's adversary register is a *different* system, an identification of it with the physical one has to be declared and the criterion has to be invariant under that identification; this contract declares none and needs none, the adversary's registers being literally the same system in both of its instances;

(R2) *(exact equality)* (2.3) is equality of the two states on the nose: not of a marginal, not up to a distance, and not up to any map on W;

(R3) *(totality and typing)* r is defined at every strategy of $\mathbb{A}$ and takes its values in $\mathbb{A}'$. That its values lie in $\mathbb{A}'$ is a condition on r and on $\mathbb{A}'$ together, and it is the only place where a restriction of the analysed set has to be earned.

What is not required, each item being used by an instance in this contract. r need not be injective, need not be surjective onto $\mathbb{A}'$, and need not preserve any structure of $\mathbb{A}$; the analysed set $\mathbb{A}'$ may be, and in this contract typically is, strictly larger than the image, because what a numerical analysis can optimize over is a set cut out by finitely many conditions and not an image (Definition 6.3); no metric, topology, order, linear or convex structure on $\mathbb{A}$, on $\mathbb{A}'$ or on the output states is used; out and out' need not be linear, affine,

continuous or given by channels, and a strategy need not be a channel or a tuple of channels (in Theorem 3.11 a strategy is a state); and nothing is asserted in the other direction, a bound over $\mathbb{A}$ transferring to $\mathbb{A}'$ under no hypothesis of this definition.

Two presentations, one per adversary system. A security criterion in the sense of Definition 2.2 is a *family* $(f_E)_E$, one function per admissible adversary system. Fixing E fixes W and turns each f_E into one function on $S_{\leq}(W)$, so the quantification over E in that definition is a quantification over presentations: a statement of this contract that holds "for every admissible E" exhibits one presentation pair and one reduction per E, and the transfer of Lemma 2.4 is applied to each.

Used by Lem. 2.4, Rem. 2.5, Cor. 3.3, Thm. 3.11, Cor. 3.12, Lem. 8.3, Thm. 8.6 · newly derived · accepted

An exact operational reduction carries every bound on every function of the full output from the analysed to the physical strategy set with the same ε, and reductions restrict, enlarge and compose. A bound over an intersection of analysed sets needs one map into all of them.

The lemma is elementary set arithmetic. Its part (iv) matters in this contract because two channels realising one and the same squashing relation give two different strategy maps, whose reachable sets are in general incomparable; that is why Corollary 3.12 (ii)–(iii) are stated for one common channel and one common family of physical inputs (Remark 3.15). Part (vi) says that an analysed set larger than the image costs tightness and never soundness, the same trade the reachable sets of Definition 3.10 make.

Lemma 2.4 (Transfer and composition of exact operational reductions)**.** Let r be an exact operational reduction from $\mathsf{P} = (\mathbb{A}, W, \mathsf{out})$ to $\mathsf{P}' = (\mathbb{A}', W, \mathsf{out}')$ in the sense of Definition 2.3. Write $\mathcal{F}(W)$ for the set of all functions $S_{\leq}(W) \to [-\infty, +\infty]$; a security criterion in the sense of Definition 2.2, at a fixed adversary system, is one such function.

(i) *(transfer, with the same ε and for every criterion at once)* for every $f \in \mathcal{F}(W)$ and every $a \in \mathbb{A}$, $f(\mathsf{out}(a)) = f(\mathsf{out}'(r(a)))$, and consequently

$$\sup_{a\in\mathbb{A}} f\big(\mathsf{out}(a)\big) \;=\; \sup_{b\in r(\mathbb{A})} f\big(\mathsf{out}'(b)\big) \;\leq\; \sup_{b\in\mathbb{A}'} f\big(\mathsf{out}'(b)\big). \tag{2.4}$$

In particular, if $f(\mathsf{out}'(b)) \leq \varepsilon$ for every $b \in \mathbb{A}'$, then $f(\mathsf{out}(a)) \leq \varepsilon$ for every $a \in \mathbb{A}$, with the *same* ε, and simultaneously for every $f \in \mathcal{F}(W)$, no property of f being used;

(ii) *(enlargement of the analysed set)* if $\mathbb{A}' \subseteq \mathbb{A}''$ and $\mathsf{out}''|_{\mathbb{A}'} = \mathsf{out}'$, then the same r is an exact operational reduction from P to $\mathsf{P}'' = (\mathbb{A}'', W, \mathsf{out}'')$. So it is enough that the image $r(\mathbb{A})$ lie in the analysed set, and the last inequality of (2.4) is where that slack is spent;

(iii) *(restriction of the physical set)* if $\mathbb{A}_0 \subseteq \mathbb{A}$, then $r|_{\mathbb{A}_0}$ is an exact operational reduction from $(\mathbb{A}_0, W, \mathsf{out}|_{\mathbb{A}_0})$ to P';

(iv) *(one map, many analysed sets)* let $(\mathbb{A}'_k)_{k\in K}$ be a family of subsets of $\mathbb{A}'$ such that *one and the same* r satisfies $r(\mathbb{A}) \subseteq \mathbb{A}'_k$ for every $k \in K$. Then r is an exact operational reduction from P to $\big(\bigcap_{k\in K} \mathbb{A}'_k, W, \mathsf{out}'\big)$, and by (i) a bound over that intersection transfers. The hypothesis cannot be weakened to the existence, for each k, of *some* exact operational reduction into $\mathbb{A}'_k$: let W be one classical bit, $\mathbb{A} = \{a\}$, $\mathbb{A}' = \{b_1, b_2\}$ with $b_1 \neq b_2$ and $\mathsf{out}(a) = \mathsf{out}'(b_1) = \mathsf{out}'(b_2) = |0\rangle\langle 0|$, and $\mathbb{A}'_k = \{b_k\}$. Each $r_k \colon a \mapsto b_k$ is an exact operational reduction into $\mathbb{A}'_k$, the intersection is empty, so "$f(\mathsf{out}'(b)) \leq 0$ for every b in the intersection" holds vacuously; yet $f(\mathsf{out}(a)) = 1$ for $f(\rho) \coloneqq \mathrm{tr}[|0\rangle\langle 0|\,\rho]$, so the conclusion of (i) is false for the intersection;

(v) *(composition, and how the analysed sets compose)* let $\mathbb{A}'_0 \subseteq \mathbb{A}'$ contain $r(\mathbb{A})$ and let r' be an exact operational reduction from $\big(\mathbb{A}'_0, W, \mathsf{out}'|_{\mathbb{A}'_0}\big)$ to $\mathsf{P}'' = (\mathbb{A}'', W, \mathsf{out}'')$. Then $r' \circ r$ is an exact operational reduction from P to P'', with image $r'(r(\mathbb{A}))$. The analysed set the composite needs is the second reduction's, or any subset of $\mathbb{A}''$ containing $r'(r(\mathbb{A}))$; the first reduction's analysed set enters only as a domain the second must cover, which is what $r(\mathbb{A}) \subseteq \mathbb{A}'_0$ says. Composition is associative and the identity map on $\mathbb{A}$ is an exact operational reduction from P to itself;

(vi) *(the direction, and what the slack costs)* nothing above bounds $\sup_{\mathbb{A}'} f \circ \mathsf{out}'$ by $\sup_{\mathbb{A}} f \circ \mathsf{out}$, and the first can be strictly larger: take $\mathbb{A} = \{a\}$, $\mathbb{A}' = \{r(a), b\}$ with $\mathsf{out}'(b) \neq \mathsf{out}(a)$ and an f separating the two states. An analysed set larger than the image therefore costs *tightness* and never soundness.

Uses Def. 2.2, Def. 2.3 · *Assumes* A-MEAS · *Used by* Rem. 2.5, Cor. 3.3, Thm. 3.11, Cor. 3.12, Lem. 8.3, Thm. 8.6 · newly derived · accepted

Proof. (i) The pointwise identity is f applied to both sides of (2.3). Hence $\{\mathsf{out}(a) : a \in \mathbb{A}\} = \{\mathsf{out}'(r(a)) : a \in \mathbb{A}\} = \{\mathsf{out}'(b) : b \in r(\mathbb{A})\}$ as subsets of $S_{\leq}(W)$, so the two suprema of f over them agree, which is the equality in (2.4); the inequality is $r(\mathbb{A}) \subseteq \mathbb{A}'$ and monotonicity of the supremum under enlarging the index set. The last sentence is (2.4) read with the right-hand supremum bounded by ε, and f was arbitrary.

(ii) The map is unchanged and (2.3) is unchanged, because $\mathsf{out}''(r(a)) = \mathsf{out}'(r(a))$ for every a by $r(\mathbb{A}) \subseteq \mathbb{A}'$ and the hypothesis on out''; the typing condition (R3) of Definition 2.3 holds because $\mathbb{A}' \subseteq \mathbb{A}''$, and (R1), (R2) are conditions on W and on the values, which did not move.

(iii) (2.3) is a condition at each strategy separately, so it survives restricting the set of strategies it is required at.

(iv) $r(\mathbb{A}) \subseteq \bigcap_k \mathbb{A}'_k$ is exactly the hypothesis, and (2.3) does not read the analysed set, so r is an exact operational reduction into the intersection; the transfer is (i). For the counterexample: each r_k satisfies (2.3) by construction and $r_k(\mathbb{A}) = \{b_k\} = \mathbb{A}'_k$, so each is an exact operational reduction into $\mathbb{A}'_k$; $\mathbb{A}'_1 \cap \mathbb{A}'_2 = \varnothing$ because $b_1 \neq b_2$; a supremum over the empty set is $-\infty$, so every bound holds over it; and $f(\mathsf{out}(a)) = \mathrm{tr}[|0\rangle\langle 0|\,|0\rangle\langle 0|] = 1$. What fails is not (i) but its hypothesis: no map has its image in the intersection.

(v) For every $a \in \mathbb{A}$, $r(a) \in \mathbb{A}'_0$ by hypothesis, so $r'(r(a))$ is defined and lies in $\mathbb{A}''$, and $\mathsf{out}(a) = \mathsf{out}'(r(a)) = \mathsf{out}''(r'(r(a)))$ by (2.3) for r and then for r'. The image statement is the definition of a composite's image. Associativity and the identity are those of maps of sets; the identity satisfies (2.3) trivially.

(vi) The two suprema are unrelated by anything proved above, and the displayed instance has $\sup_{\mathbb{A}'} f \circ \mathsf{out}' = \max\{f(\mathsf{out}(a)), f(\mathsf{out}'(b))\} > f(\mathsf{out}(a)) = \sup_{\mathbb{A}} f \circ \mathsf{out}$ for an f with $f(\mathsf{out}'(b)) > f(\mathsf{out}(a))$, which exists because the two states differ. □

Equality of full outputs lets one argument transfer every criterion with the same ε, while a distance bound would need a modulus of continuity no criterion is given. Every reduction and link of this contract, the domination link included, is exact.

Remark 2.5 (Exactness of the reductions and generality of the transfer). Definition 2.3 demands equality of the *full* output state, and that demand is what pays for everything Lemma 2.4 delivers. A security criterion in the sense of Definition 2.2 is an arbitrary function of the output state: no linearity, no monotonicity and no continuity is assumed of it anywhere in this contract. Under exact equality that costs nothing, because part (i) of the lemma is f applied to two equal states; under anything weaker it costs everything, because a hypothesis $\|\mathsf{out}(a) - \mathsf{out}'(r(a))\|_1 \leq \delta$ bounds $f(\mathsf{out}(a))$ in terms of $f(\mathsf{out}'(r(a)))$ only through a modulus of continuity for f, which the definition of a criterion does not give and which the consuming engine's criterion would have to supply case by case. So the generality of the transfer, one argument for every criterion with the same ε, and the exactness are one fact, not two.

Both instances are exact, for a checkable reason. On the receiver side the substitution rests on the channel identity (3.1), an identity of maps with no error term (Theorem 3.1). On the source side it rests on the source identity (S3) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment), an exact matrix identity on declared numbers: it is decided entry by entry where every source link supplies its channel as data, and concluded from a named claim where a link supplies its channel through a theorem constructor (Lemma 8.3).

What an approximate reduction would need. It would need a theorem converting a distance between output states into a security penalty, and this contract has none. The security transfer, Theorem 3.11, rests on an equality and has no penalty slot, so there is no parameter for such a term to enter; accordingly a declared relation carrying an error term is refused (Requirement 14.4), and a reduction chain mixing an exact link with one carrying an error term is refused rather than given a composite status, a status for the chain as a whole (Theorem 7.4 (iv), Require-

ment 15.4). The nearest thing to a continuity statement in these contracts is a refusal of one: Proposition "Instability of the rate under measurement perturbations" (meat-core/prop:optol-nobound) exhibits two exactly admissible target families within a declared distance of each other whose rate bounds per round differ by an amount limited only by the number model. The limit is that proposition's own representability condition, which binary64 arithmetic imposes and which caps the integer N measuring the gap at $N \leq 52$; so proximity of models licenses nothing about rates there either.

The emission displacement is not an error term of a reduction. The displacement Δ_s of the emission normalizer, the largest change it makes to an entry of an instance's datum in the data sector s, does not make a reduction approximate. It moves the declared model the reduction is stated about, and the reduction's identity then holds of that model with no error term at all (Theorem 11.2, Remark 11.7). Nor does the step back to the handed model, the construction before the normalizer's snap, need an error term. No bound relates a key length computed from the declared model alone to one computed from the handed model, and the displacement is not what relates them (Theorem 11.2 (v)). The gap is not bridged at all where a run can avoid it: a run that forms the number its key length is formed from on the reference companion (Corollary 12.1), built with no normalizer from the handed reference, or from the instance's own snapped blocks where none is handed, reads the declared model only to find points. Where a run does read it, the gap is closed by a disclosed premise about the device, never by an error term. The domination link, which no requirement emits, is exact as well: a noise channel followed by its target with classical flags (Theorem 23.2), so that a squashing link followed by it is exact as a whole (Corollary 24.1).

Uses Def. 2.2, Def. 2.3, Lem. 2.4, Thm. 3.1, Thm. 3.11, Thm. 7.4, Thm. 11.2 · *Used by* Req. 14.4 · newly derived · accepted

3 Output-state equality and security transfer

Replacing each round's physical measurement by the squashing channel followed by the target measurement leaves the protocol's whole output state unchanged (Theorem 3.1), also against adaptive attack sequences in either source mode (Corollary 3.3). The main result, Theorem 3.11, is that security of the squashed protocol over analysed inputs that contain the squashed image of every physical input transfers to the physical protocol with the same ε, for every criterion. A constraint is sound when every reachable squashed state satisfies it; sound constraints may be intersected when they are sound for one common channel and one family of physical inputs, and carried operators supply them (Corollary 3.12, Corollary 3.13). The remarks give the variants of the factorisation, the criteria that transfer, and the standard way a transplanted operator identity makes a constraint unsound.

Imports foundations/def:round-model

Replacing each round's physical measurement by the squashing channel followed by the target measurement leaves the protocol's whole output state unchanged. Every security criterion therefore takes the same value on the physical and the squashed protocol, with no error term.

A squashing relation is a statement about one round's measurement. Through the factorization of the measurement channel (Lemma 1.11) it becomes a statement about the whole protocol, and the proof uses no property of the criterion f_E beyond its being a function of the output state: not linearity, not monotonicity, not continuity. The one structural hypothesis is the factorization $\mathsf{G}_j = \bar{\mathsf{G}}_j \circ (\mathrm{id} \otimes \mathcal{N}_M)$ of Definition 2.1, and it rests on the measuring devices entering only through one POVM per round (Assumption 13.1) together with the round order and the one POVM for all rounds that the definition declares. The round's order is not part of Assumption 13.1: that every public message of the round follows every measurement of the round is A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)).

Theorem 3.1 (Equality of the physical and squashed output states)**.** Let $M \preceq_\Lambda F$ over $\mathcal{V}$ and let $\mathcal{E}_M, \mathcal{E}_F$ be as in Definition 2.1. Then

$$\mathcal{E}_M \;=\; \mathcal{E}_F \circ \big(\mathrm{id} \otimes \Lambda^{\otimes n}\big), \tag{3.1}$$

the identity acting on every register other than $B_1 \dots B_n$; and consequently, for every finite-dimensional adversary system E and every input ρ,

$$(\mathcal{E}_M \otimes \mathrm{id}_E)(\rho) = (\mathcal{E}_F \otimes \mathrm{id}_E)\big(\sigma\big), \qquad \sigma \coloneqq \big(\mathrm{id} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E\big)(\rho). \tag{3.2}$$

In particular $\mathsf{sec}(\mathcal{E}_M, \rho) = \mathsf{sec}(\mathcal{E}_F, \sigma)$ for every security criterion in the sense of Definition 2.2.

Uses Lem. 1.11, Def. 2.1, Def. 2.2 · *Assumes* A-MEAS, A-ROUND-SQ, A-SEQ · *Used by* Rem. 2.5, Def. 3.2, Rem. 3.6, Rem. 3.7, Rem. 3.9, Def. 3.10, Thm. 3.11, Cor. 4.2, Thm. 7.4, Cor. 24.1 · newly derived · accepted · *Oracles* registry_reductions.py

Proof. By Lemma 1.11, $\mathcal{N}_M = \mathcal{N}_F \circ \Lambda$. Substituting into (2.1),

$$\mathsf{G}_j^M = \bar{\mathsf{G}}_j \circ \big(\mathrm{id} \otimes \mathcal{N}_F \circ \Lambda\big) = \bar{\mathsf{G}}_j \circ \big(\mathrm{id} \otimes \mathcal{N}_F\big) \circ \big(\mathrm{id} \otimes \Lambda_j\big) = \mathsf{G}_j^F \circ \big(\mathrm{id} \otimes \Lambda_j\big),$$

where Λ_j denotes Λ acting on B_j and the identity elsewhere. For $i \neq j$ the maps G_i^F and Λ_j act on disjoint tensor factors — G_i^F touches A_i, Q_i and classical registers, while Λ_j touches only B_j — so they commute. Composing the n rounds therefore moves every Λ_j to the front, and the Λ_j assemble into $\Lambda^{\otimes n}$ because they act on disjoint systems. Applying K, which touches no B_j, gives (3.1). Equation (3.2) is (3.1) tensored with id_E, and the equality of the criteria is (2.2) evaluated on the two equal states. □

An attack sequence is the round model's, at the measured system; its squashed sequence applies the squashing channel to the measured system alone, leaving Alice's system and Eve's memory untouched. Where several parties measure, that channel is a product of one channel per party.

Equation (3.2) is stronger than it may look: the two protocols do not merely have close output states, they have the *same* one. This is why the transfer costs nothing and holds simultaneously for every criterion of the form (2.2), with the same ε on both sides. To state the equality against an adversary who acts round by round, we need the attack sequences of the round model and their squashed counterparts.

Definition 3.2 (Round model and attack sequences, typed by source mode)**.** The round model of the protocol (2.1) is that of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) with per-round system B, Alice's system A and the declared source mode. An attack sequence for the protocol is thus a family $(\mathcal{A}_j)_{j=1}^{n}$ of channels whose output carries the systems the adversary supplies in the declared source mode, the display being that definition's at $S = B$:

$$\mathcal{A}_j \in \begin{cases} \mathrm{CPTP}\big(E_{j-1},\ B_j\, E'_j\big), & \text{trusted-source mode,} \\ \mathrm{CPTP}\big(E_{j-1},\ A_j\, B_j\, E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{3.3}$$

where E_0 is Eve's initial register, E'_j her outgoing memory, and E_j collects E'_j together with whatever public data of round j she retains. In trusted-source mode A_j is *not* an output of $\mathcal{A}_j$: it is supplied by Alice and carried into the round as a separate input. The protocol's output under the sequence is obtained from (2.1) by supplying B_j — and, in untrusted-source mode, A_j as well — as the output of $\mathcal{A}_j$ in round j. The corresponding *squashed sequence* is

$$\mathcal{A}_j^{\mathrm{sq}} \coloneqq \begin{cases} \big(\Lambda \otimes \mathrm{id}_{E'_j}\big) \circ \mathcal{A}_j \in \mathrm{CPTP}\big(E_{j-1}, Q_j E'_j\big), & \text{trusted-source mode,} \\ \big(\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}_{E'_j}\big) \circ \mathcal{A}_j \in \mathrm{CPTP}\big(E_{j-1}, A_j Q_j E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{3.4}$$

the squashing channel acting on the B_j factor in both cases and on nothing else.

Several measuring parties. Where the measured system of a round is the product $B_j = \bigotimes_f B_{f,j}$ of party-scoped parts (Definition 2.1), the squashing channel is the product $\Lambda = \bigotimes_f \Lambda_f$ of one channel per part, $Q_j = \bigotimes_f Q_{f,j}$, and the alphabet $\mathcal{V}$ of M and F is the product of the parts' alphabets. Neither line of (3.3) changes: B_j is one output system of $\mathcal{A}_j$ whatever its factorisation, so the adversary supplies all of its parts in one channel and is given the same freedom as with one measuring party; in particular the parts may be arbitrarily correlated, and the adversary is not required to act on them independently. Nor does (3.4) change: the product channel still acts on the B_j factor alone, leaving A_j and E'_j untouched. That the product channel

realizes a squashing relation for the product family is Proposition 4.8, which this definition does not assert. With one measuring party the product has one factor.

Uses Def. 2.1, Def. 2.2, Thm. 3.1, foundations/def:round-model · *Used by* Cor. 3.3, Thm. 6.7, Lem. 8.3, Thm. 8.6 · newly derived · accepted

In either source mode, the physical protocol under an attack sequence and the squashed protocol under its squashed sequence have the same output state. Security against squashed attack families containing every squashed physical attack therefore transfers with the same ε.

Theorem 3.1 fixes the adversary's input before the protocol runs; an adversary who acts round by round, with a memory, is an attack sequence of Definition 3.2. The same substitution handles it, one squashing channel per round. The corollary is the receiver-side instance of an exact operational reduction, which Theorem 8.6 composes with the source-side one in trusted-source mode.

Corollary 3.3 (Output-state equality against adaptive attack sequences)**.** Let $M \preceq_\Lambda F$ and fix a source mode. For every attack sequence $(\mathcal{A}_j)_{j=1}^{n}$ of the corresponding type (3.3), the output state of the physical protocol $\mathcal{E}_M$ under $(\mathcal{A}_j)_j$ equals the output state of the squashed protocol $\mathcal{E}_F$ under $(\mathcal{A}_j^{\mathrm{sq}})_j$ of the same branch of (3.4). Consequently every security criterion takes the same value on the two, and if $\mathcal{E}_F$ is ε-secure against every sequence drawn from families $\mathfrak{A}_j^{\mathrm{sq}}$ with

$$\mathfrak{A}_j^{\mathrm{sq}} \supseteq \begin{cases} (\Lambda \otimes \mathrm{id}) \circ \mathrm{CPTP}(E_{j-1}, B_j E_j'), & \text{trusted-source mode,} \\ (\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}) \circ \mathrm{CPTP}(E_{j-1}, A_j B_j E_j'), & \text{untrusted-source mode,} \end{cases} \tag{3.5}$$

then $\mathcal{E}_M$ is ε-secure against *all* attack sequences of that type.

Uses Lem. 1.11, Def. 2.1, Def. 2.3, Lem. 2.4, Def. 3.2 · *Used by* Prop. 3.4, Rem. 3.7, Thm. 6.7, Rem. 6.10, Thm. 7.4, Thm. 8.6 · newly derived · accepted

Proof. Step 1 (equal output states). Both branches are the same computation; only the register the identity acts on differs. In either mode, each occurrence of $\mathcal{N}_M$ in (2.1) acts on the B_j-output of $\mathcal{A}_j$ and on nothing else that $\mathcal{A}_j$ produces — in untrusted-source mode it does not touch the A_j-output either, which is measured by Alice's POVM inside $\bar{\mathsf{G}}_j$ (Definition 2.1). Writing ι_j for $\mathrm{id}_{E_j'}$ in trusted mode and for $\mathrm{id}_A \otimes \mathrm{id}_{E_j'}$ in untrusted mode, Lemma 1.11 gives

$$(\mathrm{id}\otimes\mathcal{N}_M\otimes\mathrm{id})\circ\mathcal{A}_j = (\mathrm{id}\otimes\mathcal{N}_F\otimes\mathrm{id})\circ(\Lambda \text{ on } B_j,\ \iota_j \text{ elsewhere})\circ\mathcal{A}_j = (\mathrm{id}\otimes\mathcal{N}_F\otimes\mathrm{id})\circ\mathcal{A}_j^{\mathrm{sq}},$$

the middle map being exactly the branch of (3.4) for that mode. Substituting round by round replaces every $(\mathcal{N}_M, \mathcal{A}_j)$ pair by $(\mathcal{N}_F, \mathcal{A}_j^{\mathrm{sq}})$ and changes nothing else in (2.1), so the two output states are equal.

Step 2 (security, as the transfer of an exact operational reduction). Fix an admissible adversary system and let P be the presentation whose strategies are the attack sequences of the declared type against $\mathcal{E}_M$, with the output map of Step 1, and P' the presentation whose strategies are the sequences drawn from the families $\mathfrak{A}_j^{\mathrm{sq}}$ against $\mathcal{E}_F$. The output registers are the same in both — those of K together with the adversary's final register — and the map $r_{\mathrm{sq}}\colon (\mathcal{A}_j)_j \mapsto (\mathcal{A}_j^{\mathrm{sq}})_j$ of (3.4) takes its values in P''s strategy set precisely because (3.5) holds round by round. Step 1 is then (2.3) for r_{sq}, so r_{sq} is an exact operational reduction (Definition 2.3), and the security statement is Lemma 2.4 (i): the supremum over the physical sequences equals the supremum over the image and is at most the supremum over the families $\mathfrak{A}_j^{\mathrm{sq}}$, which is where the slack of (3.5) is spent. That any larger families serve as well is part (ii) of the same lemma. No property of the criterion is used, which is why the conclusion holds for all of them at once. □

For a well-formed declaration with one sector on A, one measured system, POVM form, two messages neither reading the other and no party-scoped entry, the round channel and its physical counterpart factor as in Definition 2.1. A squashing relation carries the one output to the other.

The consuming engine's protocol channel is generated from a declaration. For a declaration in the format its hypothesis fixes, the following proposition shows that this channel, and the physical one defined by the same closed form on the physical system, have the measurement factorization of Definition 2.1, so that Corollary 3.3 applies to them: an adversary who chooses round j after seeing the announcements of rounds $1, \dots, j-1$ is covered, since E_{j-1} may carry the public record, and the substitution is unchanged — one Λ per round, absorbed round by round.

Proposition 3.4 (Measurement factorization of the platform's protocol channel)**.** Let $\mathfrak{D}$ be a protocol declaration in the sense of Definition "The protocol declaration as the MEAT core reads it" (meat-core/def:declaration) that satisfies the conditions of Definition "Admissible declaration" (meat-core/def:admissible-declaration), in the format in which A carries one sector, one system is measured, Alice's measurement is declared as a POVM $\{M_x\}_{x\in\mathcal{X}}$, the public discussion is two messages neither of which reads the other, and no entry is scoped to a party. Write a for its joint announcement map, ϕ for its acceptance-statistic map and r for its raw-key map, and let $\mathcal{M}^{\mathrm{sq}}$ be the single-round protocol channel it generates, on AQ with the declared POVM $F = \{F_v\}_{v\in\mathcal{V}}$ on the measured system Q, the system the declaration's measuring parties hold (Definition 2.1). Let $M = \{M_v\}_{v\in\mathcal{V}}$ be a POVM on a separable B, and let $\mathcal{M}^{\mathrm{ph}}$ be the map that the same components define with (B, M) in place of (Q, F),

$$\mathcal{M}^{\mathrm{ph}}(\omega) := \sum_{x,v} |r(x,v)\rangle\langle r(x,v)|_Z \otimes |\phi(a(x,v))\rangle\langle \phi(a(x,v))|_{\widehat{C}} \otimes |a(x,v)\rangle\langle a(x,v)|_I \ \operatorname{tr}\left[(M_x \otimes M_v)\,\omega\right], \qquad \omega \tag{3.6}$$

Then:

(i) both maps satisfy the hypothesis of Definition 2.1 with one remaining processing $\bar{\mathsf{G}}$,

$$\mathcal{M}^{\mathrm{sq}} = \bar{\mathsf{G}} \circ \big(\mathrm{id}_A \otimes \mathcal{N}_F\big), \qquad \mathcal{M}^{\mathrm{ph}} = \bar{\mathsf{G}} \circ \big(\mathrm{id}_A \otimes \mathcal{N}_M\big),$$

where $\bar{\mathsf{G}}$ measures A with Alice's POVM $\{M_x\}$ and applies to the pair (x, v) the classical map $|x, v\rangle\langle x, v| \mapsto |r(x, v)\rangle\langle r(x, v)|_Z \otimes |\phi(a(x, v))\rangle\langle \phi(a(x, v))|_{\widehat{C}} \otimes |a(x, v)\rangle\langle a(x, v)|_I$. Alice's measurement is thus a step of $\bar{\mathsf{G}}$, where Definition 2.1 places the measurement of a party the declaration does not name as measuring, and not of the measurement channel;

(ii) if $M \preceq_\Lambda F$, then for every finite-dimensional E and every $\omega \in S_\leq(ABE)$

$$\big(\mathcal{M}^{\mathrm{ph}} \otimes \mathrm{id}_E\big)(\omega) = \big(\mathcal{M}^{\mathrm{sq}} \otimes \mathrm{id}_E\big)\big((\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}_E)(\omega)\big). \tag{3.7}$$

Uses Lem. 1.2, Def. 1.10, Def. 2.1, Cor. 3.3 · newly derived · accepted

Proof. Step 1 (the two closed forms). Under the conditions of Definition "Admissible declaration" (meat-core/def:admissible-declaration), the closed form of Theorem "The declaration generates a protocol channel" (meat-core/thm:protocol-channel) (ii) reads, verbatim, with $\mathcal{M} = \mathcal{M}^{\mathrm{sq}}$,

$$\begin{aligned}(\mathcal{M} \otimes \mathrm{id}_E)(\omega) = \sum_{x,v} &|r(x, v)\rangle\langle r(x, v)|_Z \otimes |\phi(a(x, v))\rangle\langle \phi(a(x, v))|_{\widehat{C}} \otimes \\ &|a(x, v)\rangle\langle a(x, v)|_I \otimes \mathrm{tr}_{AQ}\big[(M_x \otimes F_v \otimes \mathbb{1}_E)\,\omega\big]\end{aligned} \tag{3.8}$$

for every system E and every $\omega \in S_\leq(AQE)$. Index letters carry the whole distinction here: in (3.8), (3.6) and (3.10), M_x is Alice's element on A, quoted from the declaration, while M_v is the physical element on the measured system B as defined in this contract; they act on different tensor factors and the squashing channel touches only the second. The map (3.6) has the same form with (B, M) in place of (Q, F): for a finite-dimensional E with orthonormal basis $\{|i\rangle\}_i$, every $\omega \in \mathcal{T}(ABE)$ is $\sum_{i,j} \omega_{ij} \otimes |i\rangle\langle j|_E$ with $\omega_{ij} \in \mathcal{T}(AB)$, and $\sum_{i,j} \mathrm{tr}[(M_x \otimes M_v)\,\omega_{ij}]\ |i\rangle\langle j|_E = \mathrm{tr}_{AB}[(M_x \otimes M_v \otimes \mathbb{1}_E)\,\omega]$, so by linearity

$$\begin{aligned}(\mathcal{M}^{\mathrm{ph}} \otimes \mathrm{id}_E)(\omega) = \sum_{x,v} &|r(x, v)\rangle\langle r(x, v)|_Z \otimes |\phi(a(x, v))\rangle\langle \phi(a(x, v))|_{\widehat{C}} \otimes \\ &|a(x, v)\rangle\langle a(x, v)|_I \otimes \mathrm{tr}_{AB}\big[(M_x \otimes M_v \otimes \mathbb{1}_E)\,\omega\big].\end{aligned} \tag{3.9}$$

Step 2 (the factorization). Let (S, G) be (Q, F) or (B, M). For $\omega \in \mathcal{T}(AS)$, (1.4) gives $(\mathrm{id}_A \otimes \mathcal{N}_G)(\omega) = \sum_v \mathrm{tr}_S\left[(\mathbb{1}_A \otimes G_v)\,\omega\right] \otimes |v\rangle\langle v|_Y$, and $\bar{\mathsf{G}}$

maps $\tau\otimes|v\rangle\langle v|_Y$, $\tau\in\mathcal{T}(A)$, to $\sum_x \mathrm{tr}[M_x\tau]\ |r(x,v)\rangle\langle r(x,v)|_Z\otimes|\phi(a(x,v))\rangle\langle\phi(a(x,v))|_{\widehat{C}}\otimes |a(x,v)\rangle\langle a(x,v)|_I$. Hence

$$\bar{\mathsf{G}}\big((\mathrm{id}_A\otimes\mathcal{N}_G)(\omega)\big)=\sum_{x,v}|r(x,v)\rangle\langle r(x,v)|_Z\otimes|\phi(a(x,v))\rangle\langle\phi(a(x,v))|_{\widehat{C}}\otimes|a(x,v)\rangle\langle a(x,v)|_I\ \mathrm{tr}\left[(M_x\otimes G_v)\,\omega\right.$$

which for (B,M) is (3.6) and for (Q,F) is (3.8) at trivial E, extended from $S_{\le}(AQ)$ to $\mathcal{T}(AQ)$ by linearity, the finite-dimensional AQ being spanned by its states. $\bar{\mathsf{G}}$ is a channel, a measurement of A followed by a classical map, and it touches neither Q nor B; that is the hypothesis of Definition 2.1 for both maps.

Step 3 (the substitution). For (3.7) it therefore suffices, by (3.9) and (3.8), that for all x and v

$$\mathrm{tr}_{AB}\big[(M_x\otimes M_v\otimes\mathbb{1}_E)\,\omega\big]=\mathrm{tr}_{AQ}\big[(M_x\otimes F_v\otimes\mathbb{1}_E)\,(\mathrm{id}_A\otimes\Lambda\otimes\mathrm{id}_E)(\omega)\big]\quad(3.10)$$

as operators on E, the state $(\mathrm{id}_A\otimes\Lambda\otimes\mathrm{id}_E)(\omega)$ lying in $S_{\le}(AQE)$. Test both sides against an arbitrary $X\in\mathcal{L}(E)$. By (1.1) applied to $\mathrm{id}_A\otimes\Lambda\otimes\mathrm{id}_E$ and Lemma 1.2 (ii),

$$\mathrm{tr}\left[(M_x\otimes F_v\otimes X)(\mathrm{id}_A\otimes\Lambda\otimes\mathrm{id}_E)(\omega)\right]=\mathrm{tr}\left[\big(M_x\otimes\Lambda^\dagger(F_v)\otimes X\big)\,\omega\right]=\mathrm{tr}\left[(M_x\otimes M_v\otimes X)\,\omega\right],$$

and operators on the finite-dimensional E separate, so the two partial traces agree. Substituting into (3.9) and (3.8) gives (3.7). □

The reduction of Corollary 3.3 and Proposition 3.4 is published for narrower protocol models, with a restricted or an unrestricted squashed attack set, and in a non-adaptive single-channel version. No proof here uses those statements.

Remark 3.5 (Published forms of the sequential reduction). The published statement of the reduction of Corollary 3.3 and Proposition 3.4, for a narrower protocol model, is [8, Lemma 9.3, Eqs. (146)–(148)]; the case in which the squashed attack set is left unrestricted is its Rem. 9.1, and the non-adaptive, single-channel version with an unrestricted set is [6, Lemma 6]. Neither is used in the proofs of Corollary 3.3 and Proposition 3.4.

newly derived · accepted

The factorization holds verbatim for round-dependent squashers, for a private per-round setting squashed by one common channel, and for a correlated n-round device. A setting-dependent squashing channel does not fit, and hands the setting register to the adversary.

Remark 3.6 (Variants of the factorization). The proof of Theorem 3.1 uses only linearity and the single-round identity, so it applies verbatim to round-dependent squashers ($M^{(j)}\preceq_{\Lambda_j}F^{(j)}$, composite channel $\bigotimes_j\Lambda_j$); to a private per-round setting drawn from a family $\{M^{(s)}\}_s$ squashed by a *common* channel with $\Lambda^\dagger(F_v^{(s)})=M_v^{(s)}$, where the controlled measurement channel factorizes through $\mathrm{id}_S\otimes\Lambda$; and to a genuinely correlated n-round device, where a joint POVM $\{M_{\boldsymbol{v}}\}$ on B^n and a channel $\Lambda^{(n)}$ with $\Lambda^{(n)\dagger}(F_{v_1}\otimes\cdots\otimes F_{v_n})=M_{\boldsymbol{v}}$ give the same conclusion. A device

with detector memory across rounds, which Assumption 13.2 excludes, is a joint POVM on B^n, so where a channel $\Lambda^{(n)}$ as above exists, (3.2) holds for it at a fixed input; nothing is asserted of it in the round model. What Assumption 13.2 fixes is that one physical POVM describes every round, with no memory across rounds; that the declared inputs describe one round and are applied to all n, with one squashing channel for every round, is the case the engine emits (Requirement 14.5). A setting-*dependent* family $\Lambda^{(s)}$ does *not* fit this scheme and yields only the weaker model in which the setting register is handed to the adversary; that is why the probe setting enters this contract as a joint outcome under one common channel (Definition 4.4, Lemma 4.3 (ii)) and never as a setting-dependent squasher. If the setting is not drawn as Assumption 13.5 says, the joint family $\{q_t M_{y|t}\}$ is not a POVM with a common squashing channel and the setting register would have to be handed to the adversary; a probe setting the adversary can predict or influence also breaks the countermeasure it models, outside this contract.

Uses Thm. 3.1 · *Assumes* A-ROUND-SQ · *Used by* Rem. 3.7, Req. 14.5 · newly derived · accepted

The output-state equality holds for an arbitrary adversary input and an arbitrary attack sequence, so it covers an adversary entangling all n rounds. The tensor power is a hypothesis about Bob's device, not about the adversary.

Remark 3.7 (Coherent attacks covered by the output-state equality). Nothing in Theorem 3.1 or Corollary 3.3 refers to the structure of the adversary's input: ρ is an arbitrary state on $A^n B^n E$ and $(\mathcal{A}_j)_j$ an arbitrary sequence of channels, so the conclusion covers Eve entangling all n rounds and holding E correlated with everything. The tensor power in (3.1) is a hypothesis about Bob's *device* — the same POVM every round, with no memory between rounds (Assumptions 13.1, 13.2) — and is itself relaxed in Remark 3.6. It is not a hypothesis about the adversary.

Uses Thm. 3.1, Cor. 3.3, Rem. 3.6 · *Assumes* A-MEAS, A-ROUND-SQ · newly derived · accepted

Correctness and secrecy in the composable sense are functions of the output state, so equality of output states transfers them. Completeness compares the protocol against an honest reference channel, is not of that form and is not transferred.

Remark 3.8 (Correctness, secrecy and completeness as criteria). Correctness and secrecy in the composable sense — $\Pr[K_A \neq K_B] \leq \varepsilon_{\mathrm{cor}}$ and $(1-p_{\mathrm{abort}})\, D(\rho_{AE}, \pi_A \otimes \rho_E) \leq \varepsilon_{\mathrm{sec}}$ of [7, Eqs. (13)–(14)], in which π_A is the *fully mixed key state* (that source writes τ_A for it; it is *not* the trusted marginal τ_A of Definition 6.5, and the two never occur in one formula here) — are both functions of the output state in the sense of Definition 2.2, because the ideal object each compares against is a fixed channel applied to the protocol's own output. What is *not* of that form, and is therefore not transferred by Theorem 3.11, is a statement comparing the protocol against an *honest reference channel*: the completeness criterion of [7, Eq. (9)] — the second construction condition, evaluated against a blank adversarial interface, from which that paper derives robustness at its [7, Eq. (15)] — and completeness generally. Those are the concern of a different module, and Remark 3.9 records why no gap opens.

Uses Def. 2.2 · *Assumes* A-MEAS · newly derived · accepted

A completeness criterion compares the protocol against an honest reference channel and is outside the form that transfers. No gap opens: the squashed model reproduces the physical statistics exactly, so a completeness statement computed in one model holds in the other.

Remark 3.9 (Completeness statements in the physical and squashed models). A criterion comparing the protocol against an *honest reference channel* — "on the honest channel the protocol aborts with probability at most ε", the completeness criterion of [7, Eq. (9)] (robustness is its phrasing at that paper's Eq. (15)) — is not of the form (2.2) evaluated on an adversarial input, and Theorem 3.1 says nothing about it. No gap opens: the squashed model reproduces the physical outcome statistics *exactly* (Lemma 1.4 (ii)), so the honest distribution on $\mathcal{V}$ is the same object in both models, and a completeness statement computed in one is literally the statement in the other. Completeness budgets belong to the statistics module, and nothing in this contract carries one.

Uses Lem. 1.4, Def. 2.2, Thm. 3.1 · newly derived · accepted

A squashed state is reachable when it is the squashed image of a physical input, and a family of squashed states, or a constraint, is sound when it contains every reachable state. Soundness is what keeps a constrained analysis from excluding a physical attack.

Theorem 3.1 leaves exactly one thing to check, and the following definition names it: the squashed analysis must run over a set of inputs containing the squashed images of the physical ones. Everything a constrained squashed analysis can legitimately assume is the supply of such sets, and every way such an analysis can fail is a constraint some *reachable* state violates. Both notions depend on the channel and on the physical inputs; Remark 3.15 says what has to be watched when these are not fixed.

Definition 3.10 (Reachable states and sound constraints)**.** Let $M \preceq_\Lambda F$ and let $S^{\mathrm{ph}} = (S^{\mathrm{ph}}_E)_E$ be a family of *physical inputs*, one set $S^{\mathrm{ph}}_E \subseteq S_=(A^n B^n E)$ per admissible adversary system E. A state $\sigma \in S_=(A^n Q^n E)$ is *reachable* if

$$\sigma = \big(\mathrm{id}_{A^n} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E\big)(\rho) \qquad \text{for some } \rho \in S^{\mathrm{ph}}_E, \tag{3.11}$$

and S^{rch}_E denotes the set of reachable states. A family $C = (C_E)_E$ of subsets $C_E \subseteq S_=(A^n Q^n E)$, and any constraint whose solution set it is, is *sound* if $S^{\mathrm{rch}}_E \subseteq C_E$ for every admissible E. Both notions depend on Λ and on S^{ph}, which are suppressed in the notation and fixed throughout any one analysis.

Uses Thm. 3.1 · *Used by* Cor. 3.12, Cor. 3.13, Cor. 4.7, Def. 7.1, Cor. 21.6, Def. 23.1 · newly derived · accepted

If the analysed squashed inputs contain the squashed image of every physical input, a bound ε on any criterion of the output state of the squashed protocol holds for the physical protocol with the same ε. No property of the criterion is used.

With the output state unchanged (Theorem 3.1), security of the squashed protocol carries over as soon as the squashed analysis runs over enough inputs. Theorem 8.6 composes the receiver-side reduction with the source map in trusted-source mode.

Theorem 3.11 (Security transfer from a constrained squashed model). Let $M \preceq_\Lambda F$ and let $\mathcal{E}_M, \mathcal{E}_F$ be as in Definition 2.1. For every admissible adversary system E let $S_E^{\mathrm{ph}} \subseteq S_=(A^n B^n E)$ and $S_E^{\mathrm{sq}} \subseteq S_=(A^n Q^n E)$ satisfy

$$\big(\mathrm{id}_{A^n} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E\big)\big(S_E^{\mathrm{ph}}\big) \;\subseteq\; S_E^{\mathrm{sq}}. \tag{3.12}$$

If $\mathsf{sec}(\mathcal{E}_F, \sigma) \le \varepsilon$ for all $\sigma \in S_E^{\mathrm{sq}}$ and all E, then $\mathsf{sec}(\mathcal{E}_M, \rho) \le \varepsilon$ for all $\rho \in S_E^{\mathrm{ph}}$ and all E — with the *same* ε, and simultaneously for every security criterion in the sense of Definition 2.2.

Uses Def. 2.1, Def. 2.2, Def. 2.3, Lem. 2.4, Thm. 3.1 · *Used by* Rem. 2.5, Cor. 3.12, Rem. 3.14, Rem. 3.15, Thm. 7.4, Thm. 11.2, Cor. 12.1, Rem. 19.6, Cor. 24.1, protocol-compiler/req:link-row-provenance · newly derived · accepted · *Oracles* registry_reductions.py

Proof. Fix an admissible adversary system E and put $r(\rho) := (\mathrm{id}_{A^n} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E)(\rho)$. Let P be the presentation whose strategies are the states $\rho \in S_E^{\mathrm{ph}}$ — a strategy here is an input state and not a channel, which Definition 2.3 allows and says — with $\mathsf{out}(\rho) = (\mathcal{E}_M \otimes \mathrm{id}_E)(\rho)$, and P' the presentation with strategies S_E^{sq} and $\mathsf{out}'(\sigma) = (\mathcal{E}_F \otimes \mathrm{id}_E)(\sigma)$; both have the output registers of K together with E, which is condition (R1). The hypothesis (3.12) says exactly that r takes its values in S_E^{sq}, which is condition (R3), and (3.2) of Theorem 3.1 is (2.3) for r. So r is an exact operational reduction from P to P', and Lemma 2.4 (i), applied to the criterion f_E at this E, gives $\mathsf{sec}(\mathcal{E}_M, \rho) = \mathsf{sec}(\mathcal{E}_F, r(\rho)) \le \varepsilon$ for every $\rho \in S_E^{\mathrm{ph}}$, with no property of f_E used. The quantification over E is one presentation pair and one reduction per admissible E (Definition 2.3, closing paragraph). □

Analysed squashed inputs satisfy the inclusion of Theorem 3.11 exactly when they form a sound family, and any intersection of sound families for one channel and one physical input family is sound. Enlarging a sound family costs tightness, never soundness.

Corollary 3.12 (Imposing and intersecting sound constraints)**.** Let $M \preceq_\Lambda F$ and fix a family S^{ph} of physical inputs.

(i) A family S^{sq} satisfies (3.12) if and only if it is sound.

(ii) An arbitrary intersection of sound families is sound.

(iii) Consequently, if $(C^{(i)})_{i\in I}$ is any collection of sound families and $\mathsf{sec}(\mathcal{E}_F, \sigma) \le \varepsilon$ for every admissible E and every $\sigma \in \bigcap_{i\in I} C_E^{(i)}$, then $\mathsf{sec}(\mathcal{E}_M, \rho) \le \varepsilon$ for every admissible E and every $\rho \in S_E^{\mathrm{ph}}$, with the same ε and for every criterion of the form (2.2).

(iv) The tightest sound family is S^{rch} itself and the loosest is $S_E^{\mathrm{sq}} = S_=(A^nQ^nE)$; enlarging a sound family costs tightness and never soundness.

Uses Def. 2.2, Def. 2.3, Lem. 2.4, Def. 3.10, Thm. 3.11 · *Used by* Rem. 3.15, Cor. 4.7 · newly derived · accepted

Proof. (i) The left-hand side of (3.12) is S_E^{rch} by (3.11), so the inclusion says exactly that S_E^{sq} contains every reachable state — equivalently, that the strategy map $r\colon \rho \mapsto (\mathrm{id}_{A^n} \otimes \Lambda^{\otimes n} \otimes \mathrm{id}_E)(\rho)$ of the proof of Theorem 3.11, the one belonging to *this* Λ and *this* S^{ph}, has its image in S_E^{sq}, which is condition (R3) of Definition 2.3. (ii) A reachable state lies in every member of the collection, hence in the intersection. (iii) By (i), soundness of each $C^{(i)}$ says that *one and the same* map r has its image in $C_E^{(i)}$, for every i and every admissible E; Lemma 2.4 (iv) therefore makes r an exact operational reduction into $\bigcap_i C_E^{(i)}$, and its part (i) gives the bound with the same ε for every criterion. The hypothesis is one map and not one map per i, and this matters: part (iv) of that lemma exhibits analysed sets, each of them reached by some exact operational reduction, whose intersection is empty and over which every bound holds vacuously; the concrete form of that failure here is two channels realizing one squashing relation, which is why this corollary fixes one Λ and one S^{ph} (Remark 3.15). (iv) Immediate from (i); the loosest family satisfies (3.12) trivially, which is the unconstrained squashing statement. □

On the n-round relation, a transported operator tested on a squashed image gives what its pull-back gives on the physical input. Certified operator facts, and bounds assumed on the physical input family, therefore give sound constraints on the squashed states.

Part (iii) of Corollary 3.12 is what a constrained analysis runs on: soundness is established once per constraint and never again for the combination, so constraints from unrelated sources — a device property, a calibration bound, an assumption on Eve's input — may be imposed simultaneously, *provided they are sound for one and the same squashing channel* (Remark 3.15). What a squashed analysis can do is therefore fixed by the supply of sound constraints, and that supply is Corollary 1.7 read on the n-round relation.

Corollary 3.13 (Sound constraints from transported operators)**.** Let $(M;O) \preceq_\Lambda (F;\hat{O})$, fix a family S^{ph} of physical inputs and write $R := A^nE$. Let $\hat{O}^{(n)} \in \mathcal{L}(Q^n)$ be Hermitian and let $O^{(n)} := (\Lambda^\dagger)^{\otimes n}(\hat{O}^{(n)})$ be its pull-back.

Then

$$\operatorname{tr}_{Q^n}\left[(\hat{O}^{(n)} \otimes \mathbb{1}_R)\,\sigma\right] = \operatorname{tr}_{B^n}\left[(O^{(n)} \otimes \mathbb{1}_R)\,\rho\right] \tag{3.13}$$

for every $\rho \in S_E^{\mathrm{ph}}$ and its image σ, and each of the following families of squashed states is sound:

(i) *(certified)* for each certificate case of Corollary 1.7 — identity, positivity, spectral bounds — read with $B^n, Q^n, \Lambda^{\otimes n}, R$ in place of B, Q, Λ, R, the family of squashed states satisfying its conclusion;

(ii) *(assumed)* $\{\sigma : \operatorname{tr}\left[(\hat{O}^{(n)} \otimes X)\sigma\right] \bowtie c\}$ for a fixed $\bowtie \in \{\leq, =, \geq\}$, a fixed $X \in \mathcal{L}(R)$, Hermitian unless $\bowtie$ is $=$, and a fixed c, provided $\operatorname{tr}\left[(O^{(n)} \otimes X)\rho\right] \bowtie c$ holds for every $\rho \in S_E^{\mathrm{ph}}$.

Operators with a prescribed pull-back are supplied round by round: if $\hat{O}^{(n)}$ is $\hat{O}_k$ on Q_j and $\mathbb{1}_Q$ on the remaining rounds, then $O^{(n)}$ is O_k on B_j and $\mathbb{1}_B$ elsewhere; and a product $\hat{O}_{k_1} \otimes \cdots \otimes \hat{O}_{k_n}$ pulls back to $O_{k_1} \otimes \cdots \otimes O_{k_n}$.

Uses Lem. 1.2, Cor. 1.7, Def. 1.8, Def. 3.10 · *Used by* Rem. 3.15, Rem. 5.4, Thm. 6.7, Thm. 7.4, Req. 16.1, Rem. 19.9 · newly derived · accepted

Proof. $\Lambda^{\otimes n}$ is a channel $\mathcal{T}(B^n) \to \mathcal{T}(Q^n)$ whose adjoint is $(\Lambda^\dagger)^{\otimes n}$ by Lemma 1.2 (ii). Reordering $A^n B^n E$ as $B^n R$, the channel of (3.11) is $\Lambda^{\otimes n} \otimes \mathrm{id}_R$, so Corollary 1.7 applies with $B^n, Q^n, \Lambda^{\otimes n}, R$ in place of B, Q, Λ, R and with the single carried pair $(O^{(n)}, \hat{O}^{(n)})$. Equation (3.13) is its (1.3), and each conclusion of Corollary 1.7 (i)–(iii) holds on *every* reachable σ, which is soundness of the families in (i). For (ii), test (1.3) against X and bound the right-hand side by hypothesis. For the round-by-round supply, $(\Lambda^\dagger)^{\otimes n}$ acts factorwise and $\Lambda^\dagger$ is unital. □

Imposed as an operator identity on a target that fails it, the physical class-marginal identity would exclude reachable states and void Theorem 3.11. Its sound squashed form is the expectation identity, whose certificate is available exactly where the reduction instance discharges it.

Case (i) of Corollary 3.13 carries the identity and inequality certificates: they hold on every reachable state whatever the adversary is allowed to send, so the resulting families are sound for *every* choice of S^{ph} and may be adjoined to any analysis. Its case (ii) is an assumption about the input and is sound only for the S^{ph} it was assumed on. The following remark describes the standard way a constraint fails to be sound at all.

Remark 3.14 (The standard way to make a squashed constraint unsound). Equation (3.12) is the whole of soundness, and the standard way to violate it is to transplant an operator identity that holds in the *physical* model. Take a private setting of a measuring party's. Where the physical model *is* a joint family (Definition 4.4 (b)) — one branch, B the system the conditional families live on — the physical POVM satisfies the operator identity

$$\sum_{v \in \mathcal{V}_t} M_v \;=\; q_t\,\mathbb{1}_B \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}}, \tag{3.14}$$

which is the class-marginal identity (4.3): the weight q_t is that party's own and (4.2) *builds* $M_{(y,t)} = q_t \widetilde{M}_{y|t}$ out of a conditional family that is complete at each fixed t (Lemma 4.3 (ii), a statement about conditional families).

When (3.14) *is available, and when it is not.* It is a theorem of the construction where the model is a joint family outright, and it is a property of a reduction instance's model only where the instance discharges it (Definition 10.1 (I4)). An instance built by the branching rule inherits it from its branches and from nothing else (Theorem 10.4, Corollary 10.5): a branch whose POVM is an arbitrary one — the flag instance of an uncharacterised branch, Corollary 10.3, on which nothing is assumed and which may be undeclared — contributes to the class sum merely *some* operator between 0 and its branch weight, and the identity on B is then neither a theorem of this contract nor a validated datum unless the branch's declaration establishes it. Whether a declaration can establish it for such a branch is a question about what the declaration presents (Remark 11.9). The identity is discharged only where every branch establishes it; an undeclared branch, a branch missing an element and a branch that does not establish the identity alike leave it undischarged, and the entry that rests on it is then refused and never emitted (Requirement 16.5). Where it is not discharged, (3.14) may still hold of the device, but nothing declared establishes it. The squashed operators, where the identity is discharged, satisfy $\Lambda^\dagger\big(\sum_{v\in\mathcal{V}_t} F_v\big) = q_t\,\mathbb{1}_B$, while $\sum_{v\in\mathcal{V}_t} F_v \neq q_t\,\mathbb{1}_Q$ on every scalar sector of a flag-shaped target whose values do not sum to q_t over $\mathcal{V}_t$. The sound squashed constraint is therefore the *expectation* identity $\mathrm{tr}\big[\sum_{v\in\mathcal{V}_t} F_v\,\sigma\big] = q_t$ on reachable σ — a pull-back constraint in the sense of Corollary 1.7 (i) — and *not* the operator identity on Q. Imposing the latter would exclude reachable states and void Theorem 3.11, which is a security break rather than a rate loss.

The two forms, and the certificate of the expectation form. The operator form is refused because it is unsound (Requirement 16.2). The expectation form is sound but is not emitted by default either: the default constraint list is empty on the flag-state submodule's encoded route and carries an instance's own entries, at their constants, on its licensed route (Corollary 11.3 (iii)), and adding an entry on the encoded route would force a licence there and contradict that route's definition. The sound content of the physical identity is instead carried by the target elements themselves, which reproduce the statistics exactly (Lemma 1.4 (ii)). The expectation form remains available on request *under the condition that discharges its certificate, and not otherwise.* The entry is $\hat{O} = \sum_{v\in\mathcal{V}_t} F_v$, $\bowtie = =$, $c = q_t$ of Theorem 6.7. Its hypothesis (6.9) needs two steps, and only the first is unconditional: by the relation and linearity, $\Lambda^\dagger\big(\sum_{v\in\mathcal{V}_t} F_v\big) = \sum_{v\in\mathcal{V}_t} M_v$ always (the fourth consequence of Definition 10.1); that this in turn equals $q_t\mathbb{1}_B$ is (3.14), which holds by construction where the model is a joint family and otherwise exactly where the instance discharges it. *The certificate is therefore available exactly when the instance's class-marginal flag says the class-marginal identity is discharged*; for a model that is a joint family outright the flag is set by Lemma 4.3 (ii) alone. The constant q_t is declared before the run in either case (Assumption 13.7). The single equality entry at $c = q_t$ is what Corollary 11.3 (ii) emits, the certificate being discharged for the emitted numbers themselves. A request for the entry that meets the condition obtains it, under the licence a non-empty constraint list requires; a request that does not meet the condition is refused (Requirement 16.5), because emitting the entry there would impose $\mathrm{tr}\big[\sum_{v\in\mathcal{V}_t} F_v\,\sigma\big] = q_t$ on states an uncharacterised branch can reach and for which it is false — the security break this remark is about, arrived

at from the other direction.

Uses Lem. 1.4, Cor. 1.7, Thm. 3.11, Lem. 4.3, Def. 4.4, Def. 10.1 · *Assumes* A-CONST · *Used by* Req. 16.2, Req. 16.5, Rem. 19.6 · newly derived · accepted

Channels realizing one squashing relation can have different reachable sets, so constraints sound for different channels may not be combined. A constraint built from an operator in the carried tuple has the same pull-back for every realizing channel.

Remark 3.15 (Soundness relative to the channel, fixed by the carried tuple). The relation $M \preceq F$ asserts that *some* squashing channel exists, and different channels realizing it have different reachable sets: by Lemma 1.4, two channels realizing $(M;O) \preceq (F;\hat{O})$ agree only on $V \coloneqq \operatorname{span}\big(\{F_v\} \cup \{\hat{O}_k\}\big)$, and nothing forces $S_E^{\mathrm{rch}}(\Lambda) = S_E^{\mathrm{rch}}(\Lambda')$. Three consequences follow. *(a)* Theorem 3.11 needs only one channel, and every realizing channel gives a valid conclusion, since Λ does not appear in it. *(b)* There is something to choose between them for *tightness*: the reachable sets are in general incomparable, so which channel is used decides which constraints are available. *(c)* Corollary 3.12 (ii)–(iii) therefore require a *common* channel: constraints sound for Λ and constraints sound for Λ' may have an intersection excluding every state reachable under either, and imposing them together would void Theorem 3.11 exactly as an unsound constraint does.

What removes the hazard is the tuple of carried operators of Definition 1.3. If $\hat{O}^{(n)}$ lies in the span of the n-fold products of elements of V, then $(\Lambda^\dagger)^{\otimes n}(\hat{O}^{(n)}) = (\Lambda'^\dagger)^{\otimes n}(\hat{O}^{(n)})$, since the adjoints act factorwise and agree on V; the pull-back, and with it every constraint Corollary 3.13 builds from it, is then the same for every realizing channel. This is what the tuple is for: listing an operator in the relation promotes a constraint that would otherwise be tied to one construction into one that belongs to the relation itself, which is why the emitted constraints are built from the tuple (Requirement 14.3).

Uses Def. 1.3, Lem. 1.4, Thm. 3.11, Cor. 3.12, Cor. 3.13 · *Used by* Req. 14.2, Req. 14.3, Rem. 19.6 · newly derived · accepted

4 Operations on squashing relations

Squashing relations compose along a pre-order and survive common classical post-processing, a privately drawn setting under one common channel, mixtures and tensor products. Pinching onto the sectors of a block-diagonal target is itself a squashing; where the carried operators are block diagonal too, the squashed analysis may be restricted to block-diagonal states for the pinched channel (Corollary 4.7). A private setting with declared class weights enters as part of a joint outcome, the weight inside the element, and the joint family is defined once here (Definition 4.4). Two necessary conditions can show a declared relation impossible, and both are decided exactly on the declared numbers (Lemma 4.9, Remark 4.10).

Every POVM squashes to itself through the identity channel, and two successive squashing relations compose into one, their carried operator tuples included. A chain of squashers is therefore again a squashing relation that keeps its constraint information.

This proposition, Lemma 4.3 and Propositions 4.6 and 4.8 are the operations that build squashing relations out of squashing relations. With the branching rule of Lemma 5.3 they generate every construction this contract emits, so a squasher submodule assembles its constructions from them rather than proving each afresh. Where a rule also produces a constraint it is stated for one round, and Corollary 3.13 lifts every such statement to the n-round soundness of Definition 3.10.

Proposition 4.1 (The squashing relation as a pre-order)**.** $M \preceq_{\mathrm{id}} M$, and $M \preceq_{\Lambda_1} F_1$ together with $F_1 \preceq_{\Lambda_2} F_2$ imply $M \preceq_{\Lambda_2 \circ \Lambda_1} F_2$. The same holds for the relation with carried operators, the operator tuples composing the same way: $(M;O) \preceq_{\Lambda_1} (F_1;\hat{O}^1)$ and $(F_1;\hat{O}^1) \preceq_{\Lambda_2} (F_2;\hat{O}^2)$ give $(M;O) \preceq_{\Lambda_2\circ\Lambda_1} (F_2;\hat{O}^2)$.

Uses Lem. 1.2, Def. 1.3 · *Used by* Cor. 4.7, Thm. 7.4, Ex. 7.5 · newly derived · accepted

Proof. The identity is a channel and $\mathrm{id}^\dagger = \mathrm{id}$. Compositions of channels are channels and $(\Lambda_2 \circ \Lambda_1)^\dagger = \Lambda_1^\dagger \circ \Lambda_2^\dagger$, which is (1.1) applied twice. Hence $\Lambda_1^\dagger\big(\Lambda_2^\dagger(F_{2,v})\big) = \Lambda_1^\dagger(F_{1,v}) = M_v$, and identically for each $\hat{O}_k$. □

If one target squashes to a cleaner one, its worst-case criterion over a family of inputs equals the cleaner target's over the family's image under the connecting channel, and is at most the cleaner target's over any larger family. Moving up the pre-order can only raise the worst case.

Composing squashers costs nothing in soundness; the corollary says in which direction it costs tightness. Moving *up* the pre-order, to a cleaner target, hands the adversary more.

Corollary 4.2 (Monotonicity of the worst-case criterion along the pre-order)**.** For a POVM G and a family S of inputs write $\varepsilon^*(G,S) := \sup_E \sup_{\sigma\in S_E} \mathsf{sec}(\mathcal{E}_G,\sigma)$. Let $F_2 \preceq_{\Lambda_{12}} F_1$ and let both squashed protocols use the same processing $\bar{\mathsf{G}}_j$ and K. Then for every family S,

$$\varepsilon^*(F_2,S) = \varepsilon^*\big(F_1,(\mathrm{id}\otimes\Lambda_{12}^{\otimes n}\otimes\mathrm{id}_E)(S)\big) \;\le\; \varepsilon^*(F_1,S_1) \quad \text{for every } S_1 \supseteq (\mathrm{id}\otimes\Lambda_{12}^{\otimes n}\otimes\mathrm{id}_E)(S).$$

Uses Thm. 3.1 · *Used by* Rem. 5.12, Rem. 19.6, Prop. 19.7, Rem. 19.8, Rem. 20.10 · newly derived · accepted

Proof. The equality is Theorem 3.1 applied with (F_2,F_1) in place of (M,F), pointwise in σ and then in the suprema; the inequality is monotonicity of a supremum in the set. □

A squashing relation survives a stochastic post-processing of both sides, and conditional relations with one common channel give relations for the weighted joint family and for its mixture. A private setting therefore enters as part of the outcome and opens no block of the target space.

Linearity of $\Lambda^\dagger$ lets both sides of a relation be reprocessed together. It does so in three ways, which differ only in what happens to a classical variable: it is post-processed, kept as an outcome, or averaged out. A comma in a subscript denotes a joint outcome and a bar a conditional POVM (spec/notation.md#26-comma-vs-bar-on-povm-subscripts).

Lemma 4.3 (Linearity: post-processing, private settings and mixtures)**.** Let $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel.

(i) *(post-processing)* If $M \preceq_\Lambda F$ and $P(z|v)$ is a stochastic matrix from $\mathcal{V}$ to a finite alphabet $\mathcal{U}$, then the post-processed families $(PM)_z := \sum_v P(z|v)M_v$ and $(PF)_z := \sum_v P(z|v)F_v$ are POVMs and $PM \preceq_\Lambda PF$.

(ii) *(a private classical variable)* For each $t \in \mathcal{T}_{\mathrm{set}}$ let $\{M_{y|t}\}_{y\in\mathcal{Y}}$ be a *conditional* POVM on B — complete for each fixed t, $\sum_y M_{y|t} = \mathbb{1}_B$ — and let $\{F_{y|t}\}_{y\in\mathcal{Y}}$ be conditional POVMs on the same Q with $M_{|t} \preceq_\Lambda F_{|t}$ for *one common* Λ. Let $(q_t)_{t\in\mathcal{T}_{\mathrm{set}}}$ be a probability distribution. Then, over the *joint* alphabet $\mathcal{V} = \mathcal{Y} \times \mathcal{T}_{\mathrm{set}}$,

$$\{q_t M_{y|t}\}_{(y,t)} \preceq_\Lambda \{q_t F_{y|t}\}_{(y,t)}, \tag{4.1}$$

which in the comma notation reads $\{M_{y,t}\} \preceq_\Lambda \{F_{y,t}\}$ with $M_{y,t} = q_t M_{y|t}$ and $F_{y,t} = q_t F_{y|t}$.

(iii) *(mixtures)* In the setting of (ii), $\{\sum_t q_t M_{y|t}\}_y \preceq_\Lambda \{\sum_t q_t F_{y|t}\}_y$.

Uses Def. 1.3 · *Used by* Rem. 3.14, Def. 4.4, Rem. 4.5, Lem. 5.3, Rem. 5.5, Thm. 10.4, Def. 20.3, Rem. 20.9, Prop. 21.2, protocol-compiler/req:link-row-provenance · newly derived · accepted

Proof. (i) $(PM)_z \succeq 0$ and $\sum_z (PM)_z = \sum_v \big(\sum_z P(z|v)\big) M_v = \sum_v M_v = \mathbb{1}_B$, likewise for PF; and $\Lambda^\dagger\big((PF)_z\big) = \sum_v P(z|v)\Lambda^\dagger(F_v) = \sum_v P(z|v)M_v = (PM)_z$. (ii) $\Lambda^\dagger\big(q_t F_{y|t}\big) = q_t \Lambda^\dagger(F_{y|t}) = q_t M_{y|t}$ by linearity and the relation at each fixed t; and both joint families are POVMs since $\sum_{y,t} q_t M_{y|t} = \sum_t q_t\big(\sum_y M_{y|t}\big) = \sum_t q_t \mathbb{1}_B = \mathbb{1}_B$, using completeness of each conditional POVM, and likewise for F. (iii) is (ii) followed by (i) for the stochastic matrix that forgets t. □

A private setting with exact class weights joins the outcome in one joint alphabet, partitioned into one class per setting, and its conditional POVMs form one joint family with each weight inside the elements. The family satisfies the class-marginal identity and opens no target block.

A receiver with a private per-round setting, such as a probe setting or an internal coin of known bias, is described by one conditional POVM per value of the setting, and the analysis reads the pair (outcome, setting) as one joint outcome with the setting's weight inside the element (`spec/notation.md#26-comma-vs-bar-on-povm-subscripts`). The definition names that object once and in two layers. The combinatorial layer is the joint alphabet with its classes and their weights; the abstract reduction instance of Definition 10.1 is stated over it, and the branching instance of Theorem 10.4 extends it by a branch label. The POVM layer is what a device with a private setting hands this module. The weights are the measuring party's own (Assumption 13.5) and enter only through the statistics of the joint outcome (Assumption 13.6). The identity (4.3) is a fact about the *physical* model; which squashed form of it is sound is the subject of Remark 3.14, and why the joint alphabet is built *before* the first squashing link, and what may still be relabelled after one, is Remark 4.5.

Definition 4.4 (Joint alphabet, classes and the joint family of a private setting)**.** Let $\mathcal{Y}$ be a finite non-empty ordered *outcome alphabet*, let $\mathcal{T}_{\mathrm{set}}$ be a finite non-empty ordered set of *settings*, and let $(q_t)_{t\in\mathcal{T}_{\mathrm{set}}}$ be *class weights*, each a declared dyadic rational on the declared lattice (Definition 9.1 (b)), with $q_t > 0$ and $\sum_{t\in\mathcal{T}_{\mathrm{set}}} q_t = 1$ *exactly*.

(a) *(the joint alphabet, its classes and weights)* The *joint alphabet* is $\mathcal{V} := \mathcal{Y} \times \mathcal{T}_{\mathrm{set}}$, ordered with y varying fastest, which is its declared order; its *classes* are $\mathcal{V}_t := \mathcal{Y} \times \{t\}$, $t \in \mathcal{T}_{\mathrm{set}}$, a partition $\mathcal{V} = \bigsqcup_{t\in\mathcal{T}_{\mathrm{set}}} \mathcal{V}_t$, each class ordered as $\mathcal{Y}$ is; and the triple $\big(\mathcal{V}, \{\mathcal{V}_t\}_{t\in\mathcal{T}_{\mathrm{set}}}, \{q_t\}_{t\in\mathcal{T}_{\mathrm{set}}}\big)$ is the *joint alphabet with classes and weights*. When no private setting is declared, $\mathcal{T}_{\mathrm{set}}$ is a singleton $\{*\}$ with $q_* = 1$, and $\mathcal{V} = \mathcal{Y} \times \{*\}$ is identified with $\mathcal{Y}$: one class, of weight one.

(b) *(the joint family)* Let B be a system and, for each $t \in \mathcal{T}_{\mathrm{set}}$, let $\{\widetilde{M}_{y|t}\}_{y\in\mathcal{Y}}$ be a *conditional* POVM on B — complete at each fixed t, $\sum_y \widetilde{M}_{y|t} = \mathbb{1}_B$. Their *joint family* is

$$M_{(y,t)} \;:=\; q_t\, \widetilde{M}_{y|t}, \qquad (y,t) \in \mathcal{V}, \tag{4.2}$$

formed from the conditional elements and the weights by one product each, exactly. It is a POVM on B over $\mathcal{V}$ by Lemma 4.3 (ii) — $\sum_{(y,t)} q_t \widetilde{M}_{y|t} = \sum_t q_t \mathbb{1}_B = \mathbb{1}_B$ — and it satisfies the *class-marginal identity*

$$\sum_{v\in\mathcal{V}_t} M_v \;=\; q_t\, \mathbb{1}_B \qquad \text{for every } t \in \mathcal{T}_{\mathrm{set}}, \tag{4.3}$$

an operator identity on B, by completeness of the conditional POVM at t.

Two consequences are immediate and are used throughout. First, given conditional relations $\widetilde{M}_{|t} \preceq_\Lambda \widetilde{F}_{|t}$, $t \in \mathcal{T}_{\mathrm{set}}$, with one common channel Λ, Lemma 4.3 (ii) gives the relation $\{q_t \widetilde{M}_{y|t}\} \preceq_\Lambda \{q_t \widetilde{F}_{y|t}\}$ for the joint families: the setting never enters the target space, no block is opened for it, and the weights are carried by the target elements themselves. A relation for a joint family need not have a target of this form. Second, where every $\widetilde{M}_{y|t}$ is block diagonal for an orthogonal decomposition of $\mathcal{H}(B)$, so is every $M_{(y,t)}$, and (4.3) holds block by block; where such a joint family is itself a target, the decomposition being finite and the carried operators block diagonal for it as well, Corollary 4.7 applies to it.

Uses Lem. 4.3, Def. 9.1 · *Used by* Rem. 3.14, Rem. 4.5, Def. 10.1, Cor. 10.2, Cor. 10.3, Thm. 10.4, Def. 11.1, Thm. 11.2, Cor. 11.3, Def. 20.3, Prop. 20.4, Rem. 20.9, Rem. 20.10, Rem. 20.11, Prop. 21.2 · newly derived · accepted

A noise or coarse-graining map changes the physical family a squasher is built from, so it acts on the physical POVM before the first link. A stochastic relabelling of both sides by the private setting after a link keeps the channel, the sectors, every pull-back and every certificate.

Lemma 4.3 post-processes both sides of a relation at once (part (i)) and folds a private variable into the outcome (part (ii)). This remark says where in a chain of squashers each of the two may be used.

Remark 4.5 (Classical maps before a squasher, and relabelling after one). *A private variable inside the alphabet.* In Lemma 4.3 (ii) the private variable never enters Q: the target space is unchanged, no block is opened, and no constraint is produced, the weights q_t being carried by the target effects themselves. A variable the measuring party draws privately, such as a probe setting or an internal coin of known bias, is therefore squashed into the outcome alphabet at no cost *by (ii) alone*. That is the treatment the output-state test-statistics model of Assumption 13.6 requires, and the reason Definition 4.4 builds the joint alphabet before squashing rather than after. The no-cost statement concerns (ii) alone and need not survive a squasher composed with it: once (ii) is composed with a flag-shaped squasher, every value of the private variable opens flag sectors of its own. A variable the squasher's channel does not depend on, such as a test-round designation drawn against the setting, is then carried more cheaply *after* the link, by (i), than inside the alphabet before it (the relabelling (4.4)); only a variable the channel depends on, as a probe setting does through the conditional families, has to be in the alphabet before the first link. The branching rule of Lemma 5.3, by contrast, files its classical variable in Q and does cost something (Remark 5.5).

Part (i) applies a post-processing to *both* sides at once, and two uses of it differ: one is refused across a link and the other is not.

A noise or coarse-graining map changes the family a squasher is built from. A dark-count channel or a coarse-graining applied to the physical POVM produces a different physical family, and a squasher constructed for the original one is not a squasher to the same target for it: $M \preceq_\Lambda F$ gives $\Lambda^\dagger(F_v) = M_v$, which is not $(PM)_v$

in general. That is why such a map is applied to the physical POVM *before* the first squashing link (in the weight route of the flag-state submodule, the descended dark-count matrix acts before the joint family of Definition 4.4 is formed). Part (i) does move a stochastic P across a fixed squasher, but only together with the target: from $M \preceq_\Lambda F$ it gives $PM \preceq_\Lambda PF$, with the same channel and the target PF in place of F, whose elements, and with them its scalar values, its class sums and the entries built from them, are not the ones the chain was built and certified for. Where the map acts on a finer alphabet than the squasher's — a dark-count channel P on raw click patterns, applied before the declared coarse-graining K that produces the squasher's family KM_{raw} from the raw click-pattern POVM M_{raw} — moving it across the squasher asks for a stochastic P' on the coarse alphabet with $P'KM_{\mathrm{raw}} = KPM_{\mathrm{raw}}$, a linear feasibility problem in P' with no solution in general. For two detectors whose single-click patterns K merges into one letter, with dark-count probabilities $d_1 \neq d_2$ acting independently, the double-click element of KPM_{raw} carries the element of the pattern in which only detector 1 clicked with the coefficient d_2 and that of the pattern in which only detector 2 clicked with the coefficient d_1, while every element of $P'KM_{\mathrm{raw}}$ carries the two with one common coefficient; so no P' exists when the four elements of M_{raw} are linearly independent. This contract does neither: the order is fixed (Requirement 14.6).

A stochastic relabelling applied to both sides after a link is part (i) itself, and it is sound. Let $M \preceq_\Lambda F$ be a link's relation over the joint alphabet $\mathcal{V}$ of Definition 4.4, and let $P(f \,|\, t)$ be a stochastic matrix from the settings $\mathcal{T}_{\mathrm{set}}$ to a finite alphabet of values f, depending on the private component $t = t(v)$ of the outcome only and never on the device outcome y, since otherwise f would be inferred from an outcome. Then, over the support of P,

$$\big\{P(f \,|\, t(v))\, M_v\big\}_{(v,f)} \;\preceq_\Lambda\; \big\{P(f \,|\, t(v))\, F_v\big\}_{(v,f)} \tag{4.4}$$

by Lemma 4.3 (i) for the stochastic matrix from $\mathcal{V}$ to $\mathcal{V}\times\{f\}$ with entries $\delta_{vv'}P(f \,|\, t(v'))$ — every column sums to $\sum_f P(f|t(v')) = 1$ — with the *same* channel. Nothing of the link changes: Q, its sector map and its dimension are unchanged; every element $P(f|t(v))F_v$ is a scalar multiple of an existing one, so block diagonality holds by construction; every carried operator has the same pull-back; and every entry's certificate is untouched. On a scalar sector the value becomes $P(f|t(v))\phi_s(v)$, and no sector is indexed by f. Such a relabelling is therefore not a link in the sense of Definition 7.1: a link composes a new channel, while a relabelling changes the alphabet at both ends of an existing relation. This contract declares no kind, no register and no position in a chain (Definition 7.2) for it: it is the *consumer's* to apply, after the chain, by the same lemma, which the module that compiles the protocol may use directly.

Uses Lem. 4.3, Def. 4.4 · *Assumes* A-TSM · *Used by* Req. 14.6 · newly derived · accepted

Pinching onto finitely many orthogonal sectors is a self-adjoint channel, and a target whose elements are block diagonal for those sectors squashes to itself through it. The pinching can therefore follow a squasher whose target and carried operators are block diagonal.

The sector pinching below is the pinching of Definition "Operator conventions and pinching" (foundations/def:pinching) on the target Q: it deletes the coherences between sectors. The proposition proves the two facts the squashing constructions use, that it is a self-adjoint channel and that a block-diagonal target squashes to itself through it.

Proposition 4.6 (Sector pinching as a squashing)**.** Let $\{\Pi_s\}_{s\in\mathbb{S}}$ be finitely many mutually orthogonal projections on $\mathcal{H}(Q)$ with $\sum_s \Pi_s = \mathbb{1}_Q$, and let $\mathcal{P}_{\mathbb{S}}(\sigma) \coloneqq \sum_s \Pi_s \sigma \Pi_s$. Then $\mathcal{P}_{\mathbb{S}}$ is a channel with $\mathcal{P}_{\mathbb{S}}^\dagger = \mathcal{P}_{\mathbb{S}}$, and if every F_v is block diagonal, $F_v = \sum_s \Pi_s F_v \Pi_s$, then $F \preceq_{\mathcal{P}_{\mathbb{S}}} F$.

Uses Def. 1.3 · *Used by* Cor. 4.7, Ex. 7.5, Def. 10.1, Cor. 10.2, Cor. 10.3, Lem. 10.6, Thm. 11.5, Cor. 11.6 · newly derived · accepted

Proof. $\mathcal{P}_{\mathbb{S}}$ is CP as a sum of conjugations and trace preserving by cyclicity and $\sum_s \Pi_s = \mathbb{1}_Q$; it is self-adjoint because $\operatorname{tr}[X \sum_s \Pi_s \sigma \Pi_s] = \operatorname{tr}[(\sum_s \Pi_s X \Pi_s)\sigma]$. Block diagonality is $\mathcal{P}_{\mathbb{S}}(F_v) = F_v$, which is $\mathcal{P}_{\mathbb{S}}^\dagger(F_v) = F_v$. □

If every target element and carried operator is block diagonal for a sector decomposition, the squashing channel followed by the pinching realises the same relation and makes every reachable state block diagonal. For that channel the squashed analysis may be restricted to block-diagonal states.

Composing a squasher with the pinching of Proposition 4.6 costs nothing when the target and the carried operators are block diagonal, and it makes block diagonality of the squashed states a sound constraint.

Corollary 4.7 (Restriction to block-diagonal squashed states)**.** Let $(M;O) \preceq_\Lambda (F;\hat{O})$ and suppose every F_v and every $\hat{O}_k$ is block diagonal for a finite orthogonal decomposition $\{\Pi_s\}_{s\in\mathbb{S}}$ of $\mathcal{H}(Q)$. Then

(i) $(M;O) \preceq_{\mathcal{P}_{\mathbb{S}}\circ\Lambda} (F;\hat{O})$, with the same target, the same tuple and the same pull-backs;

(ii) every state reachable through $\mathcal{P}_{\mathbb{S}} \circ \Lambda$ is block diagonal in each round, so "σ is block diagonal in every round's sector decomposition" is a sound constraint for that channel (Definition 3.10);

(iii) consequently the squashed analysis may be restricted to block-diagonal states without excluding a reachable one.

Uses Def. 3.10, Cor. 3.12, Prop. 4.1, Prop. 4.6 · *Used by* Thm. 5.13, Def. 10.1, Cor. 10.2, Thm. 10.4, Thm. 11.2, Thm. 11.5, Cor. 12.1, Req. 14.2, Prop. 21.2 · newly derived · accepted

Proof. (i) By Proposition 4.6, $F \preceq_{\mathcal{P}_\mathbb{S}} F$ carrying $\hat{O}$ as well, since $\mathcal{P}_\mathbb{S}^\dagger(\hat{O}_k) = \hat{O}_k$; compose with Proposition 4.1. (ii) The output of $\mathcal{P}_\mathbb{S}$ is block diagonal by construction, and $(\mathcal{P}_\mathbb{S} \circ \Lambda)^{\otimes n}$ is block diagonal in every factor. (iii) is (ii) with Corollary 3.12 (i). □

Two squashing relations give one for the product measurement, through the tensor product of their channels. Independent components can therefore be reduced separately and assembled without a fresh existence proof.

Squashing relations for independent components combine by tensor product, because the adjoint of a tensor product of channels is the tensor product of their adjoints, also when the physical systems are infinite-dimensional (Lemma 1.2 (ii)).

Proposition 4.8 (Tensor products of squashing relations)**.** If $M \preceq_\Lambda F$ and $M' \preceq_{\Lambda'} F'$ then $\{M_v \otimes M'_{v'}\}_{(v,v')} \preceq_{\Lambda\otimes\Lambda'} \{F_v \otimes F'_{v'}\}_{(v,v')}$.

Uses Lem. 1.2, Def. 1.3 · *Used by* Thm. 6.7, Thm. 11.5 · newly derived · accepted

Proof. $(\Lambda \otimes \Lambda')^\dagger(F_v \otimes F'_{v'}) = \Lambda^\dagger(F_v) \otimes \Lambda'^\dagger(F'_{v'})$ by Lemma 1.2 (ii). □

If M squashes to F, the spectrum of each real combination of physical elements lies between the extreme spectral values of the same combination of target elements, and the physical span is no larger and inherits every linear relation with the identity. A relation violating either is impossible.

Not every pair (M, F) admits a squashing channel, and two necessary conditions are cheap enough to check by inspection. They serve to *refute* a declared relation, never to establish one; Remark 4.10 says how both are decided on declared numbers.

Lemma 4.9 (Necessary conditions for a squashing relation)**.** Let $M \preceq F$, let $X = \sum_v a_v F_v$ be a real linear combination of target effects and let $X' \coloneqq \sum_v a_v M_v$. Then:

(i) *(spectral inclusion)* $\operatorname{spec}(X') \subseteq [\inf \operatorname{spec} X,\ \sup \operatorname{spec} X]$; in particular $\operatorname{spec}(M_v) \subseteq [\inf \operatorname{spec} F_v,\ \sup \operatorname{spec} F_v]$ and $\|M_v\| \le \|F_v\|$ for every v;

(ii) *(span monotonicity)* $\dim \operatorname{span}\{M_v\} \le \dim \operatorname{span}\{F_v\} \le (\dim \mathcal{H}(Q))^2$, and every linear relation $\sum_v a_v F_v = c\,\mathbb{1}_Q$ forces $\sum_v a_v M_v = c\,\mathbb{1}_B$.

Uses Def. 1.3 · *Used by* Req. 14.7, Prop. 19.4 · newly derived · accepted

Proof. Let Λ realize the relation, so $X' = \Lambda^\dagger(X)$ by linearity. (i) With $a \coloneqq \inf \operatorname{spec} X$ and $b \coloneqq \sup \operatorname{spec} X$ we have $a\,\mathbb{1}_Q \preceq X \preceq b\,\mathbb{1}_Q$; applying the positive unital map $\Lambda^\dagger$ preserves both inequalities, so $a\,\mathbb{1}_B \preceq X' \preceq b\,\mathbb{1}_B$. Taking $X = F_v$ gives the inclusion for M_v, and the norm bound follows since $\|Y\| = \sup \operatorname{spec} Y$ for a positive operator. (ii) $M_v = \Lambda^\dagger(F_v)$, so $\operatorname{span}\{M_v\} = \Lambda^\dagger\big(\operatorname{span}\{F_v\}\big)$ and the dimension cannot increase; the bound $(\dim \mathcal{H}(Q))^2$ is $\dim \mathcal{L}(Q)$. Unitality gives the last statement. □

On declared numbers the spectral condition is decided exactly in operator form on rational enclosures, and the span condition by exact ranks and linear relations. A loose decision would refuse more or less than the lemma says, which is safe: nothing emitted rests on the check having passed.

Remark 4.10 (Exact decision of the necessary conditions). Lemma 4.9 is an exact statement about an arbitrary relation. Applied to *declared* numbers, both of its parts are decided *exactly* in the substrate of Lemma 9.3, each in a form that differs from the lemma's wording. *(i)* The spectral enclosure is decided in its *operator* form, which is what a refusal needs and what (9.3) decides: for rationals $a \le b$ the pair $a\mathbb{1}_Q \preceq X \preceq b\mathbb{1}_Q$ is decided exactly, by any of the exact procedures of Remark 9.4, and where it holds the same pair is decided for X'. *The lemma's* $\inf\operatorname{spec} X$ *and* $\sup\operatorname{spec} X$ *are in general algebraic and not rational*, so the decision tests the rational enclosures the declaration exhibits rather than the extreme values themselves; it refuses a subset of what the lemma forbids, and no eigenvalue is computed anywhere. *(ii)* The span dimension is the *exact* rank over $\mathbb{F}_i$ of the matrix whose columns are the vectorized declared elements, decided by elimination in the field. A floating-point rank is a threshold decision, and this one is not. The linear relations are decided the same way: the solutions (a, c) of $\sum_v a_v F_v = c\,\mathbb{1}_Q$ form a subspace, an exact basis of which elimination in the field produces, and testing $\sum_v a_v M_v = c\,\mathbb{1}_B$ on each basis solution tests it on every solution, the condition being linear in (a, c).

A loose decision would have been safe here. A rank read at a threshold may count a nearly dependent family as independent, and an enclosure read at a window may fail to notice a violation just inside it. Made on the target's elements, either error refuses *less* than the lemma forbids; made on the physical elements, a loose decision can refuse a relation that exists. Neither can produce a wrong artefact, because of what the check is for: it is a cheap early refusal, not a warrant. Nothing this module emits is proved from the check's having passed; what the emitted artefact rests on are the constructions themselves, which either produce a channel or do not. A missed refusal costs the construction's time and produces no artefact, and a spurious one produces none either; neither can produce a wrong one. The check is exact all the same (Requirement 14.7). *It is also vacuous on this module's own emission path*, where the request declares no relation of its own and the only relation in play is the one the reduction instance constructs (Definition 10.1 (I3)).

Used by Req. 14.7 · newly derived · accepted

5 Branching and flag targets

Every measurement with a finite alphabet squashes to the flag PVM on its outcomes (Lemma 5.1), so the existence of a squasher is never the difficulty; its tightness is. A generalized instrument followed by a squasher in each branch gives one squashing relation whose target stacks the branch targets as blocks, with each branch weight transported as an operator (Lemma 5.3). Its instances cover a known mixture, a measurement block diagonal for an orthogonal decomposition and a register read by a POVM (Corollary 5.8, Theorem 5.13); common targets that merge the blocks again, flag-extended targets and their rate cost, and the price of a general POVM on the register

follow.

Every POVM with a finite alphabet squashes to the flag PVM on its outcomes through a measure-and-prepare channel, and to any PVM on the same alphabet that vanishes only where the POVM does. A squasher therefore always exists; the work lies in finding a target tight enough to leave a key rate.

A receiver is rarely characterized as a single POVM on a single space. It is characterized *conditionally*: below a photon-number cutoff we know what it does and above it we do not; on one value of a classical branch register it behaves as specified and on another we promise nothing. The lemma shows first that existence of a squasher is never the difficulty. The work of this module's constructions is therefore not to find a squasher but to find one whose target is tight enough to leave a key rate.

Lemma 5.1 (Flag PVM as a top element)**.** Let M be any POVM on any B with finite alphabet $\mathcal{V}$, and let $P^{\mathrm{Flg}} = \{|v\rangle\langle v|\}_{v\in\mathcal{V}}$ be the *flag PVM* on $\mathcal{H}(\mathrm{Flg}) = \mathbb{C}^{|\mathcal{V}|}$ with orthonormal basis $\{|v\rangle\}_{v\in\mathcal{V}}$. Then

$$M \preceq_{\Lambda_{\mathrm{mp}}} P^{\mathrm{Flg}}, \qquad \Lambda_{\mathrm{mp}}(\rho) \coloneqq \sum_{v\in\mathcal{V}} \mathrm{tr}[M_v\rho]\ |v\rangle\langle v|\,. \tag{5.1}$$

More generally, if P is any PVM on a space Q with the same alphabet and $P_v = 0 \Rightarrow M_v = 0$, then $M \preceq P$.

Uses Def. 1.3 · *Used by* Cor. 5.8, Cor. 5.11, Rem. 5.12, Cor. 10.3, Prop. 19.4, Thm. 23.2 · newly derived · accepted

Proof. Λ_{mp} is a measure-and-prepare channel: CP, and trace preserving because $\sum_v \mathrm{tr}[M_v\rho] = \mathrm{tr}\,\rho$. Its adjoint is ${\Lambda_{\mathrm{mp}}}^{\dagger}(X) = \sum_v \langle v|\,X\,|v\rangle\ M_v$, since $\mathrm{tr}[X\Lambda_{\mathrm{mp}}(\rho)] = \sum_v \mathrm{tr}[M_v\rho]\,\langle v|\,X\,|v\rangle$; evaluating at $X = |v'\rangle\langle v'|$ gives $M_{v'}$. For the general case pick, for each v with $M_v \neq 0$, a state ω_v with $P_v\omega_vP_v = \omega_v$ (possible since $P_v \neq 0$), and set $\Lambda(\rho) \coloneqq \sum_{v:M_v\neq 0} \mathrm{tr}[M_v\rho]\,\omega_v$. This is trace preserving because the omitted terms vanish and $\sum_v M_v = \mathbb{1}_B$, and $\Lambda^{\dagger}(P_{v'}) = \sum_{v:M_v\neq 0} \mathrm{tr}[P_{v'}\omega_v]\,M_v = M_{v'}$, because $\mathrm{tr}[P_{v'}\omega_v] = \delta_{vv'}$ by orthogonality of the ranges of a PVM, and because the identity reads $0 = 0$ when $M_{v'} = 0$. □

A generalized instrument is a finite family of completely positive maps on one input system whose traces add up to the trace. Each map is a branch, occurring with the probability its trace gives and leaving a subnormalized state on a system that may differ from branch to branch.

By Lemma 5.1 every POVM squashes to a flag PVM, so a squashing relation on its own is worth nothing: the target P^{Flg} lets the adversary prepare a flag state and fix Bob's outcome outright. What buys a key rate is a target whose elements are not all rank-one projectors, or a sound constraint that excludes the flag states an adversary would use; the squasher constructions of this module's submodules and the attack-set licence of Theorem 6.7 are

the two ways of getting one. The squasher constructions describe a receiver branch by branch, and the branching step is the following object.

Definition 5.2 (Generalized instrument)**.** A *generalized instrument* on B is a finite family $\{\mathcal{I}_j\}_{j\in J}$ of CP maps $\mathcal{I}_j\colon \mathcal{T}(B) \to \mathcal{T}(B'_j)$ whose traces add up to the trace, $\sum_j \operatorname{tr}\circ\mathcal{I}_j = \operatorname{tr}$: branch j occurs with probability $\operatorname{tr}\mathcal{I}_j(\rho)$ and leaves the subnormalized state $\mathcal{I}_j(\rho)$ on the system B'_j, which may differ from branch to branch.

Used by Lem. 5.3, Thm. 5.13, Thm. 10.4, Cor. 10.5, Thm. 23.2 · newly derived · accepted

A generalized instrument followed by a squasher in each branch squashes to the direct sum of the branch targets, with or without the branch label in the outcome. Branch operators are transported, and a branch weight is a fixed number only where its pull-back is a multiple of the identity.

A receiver described branch by branch squashes as a whole once each branch does. The lemma combines a generalized instrument with one squasher per branch into a single squashing relation whose target stacks the branch targets as diagonal blocks, and carries each branch weight along as an operator.

Lemma 5.3 (Instrument-conditioned squashing)**.** Let J be a finite set and let $\mathcal{I}_j\colon \mathcal{T}(B) \to \mathcal{T}(B'_j)$ be CP maps with $\sum_j \operatorname{tr}\circ\mathcal{I}_j = \operatorname{tr}$, that is, a generalized instrument (Definition 5.2). For each j let $M^{(j)}$ be a POVM on B'_j over a common finite alphabet $\mathcal{V}$ with $(M^{(j)};O^{(j)}) \preceq_{\Lambda_j} (F^{(j)};\hat{O}^{(j)})$, where $F^{(j)}$ is a POVM on a finite-dimensional Q_j. Each $\mathcal{I}_j$ is trace non-increasing; write $\mathcal{I}_j^\dagger$ for its adjoint in the sense of Lemma 1.2, the completely positive map fixed by $\operatorname{tr}[X\mathcal{I}_j(\rho)] = \operatorname{tr}\left[\mathcal{I}_j^\dagger(X)\rho\right]$ for $X \in \mathcal{L}(B'_j)$ and $\rho \in \mathcal{T}(B)$. Then:

(i) $M_{v,j} \coloneqq \mathcal{I}_j^\dagger\big(M_v^{(j)}\big)$ defines a POVM on B with joint outcome $(v,j) \in \mathcal{V}\times J$;

(ii) with $Q \coloneqq \bigoplus_{j\in J} Q_j$ and $F_{v,j}$ the operator equal to $F_v^{(j)}$ in the j-th block and zero elsewhere,

$$\{M_{v,j}\} \preceq_\Lambda \{F_{v,j}\}, \qquad \Lambda(\rho) \coloneqq \bigoplus_{j\in J} \Lambda_j\big(\mathcal{I}_j(\rho)\big); \tag{5.2}$$

(iii) *(coarse graining)* with the same channel, $\big\{\sum_j M_{v,j}\big\}_v \preceq_\Lambda \big\{\bigoplus_j F_v^{(j)}\big\}_v$;

(iv) *(branch weights)* $\Lambda^\dagger(\Pi_j) = \mathcal{I}_j^\dagger(\mathbb{1}_{B'_j})$, where Π_j projects onto the j-th block. In particular, if $\mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = q_j\,\mathbb{1}_B$ for constants q_j, then for every finite-dimensional side system R and every reachable σ, $\operatorname{tr}_Q[(\Pi_j\otimes\mathbb{1}_R)\sigma] = q_j\,\sigma_R$, and in particular $\operatorname{tr}[\Pi_j\sigma] = q_j$;

(v) *(transported operators)* the composite carries the operator equal to $\hat{O}_k^{(j)}$ in block j and zero elsewhere, with pull-back $\mathcal{I}_j^\dagger(O_k^{(j)})$.

Uses Lem. 1.2, Cor. 1.7, Lem. 4.3, Def. 5.2 · *Used by* Rem. 5.4, Rem. 5.5, Rem. 5.6, Rem. 5.7, Cor. 5.8, Prop. 5.9, Thm. 5.13, Thm. 10.4, Cor. 10.5, Prop. 19.4, Thm. 23.2 · newly derived · accepted

Proof. For $\rho \succeq 0$ every term of $\sum_{j'} \operatorname{tr} \mathcal{I}_{j'}(\rho) = \operatorname{tr} \rho$ is non-negative, so $\operatorname{tr} \mathcal{I}_j(\rho) \leq \operatorname{tr} \rho$: each $\mathcal{I}_j$ is trace non-increasing, and Lemma 1.2 gives its adjoint $\mathcal{I}_j^\dagger$, completely positive and so positive. The hypothesis $\sum_j \operatorname{tr} \circ \mathcal{I}_j = \operatorname{tr}$ says exactly $\sum_j \mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = \mathbb{1}_B$, by Lemma 1.2 (i) applied to $\sum_j \mathcal{I}_j^\dagger(\mathbb{1}) - \mathbb{1}_B$.

(i) $M_{v,j} \succeq 0$, and $\sum_{v,j} \mathcal{I}_j^\dagger(M_v^{(j)}) = \sum_j \mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = \mathbb{1}_B$.

(ii) Λ is CP as a direct sum of compositions of CP maps, and trace preserving: $\operatorname{tr} \Lambda(\rho) = \sum_j \operatorname{tr} \Lambda_j(\mathcal{I}_j(\rho)) = \sum_j \operatorname{tr} \mathcal{I}_j(\rho) = \operatorname{tr} \rho$. Its output is block diagonal, so only the diagonal blocks of an operator on Q contribute to the pairing (1.1); for $X = \bigoplus_j X_j$,

$$\operatorname{tr}\left[X\, \Lambda(\rho)\right] = \sum_j \operatorname{tr}\left[X_j\, \Lambda_j(\mathcal{I}_j(\rho))\right] = \sum_j \operatorname{tr}\left[\mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(X_j)\big)\, \rho\right], \quad \text{so} \quad \Lambda^\dagger\Big(\bigoplus_j X_j\Big) = \sum_j \mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(X_j)\big). \tag{5.3}$$

Taking $X = F_{v,j}$, with only the j-th block non-zero, gives $\Lambda^\dagger(F_{v,j}) = \mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(F_v^{(j)})\big) = \mathcal{I}_j^\dagger(M_v^{(j)}) = M_{v,j}$. That $\{F_{v,j}\}$ is a POVM on Q follows blockwise from $\sum_v F_v^{(j)} = \mathbb{1}_{Q_j}$.

(iii) is Lemma 4.3 (i) for the stochastic map that forgets j.

(iv) Take $X = \Pi_j$, that is $X_j = \mathbb{1}_{Q_j}$ and zero elsewhere, in (5.3): $\Lambda^\dagger(\Pi_j) = \mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\mathbb{1}_{Q_j})\big) = \mathcal{I}_j^\dagger(\mathbb{1}_{B'_j})$ by unitality of $\Lambda_j^\dagger$. The constraint claim is Corollary 1.7 (i).

(v) is (5.3) evaluated on the stated operator. □

The branch weights of Lemma 5.3 (iv) are the only constraint the branching rule produces, and the operator form is the one to impose: it fixes each block's weight conditioned on everything Alice and Eve hold, while its trace alone admits a block index perfectly correlated with Eve.

Remark 5.4 (The operator form of the branch-weight constraint). Part (iv) of Lemma 5.3 is the only place the branching rule produces a constraint, and the operator form is the one worth imposing: it says the weight of block j is q_j *conditioned on everything Alice and Eve hold*, and Corollary 3.13 (i) turns it into a sound constraint, one per round. The scalar $\operatorname{tr}[\Pi_j \sigma] = q_j$ is only its trace and is strictly weaker: it still admits a squashed state whose block index is perfectly correlated with Eve's register. Nothing is averaged anywhere in the rule, because averaging over j before squashing would lose the branch structure and with it the relation.

Uses Cor. 3.13, Lem. 5.3 · newly derived · accepted

The branching rule files the branch label as a block index of the system handed to the adversary, so it models branches Eve may know. A setting the measuring party draws privately belongs to the common-channel rule of Lemma 4.3 (ii), which opens no block and produces no constraint.

Remark 5.5 (Branch labels as information the adversary may hold). The branch label ends up as a block index of Q, and Q is the system the squashed model hands to the adversary. Lemma 5.3 therefore models a branch Eve may as well know: a detector-response mode she selected, a test on the light she sent, a register that arrived with it. A setting the measuring party draws *privately* can be pushed through the rule, taking $\mathcal{I}_j = q_j \mathrm{id}$, but the resulting model files that setting in Eve's state, and only part (iv) of Lemma 5.3 in its operator form ties it back. Settings meant to stay on the measuring party's side belong to Lemma 4.3 (ii) instead: one common channel, no block opened, no constraint produced. The probe setting therefore enters this contract through Lemma 4.3 (ii), and a mode register that arrived with the light through Lemma 5.3; swapping the two would be a modelling error, giving Eve a register she does not have or withholding one she does.

Uses Lem. 4.3, Lem. 5.3 · newly derived · accepted

The branching rule squashes a conditionally characterized receiver as soon as each branch squashes, stacking the branch targets as blocks. A branch that cannot be modelled gets a block of its own, where the adversary fixes the outcome only in proportion to the branch's probability.

Remark 5.6 (Conditionally characterized receivers under the branching rule). The branching rule of Lemma 5.3 turns any conditional description of a receiver into a squashing relation. The device is squashable as soon as *each branch* is, and the squashed target is the branch targets stacked as diagonal blocks. A branch one cannot model is not fatal: it gets a block of its own, on which the squashed adversary rather than Bob's detector fixes the outcome, but only in proportion to how likely that branch was.

Uses Lem. 5.3 · newly derived · accepted

In the branching rule only the branch label is classical: the probability of a branch is in general an operator, not a number. That operator is what the squashed model retains about the branching, and it is the only constraint the rule produces.

Remark 5.7 (The branch probability as an operator). In Lemma 5.3 the branch label is classical, but nothing else about the branching is: the step that produces the label is an arbitrary CP map, and the probability of branch j is in general an operator, not a number. That operator is what the squashed model retains about the branching, and it is the only constraint the rule produces.

Uses Lem. 5.3 · newly derived · accepted

The branching rule applies to a convex mixture with a known weight, to a measurement block diagonal for an orthogonal decomposition and to a register read by a POVM. Only the known mixture fixes the branch weight as a number without further information.

Three choices of the generalized instrument recur in this module's constructions: a known mixture, a split of the input space into orthogonal blocks, and a POVM measurement of a register that arrives with the light.

Corollary 5.8 (Three special cases of instrument-conditioned squashing)**.** Let Λ_j and $F^{(j)}$ be branch channels and branch targets as in Lemma 5.3.

(i) *(convex mixture with a known weight)* If $M_v = (1-q)G_v + qR_v$ for POVMs G, R on B and $q \in [0,1]$, with $G \preceq_{\Lambda_G} F^{(G)}$ and R *arbitrary*, then $M \preceq \{F_v^{(G)} \oplus |v\rangle\langle v|_{\mathrm{Flg}}\}_v$, with the sound branch constraints $\mathrm{tr}[\Pi_1\sigma] = 1-q$ and $\mathrm{tr}[\Pi_{\mathrm{Flg}}\sigma] = q$, in the operator form of Lemma 5.3 (iv).

(ii) *(block-wise squashing)* Let $\mathcal{H}(B) = \bigoplus_j \mathcal{H}(B_j)$ be a finite orthogonal decomposition and let every M_v be block diagonal, $M_v = \bigoplus_j M_v^{(j)}$, with $M^{(j)} \preceq_{\Lambda_j} F^{(j)}$. Then $M \preceq \bigoplus_j F^{(j)}$.

(iii) *(POVM-conditioned squashing)* If $B = B_0 D$ with a POVM $\{N_j\}_{j\in J}$ on D and $M_v = \sum_j M_{v|j} \otimes N_j$ with $M_{|j} \preceq_{\Lambda_j} F_{|j}$ on Q_j, then $\{M_v\}_v \preceq_\Lambda \{\bigoplus_j F_{v|j}\}_v$ with

$$\Lambda(\rho) = \bigoplus_{j\in J} \Lambda_j\Big(\mathrm{tr}_D\left[(\mathbb{1}_{B_0} \otimes N_j^{1/2})\,\rho\,(\mathbb{1}_{B_0} \otimes N_j^{1/2})\right]\Big). \tag{5.4}$$

Uses Lem. 5.1, Lem. 5.3 · *Used by* Thm. 5.13, Prop. 19.4 · newly derived · accepted

Proof. Apply Lemma 5.3 with, respectively:

(i) $\mathcal{I}_1 = (1-q)\mathrm{id}$ and $\mathcal{I}_2 = q\,\mathrm{id}$, branch targets $F^{(G)}$ and the flag PVM of Lemma 5.1 (which exists for the unknown R), so that $M_{v,1} = (1-q)G_v$ and $M_{v,2} = qR_v$ and part (iii) gives $\sum_j M_{v,j} = M_v$. Here $\mathcal{I}_j^\dagger(\mathbb{1})$ is a multiple of $\mathbb{1}_B$, which is the hypothesis of part (iv).

(ii) $\mathcal{I}_j(\rho) \coloneqq P_j \rho P_j$ read as an operator on B_j, for P_j the projection onto $\mathcal{H}(B_j)$; then $\mathcal{I}_j^\dagger(X_j)$ is X_j embedded in the j-th block, so $M_{v,j}$ is $M_v^{(j)}$ so embedded and part (iii) gives $\sum_j M_{v,j} = M_v$. The instrument condition is $\sum_j P_j = \mathbb{1}_B$.

(iii) $\mathcal{I}_j(\rho) \coloneqq \mathrm{tr}_D\left[(\mathbb{1}_{B_0} \otimes N_j^{1/2})\rho(\mathbb{1}_{B_0} \otimes N_j^{1/2})\right]$, which is CP as a Kraus map followed by a partial trace, and satisfies $\sum_j \mathrm{tr}\,\mathcal{I}_j(\rho) = \sum_j \mathrm{tr}[(\mathbb{1}\otimes N_j)\rho] = \mathrm{tr}\,\rho$. Its adjoint is $\mathcal{I}_j^\dagger(X) = (\mathbb{1}\otimes N_j^{1/2})(X\otimes \mathbb{1}_D)(\mathbb{1}\otimes N_j^{1/2}) = X \otimes N_j$, the two factors commuting because they act on different tensor factors; hence $M_{v,j} = M_{v|j} \otimes N_j$ and part (iii) gives $\sum_j M_{v,j} = \sum_j M_{v|j} \otimes N_j = M_v$. □

If every branch target squashes to one POVM, the coarse-grained measurement of Lemma 5.3 squashes to that POVM directly, through the sum of the composed branch channels. The branch blocks then disappear from the target.

Only in Corollary 5.8 (i) is the branch weight state independent, hence the constraint a fixed number. In its case (ii) the weight is the projection onto the block the input fell in, and that operator-valued case is the one a flag-state squasher runs on. Its case (iii) is stated for a general POVM $\{N_j\}$ rather than a PVM, and it is what the register-conditioned squashing of Theorem 5.13 runs on: a *classical branch register* read by a POVM rather than a PVM, which Corollary 10.5 lifts to reduction instances. The price of the generality is that each branch gets its own block of Q rather than sharing one register, which is exactly the block structure of Q the consuming declaration wants anyway. Where every branch target squashes to one common target, the blocks can be merged again.

Proposition 5.9 (Common and dominating targets)**.** In the setting of Lemma 5.3, suppose additionally $F^{(j)} \preceq_{\Psi_j} F$ for a single POVM F on a space Q_* (a *dominating target*; $\Psi_j = \mathrm{id}$ if all branch targets already coincide). Then

$$\Big\{\sum_j M_{v,j}\Big\}_v \preceq_{\Lambda'} F, \qquad \Lambda' \coloneqq \sum_{j\in J} \Psi_j \circ \Lambda_j \circ \mathcal{I}_j.$$

Uses Lem. 5.3 · *Used by* Rem. 5.10, Req. 14.8, Thm. 23.2 · newly derived · accepted

Proof. Λ' is CP as a sum of compositions of CP maps and trace preserving by the computation in Lemma 5.3 (ii). Then $\Lambda'^{\dagger}(F_v) = \sum_j \mathcal{I}_j^{\dagger}\Lambda_j^{\dagger}\Psi_j^{\dagger}(F_v) = \sum_j \mathcal{I}_j^{\dagger}\Lambda_j^{\dagger}\big(F_v^{(j)}\big) = \sum_j \mathcal{I}_j^{\dagger}\big(M_v^{(j)}\big) = \sum_j M_{v,j}$. □

A detector whose POVM dominates a characterized one is the characterized device plus a weighted flag, as in the published block-resolved construction. By Proposition 5.9, a family of devices whose members squash to a common target becomes one squashed model with the label a branch register.

Remark 5.10 (Dominated detectors and device families). A device has the form (i) of Corollary 5.8 whenever its POVM dominates a characterized one: if $M_v \succeq (1-q)G_v$ for all v with $q \in (0,1]$, then $R_v \coloneqq \big(M_v - (1-q)G_v\big)/q$ is positive and sums to $\mathbb{1}_B$, so any detector that is q-close to a characterized detector *in this operator sense* may be analysed as the characterized device plus a q-weighted flag. The published block-resolved form of that construction is [5, Thm. 3, Eqs. (20)–(23)], which builds exactly this channel from the block dominations $\Gamma_{m,\mathrm{noise}} - (1-q_m)\Gamma_{m,\mathrm{ideal}} \succeq 0$ and records the weight $\Lambda^{\dagger}[\Pi_m] = (1-q_m)\Pi_m$ kept on block m, the flag receiving q_m.

By Proposition 5.9, a family of devices indexed by a label ξ that the adversary selects becomes one squashed model as soon as each member squashes to a *common* target, with the label a branch register carrying its own constraints from Lemma 5.3 (iv). The operator factorization such a construction rests on (range inclusion, majorization and factorization are equivalent for bounded operators) is [3, Thm. 1]; no proof of this contract uses it.

Uses Prop. 5.9 · newly derived · accepted

In a flag-extended target, the block the squashing channel never populates may carry any POVM, and the target does not depend on the device. An uncharacterized branch can therefore be represented without a device model for it.

Lemma 5.1 says every POVM squashes to a flag PVM. The following sharpening is what a construction needs: in a *flag-extended* target, the block the squashing channel never populates may be replaced by any POVM at all. An uncharacterized branch therefore needs no model, neither an approximate nor a conservative one, and one target covers every device with the same flag structure. The branch blocks of a composite target *may* accordingly be padded to a common size; a reduction instance emits them as its branches supply them, and a padded variant only where a submodule's interface declares it (Requirement 14.9).

Corollary 5.11 (Arbitrary unpopulated block in a flag-extended target)**.** Let $\{G_v\}_{v\in\mathcal{V}}$ be any POVM on a finite-dimensional space B_0 and let $\{H_v\}_{v\in\mathcal{V}}$ be any POVM on B_0. Then, on $\mathcal{H}(B_0)\oplus\mathcal{H}(\mathrm{Flg})$,

$$\left\{G_v\oplus|v\rangle\langle v|_{\mathrm{Flg}}\right\}_v \preceq_\Lambda \left\{H_v\oplus|v\rangle\langle v|_{\mathrm{Flg}}\right\}_v, \qquad \Lambda(\rho)\coloneqq 0_{B_0}\oplus\sum_v \mathrm{tr}\left[(G_v\oplus|v\rangle\langle v|)\rho\right]\ |v\rangle\langle v|_{\mathrm{Flg}}\,. \tag{5.5}$$

In particular $H_v=\frac{1}{|\mathcal{V}|}\mathbb{1}_{B_0}$ is admissible, and the squashing channel depends on G while the target does not.

Uses Lem. 5.1 · *Used by* Rem. 5.12, Req. 14.9 · newly derived · accepted · *Oracles* registry_reductions.py

Proof. Λ is a measure-and-prepare channel, hence CP; it is trace preserving because $\sum_v(G_v\oplus|v\rangle\langle v|)=\mathbb{1}_{B_0}\oplus\mathbb{1}_{\mathrm{Flg}}$. Its output is supported in the flag block, so $\Lambda^\dagger\big(X_{B_0}\oplus X_{\mathrm{Flg}}\big)=\sum_v\langle v|\,X_{\mathrm{Flg}}\,|v\rangle\ \big(G_v\oplus|v\rangle\langle v|\,\big)$, the B_0-part of the observable contributing nothing, and evaluating at $X_{B_0}=H_v$, $X_{\mathrm{Flg}}=|v\rangle\langle v|$ returns $G_v\oplus|v\rangle\langle v|$. That $\{H_v\oplus|v\rangle\langle v|\}$ is a POVM follows from $\sum_v H_v=\mathbb{1}_{B_0}$. □

Replacing the unpopulated block by an arbitrary POVM gives a cleaner target, which hands the adversary more and costs rate. What it buys is one target, independent of the device, for a branch that is unknown or adversarially chosen; an infinite-dimensional branch takes the flag PVM.

Remark 5.12 (Rate cost of a flag-extended target). The replacement of Corollary 5.11 is not free in rate: by Corollary 4.2 the target of (5.5) is cleaner than the source, so it hands the adversary more, and every squashed state it admits is a flag state as far as the B_0 block is concerned. What it buys is that G appears nowhere in the target: a branch whose device is unknown or adversarially chosen is covered by one target that does not depend on the device, and an infinite-dimensional branch by the finite flag PVM of Lemma 5.1. The instance of such a branch is the flag instance of Corollary 10.3, on which by construction nothing is assumed and nothing may be.

Uses Cor. 4.2, Lem. 5.1, Cor. 5.11 · newly derived · accepted

A receiver conditioned by a POVM measurement of an incoming register squashes to the direct sum of its branch targets, with branch operators and branch weights transported as operators. A branch weight constrains the squashed states only where it is a declared multiple of the side system's state.

When the branch is selected by a measurement on a register that arrives with the light, the per-branch squashers combine into one map. The theorem is Corollary 5.8 (iii) with the transported operators and the branch weights carried along, and it is stated for a general POVM on the register rather than a projective measurement.

Theorem 5.13 (Register-conditioned squashing)**.** Let $B = B_0 D$, let $\{N_j\}_{j\in J}$ be a POVM on D with J finite, and let

$$M_v = \sum_{j\in J} M_{v|j} \otimes N_j, \qquad v \in \mathcal{V}, \tag{5.6}$$

where $\{M_{v|j}\}_v$ is a POVM on B_0 for each j. Suppose $(M_{|j}; O^{(j)}) \preceq_{\Lambda_j} (F_{|j}; \hat{O}^{(j)})$ for each j, with $F_{|j}$ a POVM on a finite-dimensional Q_j. Put $\mathcal{H}(Q) = \bigoplus_{j\in J} \mathcal{H}(Q_j)$ with block projections Π_j and

$$\Lambda(\rho) := \bigoplus_{j\in J} \Lambda_j\Big(\operatorname{tr}_D \left[(\mathbb{1}_{B_0} \otimes N_j^{1/2})\, \rho\, (\mathbb{1}_{B_0} \otimes N_j^{1/2}) \right] \Big). \tag{5.7}$$

Then Λ is a channel and:

(i) $\{M_v\}_v \preceq_\Lambda \left\{ \bigoplus_j F_{v|j} \right\}_v$; in particular $\{M_v\}$ is a POVM on B;

(ii) the composite transports the operator $\hat{O}_k^{(j)}$ placed in block j to $O_k^{(j)} \otimes N_j$;

(iii) $\Lambda^\dagger(\Pi_j) = \mathbb{1}_{B_0} \otimes N_j$, so on every reachable σ and every finite-dimensional side system R,

$$\operatorname{tr}_Q \left[(\Pi_j \otimes \mathbb{1}_R)\, \sigma \right] = \operatorname{tr}_B \left[\left((\mathbb{1}_{B_0} \otimes N_j) \otimes \mathbb{1}_R \right) \rho \right]; \tag{5.8}$$

this is an identity, not a constraint, unless the right-hand side is a declared multiple of ρ_R;

(iv) if every $F_{|j}$ and every $\hat{O}_k^{(j)}$ is block diagonal for a finite orthogonal decomposition of $\mathcal{H}(Q_j)$, then every $\bigoplus_j F_{v|j}$ and every carried operator is block diagonal for the union decomposition of $\mathcal{H}(Q)$, and $\mathcal{P}_{\mathbb{S}} \circ \Lambda$ realizes the same relation with block-diagonal reachable states (Corollary 4.7).

Uses Lem. 1.2, Cor. 1.7, Cor. 4.7, Def. 5.2, Lem. 5.3, Cor. 5.8 · *Used by* Rem. 5.14, Cor. 10.5 · newly derived · accepted · *Oracles* registry_reductions.py

Proof. Put $\mathcal{I}_j(\rho) \coloneqq \mathrm{tr}_D[(\mathbb{1}_{B_0} \otimes N_j^{1/2})\rho(\mathbb{1}_{B_0} \otimes N_j^{1/2})]$. Each $\mathcal{I}_j$ is CP, being a single-Kraus-operator map followed by a partial trace, and $\sum_j \mathrm{tr}\,\mathcal{I}_j(\rho) = \sum_j \mathrm{tr}[(\mathbb{1} \otimes N_j)\rho] = \mathrm{tr}\,\rho$, so $\{\mathcal{I}_j\}$ is a generalized instrument (Definition 5.2). Its adjoint is

$$\mathcal{I}_j^\dagger(X) = (\mathbb{1} \otimes N_j^{1/2})(X \otimes \mathbb{1}_D)(\mathbb{1} \otimes N_j^{1/2}) = X \otimes N_j \qquad (X \in \mathcal{L}(B_0)),$$

the two factors commuting because they act on different tensor factors; the first equality is (1.1) for $\mathcal{I}_j$, since $\mathrm{tr}[X\ \mathrm{tr}_D(K\rho K)] = \mathrm{tr}[(X \otimes \mathbb{1}_D)\,K\rho K] = \mathrm{tr}[K(X \otimes \mathbb{1}_D)K\,\rho]$ for the Hermitian $K = \mathbb{1}_{B_0} \otimes N_j^{1/2}$, by cyclicity.

Now apply Lemma 5.3 with these $\mathcal{I}_j$, branch POVMs $M_v^{(j)} \coloneqq M_{v|j}$ and branch channels Λ_j. Part (i) of that lemma gives that $\{M_{v,j} = M_{v|j} \otimes N_j\}$ is a POVM on B over $\mathcal{V} \times J$, and part (iii), which coarse-grains the branch label away, gives $\sum_j M_{v,j} = M_v$ and the relation (i) here, with the channel (5.7), which is (5.2). Part (ii) here is Lemma 5.3 (v) with $\mathcal{I}_j^\dagger(O_k^{(j)}) = O_k^{(j)} \otimes N_j$. Part (iii) here is Lemma 5.3 (iv) with $\mathcal{I}_j^\dagger(\mathbb{1}_{B_0}) = \mathbb{1}_{B_0} \otimes N_j$, followed by Corollary 1.7 read on the carried operator Π_j, whose pull-back is that operator. Part (iv) is immediate: a direct sum of block-diagonal operators is block diagonal for the union of the decompositions, and Corollary 4.7 applies. □

Keeping the register intact in the target would need a projective measurement of it; filing each branch as a block of the squashed system removes that hypothesis. The price is bookkeeping only: a direct sum instead of a tensor product, the branch weight still transported as an operator.

Remark 5.14 (A general POVM on the register, and its price). Theorem 5.13 assumes only that $\{N_j\}$ is a POVM. A construction that keeps the register D intact in the target, squashing $\sum_j M_{v|j} \otimes N_j$ to $\sum_j F_{v|j} \otimes N_j$ on $Q_0 D$, does need $\{N_j\}$ to be projective, because its proof turns on $\mathcal{P}_i(X \otimes N_j) = \delta_{ij} X \otimes N_j$ for the projections $\mathcal{P}_i = (\mathbb{1} \otimes N_i)(\cdot)(\mathbb{1} \otimes N_i)$, and that identity fails for a general POVM. Filing each branch as a *block* of Q instead of a value of a retained register removes the hypothesis. The price is bookkeeping and nothing else: the target is a direct sum rather than a tensor product, which is the block-list form of Definition "Sector maps and block-list operators" (foundations/def:sector-map) and the form every reduction instance emits (Definition 10.1 (I2)); and Theorem 5.13 (iii) shows the branch weight is still transported, as the operator $\mathbb{1}_{B_0} \otimes N_j$ rather than as the number it is not.

Uses Thm. 5.13 · newly derived · accepted

6 Event-level device facts, attack sets and the licence

A device fact about a union of outcomes cannot in general be kept by a classical target (Proposition 6.1), so it travels as a constraint entry, which is sound only on the reachable set. The consuming engine restricts the

squashed attacks to the admissible set the entries cut out. The main result, the attack-set licence Theorem 6.7, proves that every squashed physical attack lies in that set whenever each carried operator's pull-back satisfies a spectral certificate, in either source mode and, for party-local entries, on a product measured system with a product squashing channel and each certificate on its factor's own channel. The remarks say how the relation symbols read at the channel and the declaration level, what the licence does and does not license, and where its format is published.

Imports foundations/def:round-model

Some three-outcome qubit POVM with a lower bound on the sum of two of its elements has no squashing to a classical target that keeps that bound as an operator inequality. A device fact about a union of outcomes therefore travels as a constraint entry rather than inside a flag target.

The weight-preserving flag-state target of the flag-state submodule encodes one certified weight *per outcome*. For an *event* $\mathcal{O}_* \subseteq \mathcal{V}$, a union of outcomes that must not be merged (for instance because they carry key information), a device may certify only the sum, $\sum_{v\in\mathcal{O}_*} M_v \succeq p_* \mathbb{1}_B$. The following proposition shows that no classical target can absorb such a bound, whatever the construction.

Proposition 6.1 (Obstruction to encoding event bounds in classical targets)**.** Call a target POVM F on Q *classical* if every F_v is diagonal in one fixed orthonormal basis $\{|f\rangle\}_f$ of $\mathcal{H}(Q)$. There exist a POVM M on $\mathcal{H}(B) = \mathbb{C}^2$, an event $\mathcal{O}_* \subseteq \mathcal{V}$ and a constant $p_* > 0$ with $\sum_{v\in\mathcal{O}_*} M_v \succeq p_* \mathbb{1}_B$ such that *no* pair (F, Λ) exists with F classical, $M \preceq_\Lambda F$ and $\sum_{v\in\mathcal{O}_*} F_v \succeq p_* \mathbb{1}_Q$. Concretely,

$$M_1 = \tfrac{1}{2}\,|0\rangle\langle 0|\,, \quad M_2 = \tfrac{1}{2}\,|+\rangle\langle +|\,, \quad M_3 = \mathbb{1}_B - M_1 - M_2, \quad \mathcal{O}_* = \{1,2\}, \quad p_* = \tfrac{1}{2}\Big(1-\tfrac{1}{\sqrt{2}}\Big), \tag{6.1}$$

with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Uses Def. 1.3 · *Used by* Rem. 6.2, Req. 15.1 · newly derived · accepted · *Oracles* registry_reductions.py

Proof. The data are admissible. For two rank-one projections with overlap $|\langle 0|+\rangle| = 1/\sqrt{2}$, the operator $|0\rangle\langle 0| + |+\rangle\langle +|$ has trace 2 and eigenvalues $1 \pm 1/\sqrt{2}$; hence $\lambda_{\min}(M_1 + M_2) = \tfrac{1}{2}(1 - 1/\sqrt{2}) = p_* > 0$ and $\lambda_{\max}(M_1 + M_2) = \tfrac{1}{2}(1 + 1/\sqrt{2}) < 1$, so $M_3 \succeq 0$ and $\{M_1, M_2, M_3\}$ is a POVM with $M_1 + M_2 \succeq p_* \mathbb{1}_B$.

No classical target encodes the bound. Suppose (F, Λ) existed. Put $T_f := \Lambda^\dagger(|f\rangle\langle f|)$. Then $\{T_f\}_f$ is a POVM on B: each $T_f \succeq 0$ because $\Lambda^\dagger$ is positive, and $\sum_f T_f = \Lambda^\dagger(\mathbb{1}_Q) = \mathbb{1}_B$ by unitality. Writing $F_v = \sum_f F_v(f)\,|f\rangle\langle f|$ with $F_v(f) \geq 0$ and $\sum_v F_v(f) = 1$ (POVM completeness read on the diagonal), linearity gives

$$M_v = \Lambda^\dagger(F_v) = \sum_f F_v(f)\, T_f.$$

The encoded event bound $\sum_{v\in\mathcal{O}_*} F_v \succeq p_* \mathbb{1}_Q$ reads, on the diagonal, $F_1(f)+F_2(f) \geq p_* > 0$ for every f, so every f has $F_1(f) > 0$ or $F_2(f) > 0$.

Suppose $F_1(f) > 0$. Every summand of $M_1 = \sum_{f'} F_1(f')T_{f'}$ is positive semidefinite, so $0 \preceq F_1(f)T_f \preceq M_1 = \frac{1}{2}|0\rangle\langle 0|$. A positive operator dominated by a rank-one positive operator is a non-negative multiple of it: if $X \preceq c\,|0\rangle\langle 0|$ with $X \succeq 0$, then for every $|\psi\rangle \perp |0\rangle$ we get $\langle\psi| X |\psi\rangle \leq 0$, hence $\langle\psi| X |\psi\rangle = 0$ and $X|\psi\rangle = 0$, so $\operatorname{supp} X \subseteq \operatorname{span}|0\rangle$. Thus $T_f \propto |0\rangle\langle 0|$. Symmetrically $F_2(f) > 0$ forces $T_f \propto |+\rangle\langle +|$. A non-zero operator cannot be proportional to both, since $|0\rangle\langle 0|$ and $|+\rangle\langle +|$ are distinct rank-one projections; so $F_1(f)F_2(f) > 0$ forces $T_f = 0$, and in every case each T_f is 0, or a non-negative multiple of $|0\rangle\langle 0|$, or a non-negative multiple of $|+\rangle\langle +|$.

Summing, $\mathbb{1}_B = \sum_f T_f = \alpha\,|0\rangle\langle 0| + \beta\,|+\rangle\langle +|$ for some $\alpha, \beta \geq 0$ (two non-negative scalars local to this proof, unrelated to the Rényi order α). Testing against $|1\rangle$ gives $1 = \langle 1|\left(\alpha\,|0\rangle\langle 0| + \beta\,|+\rangle\langle +|\right)|1\rangle = \beta/2$, so $\beta = 2$; testing the off-diagonal entry gives $0 = \langle 0|\left(\alpha\,|0\rangle\langle 0| + \beta\,|+\rangle\langle +|\right)|1\rangle = \beta/2$, so $\beta = 0$ — a contradiction. □

A general quantum target encodes an event bound trivially; the obstruction concerns targets classical on the whole target space. Such a bound travels as a constraint entry under the licence, and merging the outcomes is no remedy when they carry key information.

Remark 6.2 (Scope of the obstruction, and what it forces). With a general *quantum* target the encoding is trivial: take $F = M$ and $\Lambda = \mathrm{id}$, and the physical bound is itself an operator inequality on the target. The content of Proposition 6.1 is that a target classical on the whole of Q cannot in general carry an event-level bound, whatever the channel. Per-outcome bounds, by contrast, are encodable: a weight-preserving flag-state target of the flag-state submodule carries one weight per outcome as an operator inequality on the target. Event-level information that the target cannot carry must travel as a pull-back constraint, and a pull-back constraint is sound only on the reachable set, which is what the licence of Theorem 6.7 certifies. Three consequences follow.

(a) A record without a constraint list and a licence slot excludes a class of device knowledge that no classical target can carry; this is why the artefact carries both even when both are empty (Requirement 15.1).

(b) Re-declaring the coarse-graining does not help. Merging the outcomes of $\mathcal{O}_*$ into one letter turns the event bound into a per-outcome bound and makes it encodable, but only by destroying the distinction between the merged outcomes, which is precisely what "must not be merged" forbids when they carry key information (the fibre condition of the flag-state submodule's weight route).

(c) The counterexample is two-dimensional and uses only three outcomes, so it is not an artefact of infinite dimension, of photon number, or of the optical setting. It applies to every squashed model whose target is classical on the whole of Q, and it says nothing of a target with a block on which it is not classical, which can carry the bound as the trivial encoding above does.

Uses Prop. 6.1 · *Used by* Req. 15.1 · newly derived · accepted

For a constraint list with real constants, the admissible attack channels are the round model's attack sets: adjoint conditions in each source mode and, for party-local entries, a condition with a reference system. The licence Theorem 6.7 says when every squashed physical attack lies there.

The consuming engine accepts a declared constraint list inside its security chain only under an assumption it never verifies: that security of the physical protocol reduces to the squashed round model with attacks restricted to the declared admissible set. This definition fixes that set in the consumer's format, and Theorem 6.7 proves the reduction from one hypothesis.

Definition 6.3 (Admissible attack sets in the consumer's format)**.** Let $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, \Theta_k)\}_{k=1}^{m}$ be a declared list of Hermitian $\hat{O}_k \in \mathcal{L}(Q)$, relations $\bowtie_k \in \{\preceq, =, \succeq\}$ and right-hand sides typed by the declared source mode. For a list whose entries carry real constants c_k, as every list this module emits does (Definition 6.5), the admissible sets are the attack sets of $\mathfrak{L}$ of Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) at the per-round system $S = Q$, which the consuming contract states at the same system (Definition "Round model and licensed attack sets" (meat-core/def:roundmodel)); their displays at that system are, verbatim,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, QE'') := \Big\{ \mathcal{A} \in \mathrm{CPTP}(E', QE'') :\ \mathcal{A}^{\dagger}\big(\hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'}\ \ \forall k \Big\} \qquad \text{(trusted-source mode)}, \tag{6.2}$$

and, with the constraint operator lifted to the larger supplied system,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, AQE'') := \Big\{ \mathcal{A} \in \mathrm{CPTP}(E', AQE'') :\ \mathcal{A}^{\dagger}\big(\mathbb{1}_{A} \otimes \hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'}\ \ \forall k \Big\} \tag{6.3}$$

in untrusted-source mode, the two distinguished by the output system named in the argument. This contract writes $\bowtie_k$ in the operator spelling of (6.2) and (6.3) in both source modes, and everything it emits is admitted under either spelling. An entry that declares a Hermitian right-hand side Θ_k on Alice's system, rather than a real constant, is a *general* entry: it is admissible input to the consuming contract but is never licensed (Definition "Licensed constraint list" (protocol-compiler/def:homog)). Every entry this module emits carries a real scalar c_k and is written $(\hat{O}_k, \bowtie_k, c_k)$ (Definition 6.5, Theorem 6.7 (iv)).

Party-local entries. Where the measured system is a product $Q = \bigotimes_f Q_f$ and $\mathfrak{L}$ carries entries tagged party-local (Definition 6.5), the admissible set is cut out, for each such entry $(\hat{O}_k, \bowtie_k, c_k; f)$, by the condition

$$\big(\mathrm{id}_{R'} \otimes \mathcal{A}\big)^{\dagger}\Big(X \otimes \big(\hat{O}_k - c_k\, \mathbb{1}_{Q_f}\big) \otimes \mathbb{1}_{E''}\Big) \ \bowtie_k\ 0 \qquad \text{for every finite-dimensional } R' \text{ and every } X \in \mathrm{Pos}(R' \tag{6.4}$$

in place of the scalar condition of (6.2) or (6.3), where Q_{f^c} is the product of the other factors and R is A in untrusted-source mode and trivial in trusted-source mode; the untagged entries keep their conditions. The reference system R' is what carries Alice's retained system in trusted-source mode, where

it is no output of $\mathcal{A}$. Taking R' trivial and $X = \mathbb{1}$ in (6.4) gives the scalar condition for the lifted operator $\mathbb{1}_{Q_{fc}} \otimes \hat{O}_k$, so the set with party-local entries is contained in the set with the same entries read as scalars. The condition (6.4) is that of an entry tagged with the factor f in Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) (iii) at $S = Q$, which the consuming contract states at the same system. An attack sequence is $\mathfrak{L}$-admissible (an admissible attack sequence for $\mathfrak{L}$) if $\mathcal{A}_j$ lies in the corresponding set with $E' = E_{j-1}$ and $E'' = E'_j$ for every j.

Uses foundations/def:round-model · *Used by* Rem. 6.4, Thm. 6.7, Rem. 6.8, Rem. 6.10, Def. 7.2, Rem. 7.3, Lem. 8.1, Thm. 8.6, Rem. 8.8, Lem. 12.3 · newly derived · accepted

The consuming contract writes the relations operator-valued at the channel level and as comparisons of real numbers at the untrusted declaration level; both spellings denote the same three relations. Every entry this module emits carries a real constant and is admitted under either.

Remark 6.4 (Relation symbols at the channel and declaration levels). The consuming contract splits the relation symbols between two *levels*, not between the two source modes. At the *channel* level, the adjoint-form sets themselves, Definition "Round model and licensed attack sets" (meat-core/def:roundmodel) writes $\bowtie_k$ operator-valued in *both* modes, comparing $\mathcal{A}^\dagger(\cdot)$ against $c_k \mathbb{1}_{E'}$; that is the spelling the two displays of Definition 6.3 reproduce, with $\bowtie_k \in \{\preceq, =, \succeq\}$. At the *declaration and state* level, Definition "Admissible declaration" (meat-core/def:admissible-declaration) (vi) gives $\bowtie_k \in \{\preceq, =, \succeq\}$ with Θ_k Hermitian on A in trusted-source mode, and $\bowtie_k \in \{\leq, =, \geq\}$ with $\Theta_k = c_k \in \mathbb{R}$ in untrusted-source mode, where the entry is a comparison of real numbers; Definition "Licensed constraint list" (protocol-compiler/def:homog) is on the same side, with the clause letters *(a) trusted-source* and *(b) untrusted-source*. That definition also fixes what *licensed* means there: a licensed trusted entry declares a real c_k and the right-hand side $\Theta_k := c_k \tau_A$ is *constructed* from it, while an entry that declares a Hermitian Θ_k is a general entry, admissible *input* but never licensed. Nothing this module emits is touched by that: every entry it emits carries a real scalar c_k and is written $(\hat{O}_k, \bowtie_k, c_k)$ (Definition 6.5, Theorem 6.7 (iv)). The two spellings denote the same three relations at the two levels; this contract states operator conditions throughout and so uses the operator spelling, and everything it emits is admitted under either.

The entries it emits are those of Corollary 11.3: $\bowtie = \succeq$ with $c = 0$ for every sector projector (its part (i)), the single equality entry $(\hat{O}, =, q_t)$ for a class-marginal entry where the instance discharges it (part (ii)), and the constant at which its certificate is discharged for an instance's own or a caller's entry (part (iii)). Every one of those constants is exact as declared, and the only rounding that can touch one is a caller's own constant off the declared lattice, which is rounded in the direction that *enlarges* the admissible set (Requirement 16.6, Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction)).

Uses Def. 6.3 · newly derived · accepted

A constraint list is licensed when every right-hand side is a real constant: untagged, as a multiple of Alice's declared marginal in trusted mode and a real number in untrusted mode, or party-local. A party-local entry acts on one factor of a product measured system, against the marginal present.

Definition 6.5 (Licensed entries and the party-local form)**.** The list $\mathfrak{L}$ is *licensed* for the declared source mode when each of its entries either is untagged with its right-hand side in the form that mode admits, or is in the party-local form below. The untagged forms are $\Theta_k = c_k\,\tau_A$ with $c_k \in \mathbb{R}$ in trusted-source mode, the state-level entry reading $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega] \bowtie_k c_k\,\tau_A$; and $\Theta_k = c_k \in \mathbb{R}$ in untrusted-source mode, the entry reading $\mathrm{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big] \bowtie_k c_k$, a comparison of real numbers (Definition "Licensed constraint list" (protocol-compiler/def:homog)). In both cases the untagged entries are written $(\hat{O}_k, \bowtie_k, c_k)$.

The *party-local form*. Where the measured system is declared as a product $Q = \bigotimes_f Q_f$ of party-scoped factors, an entry may in addition be *scoped to a factor* f. It is written $(\hat{O}_k, \bowtie_k, c_k; f)$ with $\hat{O}_k \in \mathcal{L}(Q_f)$ Hermitian and $c_k \in \mathbb{R}$, it carries the party-local tag, and its state-level reading, in either source mode, is

$$\mathrm{tr}_{Q_f}\Big[\big(\mathbb{1}\otimes(\hat{O}_k - c_k\,\mathbb{1}_{Q_f})\big)\,\omega\Big] \;\bowtie_k\; 0 \qquad \text{as an operator on the systems of } \omega \text{ other than } Q_f, \tag{6.5}$$

those systems being A, where the declaration carries one, and the factors $Q_{f'}$, $f' \neq f$. Equivalently $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{O}_k)\omega] \bowtie_k c_k\,\mathrm{tr}_{Q_f}\,\omega$: the entry is homogeneous with respect to the marginal that is actually there. The attack-set licence of Theorem 6.7 covers a party-local entry only through a certificate on the factor's own squashing channel (its part (v)); the product structure that makes it sound is Lemma 6.6.

What the tag does not change. An entry without the tag is read in the form of its source mode, exactly as above. With one factor, the party-local reading in trusted-source mode is $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega] \bowtie_k c_k\,\omega_A$, which under the consumer's marginal condition $\omega_A = \tau_A$ is the trusted-source form; with one factor and $d_A = 1$ it is the untrusted-source comparison of real numbers. *With one factor, in untrusted-source mode and $d_A > 1$ it is neither*: it reads $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega] \bowtie_k c_k\,\omega_A$, an operator inequality on A against the marginal that is actually there, and the untrusted-source scalar entry is its trace (Lemma 6.6 (iii)), implied by it and not equivalent to it. With more than one factor it implies the untagged reading of the same $(\hat{O}_k, \bowtie_k, c_k)$ for the lifted operator $\mathbb{1} \otimes \hat{O}_k$, which is obtained from it by the partial trace over the other factors (and over A in untrusted-source mode; in trusted-source mode under the consumer's marginal condition $\omega_A = \tau_A$), and in general it is not implied by it.

Used by Thm. 6.7, Def. 7.2, Lem. 8.1, Thm. 8.6, Rem. 8.8, Cor. 11.3, Rem. 11.4, Thm. 11.5 · newly derived · accepted

For a product squashing channel, a spectral bound certified on one part's channel holds of every squashed attack as an operator inequality on all other systems, whose trace is the scalar entry. Without the product structure the operator form can fail while the scalar form holds.

A party-local entry (Definition 6.5) is read against the marginal that is actually present rather than a declared one. The lemma shows that a certificate on the part's own channel makes that reading hold of every squashed attack, and that it can fail once the channel correlates the parts.

Lemma 6.6 (Party-local transport)**.** Let the measured system be a tensor product $B = B_f \otimes B_{f^c}$ of a *part* B_f and the product B_{f^c} of all other parts, let $\Lambda_f \colon \mathcal{T}(B_f) \to \mathcal{T}(Q_f)$ and $\Lambda_{f^c} \colon \mathcal{T}(B_{f^c}) \to \mathcal{T}(Q_{f^c})$ be channels, and put $\Lambda \coloneqq \Lambda_f \otimes \Lambda_{f^c}$ and $Q \coloneqq Q_f \otimes Q_{f^c}$. Let $\hat{O} \in \mathcal{L}(Q_f)$ be Hermitian with a spectral pull-back certificate *on the part*,

$$\Lambda_f^\dagger(\hat{O}) \;\bowtie\; c\,\mathbb{1}_{B_f}, \qquad \bowtie \in \{\preceq, =, \succeq\}, \quad c \in \mathbb{R}. \tag{6.6}$$

Let R and R' be finite-dimensional systems, E', E'' finite-dimensional, and $\mathcal{A} \in \mathrm{CPTP}(E', R\,B\,E'')$ *any* channel, with $\mathcal{A}^{\mathrm{sq}} \coloneqq (\mathrm{id}_R \otimes \Lambda \otimes \mathrm{id}_{E''}) \circ \mathcal{A}$. Write $\hat{D} \coloneqq \hat{O} - c\,\mathbb{1}_{Q_f}$.

(i) *(channel level)* for every $X \in \mathrm{Pos}(R'R\,Q_{f^c})$,

$$\big(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}}\big)^\dagger\big(X \otimes \hat{D} \otimes \mathbb{1}_{E''}\big) \;\bowtie\; 0 \qquad \text{as an operator on } R'E', \tag{6.7}$$

the tensor factors written in the order $R'R\,Q_{f^c}$, Q_f, E''.

(ii) *(state level)* for every $\varrho \in \mathrm{Pos}(R'E')$, the operator $\omega \coloneqq \mathrm{tr}_{E''}\big[(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}})(\varrho)\big]$ on $R'R\,Q$ satisfies

$$\mathrm{tr}_{Q_f}\big[(\mathbb{1}_{R'R\,Q_{f^c}} \otimes \hat{D})\,\omega\big] \;\bowtie\; 0 \qquad \text{as an operator on } R'R\,Q_{f^c}, \tag{6.8}$$

equivalently $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{O})\,\omega] \bowtie c\,\mathrm{tr}_{Q_f}\,\omega$: the right-hand side is c times the marginal of ω *that is actually there*, and no marginal is assumed.

(iii) *(the scalar entry is its trace)* taking the trace of (6.8) gives $\mathrm{tr}\Big[(\mathbb{1} \otimes \hat{O})\,\omega\Big] \bowtie c\,\mathrm{tr}\,\omega$, the scalar form. With R' trivial and $R = A$, Q_{f^c} trivial (one factor), (6.8) reads $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O})\omega] \bowtie c\,\omega_A$.

(iv) *(the product structure is needed)* if Λ is not a product the conclusion can fail while the scalar one holds. Let Q_f and Q_{f^c} be qubits, R, R' trivial, $\hat{O} = |0\rangle\langle 0|$, $c = \frac{1}{2}$, and let Λ discard its input and prepare $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Then $\Lambda^\dagger(\mathbb{1} \otimes \hat{O}) = \frac{1}{2}\mathbb{1}_B \succeq c\,\mathbb{1}$, so every squashed attack satisfies the scalar entry with $\succeq$; but $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{D})\omega] = \frac{1}{4}\,|0\rangle\langle 0| - \frac{1}{4}\,|1\rangle\langle 1|$, which is not positive semidefinite.

Uses Lem. 1.2 · *Used by* Thm. 6.7, Rem. 11.4, Thm. 11.5 · newly derived · accepted

Proof. (i) Let $\bowtie$ be $\succeq$. By Lemma 1.2 (ii), applied to the product $\mathrm{id}_{R'R} \otimes \Lambda_{f^c} \otimes \Lambda_f \otimes \mathrm{id}_{E''}$,

$$\big(\mathrm{id}_{R'R} \otimes \Lambda \otimes \mathrm{id}_{E''}\big)^\dagger \big(X \otimes \hat{D} \otimes \mathbb{1}_{E''}\big) = \big(\mathrm{id}_{R'R} \otimes \Lambda_{f^c}\big)^\dagger (X) \;\otimes\; \Lambda_f^\dagger(\hat{D}) \;\otimes\; \mathbb{1}_{E''}.$$

The first factor is positive semidefinite: $X \succeq 0$ and the adjoint of the channel $\mathrm{id}_{R'R} \otimes \Lambda_{f^c}$ is a positive map, the adjoint of a completely positive map being completely positive. The second is $\Lambda_f^\dagger(\hat{O}) - c\,\Lambda_f^\dagger(\mathbb{1}_{Q_f}) = \Lambda_f^\dagger(\hat{O}) - c\,\mathbb{1}_{B_f} \succeq 0$ by unitality of $\Lambda_f^\dagger$ and (6.6). A tensor product of positive semidefinite operators is positive semidefinite. Finally $(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}})^\dagger = (\mathrm{id}_{R'} \otimes \mathcal{A})^\dagger \circ (\mathrm{id}_{R'R} \otimes \Lambda \otimes \mathrm{id}_{E''})^\dagger$, and $(\mathrm{id}_{R'} \otimes \mathcal{A})^\dagger$ is a positive map because $\mathrm{id}_{R'} \otimes \mathcal{A}$ is a channel; this gives (6.7). For $\preceq$ apply the argument to $-\hat{D}$, and for $=$ both hold, the left-hand side of (6.7) then vanishing for every $X \succeq 0$ and hence, by linearity, for every X.

(ii) For $X \in \mathrm{Pos}(R'R\,Q_{f^c})$, by the definition of the adjoint,

$$\mathrm{tr}\Big[X\ \mathrm{tr}_{Q_f}\big[(\mathbb{1} \otimes \hat{D})\,\omega\big]\Big] = \mathrm{tr}\big[(X \otimes \hat{D} \otimes \mathbb{1}_{E''})\,(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}})(\varrho)\big] = \mathrm{tr}\Big[\big(\mathrm{id}_{R'} \otimes \mathcal{A}^{\mathrm{sq}}\big)^\dagger \big(X \otimes \hat{D} \otimes \mathbb{1}_{E''}\big)\,\varrho\Big],$$

which by (i) and $\varrho \succeq 0$ is ≥ 0 for $\succeq$, ≤ 0 for $\preceq$ and 0 for $=$. A Hermitian operator Y with $\mathrm{tr}[XY] \geq 0$ for every $X \succeq 0$ is positive semidefinite, and $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{D})\omega]$ is Hermitian because $\hat{D}$ is and acts on the traced factor alone. The equivalent form is linearity: $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes c\,\mathbb{1}_{Q_f})\omega] = c\ \mathrm{tr}_{Q_f}\,\omega$.

(iii) The trace is a positive functional, and $\mathrm{tr}\,\big[\,\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{O})\omega]\big] = \mathrm{tr}\Big[(\mathbb{1} \otimes \hat{O})\omega\Big]$. The special case is a substitution.

(iv) $\Lambda^\dagger(Y) = \mathrm{tr}[Y\varsigma]\,\mathbb{1}_B$ for the prepared state $\varsigma = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$, so $\Lambda^\dagger(\mathbb{1} \otimes |0\rangle\langle 0|) = \frac{1}{2}\mathbb{1}_B$. Every squashed attack outputs $\omega = \varsigma$ on Q, and $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes (|0\rangle\langle 0| - \frac{1}{2}\mathbb{1}))\varsigma] = \frac{1}{2}\big(\frac{1}{2}\,|0\rangle\langle 0|\,\big) + \frac{1}{2}\big(-\frac{1}{2}\,|1\rangle\langle 1|\,\big)$, with the traced qubit written second. □

If each operator the squashing carries pulls back to a physical operator $\leq$, $=$ or $\geq$ a multiple of the identity, every squashed physical attack respects the resulting constraint list in either source mode, factor by factor for per-factor bounds. Security against such attacks transfers with the same ε.

The consuming engine may restrict the squashed attacks to the admissible set of Definition 6.3 only if every squashed physical attack lies in it. The theorem proves this from one hypothesis about the device, a spectral bound on each carried operator's physical pull-back, and uses nothing about the attack beyond its being a channel.

Theorem 6.7 (The attack-set licence). Let $(M;O) \preceq_\Lambda (F;\hat{O})$ with $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$, and suppose each carried operator has a *spectral pull-back certificate*: for each k there are $\bowtie_k \in \{\preceq, =, \succeq\}$ and $c_k \in \mathbb{R}$ with

$$O_k \;=\; \Lambda^\dagger(\hat{O}_k) \;\bowtie_k\; c_k\,\mathbb{1}_B \qquad \text{as operators on } \mathcal{H}(B). \tag{6.9}$$

Then, with $\mathfrak{L} := \{(\hat{O}_k, \bowtie_k, c_k)\}_{k=1}^m$:

(i) *(trusted-source mode)* for all finite-dimensional E', E'' and *every* $\mathcal{A} \in \mathrm{CPTP}(E', BE'')$, the composite $\mathcal{A}^{\mathrm{sq}} := (\Lambda \otimes \mathrm{id}_{E''}) \circ \mathcal{A}$ lies in $\mathfrak{A}_{\mathfrak{L}}(E', QE'')$ of (6.2);

(ii) *(untrusted-source mode)* for every $\mathcal{A} \in \mathrm{CPTP}(E', ABE'')$, the composite $\mathcal{A}^{\mathrm{sq}} := (\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}_{E''}) \circ \mathcal{A}$ lies in $\mathfrak{A}_{\mathfrak{L}}(E', AQE'')$ of (6.3);

(iii) *(the licensed reduction, in the declared mode)* consequently, for every attack sequence $(\mathcal{A}_j)_{j=1}^{n}$ against the physical protocol of the type (3.3) for the declared source mode, the sequence $(\mathcal{A}_j^{\mathrm{sq}})_j$ of the *same* branch of (3.4) is $\mathfrak{L}$-admissible — by (i) with the trusted branch of (3.3), by (ii) with the untrusted branch — and produces the *same* protocol output state, by the correspondingly named branch of Corollary 3.3; so security of the squashed round model against $\mathfrak{L}$-admissible sequences implies security of the physical protocol against all sequences of that type, with the same ε;

(iv) *(licensed entries)* $\mathfrak{L}$ is licensed for either source mode in the sense of Definition 6.5, with the same constants c_k; on every reachable state σ the transported entries hold in the form with the marginal that is present, $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\sigma] \bowtie_k c_k\, \sigma_A$, and so in the untrusted-source form $\mathrm{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\sigma\Big] \bowtie_k c_k$; and the trusted-source form $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\sigma] \bowtie_k c_k\, \tau_A$ holds on every reachable state whose marginal on A is τ_A.

(v) *(party-local entries, product measured system) in place of the hypotheses above*, which type $\hat{O}_k$ on Q and certify it against Λ, let $B = \bigotimes_f B_f$, let $M^f \preceq_{\Lambda_f} F^f$ for every factor with $\Lambda_f \colon \mathcal{T}(B_f) \to \mathcal{T}(Q_f)$, and put $M := \{\bigotimes_f M^f_{v_f}\}$, $F := \{\bigotimes_f F^f_{v_f}\}$ and $\Lambda := \bigotimes_f \Lambda_f$, so that $M \preceq_{\Lambda} F$ (Proposition 4.8). Let each entry of $\mathfrak{L}$ be scoped to a factor f_k with $\hat{O}_k \in \mathcal{L}(Q_{f_k})$ and a certificate *on that factor's channel*, $\Lambda^{\dagger}_{f_k}(\hat{O}_k) \bowtie_k c_k\, \mathbb{1}_{B_{f_k}}$. Then for every attack $\mathcal{A}$ of either type of (3.3), the squashed attack of the same branch of (3.4) satisfies (6.4) for every k, and also the scalar condition of (6.2) or (6.3) for the lifted operator $\mathbb{1}_{Q_{f_k^c}} \otimes \hat{O}_k$ with the same c_k; every squashed sequence is admissible for $\mathfrak{L}$ read with any subset of its entries tagged party-local, it produces the same protocol output state as the physical sequence, and the security implication of (iii) holds with the same ε. On every reachable state the entries hold in the form (6.5).

Uses Lem. 1.2, Cor. 1.7, Def. 3.2, Cor. 3.3, Cor. 3.13, Prop. 4.8, Def. 6.3, Def. 6.5, Lem. 6.6 · *Assumes* A-CONST · *Used by* Rem. 6.8, Rem. 6.9, Rem. 6.10, Thm. 7.4, Thm. 8.6, Rem. 8.8, Def. 10.1, Thm. 10.4, Cor. 11.3, Rem. 11.4, Thm. 11.5, Cor. 12.1, Lem. 12.3, Req. 15.2, Req. 16.3, Cor. 21.1, Cor. 21.4, Cor. 21.5, Cor. 24.1, protocol-compiler/req:link-row-provenance · newly derived · accepted · *Oracles* registry_reductions.py

Proof. (i) Fix k and write $\mathcal{A}^{\mathrm{sq}\dagger} = \mathcal{A}^{\dagger} \circ (\Lambda \otimes \mathrm{id}_{E''})^{\dagger}$, which is (1.1) applied twice. By Lemma 1.2 (ii),

$$(\Lambda \otimes \mathrm{id}_{E''})^{\dagger}\big(\hat{O}_k \otimes \mathbb{1}_{E''}\big) = \Lambda^{\dagger}(\hat{O}_k) \otimes \mathbb{1}_{E''} = O_k \otimes \mathbb{1}_{E''}. \tag{6.10}$$

Suppose $\bowtie_k$ is $\succeq$. From $O_k \succeq c_k \mathbb{1}_B$ we get $O_k \otimes \mathbb{1}_{E''} \succeq c_k\, \mathbb{1}_{BE''}$, and $\mathcal{A}^{\dagger}$ is positive and unital because $\mathcal{A}$ is completely positive and trace preserving; hence

$$\mathcal{A}^{\mathrm{sq}\dagger}\big(\hat{O}_k \otimes \mathbb{1}_{E''}\big) - c_k\, \mathbb{1}_{E'} = \mathcal{A}^{\dagger}\big(O_k \otimes \mathbb{1}_{E''} - c_k\, \mathbb{1}_{BE''}\big) \;\succeq\; 0,$$

using $\mathcal{A}^{\dagger}(\mathbb{1}_{BE''}) = \mathbb{1}_{E'}$. For $\bowtie_k = \preceq$ apply the same argument to $c_k \mathbb{1}_B - O_k \succeq 0$; for $\bowtie_k = =$ both directions hold, or directly $\mathcal{A}^{\dagger}(c_k \mathbb{1}) = c_k \mathbb{1}_{E'}$. Since k was arbitrary, $\mathcal{A}^{\mathrm{sq}} \in \mathfrak{A}_{\mathfrak{L}}(E', QE'')$.

(ii) Identical, with (6.10) replaced by $(\mathrm{id}_A \otimes \Lambda \otimes \mathrm{id}_{E''})^{\dagger}\big(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E''}\big) = \mathbb{1}_A \otimes O_k \otimes \mathbb{1}_{E''}$, again Lemma 1.2 (ii), and with $\mathbb{1}_A \otimes O_k \otimes \mathbb{1}_{E''} \bowtie_k c_k\, \mathbb{1}_{ABE''}$ following from (6.9) by tensoring with identities.

(iii) In trusted-source mode, admissibility is (i) applied per round with $E' = E_{j-1}$, $E'' = E'_j$ and the trusted branch of (3.4); in untrusted-source mode it is (ii) with the untrusted branch, the lifted operator $\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E'_j}$ of (6.3) being exactly the one that branch's composite is tested against. The equality of output states is the correspondingly typed branch of Corollary 3.3, whose hypothesis is exactly $M \preceq_{\Lambda} F$ in both modes; and the security implication is the last sentence of that corollary, the supremum over the $\mathfrak{L}$-admissible sequences dominating the value at $(\mathcal{A}^{\mathrm{sq}}_j)_j$. The two branches of (3.5) are $\mathfrak{A}_{\mathfrak{L}}(E_{j-1}, QE'_j)$ and $\mathfrak{A}_{\mathfrak{L}}(E_{j-1}, AQE'_j)$ respectively, by (i) and (ii).

(iv) The right-hand sides are the scalars c_k of (6.9), which are declared constants of the reduction and not functions of any observed quantity (Assumption 13.7); reading c_k as $c_k \tau_A$ in trusted-source mode and as the real number c_k in untrusted-source mode is Definition 6.5 in the two modes. That the entries hold on every reachable state σ in the form with the marginal present is Corollary 3.13 (i) applied to the certificate (6.9), which is case (i) or (iii) of Corollary 1.7, read on the side it bounds, with $R = A$: that corollary delivers $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\sigma] \bowtie_k c_k\, \sigma_A$, and its scalar form, the trace of both sides with $\mathrm{tr}\, \sigma_A = 1$, is the untrusted-source form. In trusted-source mode the identification of the right-hand side with $c_k \tau_A$ is the *declaration's* and not this contract's: it holds where $\sigma_A = \tau_A$, which is the consumer's marginal condition $\omega_A = \tau_A$ (defined in Definition "Constrained target set and single-round rate" (meat-core/def:hqkd-list) and asserted by the consuming contract under its own source-replacement assumption), and nothing in the hypotheses of this theorem restricts the marginal on A.

(v) Fix k, write $f = f_k$, $\Lambda_{f^c} := \bigotimes_{f' \neq f} \Lambda_{f'}$, and reorder tensor factors so that $\Lambda = \Lambda_f \otimes \Lambda_{f^c}$. Lemma 6.6 (i), with $R = A$ in untrusted-source mode and R trivial in trusted-source mode, and with the certificate on the

factor as its hypothesis (6.6), is (6.4) for the squashed attack, for every reference system R'. Its part (i) at R' trivial and $X = \mathbb{1}$, with unitality of the adjoint of the squashed attack, a channel, gives the scalar condition for the lifted operator; alternatively $\Lambda^\dagger(\mathbb{1} \otimes \hat{O}_k) = \mathbb{1}_{B_{fc}} \otimes \Lambda_f^\dagger(\hat{O}_k) \bowtie_k c_k \mathbb{1}_B$ by Lemma 1.2 (ii) and unitality, which is a certificate (6.9) for Λ, so (i) and (ii) apply to it. Admissibility of the squashed sequence is this per round, with $E' = E_{j-1}$ and $E'' = E'_j$. The equality of output states and the security implication are Corollary 3.3 for the channel Λ, whose hypothesis $M \preceq_\Lambda F$ is Proposition 4.8; the supremum over the admissible sequences dominates the value at the squashed one, whichever entries are tagged. The state-level form is Lemma 6.6 (ii), with $R' = A$ carrying Alice's retained system in trusted-source mode. □

The licence permits one thing, restricting the squashed attacks to the declared admissible set. It licenses no trusted marginal, no constraint on Eve's side information, no data-dependent or multi-round constraint, no reuse under another proof technique, and nothing about the device.

Remark 6.8 (Scope of the attack-set licence). Theorem 6.7 licenses exactly one thing: restricting the squashed round model's attacks to the declared admissible set of Definition 6.3. It does *not* license, and this contract asserts none of the following.

(a) *A trusted marginal.* Nothing here licenses declaring τ_A. That is a separate input with a separate attestation on the consumer's side, and a licensed $\mathfrak{L}$ never substitutes for it: a constraint entry restricts what *Eve's channel* may do, whereas τ_A asserts that a register never reached her at all.

(b) *Constraints on the purifying register.* Every entry is an operator on Q (resp. AQ); no constraint on Eve's side information is expressible in this format or admitted by it.

(c) *Data-dependent constraints.* The constants c_k are declared before the run (Assumption 13.7); an entry whose c_k encoded a statistic of the current run would be unsound here regardless of provenance, and such an entry is refused (Requirement 15.3).

(d) *Joint multi-round constraints.* Admissibility is per round (Definition 6.3); Theorem 6.7 is proved per round, and no constraint on the joint n-round state is licensed or expressible.

(e) *Reuse under another proof technique.* The restriction is compatible with the accumulation route the consuming contract imports. It is *not* automatically compatible with a postselection-based reduction: the published framework records that the analogous restriction "cannot be straightforwardly used with the postselection technique" ([8, Rem. 9.4]). A module importing $\mathfrak{L}$ into a different technique must re-establish the licence there; the artefact names the technique the licence was issued for (Requirement 15.2).

(f) *Anything about the device.* Whether the receiver actually satisfies the hypothesis (6.9) — concretely, whether $M_o^{(>N)} \succeq p_o \mathbb{1}$ holds of the physical detector — is *not* proved here and cannot be: it is Assumption 13.3, discharged by

the accepted receiver contract for the device that contract describes, under its own ledger.

Uses Def. 6.3, Thm. 6.7 · *Assumes* A-CONST, A-RECV · *Used by* Req. 15.2, Req. 15.3 · newly derived · accepted

In untrusted-source mode, or where Alice's system has dimension one, the consumer's mechanical check of the licensed class degenerates and its gate rests entirely on the supplier's licence. The licence is therefore a theorem with a named hypothesis rather than an attestation.

Remark 6.9 (Weight of the licence in the consumer's gate). The consuming contract gates a key-length claim on a non-empty $\mathfrak{L}$ behind two checks: (i) that the list lies in the licensed class of the declared source mode, which is mechanical; and (ii) that the supplier's attack-set licence is present, which no program can verify. It also records that check (i) *degenerates*: at $d_A = 1$ every 1×1 Hermitian right-hand side is a multiple of τ_A, and in untrusted-source mode the type of an entry is scalar by construction, so part (i) rejects nothing in untrusted-source mode and nothing a producer cannot trivially avoid at $d_A = 1$, and in both cases the gate rests entirely on part (ii) (Remark "Degeneracy of the licensed-class half of the gate" (meat-core/rem:cgate)). This is the reason Theorem 6.7 is stated as a theorem with an explicit hypothesis rather than as an attestation: in the cases that matter, the attestation *is* the security argument, and the only defensible form for it is a proof plus a named hypothesis that some other accepted contract discharges. The licence record names both, and the artefact that discharges the hypothesis (Requirement 15.2).

Uses Thm. 6.7 · *Used by* Req. 15.2 · newly derived · accepted

The adjoint format of the admissible set is the published squashed-attack-set format, and the published inclusion is the special case of Theorem 6.7 (i) at one flag-state operator with constant zero. The licence generalises it to any certified carried operator in either source mode.

Remark 6.10 (The published squashed-attack-set format). The adjoint format of (6.2) is the published squashed-attack-set format: [8, Lemma 9.4, Eq. (151)] constructs, for a flag-state-squashed measurement, the set $\mathfrak{A}_j := \{\mathcal{A}_j^{\mathrm{Sq}} : \mathcal{A}_j^{\mathrm{Sq}\dagger}(F_o^{(Q)} \otimes \mathbb{1}_{E_j}) \succeq \lambda_{\min}\mathcal{A}_j^{\mathrm{Sq}\dagger}(\Pi_{Sq} \otimes \mathbb{1}_{E_j})\}$ from the physical operator inequality $M_o^{(B)} \succeq \lambda_{\min}\Pi_{m>N_{\mathrm{FSS}}}$, the projector being onto the photon numbers above the cutoff N_{FSS} of the flag-state squasher, and proves the inclusion $\mathfrak{A}_j \supseteq \Lambda \circ \mathrm{CPTP}(E_{j-1}, B_j E_j)$; its Eq. (152) is the dimension-independent marginal form of the same set, and the modelling step of restricting the attack-channel set at all is its Def. 8.3 with Rem. 8.4. That inclusion is the special case of Theorem 6.7 (i) at $\hat{O} = F_o - \lambda_{\min}\Pi_{Sq}$, $c = 0$; that theorem is the general statement, for an arbitrary carried operator with a spectral pull-back certificate, in either source mode, with the round-model bookkeeping of Corollary 3.3 attached. It is proved there from (1.1), positivity and unitality, and appeals to no published result.

Uses Lem. 1.2, Cor. 3.3, Def. 6.3, Thm. 6.7 · newly derived · accepted

7 Typed links and reduction chains

The artefact is a chain of typed links, each carrying enough to be re-checked: its channel witness, carried operators, reachable-set statement, transported constraint entries and, where there are entries, a licence record. A chain is a source segment of the transmitter followed by composable receiver-stage links, and its receiver composite is again a squashing relation to which output-state equality, security transfer and the licence apply (Theorem 7.4). A declaration with a non-empty source segment and constraint entries is admitted only with licensed entries and in trusted-source mode. An already-finite receiver reduces by one identity link.

Imports transmitter/def:source-link, transmitter/def:source-segment

A typed link is a source link of the transmitter's type or a receiver-stage tuple of ten slots: stage, kind, source family, target, channel witness, carried operators, reachable-set statement, transported entries, licence record and status. Each link carries enough to be re-checked.

What this module emits is not a squashed POVM but a chain of typed links, each carrying enough to re-check it, in an order that is data rather than convention. This definition states a link. How source links and receiver-stage links compose is Definition 7.2, with its condition (C3) for the order the record carries and its admission rule for which such chains are admitted at all; Remark 8.8 gives the reason, Theorem 7.4 what the receiver stage composes to, and Theorem 8.6 what the two stages compose to.

Definition 7.1 (Typed link)**.** A *typed link* carries a stage $\mathsf{stage} \in \{\mathsf{source}, \mathsf{receiver}\}$. A link at the stage source is a source link in the sense of Definition "Source link" (transmitter/def:source-link): its descriptors are Alice-side, a transmitted system with a signal ensemble on either end and a channel witness between them, and no condition on it mentions the receiver's systems, outcome alphabets or measurements. *This module reads that type as the transmitter states it*: the channel witness is supplied either *as data*, in which case the source identity is decided on declared numbers, or *as a theorem constructor*, in which case the link carries an obligation record, namely the triple of the kind, the id of the soundness claim and the assumptions the obligation carries, which Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (A1) defines and which this module neither computes nor re-decides, together with the member of its declared descriptor family that its descriptors are; and a source link carries in addition that family, an optional retained-register sector map, its transported constraint entries with its licence slot, and its status. Which kinds are admitted at all, and which components a consumer decides *on data* (the emitted soundness registry, the triple compared against it entry by entry with those assumptions demanded in the ledger, and the provenance rule for a non-empty segment), is Definition "Source segment and its interface conditions" (transmitter/def:source-segment), conditions (A1) to (A3); this module re-decides none of them, adds none, and adds no source-stage kind. A

receiver-stage typed link ℓ is a tuple

$$\ell = \Big(\mathsf{stage},\ \mathsf{kind},\ \big(B^{\mathrm{in}}, M^{\mathrm{in}}\big),\ \big(B^{\mathrm{out}}, M^{\mathrm{out}}\big),\ \Lambda_\ell,\ \big(\hat{O}^{\mathrm{out}}, O^{\mathrm{in}}\big),\ S^{\mathrm{rch}}_\ell,\ \mathfrak{L}_\ell,\ \mathsf{licence},\ \mathsf{status}\Big) \tag{7.1}$$

in which: $\mathsf{stage} = \mathsf{receiver}$; kind is one of the receiver-stage link kinds the module's interface declares (the core's kinds, and those a submodule's interface adds); $(B^{\mathrm{in}}, M^{\mathrm{in}})$ is the *source family* — the system and POVM the link starts from — and $(B^{\mathrm{out}}, M^{\mathrm{out}})$ its *target*; Λ_ℓ is the *channel witness*, with $\big(M^{\mathrm{in}}; O^{\mathrm{in}}\big) \preceq_{\Lambda_\ell} \big(M^{\mathrm{out}}; \hat{O}^{\mathrm{out}}\big)$; S^{rch}_ℓ is the *reachable-set statement* of Definition 3.10 for Λ_ℓ and the declared family of inputs; $\mathfrak{L}_\ell$ is the list of transported constraint entries $(\hat{O}_k, \bowtie_k, c_k)$ with, per entry, the certificate case of Corollary 1.7 that discharges it; $\mathsf{licence}$ is the *licence record* of the module's outputs, present exactly when $\mathfrak{L}_\ell \neq \varnothing$; and $\mathsf{status} \in \{\mathsf{exact}, \mathsf{certified_outer}, \mathsf{exploratory_numeric}\}$, with exact meaning that $\Lambda^\dagger_\ell(M^{\mathrm{out}}_v) = M^{\mathrm{in}}_v$ holds with no error term.

The count. A receiver-stage link has ten slots. Six of them are the required fields — source family, target, channel witness, reachable-set statement, transported constraint entries, licence record — and the remaining four are stage, kind, the carried operator pair, and status. Wherever this contract says "the six required fields" it means the first group; the tuple (7.1) is the whole of what an implementer binds.

Uses Cor. 1.7, Def. 1.8, Def. 3.10, transmitter/def:source-link, transmitter/def:source-segment · *Used by* Def. 7.2, Thm. 7.4, Cor. 10.2, Thm. 11.2, Cor. 12.1, Def. 23.1, Thm. 23.2, Cor. 24.1 · newly derived · accepted

A reduction chain pairs a source segment with composable, tuple-chained receiver-stage links, and its composite is the pair of their composites. A declaration with a non-empty source segment and constraints is admitted only with licensed entries and in trusted-source mode.

Remark 8.8 gives the reason for the stage order (C3), and for the admission rule of the definition the reason it gives is Lemma 8.1 and its counterexample.

Definition 7.2 (Reduction chain)**.** A *reduction chain* is a pair $\mathfrak{C} = (\mathfrak{C}_{\mathrm{src}}, \mathfrak{C}_{\mathrm{rec}})$ of a source segment and a *receiver segment*. The source segment $\mathfrak{C}_{\mathrm{src}} = (\Xi_1, \ldots, \Xi_{L_{\mathrm{s}}})$, $L_{\mathrm{s}} \geq 0$, is a source segment in the sense of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (written $\mathfrak{R}_{\mathrm{src}}$ there): a list of source links (Definition "Source link" (transmitter/def:source-link)) satisfying that definition's interface conditions (S1)–(S4) — descriptor match, index preservation, the source identity where checkable, and the slots — every descriptor of which is Alice-side. The receiver segment $\mathfrak{C}_{\mathrm{rec}} = (\ell_1, \ldots, \ell_k)$, $k \geq 1$, is a list of receiver-stage typed links (Definition 7.1) that is

(C1) *composable*: $(B^{\mathrm{out}}_i, M^{\mathrm{out}}_i) = (B^{\mathrm{in}}_{i+1}, M^{\mathrm{in}}_{i+1})$ for $1 \leq i < k$;

(C2) *tuple-chained*: $\hat{O}_i^{\mathrm{out}} = O_{i+1}^{\mathrm{in}}$ for $1 \le i < k$, so that each declared operator on the final target has a pull-back at every intermediate stage;

(C3) *stage-ordered*: every source-stage link precedes every receiver-stage link, which the pair form makes structural.

Its *composite* is the *pair*

$$\big(\Xi_{\mathfrak{C}},\ \Lambda_{\mathfrak{C}}\big), \qquad \Xi_{\mathfrak{C}} \coloneqq \Xi_{L_{\mathrm{s}}} \circ \cdots \circ \Xi_1, \qquad \Lambda_{\mathfrak{C}} \coloneqq \Lambda_{\ell_k} \circ \cdots \circ \Lambda_{\ell_1}, \tag{7.2}$$

of the *source composite* — the composite the transmitter's (S3) writes Ξ, the identity when $L_{\mathrm{s}} = 0$ — and the *receiver composite*. The two act on different systems and no condition above relates them: $\Xi_{\mathfrak{C}}$ carries the virtual transmitted system to the transmitted one, on Alice's side, and $\Lambda_{\mathfrak{C}}$ the measured system to the target, on the receiver's. What the pair composes to *as a reduction* is Theorem 8.6, and Theorem 7.4 is what the receiver composite alone delivers.

Which chains are admitted. The rule is a condition on a *declaration* that carries both a non-empty source segment and a constraint list: such a declaration is *admitted* only where every entry of the list is licensed in the sense of Definition 6.5 — a real constant c_k as the right-hand side, the entry tagged or untagged — *and only in trusted-source mode*, Theorem 8.6 being stated there alone, so that a declaration pairing a non-empty source segment with untrusted-source mode is refused rather than admitted in silence. An entry declaring a Hermitian right-hand side on Alice's retained system, a *general* entry in the sense of Definition 6.3, is *refused* beside one. The rule is decided by the consumer that assembles those two.

Where the source segment is empty, $L_{\mathrm{s}} = 0$, the source composite is the identity, $A'' = A'$, the virtual ensemble is the specified one, and the pair (7.2) carries exactly the receiver composite: Alice's declared states are the ones the security statement is about, and the receiver segment is the whole of the chain's content (Theorem 7.4).

Uses Def. 6.3, Def. 6.5, Def. 7.1, transmitter/def:source-link, transmitter/def:source-segment · *Used by* Rem. 7.3, Thm. 7.4, Ex. 7.5, Rem. 8.8, Cor. 10.2, Thm. 11.2, Cor. 12.1, Req. 15.5, Cor. 24.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.links:Chain

This module states the admission rule of a chain and decides no instance of it; the protocol compiler decides both its halves. Every entry this module emits carries a real constant and every artefact an empty source segment, so no emitted artefact can violate the rule.

Remark 7.3 (The admission rule and what this module emits). This module states the admission rule of Definition 7.2 and decides no instance of it. It declares no general entry and emits none, every entry it emits carrying a real scalar (Definition 6.3),

and every artefact it emits has an empty source segment; so no artefact of this module can violate the rule. The rule is decided where a declaration's source segment is assembled with its constraint list, which is the protocol compiler's chain condition (Definition "Interface conditions on a chain" (protocol-compiler/def:chaincond)). It decides both halves of the rule: the *entry* half, a general entry beside a non-empty source segment, and the *mode* half, a declaration carrying a non-empty source segment in untrusted-source mode, each refused by name there (Requirement "Refusals of the interface conditions on a chain" (protocol-compiler/req:chain-refusals)).

Uses Def. 6.3, Def. 7.2 · newly derived · accepted

The receiver composite of a chain is again a squashing relation with the composed pull-backs, so output-state equality, security transfer and the licence apply to it at every length of the source segment. Its status is exact whenever every link's is.

The order of the source and receiver stages, which this theorem does not read, is the subject of Remark 8.8.

Theorem 7.4 (Composition of a chain)**.** Let $\mathfrak{C} = (\mathfrak{C}_{\mathrm{src}}, \mathfrak{C}_{\mathrm{rec}})$ be a reduction chain with receiver segment $\mathfrak{C}_{\mathrm{rec}} = (\ell_1, \ldots, \ell_k)$ and receiver composite $\Lambda_{\mathfrak{C}}$ of (7.2). Then:

(i) $\left(M_1^{\mathrm{in}}; O_1^{\mathrm{in}}\right) \preceq_{\Lambda_{\mathfrak{C}}} \left(M_k^{\mathrm{out}}; \hat{O}_k^{\mathrm{out}}\right)$, and $\Lambda_{\mathfrak{C}}^{\dagger} = \Lambda_{\ell_1}^{\dagger} \circ \cdots \circ \Lambda_{\ell_k}^{\dagger}$;

(ii) Theorem 3.1, Theorem 3.11 and Corollary 3.3 apply to the composite: the physical protocol with M_1^{in} and the final squashed protocol with M_k^{out} have the same output state, round by round and adaptively;

(iii) an entry $(\hat{O}_k, \bowtie_k, c_k)$ declared on the *final* target is sound for the receiver composite when the certificate (6.9) holds for its pull-back $\Lambda_{\mathfrak{C}}^{\dagger}(\hat{O}_k)$, which (C2) computes link by link; and Theorem 6.7 then applies with $\Lambda_{\mathfrak{C}}$;

(iv) the composite status is exact whenever every link's is, and no other composite value is asserted.

Nothing above reads the source segment: (i)–(iv) are statements about the receiver composite $\Lambda_{\mathfrak{C}}$ and hold at every length $L_{\mathrm{s}} \geq 0$. What the *pair* (7.2) composes to as a reduction is Theorem 8.6, under the source segment's own interface conditions and the admission rule of Definition 7.2, and this theorem does not restate it. Where the source segment is empty the source composite is the identity and the receiver composite is the whole of the chain's reduction.

Uses Thm. 3.1, Cor. 3.3, Thm. 3.11, Cor. 3.13, Prop. 4.1, Thm. 6.7, Def. 7.1, Def. 7.2 · *Used by* Rem. 2.5, Ex. 7.5, Cor. 10.2, Thm. 11.2, Cor. 11.6, Cor. 12.1, Req. 15.4, Req. 15.5, Cor. 24.1, protocol-compiler/req:link-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.links:Chain · *Oracles* registry_reductions.py

Proof. (i) is Proposition 4.1 applied $k-1$ times, the tuple chaining (C2) being exactly the hypothesis that makes the operator tuples compose. (ii) is those results applied to the relation of (i), which is all they need. (iii) is Corollary 3.13 and Theorem 6.7 applied to the composite relation; the pull-back is computed as $\Lambda^\dagger_{\ell_1}\big(\dots\Lambda^\dagger_{\ell_k}(\hat{O}_k)\dots\big)$ by (i). (iv): a composition of exact identities is an exact identity, by (i) applied to identities carrying no error term. The remaining half is not proved and is not claimed: a chain containing a link that is not exact is declined rather than assigned a composite status, by the rules of Requirements 14.4 and 15.4. The closing paragraph is Definition 7.2: the composite is defined there as the pair (7.2), whose second member is the receiver segment's and is the only member any of (i)–(iv) reads — each of them quantifies over the links $\ell_1,\dots,\ell_k$ and over nothing else — and with $L_{\mathrm{s}}=0$ its first member is the identity. □

An already-finite receiver reduces by one identity link (the identity channel or the sector pinching) with the same POVM, no constraint entries and exact status; the chain theorem returns the physical model unchanged. No input the module's interface declares carries such a receiver.

Example 7.5 (The identity link). For an already-finite receiver (an ideal qubit protocol, or any fixture whose POVM can be consumed directly), the chain has one link with stage = receiver, kind = identity, $\Lambda_\ell = \mathrm{id}$ — or the sector pinching $\mathcal{P}_{\mathbb{S}}$ of the declared decomposition, which is the identity on every block-diagonal state and is the channel of the identity instance, Corollary 10.2 —, $M^{\mathrm{out}} = M^{\mathrm{in}}$, the carried tuple empty or the identity on any declared operators (block-diagonal ones where the channel is $\mathcal{P}_{\mathbb{S}}$), S^{rch}_ℓ the whole declared input family, pinched where the channel is $\mathcal{P}_{\mathbb{S}}$, $\mathfrak{L}_\ell = \varnothing$, no licence, and status = exact. It is a squashing relation: by reflexivity (Proposition 4.1) for id, and by Proposition 4.6 for $\mathcal{P}_{\mathbb{S}}$, every declared element being block diagonal for the decomposition. So Theorem 7.4 applies and returns the physical model unchanged. The link is recorded rather than omitted (Requirement 15.5). It is the chain of the identity instance (Corollary 10.2), whose emitted family is Corollary 11.6. No input the module's interface declares carries an already-finite receiver, so no declaration reaches this link; admitting one is a new declared input with a new refusal rule.

Uses Prop. 4.1, Prop. 4.6, Def. 7.2, Thm. 7.4 · newly derived · accepted

8 The source-side reduction and the two-sided theorem

Admissibility for a list of entries with real constants survives precomposition with any channel, in either source mode (Lemma 8.1). Replacing the specified signal ensemble of a trusted-source specification by its virtual ensemble is itself an exact operational reduction, and the entanglement-based description a source-replacement certificate presents has the same full output (Lemma 8.3). The main result, Theorem 8.6, is that the source map and the squashing reduction then compose, in either order, into one exact opera-

tional reduction whose image lies in the admissible attack set, so security of the virtual squashed model transfers to the specified physical protocol with the same ε, against attacks that receive every signal in the first round and keep finite-dimensional registers; every other attack is reached by a limit in the consuming engine (Corollary "Security of the protocol and its net key" (meat-core/cor:composed-security)). In untrusted-source mode there is no source segment to transfer, and the receiver-side reduction stands alone.

Imports transmitter/def:source-segment, transmitter/def:source-spec, transmitter/def:srcert, transmitter/thm:certificate-sufficiency

An attack channel admissible for a list of entries with real constants stays admissible when any channel is composed before it, in either source mode and for party-local entries. A Hermitian right-hand side that is not a real multiple of the identity is not preserved.

A receiver-stage link hands the analysis an attack set that is a subset of the channels, in general a proper one (Theorem 6.7), while a source-stage link re-parameterises the first-round attack as a composition with a channel; applying the source-stage link after the receiver-stage link needs an inclusion (Remark 8.8). The lemma proves that inclusion for every list whose right-hand sides are real constants, which is the form of a licensed list in either source mode (Definition 6.5) and of every entry this contract emits (Theorem 6.7 (iv)), and exhibits a right-hand side for which it fails.

Lemma 8.1 (Licensed admissibility survives precomposition with a source channel)**.** Let $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, c_k)\}_{k=1}^{m}$ be a list of Hermitian $\hat{O}_k$, relations $\bowtie_k \in \{\preceq, =, \succeq\}$ and *real* constants c_k. Let $\mathcal{H}(E_{\mathrm{pre}})$ and $\mathcal{H}(E')$ be separable, let Q, E'', A, R and R' be finite-dimensional, and let $\Gamma \in \mathrm{CPTP}(E_{\mathrm{pre}}, E')$ be *any* channel. The sets below are the ones the displays (6.2), (6.3) and (6.4) of Definition 6.3 cut out, read at these registers with the adjoints of Lemma 1.2; where E_{pre} and E' are finite-dimensional they are the attack sets that definition names.

(i) *(trusted-source mode)* if $\mathcal{A} \in \mathfrak{A}_{\mathfrak{L}}(E', QE'')$ of (6.2), then $\mathcal{A} \circ \Gamma \in \mathfrak{A}_{\mathfrak{L}}(E_{\mathrm{pre}}, QE'')$;

(ii) *(untrusted-source mode)* if $\mathcal{A} \in \mathfrak{A}_{\mathfrak{L}}(E', AQE'')$ of (6.3), then $\mathcal{A} \circ \Gamma \in \mathfrak{A}_{\mathfrak{L}}(E_{\mathrm{pre}}, AQE'')$;

(iii) *(party-local entries)* let $Q = \bigotimes_f Q_f$ and let the entry $(\hat{O}_k, \bowtie_k, c_k; f_k)$ carry the party-local tag (Definition 6.5), so that its condition on $\mathcal{A}$ is (6.4). If $\mathcal{A}$ satisfies (6.4) for every finite-dimensional R' and every $X \in \mathrm{Pos}(R'RQ_{f_k^c})$, then so does $\mathcal{A} \circ \Gamma$, with E_{pre} in place of E'. Consequently the admissible set of Definition 6.3, with *any* subset of its entries tagged and the rest read as scalars, is closed under precomposition with a channel into its input register;

(iv) *(the real constant is what carries it)* replace the right-hand side $c_k\,\mathbb{1}_{E'}$ of (6.2) by a Hermitian $\Theta \in \mathcal{L}(E')$ that is not a real multiple of $\mathbb{1}_{E'}$. Then the condition $\mathcal{A}^\dagger(\hat{O}\otimes\mathbb{1}_{E''}) \succeq \Theta$ is not preserved by precomposition: take $E_{\mathrm{pre}} = E' = Q = \mathbb{C}^2$ with declared basis $\{|0\rangle, |1\rangle\}$, E'' one-dimensional, $\mathcal{A} = \mathrm{id}$ the identity channel $E' \to Q$, $\hat{O} = \Theta = |0\rangle\langle 0|$, and $\Gamma(\rho) \coloneqq \mathrm{tr}[\rho]\ |1\rangle\langle 1|$. The condition holds for $\mathcal{A}$ with equality, and $(\mathcal{A}\circ\Gamma)^\dagger(\hat{O}) = 0 \not\succeq |0\rangle\langle 0| = \Theta$.

Uses Lem. 1.2, Def. 6.3, Def. 6.5 · *Used by* Rem. 8.2, Thm. 8.6, Rem. 8.8 · newly derived · accepted

Proof. Two facts are used throughout, and nothing else. First, for a channel Φ whose output system is finite-dimensional the adjoint $\Phi^\dagger$ is the unital positive map fixed by (1.1); the composite $\mathcal{A}\circ\Gamma$ has the same output system as $\mathcal{A}$, so $(\mathcal{A}\circ\Gamma)^\dagger$ is defined on the same operators as $\mathcal{A}^\dagger$, and no adjoint of Γ is formed in (i) to (iii), which is what lets $\mathcal{H}(E')$ be separable and infinite-dimensional. Second, a bounded self-adjoint H on a separable $\mathcal{H}(K)$ with $\mathrm{tr}[H\varrho] \geq 0$ for every $\varrho \in \mathrm{Pos}(K)$ of finite trace satisfies $H \succeq 0$: take $\varrho = |\phi\rangle\langle\phi|$ for a unit vector ϕ, so that $\langle\phi|\,H\,|\phi\rangle = \mathrm{tr}[H\,|\phi\rangle\langle\phi|] \geq 0$.

(i) Fix k, write $Z_k \coloneqq \hat{O}_k \otimes \mathbb{1}_{E''}$ and suppose $\bowtie_k$ is $\succeq$, so that $\mathcal{A}^\dagger(Z_k) \succeq c_k\,\mathbb{1}_{E'}$. Let $\varrho \in \mathrm{Pos}(E_{\mathrm{pre}})$ have finite trace. By (1.1) applied to $\mathcal{A}\circ\Gamma$, then to $\mathcal{A}$,

$$\mathrm{tr}\big[(\mathcal{A}\circ\Gamma)^\dagger(Z_k)\,\varrho\big] = \mathrm{tr}\big[Z_k\,\mathcal{A}\big(\Gamma(\varrho)\big)\big] = \mathrm{tr}\big[\mathcal{A}^\dagger(Z_k)\,\Gamma(\varrho)\big] \;\geq\; c_k\ \mathrm{tr}\big[\Gamma(\varrho)\big] \;=\; c_k\ \mathrm{tr}\,\varrho,$$

the inequality because $\mathcal{A}^\dagger(Z_k) - c_k\mathbb{1}_{E'}$ is a positive bounded operator and $\Gamma(\varrho)$ a positive trace-class one, so their product has non-negative trace, and the last step because Γ is trace preserving. Hence $\mathrm{tr}\big[\big((\mathcal{A}\circ\Gamma)^\dagger(Z_k) - c_k\mathbb{1}_{E_{\mathrm{pre}}}\big)\varrho\big] \geq 0$ for every such ϱ, and the second fact gives $(\mathcal{A}\circ\Gamma)^\dagger(Z_k) \succeq c_k\,\mathbb{1}_{E_{\mathrm{pre}}}$. The right-hand side's being a real multiple of the identity is used exactly once, in the last step of the display: it makes the right-hand side a multiple of $\mathrm{tr}\,\varrho$, and therefore the same constant on the two registers. For $\bowtie_k = \preceq$ apply the argument to $-Z_k$ and $-c_k$; for $\bowtie_k = =$ both directions hold. Since k was arbitrary and $\mathcal{A}\circ\Gamma$ is a channel from E_{pre} with the output system of $\mathcal{A}$, it lies in $\mathfrak{A}_{\mathfrak{L}}(E_{\mathrm{pre}}, QE'')$.

(ii) Identical, with $Z_k \coloneqq \mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E''}$: the tested operator is the only thing that differs between (6.2) and (6.3), and Γ acts on the input register in both.

(iii) Fix k, R' and $X \in \mathrm{Pos}(R'RQ_{f_k^c})$, write $\hat{D}_k \coloneqq \hat{O}_k - c_k\,\mathbb{1}_{Q_{f_k}}$ and $Z \coloneqq X \otimes \hat{D}_k \otimes \mathbb{1}_{E''}$, and suppose $\bowtie_k$ is $\succeq$, so that $(\mathrm{id}_{R'}\otimes\mathcal{A})^\dagger(Z) \succeq 0$. Since $\mathrm{id}_{R'}\otimes(\mathcal{A}\circ\Gamma) = (\mathrm{id}_{R'}\otimes\mathcal{A})\circ(\mathrm{id}_{R'}\otimes\Gamma)$, the same two applications of (1.1) give, for every $\varrho \in \mathrm{Pos}(R'E_{\mathrm{pre}})$ of finite trace,

$$\mathrm{tr}\Big[\big(\mathrm{id}_{R'}\otimes(\mathcal{A}\circ\Gamma)\big)^\dagger(Z)\,\varrho\Big] = \mathrm{tr}\Big[(\mathrm{id}_{R'}\otimes\mathcal{A})^\dagger(Z)\ \big(\mathrm{id}_{R'}\otimes\Gamma\big)(\varrho)\Big] \;\geq\; 0,$$

because $(\mathrm{id}_{R'} \otimes \Gamma)(\varrho)$ is positive and trace class. The second fact gives (6.4) for $\mathcal{A} \circ \Gamma$ at that R' and that X, and R', X were arbitrary. For $\preceq$ apply this to $-Z$; for $=$ both directions hold. No unitality and no constant is used here, the right-hand side of (6.4) being 0. The closure statement is (i) or (ii) for the untagged entries and this for the tagged ones, each entry's condition being a separate condition on the same channel.

(iv) $\mathcal{A} = \mathrm{id}$ has $\mathcal{A}^\dagger = \mathrm{id}$, so $\mathcal{A}^\dagger(\hat{O}) = |0\rangle\langle 0| = \Theta$ and the condition holds with equality, hence in the direction $\succeq$. The map Γ is completely positive and trace preserving, and $\mathrm{tr}[Y\,\Gamma(\rho)] = \mathrm{tr}[\rho]\,\langle 1|\,Y\,|1\rangle = \mathrm{tr}\left[\left(\langle 1|\,Y\,|1\rangle\ \mathbb{1}_{E_{\mathrm{pre}}}\right)\rho\right]$ for every Y and ρ, so $\Gamma^\dagger(Y) = \langle 1|\,Y\,|1\rangle\ \mathbb{1}$; this is the one place in the proof where an adjoint of Γ is written, and here both spaces are two-dimensional. Hence $(\mathcal{A} \circ \Gamma)^\dagger(\hat{O}) = \Gamma^\dagger(|0\rangle\langle 0|) = \langle 1|0\rangle\,\langle 0|1\rangle\ \mathbb{1} = 0$, while $\Theta = |0\rangle\langle 0|$, and $0 - |0\rangle\langle 0|$ is not positive semidefinite. The scalar entry built from the same operator is not broken by the same Γ: $\mathcal{A}^\dagger(|0\rangle\langle 0|) = |0\rangle\langle 0| \succeq 0 \cdot \mathbb{1}$ is the licensed entry at $c = 0$, and $\Gamma^\dagger(|0\rangle\langle 0|) = 0 \succeq 0 \cdot \mathbb{1}$ holds. □

The precomposition lemma needs only membership in the set the attack-set displays cut out, at a separable input, and gives the inclusion of the specified-source attack set, composed with any channel, in the virtual-source one. A general entry has no reading beside a non-empty source segment.

Remark 8.2 (Scope of the precomposition lemma). *The general entry.* The right-hand side Θ of Lemma 8.1 (iv), Hermitian and not a real multiple of the identity, is the channel-level shape that the Hermitian right-hand side of a *general* entry asks for, an entry Definition 6.3 records as admissible input to the consuming contract and never licensed there. At the state level such an entry beside a source link fails earlier still, and this is separate from the counterexample. There its right-hand side is a Hermitian operator on Alice's retained system (Definition 6.3; Definition "Admissible declaration" (meat-core/def:admissible-declaration) (vi)), and the retained system a source-replacement certificate is written against is the one of the *virtual* description, at every length of the source segment (Definition "Source-replacement certificate" (transmitter/def:srcert), "One reading at every L_s"); no contract relates that system or its marginal to the specified description's. So such an entry does not merely fail to be preserved: beside a non-empty source segment it has no reading to preserve. That, and not the counterexample alone, is why the admission rule of Definition 7.2 is a refusal.

Membership, and where it comes from. The hypothesis of Lemma 8.1 (i) to (iii) is *membership* in the set the displays of Definition 6.3 cut out. The attack sets that definition names are those of the round model, whose input registers are finite-dimensional; the displays themselves carry no condition on the dimension of E', and the lemma reads them at a separable input register, so (i) to (iii) hold there. Obtaining that membership from Theorem 6.7 (i), (ii) or (v) is a different matter, because that theorem types its attack on a finite-dimensional input register. Theorem 8.6 (ii) therefore does not route the first round through the lemma: it applies Theorem 6.7 to the pre-composed attack, whose input register is the virtual transmitted system. The lemma is the inclusion Remark 8.8 needs, and the second of the two routes in the proof of Theorem 8.6 (ii).

The inclusion. Write $\mathfrak{A}_1^{\mathrm{real}} := \mathfrak{A}_{\mathfrak{L}}(E', \cdot)$ for the set the first-round attack of

the specified-source protocol is admissible in and $\mathfrak{A}_1^{\text{virtual}} := \mathfrak{A}_{\mathfrak{L}}(E_{\text{pre}}, \cdot)$ for the set its precomposition with Γ must land in. Then Lemma 8.1 (i) to (iii) say exactly $\mathfrak{A}_1^{\text{real}} \circ \Gamma \subseteq \mathfrak{A}_1^{\text{virtual}}$, *for every channel* Γ. That is the inclusion Remark 8.8 names, and Theorem 8.6 is where it is spent.

Uses Lem. 8.1 · newly derived · accepted

In trusted-source mode, replacing the specified signal ensemble by the virtual one is an exact operational reduction, and the entanglement-based description a certificate presents has the same full output. One source channel common to every letter is essential.

A source map replaces *Alice's* preparation, and only a source specification in trusted-source mode has one. In untrusted-source mode the specification carries "no certificate and no marginal" (Definition "Source specification" (transmitter/def:source-spec)), the retained system A_j is an *output* of the attack (Definition 3.2, untrusted branch), and both (SR3), that no operation of the adversary acts on A, and the register discipline that Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (iii) requires of its channel fail there; correspondingly A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) is stated of a specification in trusted-source mode. So the lemma is stated in trusted-source mode alone; in untrusted-source mode there is no source segment to transfer, and what remains is the receiver-side instance, Corollary 3.3. The specified-source input ρ^{sp} of the lemma is the state in which the physical source prepares the letters and signals of the n rounds, under the transmitter's physical-source premise that makes the rounds independent: A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)) or, for a specification declared by a source model, the premise that model declares in its place, A-WCP (Assumption "Emission of the declared weak-coherent-pulse states" (transmitter/decoy/ass:wcp)) for the decoy model. The lemma is a statement about ρ^{sp} and does not assume that premise.

Lemma 8.3 (Exactness of the source map in trusted-source mode)**.** Let a source specification in *trusted-source mode* (Definition "Source specification" (transmitter/def:source-spec)) declare an alphabet $\mathcal{X}$ with weights $(p_x)_{x\in\mathcal{X}}$, the *specified* ensemble $\{p_x, \rho_x\}$ on the transmitted system A' and the virtual ensemble $\{p_x, \rho_x^{\text{v}}\}$ on the virtual transmitted system A'', with $\mathcal{H}(A')$ separable and $\mathcal{H}(A'')$ finite-dimensional, as Definition "Source specification" (transmitter/def:source-spec) declares the virtual transmitted system, and let its source segment satisfy the interface conditions (S1)–(S4) of Definition "Source segment and its interface conditions" (transmitter/def:source-segment), so that its composite $\Xi\colon \mathcal{T}(A'') \to \mathcal{T}(A')$ satisfies the *source identity*

$$\Xi\big(\rho_x^{\text{v}}\big) = \rho_x \qquad \text{for every } x \in \mathcal{X}. \tag{8.1}$$

Fix n, a finite-dimensional adversary register E_0 and a protocol in the sense of Definition 2.1, and put

$$\rho^{\mathrm{sp}} := \bigotimes_{j=1}^{n} \Big(\sum_{x \in \mathcal{X}} p_x \, |x\rangle\langle x|_{X_j} \otimes \rho_x \Big), \qquad \rho^{\mathrm{v}} := \bigotimes_{j=1}^{n} \Big(\sum_{x \in \mathcal{X}} p_x \, |x\rangle\langle x|_{X_j} \otimes \rho_x^{\mathrm{v}} \Big), \tag{8.2}$$

the j-th factor living on $X_j A_j'$ and on $X_j A_j''$ respectively.

The two presentations. A *strategy* is a pair $a = (\zeta, (\mathcal{A}_j)_{j=1}^{n})$ of an initial adversary state $\zeta \in S_{=}(E_0)$ and channels

$$\mathcal{A}_1 \in \mathrm{CPTP}\big(A'^n E_0,\ B_1 E_1'\big), \qquad \mathcal{A}_j \in \mathrm{CPTP}\big(E_{j-1},\ B_j E_j'\big) \quad (2 \le j \le n), \tag{8.3}$$

with E_j', E_j as in Definition 3.2: the *trusted* branch of (3.3), the only branch this lemma is about, in which A_j is Alice's and is no output of $\mathcal{A}_j$. A strategy of the *virtual* type is the same with A''^n in place of A'^n in (8.3). The *specified-source* presentation P^{sp} has these strategies and the output map $\mathsf{out}^{\mathrm{sp}}(a) :=$ the state (2.1) produces on the output registers of K together with E_n, from the input $\rho^{\mathrm{sp}} \otimes \zeta$, with B_j supplied by $\mathcal{A}_j$ and the classical X_j carried into $\bar{\mathsf{G}}_j$; the *virtual-source* presentation P^{v} has the strategies of the virtual type and the output map built the same way from $\rho^{\mathrm{v}} \otimes \zeta$, the two protocols agreeing in *every* other component: the measured system and its POVM, $\bar{\mathsf{G}}_j$ and K. The output registers are literally the same, which is condition (R1) of Definition 2.3.

(i) *(the source map)* define $r_{\mathrm{src}}(\zeta, (\mathcal{A}_j)_j) := (\zeta, (\mathcal{A}_j^{\mathrm{v}})_j)$ by

$$\mathcal{A}_1^{\mathrm{v}} := \mathcal{A}_1 \circ \big(\Xi^{\otimes n} \otimes \mathrm{id}_{E_0}\big), \qquad \mathcal{A}_j^{\mathrm{v}} := \mathcal{A}_j \quad (2 \le j \le n). \tag{8.4}$$

Then r_{src} is an exact operational reduction from P^{sp} to P^{v} in the sense of Definition 2.3, with the whole of the virtual type as the admissible strategy set;

(ii) *(the two descriptions of the virtual protocol)* let the virtual protocol be described *entanglement-based* instead: Alice's retained systems $A_1, \dots, A_n$ carry, jointly with the virtual transmitted systems, the preparation $\Psi^{\otimes n}$, where Ψ is a source-replacement certificate for the declared data $(A, \mathbb{T}, \tau_A, \{M_x\})$ against $\{p_x, \rho_x^{\mathrm{v}}\}$ in any of the declared forms of Definition "Source-replacement certificate" (transmitter/def:srcert), satisfying (SR1)–(SR3); and x_j is produced by her declared measurement channel $\mathcal{N}_{\mathsf{A}}$, the measurement map of part (iii) of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency), applied to A_j at its declared time in round j. Write $\mathsf{out}^{\mathrm{eb}}$ for the resulting output map on the virtual type. Then, under A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), $\mathsf{out}^{\mathrm{eb}}(a) = \mathsf{out}^{\mathrm{v}}(a)$ for every

strategy a of that type, so the identity map on strategies is an exact operational reduction between the two presentations in either direction;

(iii) *(the composite)* consequently r_{src} is an exact operational reduction from P^{sp} to the virtual protocol in its entanglement-based description, the object the rest of this contract and the consuming engine analyse;

(iv) *(one channel for every label is essential)* (8.1) with *one* Ξ cannot be weakened to a family $(\Xi_x)_x$ with $\Xi_x(\rho_x^{\mathrm{v}}) = \rho_x$. Take $n = 1$, $\mathcal{X} = \{0,1\}$, $p_0 = p_1 = \frac{1}{2}$, $A' = A'' = \mathbb{C}^2$, $\rho_0^{\mathrm{v}} = \rho_1^{\mathrm{v}} = |0\rangle\langle 0|$, $\rho_0 = |0\rangle\langle 0|$, $\rho_1 = |1\rangle\langle 1|$, and $\Xi_x(\cdot) \coloneqq \mathrm{tr}[\cdot]\,\rho_x$. Every identity $\Xi_x(\rho_x^{\mathrm{v}}) = \rho_x$ holds, yet for the attack that measures the received signal in the basis $\{|0\rangle, |1\rangle\}$ and keeps the outcome in E_1', the marginal on X_1E_1' is perfectly correlated in the specified-source protocol and, for *every* attack of the virtual type, a product in the virtual one, so no map of strategies can make the two full outputs equal; the protocol exhibited in the proof announces nothing in round one, so that E_1 is E_1' there.

Uses Lem. 1.2, Def. 2.1, Def. 2.3, Lem. 2.4, Def. 3.2, transmitter/def:source-segment, transmitter/def:source-spec, transmitter/def:srcert, transmitter/thm:certificate-sufficiency · *Assumes* A-SEQ · *Used by* Rem. 8.4, Rem. 8.5, Thm. 8.6 · newly derived · accepted

Proof. (i) The map $\Gamma \coloneqq \Xi^{\otimes n} \otimes \mathrm{id}_{E_0}$, and with it $\mathrm{id}_{X^n} \otimes \Gamma$, is a channel that acts factor by factor on products: its inputs A'', E_0 and X_j are finite-dimensional, so Lemma 1.2 (iii) gives it, with $\Phi_i = \Xi$ on each A_j'' and the identity maps on E_0 and on the classical registers. First, $\big(\mathrm{id}_{X^n} \otimes \Xi^{\otimes n}\big)(\rho^{\mathrm{v}}) = \rho^{\mathrm{sp}}$. The map acts factor by factor on (8.2), and on the j-th factor $\sum_x p_x\,|x\rangle\langle x| \otimes \Xi(\rho_x^{\mathrm{v}}) = \sum_x p_x\,|x\rangle\langle x| \otimes \rho_x$ by (8.1), linearity and continuity of Ξ on the trace-class operators; a term with $p_x = 0$ is the zero operator on both sides whatever ρ_x^{v} and ρ_x are, which is why (8.1) is needed only where the weight is positive. Tensoring with id_{E_0} gives

$$\big(\mathrm{id}_{X^n} \otimes \Xi^{\otimes n} \otimes \mathrm{id}_{E_0}\big)\big(\rho^{\mathrm{v}} \otimes \zeta\big) = \rho^{\mathrm{sp}} \otimes \zeta. \tag{8.5}$$

Second, write $\mathsf{G}_1^{\mathcal{A}_1}$ for the round-one map of (2.1) with B_1 supplied by $\mathcal{A}_1$, and $\mathsf{C} \coloneqq \mathsf{K} \circ \mathsf{G}_n^{\mathcal{A}_n} \circ \cdots \circ \mathsf{G}_2^{\mathcal{A}_2}$ for everything that follows it. Since $\Xi^{\otimes n}$ acts on A''^n alone and $\mathcal{A}_1^{\mathrm{v}} = \mathcal{A}_1 \circ (\Xi^{\otimes n} \otimes \mathrm{id}_{E_0})$,

$$\mathsf{G}_1^{\mathcal{A}_1^{\mathrm{v}}} = \mathsf{G}_1^{\mathcal{A}_1} \circ \big(\mathrm{id}_{X^n} \otimes \Xi^{\otimes n} \otimes \mathrm{id}_{E_0}\big),$$

the inserted map being the identity on every system $\mathsf{G}_1^{\mathcal{A}_1}$ reads apart from the transmitted ones. And C is *literally the same* composite in the two executions: $\mathcal{A}_j^{\mathrm{v}} = \mathcal{A}_j$ for $j \geq 2$ by (8.4), the two protocols agree in every component other than the released signal system, and no map of C touches

that system, the transmitted systems having been consumed by $\mathcal{A}_1$. Therefore

$$\mathsf{out}^{\mathrm{v}}\big(r_{\mathrm{src}}(a)\big) = \mathsf{C} \circ \mathsf{G}_1^{\mathcal{A}_1^{\mathrm{v}}}\big(\rho^{\mathrm{v}} \otimes \zeta\big) = \mathsf{C} \circ \mathsf{G}_1^{\mathcal{A}_1}\big(\rho^{\mathrm{sp}} \otimes \zeta\big) = \mathsf{out}^{\mathrm{sp}}(a),$$

using (8.5) in the middle step. This is (2.3). Condition (R3) holds because $\mathcal{A}_1^{\mathrm{v}}$ is a composition of channels and has the input $A''^n E_0$ and the output $B_1 E_1'$ required by (8.3) of the virtual type, and (R2) is the display.

(ii) By (SR3) no operation of the protocol and no operation of the adversary acts on A_j except Alice's declared measurement, and $\mathcal{N}_{\mathsf{A}}$ acts on A_j alone and writes into the fresh classical register X_j. Every map of the execution that *reads* X_j, namely the announcement of round j inside $\bar{\mathsf{G}}_j$ and, through the public record, every later map, comes after Alice's round-j measurement, by A-SEQ: every public message of round j follows every measurement of round j, and no operation on a quantum system of round j depends on a message of round j. So each map of the execution that precedes the first map reading X_j acts on systems complementary to $A_j X_j$, and part (iii)(a) of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency), the interchange law, which reads no property of the channel it is stated for beyond the register discipline, lets Alice's round-j measurement be performed before it. Applying this for $j = 1, \ldots, n$ moves all n measurements to the start of the execution, where the state is $\Psi^{\otimes n} \otimes \zeta$. There

$$\big(\mathcal{N}_{\mathsf{A}}^{\otimes n} \otimes \mathrm{id}_{E_0}\big)\big(\Psi^{\otimes n} \otimes \zeta\big) = \bigotimes_{j=1}^{n} \Big(\sum_{x \in \mathcal{X}} |x\rangle\langle x|_{X_j} \otimes \mathrm{tr}_A \left[(M_x \otimes \mathbb{1}_{A''}) \, \Psi \right] \Big) \otimes \zeta = \rho^{\mathrm{v}} \otimes \zeta,$$

by the definition of $\mathcal{N}_{\mathsf{A}}$ and by (SR2), which is the source identification of that theorem's part (iii)(b) at the trivial attack. From that point the two executions are the same composite of the same maps applied to the same state, so the full outputs agree; the output registers agree because $\mathcal{N}_{\mathsf{A}}$ consumes A_j and produces X_j in both descriptions. Neither direction of the equality reads any property of Ψ beyond (SR2) and (SR3), so the conclusion holds in whichever declared form the certificate was given; (SR1) enters as a hypothesis of the theorem cited and as what the consuming engine reads as τ_A, and no step of this proof uses it.

(iii) By (ii) the identity map is an exact operational reduction from P^{v} to the entanglement-based presentation, whose strategy set is the same; composing it with r_{src} is Lemma 2.4 (v).

(iv) Each Ξ_x is a channel and $\Xi_x(\rho_x^{\mathrm{v}}) = \mathrm{tr}[\rho_x^{\mathrm{v}}]\rho_x = \rho_x$. In the specified-source protocol the state on $X_1 A_1'$ is $\frac{1}{2}\,|0\rangle\langle 0|_{X_1} \otimes |0\rangle\langle 0| + \frac{1}{2}\,|1\rangle\langle 1|_{X_1} \otimes |1\rangle\langle 1|$, so the stated attack leaves X_1 and E_1' perfectly correlated; in the virtual protocol the state is $\big(\frac{1}{2}\,|0\rangle\langle 0|_{X_1} + \frac{1}{2}\,|1\rangle\langle 1|_{X_1}\big) \otimes |0\rangle\langle 0|$, a product, so the same attack and, the signal carrying no dependence on x, *every* attack of the virtual type

leaves E_1' independent of X_1. Take the protocol whose $\bar{\mathsf{G}}_1$ retains x_1 as its per-round datum and *appends nothing to the public record*, and whose K is the identity on the retained data and on E_1; it is a protocol of the form (2.1), and with no round-one announcement E_1 is E_1', the adversary retaining no public datum of that round. The two outputs are then compared on the marginal on X_1E_1': for the stated attack it is perfectly correlated in the specified-source protocol and, for *every* attack of the virtual type, a product in the virtual one, so the two full outputs differ and no map of strategies can equate them. The argument reads the marginal on X_1E_1', and that is why the exhibited protocol announces nothing in round one: at a protocol whose round-one announcement is a function of x_1, E_1 would collect that datum and would not be independent of X_1 in either protocol. □

The transmitted systems enter the first-round attack as systems in transit, not as registers of the adversary, and the virtual one is finite-dimensional. A round-by-round attack with a memory of any dimension is reached from this type by a limit (Lemma "Approximating an attack by all-at-once attacks with finite memory" (meat-core/lem:memory-lift)).

Remark 8.4 (The released signals as systems in transit). *What the released signals are.* The first round's input register in (8.3) is $A'^n \otimes E_0$, of which *only* E_0 is one of the adversary's registers, finite-dimensional as every register of the round model of Definition 3.2 is. The transmitted systems are named beside it because they are systems *in transit*: each is an intermediate system of one channel, produced by the trusted preparation, consumed by $\mathcal{A}_1$, and held by no party at the end of the round. That is the standing the measured system B has on the receiver's side, which is why Definition 1.1 names B separately as separable, and neither it nor the round model names the transmitted system at all. Neither is read more widely than it is written: every finiteness that Lemma 8.3 and Theorem 8.6 use, of E_0, of each E_j', of Q and of A'', is hypothesised where it is used, and every claim they apply at a register is applied only where that claim's own typing holds.

Which attacks the type covers. Where $\mathcal{H}(A')$ is finite-dimensional, a strategy that receives all n signals before the first round subsumes one that receives them round by round: the adversary may hold the later ones untouched in a finite-dimensional E_1' large enough to hold them and pass them on, through $E_1, \dots, E_{j-1}$, to the round j that releases them. A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)) orders the round's *messages* after the round's *measurements* and says nothing about when a signal is released. Where $\mathcal{H}(A')$ is infinite-dimensional, as the transmitted system of a source model is, a finite-dimensional E_1' cannot hold the later signals untouched, and a round-by-round attack in which A_j' arrives in transit at round j is not a strategy of the type (8.3) with finite registers. The type is the one the published source map makes: its first attack channel receives all n signals [8, Lemma 9.1, Eq. (140)], and each E_j' has a fixed finite dimension [8, Rem. 8.4]. Every other attack is reached by a limit over the dimensions of the adversary's registers, taken in the consuming engine (Lemma "Approximating an attack by all-at-once attacks with finite memory" (meat-core/lem:memory-lift)), as the published analysis takes it [8, Thm. A.1, Rem. A.2]: compressing the adversary's memory, and each signal released after the first round, onto finite-dimensional subspaces turns a round-by-round attack with a memory of any dimension into a

strategy of this type with finite registers whose output state is as close as wanted, and the trace distances of the security statement pass to the limit. Lemma 8.3 and Theorem 8.6 are exact about the type they name and transfer every criterion; the limit is taken for the trace distances alone.

The virtual transmitted system is finite-dimensional. It is so in both cases of Definition "Source specification" (transmitter/def:source-spec): trivially where the ensemble is declared as matrices, and by its clause (b) where the specified system is separable, "what a consumer receives is never the model but the virtual ensemble, which is finite dimensional". So the virtual presentation's first-round input register is finite-dimensional even where the specified one is not, which is the hypothesis Theorem 8.6 (ii) spends.

Uses Lem. 8.3 · newly derived · accepted

The source-map lemma gives the specified-source protocol no entanglement-based description, uses no purification and asserts no equality of marginals. With an empty source segment the two descriptions coincide, and parts (i) and (iii) of the lemma add nothing to the definition of one protocol.

Remark 8.5 (Scope of the source-map lemma). Lemma 8.3 never gives the specified-source protocol an entanglement-based description. So no purification of a mixed ρ_x is a system of either presentation, nothing is asserted about its dimension, its marginal or its correlations, and no statement relating the purifications of the two preparations is used: where ρ_x is mixed such a purification exists, and it is a system of neither protocol. Nor is any equality of marginals asserted: τ_A is the marginal of the *virtual* description, which is the pair a certificate is written against at every length of the source segment (Definition "Source-replacement certificate" (transmitter/def:srcert), "One reading at every L_s"), and nothing in the lemma equates it with a marginal of the specified description. At $L_\mathrm{s} = 0$ the two descriptions coincide, Ξ is the identity, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$, and Lemma 8.3 (i) and (iii) say nothing beyond what Definition 2.1 already says of one protocol.

Uses Lem. 8.3 · newly derived · accepted

In trusted-source mode the source map and the squashing reduction compose, in either order, into one exact operational reduction into the admissible attack set. Security of the virtual squashed model thus transfers with the same ε to the specified physical protocol, against attacks of its type.

The theorem is about trusted-source mode, as Lemma 8.3 is and for its reasons: in untrusted-source mode there is no source segment to transfer, the specification carrying no certificate and no marginal and the retained system being an output of the attack (Definition 3.2, untrusted branch), and what remains there is the receiver-side instance alone, Corollary 3.3 with Theorem 6.7 (ii), (iii); nothing in the theorem is asserted of that mode. In part (iii) the specified-source physical protocol pairs the specified source with the physical receiver. That the physical source is the specified one is the transmitter's physical-source premise, A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), or A-WCP (Assumption "Emission of the declared weak-coherent-pulse states" (transmitter/decoy/ass:wcp)) for

the decoy model, and the theorem does not assume it.

Theorem 8.6 (The two-sided reduction in trusted-source mode). Assume the hypotheses of Lemma 8.3: a trusted-source specification with virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$, a source segment whose composite Ξ satisfies the source identity (8.1), a source-replacement certificate satisfying (SR1)–(SR3), and A-SEQ. Assume in addition $M \preceq_\Lambda F$ together with a spectral pull-back certificate (6.9) for the list $\mathfrak{L} = \{(\hat{O}_k, \bowtie_k, c_k)\}$, or the factorwise hypotheses of Theorem 6.7 (v) for a list with entries carrying the party-local tag. Let P^{sp} be the specified-source presentation of Lemma 8.3, its receiver measuring M on B_j, and let $\mathsf{P}_{\mathfrak{L}}^{\mathrm{v,sq}}$ be the presentation whose protocol has the virtual source and the target F on Q_j and whose strategies are the $\mathfrak{L}$-admissible sequences of the virtual type (Definition 6.3). Define

$$r_{\mathrm{two}}\big(\zeta, (\mathcal{A}_j)_j\big) \coloneqq \big(\zeta, (\mathcal{A}_j^{\mathrm{v,sq}})_j\big), \qquad \mathcal{A}_1^{\mathrm{v,sq}} \coloneqq \lambda_1 \circ \mathcal{A}_1 \circ \big(\Xi^{\otimes n} \otimes \mathrm{id}_{E_0}\big), \quad \mathcal{A}_j^{\mathrm{v,sq}} \coloneqq \lambda_j \circ \mathcal{A}_j \ \ (j \geq 2), \tag{8.6}$$

where $\lambda_j \coloneqq \Lambda \otimes \mathrm{id}_{E_j'}$ is the trusted-source branch of (3.4), the only branch the strategies (8.3) of P^{sp} are typed in.

(i) *(one map, and the order of the two replacements is immaterial)* $r_{\mathrm{two}} = r_{\mathrm{sq}} \circ r_{\mathrm{src}} = r_{\mathrm{src}} \circ r_{\mathrm{sq}}$, where r_{src} is the source map (8.4) and r_{sq} the squashing map $(\mathcal{A}_j)_j \mapsto (\mathcal{A}_j^{\mathrm{sq}})_j$ of (3.4); both composites are defined on the same strategies and agree there. And r_{two} is an exact operational reduction from P^{sp} to the virtual, squashed protocol taken with *all* strategies of its type;

(ii) *(the image is admissible)* let E_0, every E_j' and Q be finite-dimensional, as the round model of Definition 3.2 makes the adversary's registers, every register of it being finite-dimensional, and Definition 1.1 makes the target space, and let $\mathcal{H}(A'')$ be finite-dimensional, as it is in both cases of Definition "Source specification" (transmitter/def:source-spec): trivially for an ensemble declared as matrices, and by its clause (b) for a separable specified system. Then every $\mathcal{A}_j^{\mathrm{v,sq}}$ of (8.6) lies in the set Definition 6.3 assigns to its round, so r_{two} has its image in the strategy set of $\mathsf{P}_{\mathfrak{L}}^{\mathrm{v,sq}}$ and is an exact operational reduction into it. *No hypothesis on the specified transmitted system is needed*: $\mathcal{H}(A')$ may be separable and infinite-dimensional, as the transmitted system of a source model is, and it is not a register Definition 1.1 or the round model ranges over but a system in transit;

(iii) *(transfer, with the same ε)* consequently, if the virtual, squashed protocol is ε-secure against every $\mathfrak{L}$-admissible sequence of the virtual type, a security criterion in the sense of Definition 2.2 taking a value at most ε at each of them, then the specified-source physical protocol is ε-secure against *every* attack sequence of its type, with the *same* ε and simultaneously for every such criterion;

(iv) *(in the description the certificate presents)* (i) to (iii) hold with the virtual protocol read entanglement-based, Alice retaining A_j, which trusted-source mode is the mode that gives her, and measuring $\{M_x\}$, this being the description the consuming engine analyses;

(v) *(the empty source segment)* where the source segment is empty, Ξ is the identity, $A'' = A'$, $\rho_x^{\mathrm{v}} = \rho_x$, r_{src} is the identity map on strategies and r_{two} is r_{sq}; (i) to (iii) are then Corollary 3.3 and Theorem 6.7 (iii) word for word, and the theorem states nothing new about the case this module emits.

Uses Def. 1.1, Def. 2.2, Def. 2.3, Lem. 2.4, Def. 3.2, Cor. 3.3, Def. 6.3, Def. 6.5, Thm. 6.7, Lem. 8.1, Lem. 8.3, transmitter/def:source-segment · *Used by* Rem. 8.7, Rem. 8.8 · newly derived · accepted

Proof. (i) Both composites are defined: r_{sq} is defined at an attack sequence of either the specified or the virtual type, being the insertion of λ_j after $\mathcal{A}_j$ and reading nothing about the attack's input register; and r_{src} is defined at a sequence with either receiver, being the insertion of $\Xi^{\otimes n}\otimes \mathrm{id}_{E_0}$ before $\mathcal{A}_1$ and reading nothing about the attack's output systems. Applying them in either order gives, in round one, $\lambda_1 \circ \big(\mathcal{A}_1 \circ (\Xi^{\otimes n} \otimes \mathrm{id}_{E_0})\big) = \big(\lambda_1 \circ \mathcal{A}_1\big) \circ (\Xi^{\otimes n} \otimes \mathrm{id}_{E_0})$ by associativity of composition, and in every later round $\lambda_j \circ \mathcal{A}_j$; so both equal r_{two}. The two insertions are at disjoint places of one composite: $\Xi^{\otimes n}$ acts on the transmitted systems, which are inputs of $\mathcal{A}_1$, and λ_j on the measured system, which is an output of $\mathcal{A}_j$, and neither acts on a system the other acts on.

For exactness, apply Lemma 2.4 (v) to r_{src}, an exact operational reduction from P^{sp} to the virtual-source protocol with the physical receiver (Lemma 8.3 (i)), and to r_{sq} on that protocol. That r_{sq} is an exact operational reduction there is Corollary 3.3: its first sentence is the equality of the two output states, at every attack sequence of the type of (3.3), and the strategies of the virtual-source protocol are sequences of that type, (8.3) naming the first round's input register and nothing else. Both reductions are over the same output registers, which are those of K together with E_n in all three presentations, so (R1) holds along the composition.

(ii) Fix $j \geq 2$. Theorem 6.7 (i), which is its trusted-source part, applied at $E' = E_{j-1}$ and $E'' = E'_j$, gives $\lambda_j \circ \mathcal{A}_j \in \mathfrak{A}_{\mathfrak{L}}(E_{j-1}, QE'_j)$; for a list with tagged entries it is its part (v), read at the trusted branch of (3.3), which gives both the party-local condition (6.4) and the scalar one.

For $j = 1$ the same parts are applied *to the pre-composed attack*, and that is why no hypothesis on $\mathcal{H}(A')$ is needed. Write $\Gamma \coloneqq \Xi^{\otimes n} \otimes \mathrm{id}_{E_0}$ and $\mathcal{A}_1^{\mathrm{v}} \coloneqq \mathcal{A}_1 \circ \Gamma$, so that $\mathcal{A}_1^{\mathrm{v,sq}} = \lambda_1 \circ \mathcal{A}_1^{\mathrm{v}}$ by (8.6) and associativity of composition. Being a composition of two channels, $\mathcal{A}_1^{\mathrm{v}}$ is a channel $\mathcal{T}(A''^n E_0) \to \mathcal{T}(B_1 E'_1)$, of exactly the type Theorem 6.7 (i) and (v) quantify over, and its *input* register $A''^n E_0$ is finite-dimensional by hypothesis. Applying that theorem at $E' = A''^n E_0$ and $E'' = E'_1$ therefore gives $\mathcal{A}_1^{\mathrm{v,sq}} \in \mathfrak{A}_{\mathfrak{L}}(A''^n E_0, \cdot)$ outright.

The separable system A'^n occurs nowhere in this step: it is an intermediate system of one channel (Remark 8.4), and the trusted branch of (3.3) types an attack by the systems it *outputs*, the measured system and the outgoing memory, placing no condition on the register the round receives beyond its being that register.

The other order lands on the same set and needs a hypothesis this one does not. Applying Theorem 6.7 to $\mathcal{A}_1$ first requires $E' = A'^n E_0$, which that theorem types finite-dimensional, and then Lemma 8.1 at $E_{\mathrm{pre}} = A''^n E_0$, $E' = A'^n E_0$ and the same Γ, its part (i) and its part (iii) for the tagged entries, gives the same membership. That lemma asks only that the two input registers be separable, so what makes the second route the narrower one is Theorem 6.7's own typing and nothing else; that the two routes agree is (i) read at the level of admissible sets. The image of r_{two} therefore lies in the strategy set of $\mathsf{P}_{\mathfrak{L}}^{\mathrm{v,sq}}$. Equation (2.3) holds at every strategy by (i) and does not read the analysed set, and condition (R3) of Definition 2.3 now holds for the smaller set, so r_{two} is an exact operational reduction into it.

(iii) Lemma 2.4 (i) applied to the reduction of (ii), with f the criterion at the adversary system of the presentation: sup over the physical strategies equals sup over the image and is at most sup over the $\mathfrak{L}$-admissible sequences, which is at most ε. No property of the criterion is used, which is why the conclusion is simultaneous in all of them.

(iv) By Lemma 8.3 (ii) the identity map on strategies is an exact operational reduction between the prepare-and-measure and the entanglement-based presentations of the virtual protocol, in either direction, and the squashing replacement touches only B_j, a system neither description of the source names. Composing with the identity map changes neither the strategies nor their images, so Lemma 2.4 (v) carries (i) to (iii) across unchanged.

(v) The closing paragraph of Definition "Source segment and its interface conditions" (transmitter/def:source-segment) states that when $L_{\mathrm{s}} = 0$ the segment is empty, Ξ is the identity, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$, so $\mathcal{A}_1^{\mathrm{v}} = \mathcal{A}_1$ in (8.4) and r_{src} is the identity; then (8.6) is $\mathcal{A}_j^{\mathrm{sq}} = \lambda_j \circ \mathcal{A}_j$ of (3.4). Part (i) is then the output equality of Corollary 3.3, part (ii) is Theorem 6.7 (i) or (v) per round, and part (iii) is that corollary's last sentence as Theorem 6.7 (iii) states it. In *untrusted*-source mode there is no source segment to transfer and this theorem asserts nothing, so what is left there is exactly those two claims, read at the untrusted branch. □

The two-sided reduction relates no marginals of the specified and virtual descriptions and licenses no entry whose right-hand side is not a real constant. Its order-independence concerns soundness; the order a chain records stays data.

Remark 8.7 (Scope of the two-sided reduction). Theorem 8.6 uses and implies no relation between a marginal of the specified description and one of the virtual description, and none would follow (Remark 8.5). It licenses no entry whose right-hand side is not a real constant: Lemma 8.1 (iv) is a counterexample for such an entry, and the admission rule of Definition 7.2 refuses one beside a non-empty

source segment. And part (i) does not say that a chain may be *recorded* in either order: the record's order is data, stated as condition (C3) of Definition 7.2, and what (i) says is that for a licensed list the order is not what carries the soundness (Remark 8.8).

Uses Thm. 8.6 · newly derived · accepted

Condition (C3) records the order of the source and receiver stages as data; for a list with real constants that order does not carry soundness (Lemma 8.1, Theorem 8.6). A general entry is not preserved and is refused beside a non-empty source segment.

One of the artefact shapes the module's design fixes is that composition order is data (`spec/modules/reductions/design.md#structural-design-decisions`, the decisions on the artefact's shape and on the two-sided chain), and condition (C3) of Definition 7.2 states it. This remark says what that condition carries and what it does not.

Remark 8.8 (Recorded stage order, immaterial to soundness for licensed lists). *The record.* Condition (C3) is a statement about the *record*: the order in which a reduction was performed is carried as data, so that a later reader re-checks the path instead of inferring it from when an artefact was built.

The inclusion a source link after a receiver link needs. A *source*-stage link changes what Alice emits: it asserts that the states actually prepared are the image, under a declared channel Γ, of simpler virtual ones, and its security argument re-parameterises the first-round attack as $\mathcal{A}_1 = \mathcal{A}_1' \circ \Gamma$ ranging over *all* CPTP maps. A *receiver*-stage link, by Theorem 6.7, hands the analysis an attack set that is a subset of the CPTP maps, in general a *proper* one. Applying a source link after one is therefore not free: what must additionally hold is the inclusion $\mathfrak{A}_1^{\text{real}} \circ \Gamma \subseteq \mathfrak{A}_1^{\text{virtual}}$.

For a licensed list the inclusion holds, and the order is immaterial to soundness. Lemma 8.1 establishes exactly that inclusion, for *every* channel Γ and for every list whose right-hand sides are real constants, which is every list this module emits (Theorem 6.7 (iv)) and every list the consuming contract calls licensed (Definition 6.5), tagged entries included. For such a list the order of the two stages is immaterial to soundness, and that is proved: Theorem 8.6 (i), in trusted-source mode, the only mode in which a source stage exists, shows that replacing the source first and replacing the receiver first give one and the same map of attack strategies, because the two replacements act on different systems, the source composite at the *input* of the first round's attack and the squashing channel at the *output* of each round's attack. Condition (C3) stays as the shape of the record and the order in which the artefact presents its two segments; it does not carry the soundness.

The published framework names both resolutions, and this contract takes the second. [8, Rem. 9.2] records the same interaction and resolves it by applying its source-map lemma *before* its squashing lemma, which is the ordering resolution; the same remark names the alternative, reproving the source-map statement for restricted attack sets, and Lemma 8.1 is that alternative carried out here. No step of the lemma appeals to the paper: it is proved from positivity and trace preservation.

What is refused, and why. The inclusion is established for a licensed list and for no other. An entry whose right-hand side is a Hermitian operator rather than a real constant, a *general* entry, which Definition 6.3 records as admissible input to the consuming contract and never licensed there, is not preserved: Lemma 8.1 (iv)

exhibits a channel and an entry that hold before the composition and fail after it. At the state level such an entry has no reading beside a non-empty source segment at all, its right-hand side being an operator on a retained system the two descriptions do not share (Remark 8.2). A chain with a non-empty source segment carrying such an entry is therefore *refused*, which is the admission rule of Definition 7.2.

Where the rule is decided. The rule is a condition on a declaration that carries both a non-empty source segment and a constraint list, and it is decided where the declaration is assembled, by the chain conditions of the protocol compiler (Definition "Interface conditions on a chain" (protocol-compiler/def:chaincond)); this module states the rule and decides no instance of it. It declares no general entry and emits none, every entry it emits carrying a real scalar, so no artefact of this module can violate the rule. The rule is not vacuous: a presented declaration may carry a non-empty source segment. Both of its halves are decided by the protocol compiler's chain conditions. The *entry* half refuses a general entry beside a non-empty source segment; the *mode* half admits a non-empty source segment only in trusted-source mode, and refuses a declaration that carries one while its marginal slot holds the no-marginal value none. So neither half rests on the provenance of the artefact that carried the segment.

Uses Def. 6.3, Def. 6.5, Thm. 6.7, Def. 7.2, Lem. 8.1, Thm. 8.6 · newly derived · accepted

9 The number model and exact decisions

Every number this module reads or emits, except the recorded emission displacement, is a binary64 value read as the dyadic rational it is, every declared or emitted operator entry and declared real scalar satisfies one declared lattice condition, and every condition on declared data is decided exactly; the section states the strengthening of exact admissibility this module decides (Definition 9.1). The decisions are taken in an ordered field that the contract leaves open, since no admissible choice changes a decision (Definition 9.2). Positivity and operator-norm bounds reduce to sign comparisons of characteristic-polynomial coefficients (Lemma 9.3), and partial sums on the lattice are exact while the alphabet is small enough (Lemma 9.5).

Imports foundations/def:binary64, foundations/def:exact-admissibility, foundations/lem:binary64-lattice

Every number this module reads or emits, except the recorded emission displacement, is a binary64 value read as its dyadic rational; every declared or emitted operator entry and declared real scalar meets one lattice condition. Exact admissibility here is the foundations core plus that condition.

Every operator datum this module consumes arrives as a finite list of binary64 numbers, and under the number model those numbers are the model: a declared value *is* its binary64 datum (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)), declared operator data are exactly admissible at every module boundary (Requirement "Exact admissibility at every module boundary" (foundations/req:boundary-

admission)), and every consumer decides that exactly and refuses otherwise (Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)). This definition states the module's side of that policy, for what it reads and for what it emits: the common core of exact admissibility, restated in this module's terms so that both ends of one boundary name one condition, and strengthened by one declared lattice. On operator entries the lattice condition (9.1) of (b) is condition (D-LAT) of Requirement "The declared lattice condition" (foundations/req:declared-lattice) at $k = \kappa$; this module states it of the declared real scalars of (b) as well, as a condition of what it admits. Together with (9.1), conditions (a)–(e) of (c) are, for the family this module emits, the six conditions of Requirement "Exact verification before emission" (foundations/req:exact-emission). Condition (e) is stated on the additions because Remark "A complete computed sum from an incomplete family" (foundations/rem:computed-completeness) is a family on which a vanishing completeness defect and completeness of the numbers differ, and Remark 11.9 discusses the objects that scope clause (d) leaves to the hypotheses of an instance.

Definition 9.1 (The number model, the declared lattice and exact admissibility)**.** *(a) The model.* Every numerical datum this module reads or emits, the displacement record below aside, is a binary64 value on the wire and *is* that value: no exact object is posited behind it and none is approximated by it. A binary64 value is a dyadic rational, and *this contract reads it as that rational* (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64) (iii)). Every predicate this contract decides is decided, and every emitted number but one is formed, by finitely many operations in the substrate of Definition 9.2 and Lemma 9.3. *The exception is the emission displacement* Δ_s *of* (11.4): an entrywise maximum *modulus*, hence the square root of an exact rational and in general irrational, emitted as a binary64 rendering of a record of what the normalizer of Definition 11.1 did, in a direction nothing records; it decides nothing, and no statement of this contract reads it. That is the reading the exact island (Definition "The exact island" (foundations/def:exact-island)) gives, which contains both the admission decisions on declared blocks and a producer's construction of what it emits. The only non-numerical data are combinatorial (alphabets, orders, block identifiers, a coarse-graining map, a branch index set, a cutoff) and carry no arithmetic.

(b) The declared lattice. The request declares one integer exponent κ, and every declared and every emitted operator entry, together with every *real scalar* the request declares of the following kinds — the class weights q_t, any certified weight, the entries of a declared stochastic matrix, and any declared device constant — satisfies

$$2^{\kappa}x \in \mathbb{Z}, \qquad |x| \le 1, \qquad 1 \le \kappa \le 53. \tag{9.1}$$

Such an x is exactly a binary64 number (Lemma "Binary64 numbers on

dyadic lattices" (foundations/lem:binary64-lattice) (i)), its significand carrying at most $\kappa \leq 53$ bits. The list is exhaustive: the constant a caller declares for a requested constraint entry is not among them, and where it is off the lattice it is rounded onto it or refused (Corollary 11.3 (iii)). It is a condition on a *set* of values: it says which numbers may be declared and nothing about how near one number is to another. A complex entry satisfies (9.1) when its real and imaginary parts do.

(c) Exact admissibility on the declared lattice. Let $\{A_v\}_{v \in I}$ be a finite family of matrices presented by its blocks over declared bases, indexed by the finite set I the invoking rule names (the outcome alphabet at a fixed setting for a declared block family, a class $\mathcal{V}_t$ of the datum of a reduction instance (Definition 10.1), $\mathcal{V}$ for the family this module emits), and let c be the constant the rule names for its completeness identity: 1 for a POVM, the class weight q_t for a class of a datum. The family is *exactly admissible on the declared lattice* when

(a) *(the sector map)* the block kind, the block identifiers and their order are the ones the consumer holds, a comparison of declared labels, decided and never measured;

(b) *(Hermiticity)* $A_v = A_v^\dagger$, entry for entry;

(c) *(positivity)* $A_v \succeq 0$, decided by (9.3);

(d) *(completeness)* $\sum_{v \in I} A_v = c\,\mathbb{1}$ on each block, as an identity of the entries;

(e) *(the sum rounds nowhere)* every entrywise binary64 addition of that accumulation, taken in the declared order of I, is exact, so that the *computed* sum *is* the sum of (d);

and every entry satisfies (9.1). Conditions (a)–(e) say that the family is exactly admissible with constant c for the sector map the consumer holds (Definition "Exact admissibility of a declared operator family" (foundations/def:exact-admissibility)), each block being presented at the dimension the consumer's map declares for it. Condition (e) is a condition in its own right and is stated on the *additions*: a vanishing completeness *defect* says that the computed sum is the identity, not that $\sum_v A_v = c\,\mathbb{1}$ holds of the numbers. Lemma 9.5 shows that (e) is satisfiable on this contract's own emission, rather than a condition that would refuse every family.

(d) Scope, stated once and binding wherever this definition is invoked. This definition and the rules that invoke it are about data this contract requires to be presented as a finite list of matrices of declared dimension: the data sectors of a reduction instance (Definition 10.1 (I1)) and the declared blocks they are formed from; the declared classical scalars and matrices a submodule's route reads (a stochastic matrix and its descended form,

the class weights, each certified weight, any declared device constant); and the family this module emits. They are *not* available for the objects an instance's hypotheses (H) are about: operators on a part of $\mathcal{H}(B)$ that no finite list of matrices presents, an above-cutoff family, a register POVM, an uncharacterised branch. There is nothing on those to decide, and their conditions remain hypotheses of the instance (Definition 10.1 (I1)).

Uses foundations/def:binary64, foundations/def:exact-admissibility, foundations/lem:binary64-lattice · *Used by* Def. 4.4, Def. 9.2, Lem. 9.5, Def. 10.1, Cor. 10.2, Cor. 10.3, Thm. 10.4, Def. 11.1, Thm. 11.2, Cor. 11.3, Cor. 11.6, Def. 20.3, Prop. 21.2, Cor. 21.3 · newly derived · accepted

Every exact decision is taken in an ordered field F between $\mathbb{Q}$ and $\mathbb{R}$, with its Gaussian extension $F + iF$, and the field is left open. No admissible choice changes a decision, every number a rule reads being rational, so $\mathbb{Q}$ always suffices.

Definition 9.2 (The arithmetic substrate)**.** The *substrate* of this contract is an ordered field $\mathbb{F}$ with $\mathbb{Q} \subseteq \mathbb{F} \subseteq \mathbb{R}$ together with its Gaussian extension $\mathbb{F}_i := \mathbb{F} + i\mathbb{F}$, which is closed under complex conjugation. What is fixed is the *predicate* each rule decides and the requirement that it be decided exactly; which such field carries the arithmetic is not. The choice cannot change any decision. Binary64 values are dyadic rationals (Definition 9.1 (a)), and every number a rule reads is formed from the declared entries and from rational constants by field operations, floors of rationals and roundings of rationals to a lattice, so it is rational; since $\mathbb{Q}$ is an ordered subfield of every admissible $\mathbb{F}$ with the induced order, and $\mathbb{Q} + i\mathbb{Q}$ a subfield of $\mathbb{F}_i$, each decision, a comparison or an equality of such numbers, is the same decision taken over $\mathbb{Q}$. In particular $\mathbb{F} = \mathbb{Q}$ always suffices. The latitude is bounded: it is a choice of arithmetic substrate, not of mathematics.

Uses Def. 9.1 · *Used by* Lem. 9.3, Thm. 11.2, Cor. 12.1, Req. 17.1, Def. 20.3, Prop. 20.4, Cor. 21.3, Cor. 21.4, Cor. 21.5 · newly derived · accepted

A Hermitian matrix with entries in the substrate is positive semidefinite exactly when, for every k, the sum of its principal minors of order k is non-negative, and an operator-norm bound reduces to such a test. Both are decided by field operations and sign comparisons, with no eigenvalue computed.

An eigenvalue of a matrix with rational entries is in general irrational, so an exact decision of positivity cannot go through one. The lemma replaces it by the signs of the coefficients of the characteristic polynomial, which lie in the substrate.

Lemma 9.3 (Exact decision of positivity and of an operator-norm bound)**.** Let $\mathbb{F}, \mathbb{F}_i$ be the substrate of Definition 9.2 and let $d, d' \geq 1$.

(i) Let $X = X^\dagger \in \mathcal{L}(\mathbb{C}^d)$ have all entries in $\mathbb{F}_i$ and write

$$\det\left(\lambda\, \mathbb{1}_d + X\right) = \sum_{k=0}^{d} e_k(X)\, \lambda^{d-k}, \qquad e_0(X) = 1, \tag{9.2}$$

so that $e_k(X)$ is the sum of the $k \times k$ principal minors of X. Then every $e_k(X)$ lies in $\mathbb{F}$, and

$$X \succeq 0 \quad \Longleftrightarrow \quad e_k(X) \geq 0 \ \text{ for every } k = 1, \ldots, d. \tag{9.3}$$

(ii) Let $Z \in \mathcal{L}(\mathbb{C}^{d'}, \mathbb{C}^d)$ have all entries in $\mathbb{F}_i$ and let $\delta \in \mathbb{F}$, $\delta \geq 0$. Then

$$\|Z\| \leq \delta \quad \Longleftrightarrow \quad \delta^2 \, \mathbb{1}_{d'} - Z^\dagger Z \ \succeq \ 0, \tag{9.4}$$

the right-hand side being a test of the form (i) on a Hermitian matrix with entries in $\mathbb{F}_i$.

Both predicates are therefore decided by finitely many field operations and sign comparisons in $\mathbb{F}$: no rounding, no threshold, and *no eigenvalue is computed*.

Uses Def. 9.2 · *Used by* Rem. 9.4, Def. 10.1, Def. 11.1, Thm. 11.2, Cor. 11.6, Rem. 11.9, Def. 20.3, Prop. 20.4, Cor. 20.6, Cor. 21.3, Cor. 21.4 · newly derived · accepted

Proof. (i) A principal submatrix of a Hermitian matrix is Hermitian, so each principal minor is real; it is also a polynomial with integer coefficients in entries of $\mathbb{F}_i$, hence lies in $\mathbb{F}_i$. Being both real and in $\mathbb{F}_i$, it lies in $\mathbb{F}$, and so does each $e_k(X)$. The identification of $e_k(X)$ with the coefficient in (9.2) is the standard expansion of the characteristic polynomial by principal minors.

Let $\lambda_1, \ldots, \lambda_d \in \mathbb{R}$ be the eigenvalues of X with multiplicity, so that $\det(\lambda \mathbb{1}_d + X) = \prod_{j=1}^{d} (\lambda + \lambda_j)$ and $e_k(X)$ is the k-th elementary symmetric polynomial in them. If $X \succeq 0$ then every $\lambda_j \geq 0$ and every $e_k(X)$, a sum of products of λ_j's, is ≥ 0. Conversely suppose $e_k(X) \geq 0$ for every k and that some $\lambda_{j_0} < 0$. Put $\lambda := -\lambda_{j_0} > 0$. On the one hand $\prod_j (\lambda + \lambda_j) = 0$, because the factor j_0 vanishes. On the other hand $\sum_{k=0}^{d} e_k(X) \lambda^{d-k} \geq e_0(X) \lambda^d = \lambda^d > 0$, every omitted term being non-negative. The two sides of (9.2) at that λ are therefore 0 and something strictly positive, a contradiction. Hence no eigenvalue is negative and $X \succeq 0$.

(ii) $Z^\dagger Z$ is Hermitian with entries in $\mathbb{F}_i$, and $\|Z\|^2 = \lambda_{\max}(Z^\dagger Z)$ by definition of the operator norm. So $\|Z\| \leq \delta$ iff $\lambda_{\max}(Z^\dagger Z) \leq \delta^2$ iff $\delta^2 \mathbb{1}_{d'} - Z^\dagger Z \succeq 0$. □

Any procedure that decides the positivity test of Lemma 9.3 exactly in the substrate conforms, such as the Faddeev–Le Verrier recursion or a rational LDL factorisation with the zero-pivot test. A procedure that decides it approximately does not, whatever its error bound.

Remark 9.4 (Algorithms that decide the positivity test). Lemma 9.3 fixes *what* is decided, not *how*. The coefficients $e_k(X)$ are produced exactly by the Faddeev–Le Verrier recursion or by any division-free characteristic-polynomial method. A rational $LDL^\dagger$ factorisation with symmetric pivoting and the zero-pivot row test

decides the same predicate, often faster, in $O(d^3)$ field operations. Enumerating all $2^d - 1$ principal minors also decides it and is never necessary. Any procedure that decides (9.3) exactly in $\mathbb{F}$ conforms to this contract; *one that decides it approximately does not, whatever its error bound.* A floating-point eigenvalue is such a procedure, which is why Requirement "Exact verification before emission" (foundations/req:exact-emission) (iii) excludes one by name.

Uses Lem. 9.3 · newly derived · accepted

If $\kappa + \lceil \log_2 L \rceil \le 53$, every partial sum of L entries on the declared lattice is a binary64 number and every addition forming it is exact. A family on the lattice over such an alphabet therefore meets the rounds-nowhere condition by construction.

A sum rounds nowhere exactly when every partial sum is a binary64 number (Lemma "Exactness of a sum that rounds nowhere" (foundations/lem:exact-accumulation)). On the declared lattice that holds as long as the alphabet is small enough for the partial sums to fit in the significand.

Lemma 9.5 (Partial sums on the declared lattice are exact). Let $x_1, \dots, x_L$ satisfy (9.1) and let $\Sigma_j := \sum_{i \le j} x_i$, $1 \le j \le L$, be the partial sums of the accumulation in that order. If

$$\kappa + \lceil \log_2 L \rceil \;\le\; 53, \tag{9.5}$$

then every Σ_j is exactly a binary64 number and every one of the $L - 1$ additions forming it is exact in binary64. Consequently a family whose entries satisfy (9.1), over an alphabet satisfying (9.5), meets condition (e) of Definition 9.1 by construction.

Uses Def. 9.1 · *Used by* Thm. 11.2 · newly derived · accepted

Proof. Work one real component of one entry position at a time; the accumulation is entrywise and the real and imaginary parts are accumulated independently. Each x_i is an integer multiple of $2^{-\kappa}$, so each Σ_j is one too, and $|\Sigma_j| \le \sum_{i \le j} |x_i| \le L$. Hence $2^\kappa \Sigma_j$ is an integer with $|2^\kappa \Sigma_j| \le L\, 2^\kappa \le 2^{53}$ by (9.5). Every integer of modulus at most 2^{53} is exactly representable in binary64, and multiplication by the power of two $2^{-\kappa}$ is exact and changes no significand, so Σ_j is a binary64 number; and $2^{-\kappa} \ge 2^{-53}$ with $|\Sigma_j| \le L$ keeps it far inside the normal range, so no subnormal or overflow case arises. An IEEE-754 addition returns the correctly rounded value of the exact sum of its operands; here that exact sum is Σ_j, which is representable, so the rounding is the identity and the addition is exact. □

10 Reduction instances

A reduction instance packages what every construction emits: a handed model with a finite datum and hypotheses about the rest, a target built block by block from the datum and scalar values, a squashing relation with

a transported set, and certified entries, all holding at every admissible datum (Definition 10.1). The core proves the three instances a protocol needs regardless of device: the identity instance, the flag instance of an uncharacterised branch, and the branching instance of a generalized instrument with its register form (Theorem 10.4). An instance precomposed with a channel on its system, such as a known loss in front of the device, is again an instance with the same datum, target, entries and emission (Lemma 10.6).

A reduction instance is a handed model with a finite datum and hypotheses about the rest, a target built block by block from the datum and scalar values, a squashing relation with a transported set, and certified entries, all holding at every admissible datum. Exact emission is proved once over it.

The emission theorem of this module (Theorem 11.2) is stated over the object defined here, in the way the receiver states its emitted bound over a setting family (Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound)). The object packages what every construction this module or a submodule emits has in common: a handed model on B with a finite datum, a finite target built from that datum and from scalar values, an existence theorem giving the squashing relation with a transported set, and a list of certified entries. The obligation of exact emission is then proved once, and each construction is shown to be an instance where it lives. Because every part is established at every admissible datum, the emission theorem can apply the existence theorem to the datum the normalizer declares (Definition 11.1, Theorem 11.2 (iii)) without re-proving anything, the role the exact flag plays for the receiver's setting family. The core establishes three instances a protocol needs regardless of device: the identity instance (Corollary 10.2), the flag instance of an uncharacterised branch (Corollary 10.3) and the branching instance (Theorem 10.4) with its register form (Corollary 10.5); a squasher submodule proves that its construction is an instance and applies Theorem 11.2. The downward rounding of a scalar value in (I2) is the rounding direction of Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction), and Remark 3.14 shows that the operator form of the class-marginal identity on the target, which the fourth consequence below sets aside, is unsound.

Definition 10.1 (Reduction instance)**.** Fix the declared lattice exponent κ of Definition 9.1 (b) and a joint alphabet with classes and class weights $\big(\mathcal{V}, \{\mathcal{V}_t\}_{t\in\mathcal{T}_{\mathrm{set}}}, \{q_t\}\big)$ (Definition 4.4 (a)). Let $\mathbb{S}_{\mathrm{dat}}$ be a finite ordered list of *data sector*s, each s with a declared dimension $d_s \geq 1$ and a declared *support* $\mathcal{V}^{(s)} \subseteq \mathcal{V}$ that meets every class. A *datum* for these sectors is a family $A = \{A_v^{(s)}\}_{s\in\mathbb{S}_{\mathrm{dat}},\, v\in\mathcal{V}}$ of $d_s \times d_s$ matrices with entries in $\mathbb{F}_i$ and $A_v^{(s)} = 0$ for $v \notin \mathcal{V}^{(s)}$: these elements lie *outside the sector support*, and no step of this contract reads, rounds or repairs them. It is an *admissible datum*, written $A \in \mathcal{D}_{\mathrm{adm}}$, when for every s: (a) $A_v^{(s)} = (A_v^{(s)})^\dagger$ entry for entry; (b) $A_v^{(s)} \succeq 0$,

decided by (9.3); (c) it is *class complete*,

$$\sum_{v\in\mathcal{V}_t} A_v^{(s)} \;=\; q_t\,\mathbb{1}_{d_s} \qquad \text{exactly, for every } t\in\mathcal{T}_{\mathrm{set}}; \tag{10.1}$$

and it is an *exactly admissible datum*, written $A\in\mathcal{D}_\kappa$, when in addition (d) every entry satisfies (9.1). These are conditions (b)–(d) of Definition 9.1 (c) and the lattice invariant, read per data sector with the class weight q_t as the constant of the completeness identity.

A *reduction instance* $\mathcal{J}$ over $\big(\mathcal{V},\{\mathcal{V}_t\},\{q_t\}\big)$ consists of the following four parts.

(I1) *(the handed model and its datum)* A system B with $\mathcal{H}(B)$ separable (Definition 1.1); data sectors $\mathbb{S}_{\mathrm{dat}}$ with their declared dimensions and supports; a *handed datum* $A\in\mathcal{D}_{\mathrm{adm}}$, every entry of which is formed from the numbers the instance was handed by finitely many operations in $\mathbb{F}_i$; a finite list of *hypotheses (H)* about the rest of the model (operators without a finite presentation, facts about a part of $\mathcal{H}(B)$ the datum does not describe, a register POVM), each of which is either decided exactly on declared data, the request being refused where it fails, or carried as an assumption of the emitted ledger; and a *model map* $A'\mapsto M[A']=\{M_v[A']\}_{v\in\mathcal{V}}$ that assigns to every admissible datum $A'\in\mathcal{D}_{\mathrm{adm}}$ a POVM on B over $\mathcal{V}$, under (H). The *handed model* is $M[A]$, the model map at the handed datum. The *stability clause* requires that no hypothesis in (H) read an entry of the datum, so that (H) holds of $M[A']$ for every $A'\in\mathcal{D}_{\mathrm{adm}}$ as soon as it holds of the handed model; and that parts (I2)–(I4) be established for *every* $A'\in\mathcal{D}_{\mathrm{adm}}$ under (H), not only at the handed datum.

(I2) *(the target)* A finite ordered *sector set* $\mathbb{S}=\mathbb{S}_{\mathrm{dat}}\sqcup\mathbb{S}_{\mathrm{sc}}$, the second summand the *scalar sector*s, each $s\in\mathbb{S}$ with a declared dimension $d_s\geq 1$; for each scalar sector s and each $v\in\mathcal{V}$ a *scalar value* $\phi_s(v)\in[0,1]$ satisfying (9.1) with $\sum_{v\in\mathcal{V}}\phi_s(v)=1$ exactly, each formed from the numbers the instance was handed by finitely many operations in $\mathbb{F}$ and, where a rounding onto the lattice is needed, by the *downward* rounding of a quantity the value is a certified lower bound of, any complementary value being recomputed from the rounded ones and never rounded on its own; the target space $\mathcal{H}(Q):=\bigoplus_{s\in\mathbb{S}}\mathbb{C}^{d_s}$ with sector projectors Π_s, so that $\dim\mathcal{H}(Q)=\sum_s d_s$; and the *target family*

$$F_v[A'] \;:=\; \bigoplus_{s\in\mathbb{S}} F_v^{(s)}[A'], \qquad F_v^{(s)}[A'] \;:=\; \begin{cases} A_v'^{(s)} & s\in\mathbb{S}_{\mathrm{dat}},\\ \phi_s(v)\,\mathbb{1}_{d_s} & s\in\mathbb{S}_{\mathrm{sc}}, \end{cases} \tag{10.2}$$

assigned one sector block at a time, so that $F_v[A']$ is sector-block-diagonal *by construction*: no off-sector entry is written and none is

to be verified. The sector map of the instance is the list $\mathbb{S}$ with its order, each sector's dimension and its kind (data or scalar), and it is what a consumer compares (Definition 9.1 (c), condition (a)). The *default party partition* of the target assigns every factor of every $\mathcal{H}(Q_s)$, $s \in \mathbb{S}$, to *Bob*: *every sector of the target is Bob's*. That is the assignment the physical model makes: the target describes what reaches Bob's receiver, the handed model living on his system B. An instance may in addition be declared with *another party's label*, one label for the whole instance, and then the same assignment is made with that label in its place: every factor of every $\mathcal{H}(Q_s)$ is that party's, because the handed model lives on the system *that* party receives. Where no label is declared the label is Bob's. Nothing else in the instance depends on the label: it is a declared identifier, compared and never computed, and no condition of (I1)–(I4) reads it.

(I3) *(the existence theorem)* A channel $\Lambda[A'] = \mathcal{P}_{\mathbb{S}} \circ \Lambda_0[A']\colon \mathcal{T}(B) \to \mathcal{T}(Q)$, given by a constructor and recorded as one, with $\mathcal{P}_{\mathbb{S}}$ the sector pinching of Proposition 4.6 for $\{\Pi_s\}_{s\in\mathbb{S}}$; and a finite *transported set* $\hat{O}[A']$ of sector-block-diagonal Hermitian operators on Q containing every sector projector Π_s, with pull-backs $O[A']$; such that

$$\big(M[A'];\, O[A']\big) \preceq_{\Lambda[A']} \big(F[A'];\, \hat{O}[A']\big) \qquad \text{for every } A' \in \mathcal{D}_{\text{adm}} \text{ under (H).} \tag{10.3}$$

By linearity of $\Lambda[A']^\dagger$, every real linear combination of transported operators is transported to the same combination of their pull-backs.

(I4) *(the certified entries)* A finite list $\mathfrak{L}[A'] = \{(\hat{O}_k, \bowtie_k, c_k)\}_k$ of *certified entries*, each $\hat{O}_k$ a real linear combination of transported operators of the form

$$\hat{O}_k \;=\; \sum_{v\in\mathcal{V}} \sum_{s\in\mathbb{S}} \alpha_{k,v,s}\, \Pi_s F_v[A'] \Pi_s \;+\; Z_k, \tag{10.4}$$

with real coefficients $\alpha_{k,v,s}$ *on the lattice*, each an integer multiple of $2^{-\kappa}$, and Z_k a real combination of sector projectors with coefficients of the same kind, $\bowtie_k \in \{\preceq, =, \succeq\}$ and c_k a declared constant on the lattice, whose certificate (6.9), $\Lambda[A']^\dagger(\hat{O}_k) \bowtie_k c_k \mathbb{1}_B$, is *discharged* for every $A' \in \mathcal{D}_{\text{adm}}$ under (H); and a *class-marginal flag* recording whether the class-marginal identity

$$\sum_{v\in\mathcal{V}_t} M_v[A'] \;=\; q_t\, \mathbb{1}_B \qquad \text{for every } t \in \mathcal{T}_{\text{set}} \tag{10.5}$$

is *discharged*, that is, established by the construction for every $A' \in \mathcal{D}_{\text{adm}}$ under (H), or not. The modulus bound of (9.1) is not imposed

on a coefficient: $|x| \leq 1$ is a condition on a declared or emitted *entry* (a value of the datum, a scalar value, a declared scalar or constant), where a POVM element and a completeness identity give it, and not on the coefficients of a linear combination, which a branching instance forms as the difference $\beta_s - c_k$ of two lattice multiples (Theorem 10.4), again a lattice multiple; *being on the lattice is all that is asked of a coefficient*. What (9.1) does bind is the entries of the operator the combination produces (Corollary 11.3 (iii)).

The instance *at* a datum $A' \in \mathcal{D}_{\text{adm}}$ is the same B, (H), sectors, supports, scalar values, constructor and entry rule with A' in place of the handed datum; by the stability clause it is a reduction instance whenever $\mathcal{J}$ is.

Five consequences, each immediate. First, for every $A' \in \mathcal{D}_{\text{adm}}$ the target $\{F_v[A']\}_{v\in\mathcal{V}}$ is a POVM on Q: every sector block is Hermitian and positive semidefinite, by (a) and (b) on a data sector and by $\phi_s(v) \in [0,1]$ on a scalar one; and it is complete, on a data sector because $\sum_{v\in\mathcal{V}} A'^{(s)}_v = \sum_t \sum_{v\in\mathcal{V}_t} A'^{(s)}_v = \sum_t q_t \mathbb{1}_{d_s} = \mathbb{1}_{d_s}$ by (10.1) and $\sum_t q_t = 1$, on a scalar sector because $\sum_v \phi_s(v) = 1$. Its data sectors *are* the datum, entry by entry, so the emitted target carries the datum and nothing else of the model. *Second*, every $F_v[A']$, every Π_s and every transported operator is sector-block-diagonal, so Corollary 4.7 (i) applies: a relation established for $\Lambda_0[A']$ holds for $\mathcal{P}_{\mathbb{S}} \circ \Lambda_0[A']$ with the same target, the same transported set and the same pull-backs, which is why (I3) may be stated with the pinching in front, and the squashed analysis may be restricted to block-diagonal states without excluding a reachable one (Corollary 4.7 (iii)). *Third*, the pull-back of every sector projector is positive semidefinite, $\Lambda[A']^\dagger(\Pi_s) \succeq 0$, because the adjoint of a channel is a positive map, and $\sum_s \Lambda[A']^\dagger(\Pi_s) = \mathbb{1}_B$ by unitality; so $(\Pi_s, \succeq, 0)$ carries the certificate of Corollary 1.7 (ii) for *every* instance and needs no part of (I4) to say so. *That certificate travels with the transported set, whatever the request carries*, which is where (I3) puts every sector projector and where the artefact records the certificate case per entry; it becomes an *entry* of $\mathfrak{L}$ only where a caller requests it *as an entry* (Corollary 11.3 (i)), the emitted list being empty where the request carries no entry. *Fourth*, the relation pulls a class sum of the target back to the class sum of the model, $\Lambda[A']^\dagger\big(\sum_{v\in\mathcal{V}_t} F_v[A']\big) = \sum_{v\in\mathcal{V}_t} M_v[A']$, by linearity; so (10.5) holds of the model exactly when the pull-back identity $\Lambda[A']^\dagger\big(\sum_{v\in\mathcal{V}_t} F_v[A']\big) = q_t \mathbb{1}_B$ does, which is the expectation-form entry Corollary 11.3 (ii) emits where the flag says it is discharged, and never the operator identity $\sum_{v\in\mathcal{V}_t} F_v[A'] = q_t \mathbb{1}_Q$, which fails on every scalar sector whose values do not sum to q_t over $\mathcal{V}_t$. *Fifth*, $\dim \mathcal{H}(Q) = \sum_s d_s$ is a sum of declared dimensions, finite, and the sector map is a comparison of declared labels; both are read off the declaration with no arithmetic performed.

The link of Definition 7.1 an instance produces is the relation (10.3) at the declared datum, with the transported set and the entry list as its

slots, and its chain (Definition 7.2) is that link preceded by whatever post-processing the instance's submodule applies before it.

Uses Def. 1.3, Cor. 1.7, Def. 1.8, Def. 4.4, Prop. 4.6, Cor. 4.7, Thm. 6.7, Def. 9.1, Lem. 9.3 · *Used by* Rem. 3.14, Cor. 10.2, Cor. 10.3, Thm. 10.4, Cor. 10.5, Lem. 10.6, Def. 11.1, Thm. 11.2, Cor. 11.3, Thm. 11.5, Cor. 11.6, Rem. 11.7, Rem. 11.8, Rem. 11.9, Cor. 12.1, Cor. 12.2, Req. 16.9, Req. 17.4, Req. 17.5, Prop. 20.5, Prop. 21.2, Cor. 21.3, Cor. 21.4, Cor. 21.5, Def. 23.1, Cor. 24.1,
meat-core/cor:reference-companion-factors · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.instance:ReductionInstance

A declared block-diagonal measurement, decided exactly admissible on entry, is a reduction instance with the sector pinching as its channel, no scalar sector, no entry and the class-marginal identity discharged. Its chain is the identity link of Example 7.5.

A receiver that is already finite needs no squasher, only the sector pinching that every instance's channel carries in front. Its chain is the identity link of Example 7.5, and its emission is Corollary 11.6.

Corollary 10.2 (The identity instance)**.** Let $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$ be a joint alphabet with classes and weights, let $\mathbb{S}_0$ be a declared finite ordered sector set with dimensions d_s, and let $\{M_v\}_{v\in\mathcal{V}}$ be a POVM on $\mathcal{H}(B) = \bigoplus_{s\in\mathbb{S}_0} \mathbb{C}^{d_s}$ presented as the sector-block list $M_v = \bigoplus_s A_v^{(s)}$ of an admissible datum $A \in \mathcal{D}_{\mathrm{adm}}$ for the sectors $\mathbb{S}_0$ with the default supports $\mathcal{V}^{(s)} = \mathcal{V}$, that is, a declared family every block of which is decided exactly admissible on the declared lattice on entry with the class sums at q_t (Definition 9.1 (c)). Then the *identity instance* $\mathcal{J}_{\mathrm{id}}$, given by

(I1) B as given; $\mathbb{S}_{\mathrm{dat}} = \mathbb{S}_0$; the handed datum A; (H) empty; and the model map $M_v[A'] := \bigoplus_s A_v'^{(s)}$;

(I2) $\mathbb{S}_{\mathrm{sc}} = \varnothing$, so that $Q = B$ and $F[A'] = M[A']$;

(I3) $\Lambda = \mathcal{P}_{\mathbb{S}}$ for the decomposition $\mathbb{S}_0$ (that is, $\Lambda_0 = \mathrm{id}$), independent of the datum, with the transported set $\{\Pi_s\}_{s\in\mathbb{S}_0}$ and pull-backs Π_s;

(I4) an empty entry list, and the class-marginal identity discharged,

is a reduction instance. Its chain is the identity link: one receiver-stage link of kind identity with $\Lambda_\ell = \mathcal{P}_{\mathbb{S}}$ (the identity on every state block diagonal for $\mathbb{S}_0$, and id outright when $\mathbb{S}_0$ has one sector), with $M^{\mathrm{out}} = M^{\mathrm{in}}$, the sector projectors carried to themselves, the reachable-set statement the declared input family, pinched, no entries, no licence and status = exact.

Uses Def. 1.1, Def. 4.4, Prop. 4.6, Cor. 4.7, Def. 7.1, Def. 7.2, Thm. 7.4, Def. 9.1, Def. 10.1 · *Used by* Cor. 11.6, Cor. 12.1, Cor. 12.2 · newly derived · accepted

Proof. (I1) $M[A']$ is a POVM on B for every $A' \in \mathcal{D}_{\mathrm{adm}}$: each block is positive semidefinite by admissibility (b), and $\sum_v M_v[A'] = \bigoplus_s \sum_t \sum_{v\in\mathcal{V}_t} A_v'^{(s)} =$

$\bigoplus_s \sum_t q_t \mathbb{1}_{d_s} = \mathbb{1}_B$ by class completeness (10.1) and $\sum_t q_t = 1$. $\mathcal{H}(B)$ is finite-dimensional, hence separable (Definition 1.1). (H) is empty, so the stability clause holds vacuously and what follows is established for every admissible datum.

(I2) The target (10.2) with no scalar sector is $F_v[A'] = \bigoplus_s A_v'^{(s)} = M_v[A']$, on $Q = B$, and the sector map is $\mathbb{S}_0$ with its dimensions, every sector of data kind.

(I3) Every $F_v[A']$ is block diagonal for $\{\Pi_s\}_{s\in\mathbb{S}_0}$ by construction, so Proposition 4.6 gives $F[A'] \preceq_{\mathcal{P}_\mathbb{S}} F[A']$, which is $M[A'] \preceq_{\mathcal{P}_\mathbb{S}} F[A']$; and $\mathcal{P}_\mathbb{S}^\dagger(\Pi_s) = \mathcal{P}_\mathbb{S}(\Pi_s) = \Pi_s$, so every sector projector is transported to itself. $\Lambda = \mathcal{P}_\mathbb{S} \circ \mathrm{id}$ has the form (I3) requires.

(I4) The instance carries no entry of its own: the sector-projector certificate travels with the transported set, not in the entry list (the third consequence of Definition 10.1), and there is nothing else to certify. The class-marginal identity (10.5) reads $\sum_{v\in\mathcal{V}_t} M_v[A'] = \bigoplus_s q_t \mathbb{1}_{d_s} = q_t \mathbb{1}_B$, which is class completeness (10.1); it is therefore discharged for every admissible datum.

The chain: the link with source family $(B, M[A'])$, target $(Q, F[A']) = (B, M[A'])$, channel witness $\mathcal{P}_\mathbb{S}$, the carried pair $(\{\Pi_s\}, \{\Pi_s\})$, the reachable-set statement "the declared input family, pinched", $\mathfrak{L} = \varnothing$, no licence and $\mathsf{status} = \mathsf{exact}$ is a receiver-stage link of Definition 7.1 and, with an empty source segment, a chain of Definition 7.2; Theorem 7.4 (iv) gives the status, the identity $\mathcal{P}_\mathbb{S}^\dagger(M_v[A']) = M_v[A']$ carrying no error term. It is the identity link of Example 7.5 with $\mathcal{P}_\mathbb{S}$ in the channel slot: $\mathcal{P}_\mathbb{S}(\sigma) = \sum_s \Pi_s \sigma \Pi_s = \sigma$ for every σ block diagonal for $\mathbb{S}_0$, so on the block-diagonal states to which Corollary 4.7 (iii) restricts the analysis the two channels agree, and when $\mathbb{S}_0$ is a single sector $\Pi_s = \mathbb{1}_B$ and $\mathcal{P}_\mathbb{S} = \mathrm{id}$. □

Any POVM on any separable system is a reduction instance with one classical flag per outcome, no data sector and no entry, so there is nothing to normalise. The class-marginal identity is discharged exactly where the branch's declaration supplies it.

A branch about which nothing beyond its POVM is declared, such as one an adversary controls, still squashes, to the flag PVM of Lemma 5.1; Remark 5.12 names the resulting instance as the one for a branch whose device is unknown or adversarially chosen. The corollary shows that this squasher is a reduction instance, so that the branching instance of Theorem 10.4 can combine such a branch with characterised ones.

Corollary 10.3 (The flag instance of an uncharacterised branch)**.** Let $(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\})$ be a joint alphabet with classes and weights and let $\{M_v\}_{v\in\mathcal{V}}$ be any POVM on any system B with $\mathcal{H}(B)$ separable, about which nothing beyond that is declared. Then the *flag instance* $\mathcal{J}_{\mathrm{flag}}[M]$, given by

(I1) B as given; no data sector, $\mathbb{S}_{\mathrm{dat}} = \varnothing$, so that the datum is empty and

$\mathcal{D}_{\mathrm{adm}} = \{\varnothing\}$; (H) the single hypothesis that $\{M_v\}$ is a POVM on B; and the constant model map $M[\varnothing] \coloneqq M$;

(I2) one scalar sector (f) of dimension 1 for each $f \in \mathcal{V}$, in the order of $\mathcal{V}$, with $\phi_f(v) \coloneqq \delta_{vf}$, so that $\mathcal{H}(Q) = \mathcal{H}(\mathrm{Flg}) = \mathbb{C}^{|\mathcal{V}|}$ and $F_v = |v\rangle\langle v|_{\mathrm{Flg}}$, the flag PVM P^{Flg} of Lemma 5.1;

(I3) $\Lambda = \mathcal{P}_{\mathbb{S}} \circ \Lambda_{\mathrm{mp}} = \Lambda_{\mathrm{mp}}$, the measure-and-prepare channel (5.1) of M, with the transported set $\{\Pi_f = |f\rangle\langle f|\}_{f\in\mathcal{V}}$ and pull-backs M_f;

(I4) an empty entry list, and the class-marginal identity discharged exactly when (H) is extended by the declared fact $\sum_{v\in\mathcal{V}_t} M_v = q_t \mathbb{1}_B$ for every t, as it is when the branch is declared to be a joint family (Definition 4.4 (b)), and not otherwise,

is a reduction instance. The normalizer of Definition 11.1 has nothing to act on: the declared datum is the handed one, no Δ_s exists, and the declared model is M itself.

Uses Def. 4.4, Prop. 4.6, Lem. 5.1, Def. 9.1, Def. 10.1, Def. 11.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.sectors:flag_scalar_values

Proof. (I1) There is one datum, the empty family, and it is admissible vacuously; $M[\varnothing] = M$ is a POVM by (H); the stability clause holds because (H) reads no datum entry, there being none.

(I2) $\phi_f(v) = \delta_{vf} \in \{0, 1\}$ satisfies (9.1) for every κ, is formed with no rounding, and $\sum_v \delta_{vf} = 1$; (10.2) gives $F_v = \bigoplus_f \delta_{vf}$, the diagonal operator $|v\rangle\langle v|$ in the basis $\{|f\rangle\}_{f\in\mathcal{V}}$ of $\mathcal{H}(\mathrm{Flg})$, and $\dim \mathcal{H}(Q) = |\mathcal{V}|$.

(I3) Lemma 5.1 gives $M \preceq_{\Lambda_{\mathrm{mp}}} P^{\mathrm{Flg}}$ with ${\Lambda_{\mathrm{mp}}}^{\dagger}(X) = \sum_v \langle v| X |v\rangle M_v$, so ${\Lambda_{\mathrm{mp}}}^{\dagger}(|f\rangle\langle f|) = M_f$, which is the pull-back of the sector projector Π_f; the output of Λ_{mp} is diagonal in the flag basis, so $\mathcal{P}_{\mathbb{S}} \circ \Lambda_{\mathrm{mp}} = \Lambda_{\mathrm{mp}}$ (Proposition 4.6), and Λ has the form (I3) requires with $\Lambda_0 = \Lambda_{\mathrm{mp}}$. Since the datum is empty this holds for every admissible datum.

(I4) The instance carries no entry of its own: the sector-projector certificate travels with the transported set, not in the entry list (the third consequence of Definition 10.1); here that certificate reads $M_f \succeq 0$. The class-marginal identity (10.5) is a statement about $\sum_{v\in\mathcal{V}_t} M_v$, and (H) alone can supply it, nothing else about M being declared: where the branch is declared to be a joint family it is (4.3), and for an arbitrary POVM $\sum_{v\in\mathcal{V}_t} M_v$ is some operator with $0 \preceq \sum_{v\in\mathcal{V}_t} M_v \preceq \mathbb{1}_B$ and nothing more. So the flag is set exactly where the declared fact is part of (H).

The last sentence: Definition 11.1 acts on data sectors, of which there are none, so $A' = A = \varnothing$ and $M'^{\mathrm{ph}} = M[\varnothing] = M$. □

A generalized instrument followed by a reduction instance in each branch is a reduction instance, keeping or forgetting the branch label. A branch entry with constant c becomes a zero-constant entry with c times the branch projector moved into the operator, and the normalizer acts branch by branch.

A receiver described branch by branch, each branch a reduction instance (the flag instance of Corollary 10.3 for an uncharacterised branch, for example), is a reduction instance as a whole. The theorem builds it on Lemma 5.3 in two forms: the joint-outcome form keeps the branch label in the outcome, and the coarse-grained form forgets it. Both file the branch label as a block index of the target, so they model branches the adversary may know (Remark 5.5). Corollary 10.5 is the case of a register read by a POVM.

Theorem 10.4 (The branching instance: an instrument followed by branch instances)**.** Let J be a finite ordered set, let $\{\mathcal{I}_j\}_{j\in J}$, $\mathcal{I}_j\colon \mathcal{T}(B) \to \mathcal{T}(B'_j)$, be a generalized instrument (Definition 5.2), $\sum_j \mathrm{tr}\circ\mathcal{I}_j = \mathrm{tr}$, given independently of any datum, and for each j let $\mathcal{J}_j$ be a reduction instance on B'_j over one common joint alphabet with classes and weights $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$, with handed datum A_j, data sectors $\mathbb{S}_{j,\mathrm{dat}}$ with supports $\mathcal{V}^{(j,s)}$, scalar sectors $\mathbb{S}_{j,\mathrm{sc}}$ with values $\phi_{j,s}$, hypotheses (H_j), channel $\Lambda_j[\cdot]$, transported set $\hat{O}^{(j)}[\cdot]$ and entry list $\mathfrak{L}_j[\cdot]$. Write $\mathbb{S} \coloneqq \bigsqcup_{j\in J}\{j\}\times\mathbb{S}_j$, ordered by j and then as $\mathbb{S}_j$ is, $\Pi_{(j,s)}$ for its sector projectors, $\Pi_j \coloneqq \sum_{s\in\mathbb{S}_j}\Pi_{(j,s)}$ for the *branch projectors*, $\hat{X}|_j$ for the operator equal to $\hat{X}$ in block j and zero elsewhere, and $A \coloneqq (A_j)_{j\in J}$ for the joint datum. Then each of the two constructions (i) and (ii) is a reduction instance, with

- the channel
$$\Lambda[A'] \;\coloneqq\; \mathcal{P}_{\mathbb{S}} \circ \bigoplus_{j\in J} \Lambda_j[A'_j] \circ \mathcal{I}_j, \tag{10.6}$$
- the sector set $\mathbb{S}$, the hypotheses $(\mathrm{H}) = \bigcup_j (\mathrm{H}_j)$ together with the hypothesis that $\{\mathcal{I}_j\}$ is a generalized instrument and, for every j in a declared set $J_{\mathrm{w}} \subseteq J$ of branches with declared lattice constants q_j, the *branch-weight hypothesis* $\mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = q_j \mathbb{1}_B$,
- the transported set consisting of the sector projectors and, for every j and every $\hat{X} \in \hat{O}^{(j)}$, the operator $\hat{X}|_j$, with pull-back $\mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\hat{X})\big)$, in particular
$$\Lambda^\dagger\big(\Pi_{(j,s)}\big) = \mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\Pi_s)\big), \qquad \Lambda^\dagger\big(\Pi_j\big) = \mathcal{I}_j^\dagger\big(\mathbb{1}_{B'_j}\big), \tag{10.7}$$
- the entry list consisting of, for every branch j and every entry $(\hat{O}_k, \bowtie_k, c_k) \in \mathfrak{L}_j$, the entry
$$\big(\hat{O}_k|_j - c_k\,\Pi_j,\ \bowtie_k,\ 0\big), \tag{10.8}$$
together with the *branch-weight entry* $(\Pi_j, =, q_j)$ for every $j \in J_{\mathrm{w}}$, where that set is not empty:

(i) *(the joint-outcome form)* over the joint alphabet $\mathcal{V} \times J$ of Definition 4.4 (a) whose outcome alphabet is $\mathcal{Y} \times J$, ordered with the branch label major and the outcome minor, and whose settings are $\mathcal{T}_{\mathrm{set}}$: writing its element (y, j, t) as (v, j) with $v = (y, t) \in \mathcal{V}$, its declared order has $(v, j) < (v', j')$ iff $t < t'$, or $t = t'$ and $j < j'$, or $t = t'$, $j = j'$ and $y < y'$, so that every class $\mathcal{V}_t \times J$ is a contiguous run and the restriction of the order to $\mathcal{V} \times \{j\}$ *is* the order of $\mathcal{V}$; its classes are $\mathcal{V}_t \times J$ and its weights q_t. On it, the model map $M_{(v,j)}[A'] \coloneqq \mathcal{I}_j^{\dagger}\big(M_v^{(j)}[A'_j]\big)$; data sectors (j, s), $s \in \mathbb{S}_{j,\mathrm{dat}}$, of dimension $d_{j,s}$ and support $\mathcal{V}^{(j,s)} \times \{j\}$, with $A'^{(j,s)}_{(v,j')} \coloneqq \delta_{jj'} A'^{(s)}_{j,v}$; scalar sectors (j, s), $s \in \mathbb{S}_{j,\mathrm{sc}}$, with $\phi_{(j,s)}(v, j') \coloneqq \delta_{jj'} \phi_{j,s}(v)$; and the target $F_{(v,j)}[A'] = F_v^{(j)}[A'_j]\big|_j$, the branch target in block j and zero elsewhere;

(ii) *(the coarse-grained form)* over $\big(\mathcal{V}, \{\mathcal{V}_t\}, \{q_t\}\big)$ itself: the model map $M_v[A'] \coloneqq \sum_j \mathcal{I}_j^{\dagger}\big(M_v^{(j)}[A'_j]\big)$; the same data and scalar sectors with supports $\mathcal{V}^{(j,s)}$, $A'^{(j,s)}_v \coloneqq A'^{(s)}_{j,v}$ and $\phi_{(j,s)}(v) \coloneqq \phi_{j,s}(v)$; and the target $F_v[A'] = \bigoplus_j F_v^{(j)}[A'_j]$.

In both forms the class-marginal identity is discharged when every branch discharges its own, and the flag is set in exactly that case; and the normalizer of Definition 11.1 acts on the composite datum *branch by branch*: its output at the data sectors of branch j is the output of the normalizer of $\mathcal{J}_j$ at A_j, so that the declared datum is $A' = (A'_j)_j$ with A'_j the branches' declared data and the declared model is $M[A']$ at those.

Uses Lem. 1.2, Cor. 1.7, Lem. 4.3, Def. 4.4, Cor. 4.7, Def. 5.2, Lem. 5.3, Thm. 6.7, Def. 9.1, Def. 10.1, Def. 11.1 · *Used by* Cor. 10.5, Cor. 12.1 · newly derived · accepted

Proof. Idea. Lemma 5.3 combines the branch relations into one; each part of a reduction instance is then read off branch by branch, and a branch entry survives the instrument once its constant is moved into the operator. Throughout, $A' = (A'_j)_j$ denotes a datum of the composite.

Step 1 (the data). In the joint form, sector (j, s) has support $\mathcal{V}^{(j,s)} \times \{j\}$, which meets every class $\mathcal{V}_t \times J$ because $\mathcal{V}^{(j,s)}$ meets $\mathcal{V}_t$; its elements off the support are zero by the definition of $A'^{(j,s)}$, and its elements on the support are those of A'_j at sector s. Hermiticity and positivity of the elements are those of A'_j; class completeness over $\mathcal{V}_t \times J$ reads $\sum_{(v,j') \in \mathcal{V}_t \times J} \delta_{jj'} A'^{(s)}_{j,v} = \sum_{v \in \mathcal{V}_t} A'^{(s)}_{j,v} = q_t \mathbb{1}_{d_{j,s}}$, which is class completeness of A'_j; and the lattice condition is read entry by entry. In the coarse form the support is $\mathcal{V}^{(j,s)}$ and every condition is literally the branch's. So in both forms a composite datum is an admissible datum (resp. an exactly admissible datum) if and only if every A'_j is: $\mathcal{D}_{\mathrm{adm}} = \prod_j \mathcal{D}^{(j)}_{\mathrm{adm}}$. The handed datum $A = (A_j)_j$ is

admissible because each A_j is, and its entries are formed exactly because each branch's are.

Step 2 ((I1), the model map and stability). The joint-outcome alphabet of (i) is a joint alphabet with classes and weights in the sense of Definition 4.4 (a): its outcome alphabet $\mathcal{Y} \times J$ is finite, non-empty and ordered, its settings and weights are those of $\mathcal{V}$, and its declared order is that definition's, the outcome varying fastest and the setting slowest. For every admissible A', each $M^{(j)}[A'_j]$ is a POVM on B'_j by (I1) of $\mathcal{J}_j$ under (H_j), which holds by the branch's stability clause. Lemma 5.3 (i) with these branch POVMs gives that $\{\mathcal{I}_j^\dagger(M_v^{(j)}[A'_j])\}_{(v,j)\in\mathcal{V}\times J}$ is a POVM on B, the joint-form model; coarse-graining j away is Lemma 4.3 (i) for the stochastic matrix that forgets j, so the coarse-form model is a POVM as well. $\mathcal{H}(B)$ is separable by hypothesis. Stability: (H_j) reads no entry of A'_j by the branch's clause, and the instrument hypothesis and the branch-weight hypotheses read no datum because the $\mathcal{I}_j$ are given independently of any; so (H) reads no entry of the composite datum, and the remaining parts, established in Steps 3 to 5 for every admissible A' under (H), are established as (I1) demands.

Step 3 ((I2), the target). The scalar values $\phi_{(j,s)}$ are the branches' values, lattice values in $[0,1]$ formed as (I2) of the branch requires, with $\sum_{(v,j')} \delta_{jj'}\phi_{j,s}(v) = \sum_v \phi_{j,s}(v) = 1$ in the joint form and $\sum_v \phi_{j,s}(v) = 1$ in the coarse form. Read (10.2) at the sectors (j',s): in the joint form the block (j',s) of $F_{(v,j)}[A']$ is $\delta_{jj'}A'^{(s)}_{j,v}$ on a data sector and $\delta_{jj'}\phi_{j,s}(v)\mathbb{1}$ on a scalar one, so $F_{(v,j)}[A']$ is the branch target $F_v^{(j)}[A'_j]$ of (10.2) for $\mathcal{J}_j$ in block j and zero elsewhere, the $F_{v,j}$ of Lemma 5.3 (ii); in the coarse form the block (j,s) of $F_v[A']$ is the block s of $F_v^{(j)}[A'_j]$, so $F_v[A'] = \bigoplus_j F_v^{(j)}[A'_j]$, the target of Lemma 5.3 (iii). Block diagonality for $\mathbb{S}$ holds by construction in both forms, and $\dim \mathcal{H}(Q) = \sum_j \dim \mathcal{H}(Q_j)$ is finite.

Step 4 ((I3), the relation). By (I3) of each branch, $\big(M^{(j)}[A'_j]; O^{(j)}\big) \preceq_{\Lambda_j[A'_j]} \big(F^{(j)}[A'_j]; \hat{O}^{(j)}\big)$ with $F^{(j)}$ on the finite-dimensional Q_j. Lemma 5.3 (ii) with these branch relations gives $\{M_{(v,j)}[A']\} \preceq_{\Lambda_0} \{F_{(v,j)}[A']\}$ for $\Lambda_0 \coloneqq \bigoplus_j \Lambda_j[A'_j] \circ \mathcal{I}_j$; its part (v) transports $\hat{X}|_j$ to $\mathcal{I}_j^\dagger(\Lambda_j^\dagger(\hat{X}))$ for every $\hat{X} \in \hat{O}^{(j)}$; its part (iv) gives $\Lambda_0^\dagger(\Pi_j) = \mathcal{I}_j^\dagger(\mathbb{1}_{B'_j})$; and $\Pi_{(j,s)} = \Pi_s|_j$ with $\Pi_s \in \hat{O}^{(j)}$ by (I3) of the branch, so $\Lambda_0^\dagger(\Pi_{(j,s)}) = \mathcal{I}_j^\dagger(\Lambda_j^\dagger(\Pi_s))$, which is (10.7). Its part (iii) gives the coarse-grained relation with the same channel. Every operator involved is block diagonal for $\mathbb{S}$ (Step 3; $\hat{X}|_j$ is block diagonal for $\mathbb{S}_j$ inside block j by (I3) of the branch), so Corollary 4.7 (i) gives the same relations, transported set and pull-backs for $\mathcal{P}_{\mathbb{S}} \circ \Lambda_0 = \Lambda[A']$, which has the form (I3) requires; the transported set contains every $\Pi_{(j,s)}$. All of this for every admissible A' under (H).

Step 5 ((I4), the entries). Let $(\hat{O}_k, \bowtie_k, c_k) \in \mathfrak{L}_j[A'_j]$. By (I4) of the branch, $\hat{O}_k$ is a real combination of transported operators of $\mathcal{J}_j$ of the form

(10.4) over $\mathbb{S}_j$, with coefficients $\alpha_{k,v,s}$ and a projector combination Z_k. Then $\hat{O}_k|_j$ is the same combination of the operators $\hat{X}|_j$, hence transported, and it has the form (10.4) over $\mathbb{S}$: in the joint form $\Pi_{(j,s)}F_{(v,j')}[A']\Pi_{(j,s)} = \delta_{jj'}\big(\Pi_s F_v^{(j)}[A'_j]\Pi_s\big)\big|_j$, so $\hat{O}_k|_j = \sum_{(v,j')}\sum_s \alpha_{k,(v,j'),(j,s)}\Pi_{(j,s)}F_{(v,j')}\Pi_{(j,s)} + Z_k|_j$ with $\alpha_{k,(v,j'),(j,s)} \coloneqq \delta_{jj'}\alpha_{k,v,s}$; in the coarse form $\hat{O}_k|_j = \sum_v\sum_s \alpha_{k,v,s}\Pi_{(j,s)}F_v\Pi_{(j,s)} + Z_k|_j$ directly. $Z_k|_j$ and $c_k\Pi_j$ are real combinations of sector projectors of $\mathbb{S}$ with coefficients on the lattice: a coefficient of $Z_k|_j - c_k\Pi_j$ is $\beta_s - c_k$, a difference of two integer multiples of $2^{-\kappa}$ and hence one itself, which is all that is asked of it, since Definition 10.1 (I4) puts the band of (9.1) on an *entry* and not on a coefficient and this step uses nothing about the coefficient beyond its being on the lattice. So $\hat{O}_k|_j - c_k\Pi_j$ has the form (10.4) with $Z = Z_k|_j - c_k\Pi_j$, and it is a real combination of transported operators. Its pull-back, by (10.7) and linearity, is

$$\Lambda^\dagger\big(\hat{O}_k|_j - c_k\Pi_j\big) = \mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\hat{O}_k)\big) - c_k\,\mathcal{I}_j^\dagger\big(\mathbb{1}_{B'_j}\big) = \mathcal{I}_j^\dagger\Big(\Lambda_j^\dagger(\hat{O}_k) - c_k\,\mathbb{1}_{B'_j}\Big).$$

The branch certificate says $\Lambda_j^\dagger(\hat{O}_k) - c_k\mathbb{1}_{B'_j} \bowtie_k 0$, and $\mathcal{I}_j^\dagger$ is a positive map, being the adjoint of a completely positive one (Lemma 1.2 (i) identifies it; positivity is (1.1) tested on positive operators): the image of an operator $\succeq 0$ is $\succeq 0$, of one $\preceq 0$ is $\preceq 0$, and of 0 is 0. So the pull-back $\bowtie_k 0$, which is (6.9) for (10.8) at the constant 0, for every admissible A' under (H). The constant moves into the operator because positivity of $\mathcal{I}_j^\dagger$ bounds $\mathcal{I}_j^\dagger(\Lambda_j^\dagger(\hat{O}_k))$ against $c_k\mathcal{I}_j^\dagger(\mathbb{1})$, the branch weight, and not against $c_k\mathbb{1}_B$; the entry (10.8) is the form in which the branch certificate survives without a hypothesis on that weight. For $j \in J_{\mathrm{w}}$ the branch-weight hypothesis of (H) makes the weight a constant, $\mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = q_j\mathbb{1}_B$, and the branch-weight entry is listed only there; Lemma 5.3 (iv) gives $\Lambda^\dagger(\Pi_j) = q_j\mathbb{1}_B$, an equality certificate at $c = q_j$ for Π_j, which has the form (10.4) with every $\alpha = 0$; and then also $\Lambda^\dagger(\hat{O}_k|_j) \bowtie_k c_k q_j\mathbb{1}_B$, the branch entry at the scaled constant, which is (10.8) plus c_k times the branch-weight entry. The class-marginal identity: in the joint form, $\sum_{(v,j')\in\mathcal{V}_t\times J} M_{(v,j')}[A'] = \sum_{j'}\mathcal{I}_{j'}^\dagger\big(\sum_{v\in\mathcal{V}_t} M_v^{(j')}[A'_{j'}]\big)$, and in the coarse form $\sum_{v\in\mathcal{V}_t} M_v[A']$ is the same sum; where every branch discharges (10.5) this is $q_t\sum_{j'}\mathcal{I}_{j'}^\dagger(\mathbb{1}_{B'_{j'}}) = q_t\mathbb{1}_B$ by the instrument identity $\sum_j\mathcal{I}_j^\dagger(\mathbb{1}_{B'_j}) = \mathbb{1}_B$ established in the proof of Lemma 5.3, for every admissible A' under (H). Where some branch does not discharge it, nothing in this construction establishes the identity, and the flag is not set.

Step 6 (the normalizer acts per branch). Definition 11.1 acts per data sector and per class on the elements of the class within the sector's support. At sector (j,s) of the joint form the support is $\mathcal{V}^{(j,s)}\times\{j\}$ and the class is $\mathcal{V}_t\times J$, so the elements acted on are $\{A_{j,v}^{(s)} : v \in \mathcal{V}_t\cap\mathcal{V}^{(j,s)}\}$, exactly the elements the normalizer of $\mathcal{J}_j$ acts on at sector s and class t, with the same residue

$q_t\mathbb{1} - \sum_{v\in\mathcal{V}_t} A^{(s)}_{j,v}$, the same exact traces, the same declared order (the order of $\mathcal{V}\times J$ declared in (i) restricts on $\mathcal{V}\times\{j\}$ to the order of $\mathcal{V}$, the setting varying slowest in both), hence the same designation order, the same roundings to nearest and mixing weights of the fidelity snap (Definition 11.1 (2)), the same components and hosts, and the same last resort (Definition 11.1 (3)), which read the elements of the class, their number, the class weight, the designation order and the structural supports alone, all the same in the two runs under one run configuration and its strict option; every step of the two runs therefore coincides, and so do their outputs, their displacements and their moved classes. The elements outside the sector support are touched by neither run. In the coarse form the support is $\mathcal{V}^{(j,s)}$ and the class is $\mathcal{V}_t$, and the identification is immediate. Hence $A' = (A'_j)_j$, and the declared model is $M[A']$ at the branches' declared data by (11.3). □

When a register read by a POVM selects the branch, the branching instance is a reduction instance whose pull-backs are the branches' tensored with the register element. Branch entries hold at constant zero, and a branch-weight entry only where (H) makes that element a multiple of the identity.

A register D, read by a POVM that need not be projective (Remark 5.14), selects which branch instance acts; a branch may be the flag instance of Corollary 10.3. The corollary is the branching instance at the instrument that reads the register.

Corollary 10.5 (The register instance: register-conditioned squashing of instances)**.** Let $B = B_0 D$, let $\{N_j\}_{j\in J}$ be a POVM on the register D with J finite and ordered, and for each j let $\mathcal{J}_j$ be a reduction instance on B_0 over one common joint alphabet with classes and weights, with handed model $\{M_{v|j}\}_{v\in\mathcal{V}}$, channel Λ_j, transported set $\hat{O}^{(j)}$ and entry list $\mathfrak{L}_j$. Let $\mathcal{I}_j$ be the instrument of Theorem 5.13, $\mathcal{I}_j(\rho) = \mathrm{tr}_D\left[(\mathbb{1}_{B_0}\otimes N_j^{1/2})\rho(\mathbb{1}_{B_0}\otimes N_j^{1/2})\right]$. Then the branching instance of Theorem 10.4 for $\{\mathcal{I}_j\}$ and $\{\mathcal{J}_j\}$, the *register instance*, is a reduction instance on B in both forms, with the handed model

$$M_v = \sum_{j\in J} M_{v|j}\otimes N_j \quad \text{(coarse-grained)}, \qquad M_{(v,j)} = M_{v|j}\otimes N_j \quad \text{(joint-outcome)}, \tag{10.9}$$

the channel (5.7) pinched, $\Lambda = \mathcal{P}_{\mathbb{S}}\circ\bigoplus_j \Lambda_j\circ\mathcal{I}_j$, the target $\bigoplus_j F_v^{(j)}$ resp. $F_v^{(j)}|_j$, the hypotheses $\bigcup_j(\mathrm{H}_j)$ together with "$\{N_j\}$ is a POVM on D" and, for every j in a declared set $J_{\mathrm{w}}\subseteq J$ with declared lattice constants q_j, "$N_j = q_j\mathbb{1}_D$", and:

(i) *(transport)* for every branch j and sector s of $\mathcal{J}_j$, and for every $\hat{X}\in\hat{O}^{(j)}$,

$$\Lambda^\dagger\big(\Pi_{(j,s)}\big) = \Lambda_j^\dagger(\Pi_s)\otimes N_j, \qquad \Lambda^\dagger\big(\Pi_j\big) = \mathbb{1}_{B_0}\otimes N_j, \qquad \Lambda^\dagger\big(\hat{X}|_j\big) = \Lambda_j^\dagger(\hat{X})\otimes N_j; \tag{10.10}$$

(ii) *(the entry clause)* every entry $(\hat{O}, \bowtie, c)$ of a branch j is carried as $\big(\hat{O}|_j - c\,\Pi_j, \bowtie, 0\big)$, with pull-back

$$\Lambda^\dagger\big(\hat{O}|_j - c\,\Pi_j\big) = \big(\Lambda_j^\dagger(\hat{O}) - c\,\mathbb{1}_{B_0}\big) \otimes N_j, \tag{10.11}$$

a tensor product of the branch's certified operator with $N_j \succeq 0$, so that the certificate holds at 0 with the branch's relation; the branch-weight entry $(\Pi_j, =, q_j)$ is listed for every $j \in J_{\mathrm{w}}$;

(iii) *(the class-marginal identity)* it is discharged when every branch discharges it, and the flag is set in exactly that case;

(iv) *(the declared model)* the normalizer acts branch by branch, and the declared model is

$$M'^{\mathrm{ph}}_{v} = \sum_{j \in J} M'^{\mathrm{ph}\,(j)}_{v} \otimes N_j \quad \text{resp.} \quad M'^{\mathrm{ph}}_{(v,j)} = M'^{\mathrm{ph}\,(j)}_{v} \otimes N_j, \tag{10.12}$$

with $M'^{\mathrm{ph}\,(j)}$ the declared model of branch j.

Uses Def. 5.2, Lem. 5.3, Thm. 5.13, Def. 10.1, Thm. 10.4, Def. 11.1 · *Used by* Cor. 12.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.instance:combine_branches

Proof. The proof of Theorem 5.13 shows that $\{\mathcal{I}_j\}$ is a generalized instrument (Definition 5.2) with $\mathcal{I}_j^\dagger(X) = X \otimes N_j$ for $X \in \mathcal{L}(B_0)$, and $\mathcal{I}_j$ is given by N_j alone, independently of any datum; the hypothesis that it is an instrument is the hypothesis that $\{N_j\}$ is a POVM on D, the branch-weight hypothesis of the theorem at $j \in J_{\mathrm{w}}$ is $N_j = q_j \mathbb{1}_D$ (below), and that $\mathcal{H}(B)$ is separable is inherited from B_0 and D. Theorem 10.4 therefore applies. Its model maps are $M_{(v,j)}[A'] = \mathcal{I}_j^\dagger(M_v^{(j)}[A'_j]) = M_v^{(j)}[A'_j] \otimes N_j$ and their sum over j, which at the handed data are the two models (10.9); its channel (10.6) is (5.7) composed with the sector pinching, as in Theorem 5.13 (i) and (iv). (i) is (10.7) and the transport clause of the theorem with $\mathcal{I}_j^\dagger(X) = X \otimes N_j$ substituted, the second identity being Theorem 5.13 (iii). (ii) is (10.8) with the same substitution: $\mathcal{I}_j^\dagger\big(\Lambda_j^\dagger(\hat{O}) - c\mathbb{1}_{B_0}\big) = \big(\Lambda_j^\dagger(\hat{O}) - c\mathbb{1}_{B_0}\big) \otimes N_j$; if $Y \succeq 0$ on B_0 and $N_j \succeq 0$ then $Y \otimes N_j \succeq 0$, if $Y \preceq 0$ then $Y \otimes N_j \preceq 0$, and $Y = 0$ gives 0, so the branch's relation survives at the constant 0. The branch weight $\mathcal{I}_j^\dagger(\mathbb{1}_{B_0}) = \mathbb{1}_{B_0} \otimes N_j$ is a multiple $q_j \mathbb{1}_B$ exactly when $N_j = q_j \mathbb{1}_D$, so the theorem's branch-weight hypothesis at $j \in J_{\mathrm{w}}$ is the hypothesis $N_j = q_j \mathbb{1}_D$ stated here, and the entry is the theorem's (and that is the reason Theorem 5.13 (iii) calls the general case an identity rather than a constraint). (iii) and (iv) are the theorem's two closing sentences at this instrument, the declared model (10.12) being the model map at $A' = (A'_j)_j$ by (11.3). □

A squashing relation precomposed with a channel on the physical system is again one, with the same target and every spectral certificate at the same constant. A reduction instance so precomposed is again one, with the same datum, target, entries and emission.

Proposition 4.1 composes two squashing relations whose middle system is a finite target. The lemma below precomposes with a channel between two physical systems instead, possibly infinite-dimensional ones, as when a device is a known channel followed by a model a reduction instance is built on; the certificates it keeps are those of the form Theorem 6.7 reads.

Lemma 10.6 (Precomposition of a reduction instance with a channel)**.** Let B and B' be systems with separable Hilbert spaces, and let $\Phi\colon \mathcal{T}(B) \to \mathcal{T}(B')$ be a channel with adjoint $\Phi^\dagger\colon \mathcal{L}(B') \to \mathcal{L}(B)$ (Lemma 1.2).

(i) *(relations)* If $(M';O') \preceq_\Lambda (F;\hat{O})$ with $\Lambda\colon \mathcal{T}(B') \to \mathcal{T}(Q)$, then $\Phi^\dagger(M') := \{\Phi^\dagger(M'_v)\}_{v\in\mathcal{V}}$ is a POVM on B and

$$\big(\Phi^\dagger(M');\Phi^\dagger(O')\big) \preceq_{\Lambda\circ\Phi} \big(F;\hat{O}\big), \qquad (\Lambda\circ\Phi)^\dagger = \Phi^\dagger\circ\Lambda^\dagger. \tag{10.13}$$

If $\Lambda^\dagger(\hat{O}_k) \bowtie c\,\mathbb{1}_{B'}$ for a relation $\bowtie \in \{\preceq, =, \succeq\}$ and a real c, then $(\Lambda\circ\Phi)^\dagger(\hat{O}_k) \bowtie c\,\mathbb{1}_B$, with the same relation and the same constant.

(ii) *(instances)* Let $\mathcal{J}$ be a reduction instance on B' with model map $M[\cdot]$, channel $\Lambda[\cdot] = \mathcal{P}_\mathbb{S}\circ\Lambda_0[\cdot]$ and pull-backs $O[\cdot]$. The *precomposed instance* $\mathcal{J}\circ\Phi$ has the system B; the joint alphabet, data sectors, supports, handed datum, hypotheses (H), scalar values, sector set, target family, transported set, entry list and class-marginal flag of $\mathcal{J}$; the model map $A' \mapsto \Phi^\dagger(M[A'])$; the channel $A' \mapsto \Lambda[A']\circ\Phi = \mathcal{P}_\mathbb{S}\circ(\Lambda_0[A']\circ\Phi)$; and the pull-backs $A' \mapsto \Phi^\dagger(O[A'])$. It is a reduction instance on B.

(iii) *(at a datum, and the emission)* For every admissible datum A', $\mathcal{J}\circ\Phi$ at A' is $\mathcal{J}$ at A' precomposed with Φ. The emission normalizer produces the same declared datum from the handed datum of either, so the emitted target, scalar values, entries and constants of $\mathcal{J}\circ\Phi$ are those of $\mathcal{J}$; its emitted channel is the emitted channel of $\mathcal{J}$ precomposed with Φ, and its declared model is $\Phi^\dagger$ of the declared model of $\mathcal{J}$.

Uses Lem. 1.2, Def. 1.3, Prop. 4.6, Def. 10.1, Def. 11.1 · *Used by* Cor. 21.6, Req. 22.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.instance:precompose

Proof. (i) Φ is completely positive and trace preserving, so by Lemma 1.2 (its trace-non-increasing case, with B' in place of the target) its adjoint is completely positive, and it is unital: for every $\rho \in \mathcal{T}(B)$, $\operatorname{tr}\big[\Phi^\dagger(\mathbb{1}_{B'})\rho\big] = \operatorname{tr}\Phi(\rho) = \operatorname{tr}\rho$, so $\Phi^\dagger(\mathbb{1}_{B'}) = \mathbb{1}_B$ by Lemma 1.2 (i). Hence every $\Phi^\dagger(M'_v)$ is

positive semidefinite and $\sum_v \Phi^\dagger(M'_v) = \Phi^\dagger(\mathbb{1}_{B'}) = \mathbb{1}_B$. The composite $\Lambda \circ \Phi \colon \mathcal{T}(B) \to \mathcal{T}(Q)$ is a channel, and for $X \in \mathcal{L}(Q)$ and $\rho \in \mathcal{T}(B)$ the pairings (1.1) of Λ and of Φ, the second applied to the bounded operator $\Lambda^\dagger(X)$, give $\operatorname{tr}[X\,\Lambda(\Phi(\rho))] = \operatorname{tr}\big[\Lambda^\dagger(X)\,\Phi(\rho)\big] = \operatorname{tr}\big[\Phi^\dagger(\Lambda^\dagger(X))\,\rho\big]$; the adjoint fixed by (1.1) is unique (Lemma 1.2), so $(\Lambda \circ \Phi)^\dagger = \Phi^\dagger \circ \Lambda^\dagger$. Then $(\Lambda \circ \Phi)^\dagger(F_v) = \Phi^\dagger(M'_v)$ and $(\Lambda \circ \Phi)^\dagger(\hat{O}_k) = \Phi^\dagger(O'_k)$, which is (10.13) by Definition 1.3. For the certificate with $\succeq$: $\Lambda^\dagger(\hat{O}_k) - c\,\mathbb{1}_{B'} \succeq 0$, and $\Phi^\dagger$ is positive and unital, so $(\Lambda \circ \Phi)^\dagger(\hat{O}_k) - c\,\mathbb{1}_B = \Phi^\dagger\big(\Lambda^\dagger(\hat{O}_k) - c\,\mathbb{1}_{B'}\big) \succeq 0$; for $\preceq$ apply this to $-\hat{O}_k$ and $-c$, and for $=$ use both.

(ii) We check the four parts of Definition 10.1. (I1): $\mathcal{H}(B)$ is separable; the handed datum and its formation are those of $\mathcal{J}$; the hypotheses (H) are those of $\mathcal{J}$ and read no entry of the datum; and for every admissible A' the family $\Phi^\dagger(M[A'])$ is a POVM on B by (i), so the stability clause holds wherever it holds for $\mathcal{J}$. (I2): the sector set, the scalar values and the target family are functions of the datum and the scalar values alone, which are unchanged. (I3): $\Lambda_0[A'] \circ \Phi$ is a channel from $\mathcal{T}(B)$, given by the constructor of $\mathcal{J}$ preceded by Φ, so $\Lambda[A'] \circ \Phi$ has the required form with the same sector pinching of Proposition 4.6; the transported set is unchanged and contains every sector projector, and (10.3) for $\mathcal{J} \circ \Phi$ at every admissible A' under (H) is (i) applied to that relation for $\mathcal{J}$. (I4): every entry is the same combination (10.4) of the same target, so it is an entry of the required form, and its certificate is discharged at every admissible datum by the certificate clause of (i). Where $\mathcal{J}$ discharges the class-marginal identity (10.5), $\sum_{v \in \mathcal{V}_t} \Phi^\dagger(M_v[A']) = \Phi^\dagger(q_t \mathbb{1}_{B'}) = q_t \mathbb{1}_B$, so $\mathcal{J} \circ \Phi$ discharges it too, and the flag is the same.

(iii) The first sentence is the definition in (ii) read at A'. The emission normalizer of Definition 11.1 acts on the handed datum, class by class and sector by sector, with the class weights and the lattice exponent, and reads nothing else of the instance; the handed datum, the class weights and the lattice exponent of $\mathcal{J} \circ \Phi$ are those of $\mathcal{J}$, so it produces the same declared datum A'. The emitted target, scalar values and entries are functions of A', the scalar values and the entry rule, all unchanged, and the emitted channel and declared model are those of $\mathcal{J} \circ \Phi$ at A', namely $\Lambda[A'] \circ \Phi$ and $\Phi^\dagger(M[A'])$. □

11 Exact emission

Where the exact construction of a reduction instance leaves the declared lattice, the emission normalizer declares a nearby datum, keeping the classes already on the lattice and snapping every other class for fidelity without reviving a zero row or column, or, where no element can host its residue, by a flagged last resort, and decides all six emission conditions last (Definition 11.1). The main result, Theorem 11.2, is that the emitted family is

exactly admissible and squashes the declared model exactly, and that every claim of the instance holds of that model; where the normalizer moved the datum, nothing proved relates the two models, their distance least of all, and a run either evaluates off the snap or rests on a disclosed premise. Which constraint entries are emitted and at which constants (Corollary 11.3), and how several measuring parties' emissions combine without a product datum (Theorem 11.5), follow; the remarks say what an emitted claim is about, where exact decisions stop, and why the emitted constraints are sound by theorem rather than by assumption.

Imports foundations/def:exact-reference, foundations/def:snap

The normalizer puts an exact datum back on the lattice class by class: it keeps the classes already there and snaps the others for fidelity, rounding to nearest and placing the residue only where support allows, or, failing that, by a flagged last resort.

A reduction instance forms its handed datum by exact arithmetic; every exact step multiplies, and a product of dyadics adds significand widths, so the datum can leave the declared lattice on which this module emits. The normalizer defined here puts it back by a snap of the module's own emission, never of anything it was handed. Every class of the handed datum is Hermitian, positive semidefinite and complete at its class weight with Gaussian-rational entries, so it is an exact reference of itself (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference), every radius zero). The normalizer keeps the classes already on the lattice and snaps every other class by one procedure that stays as close to the handed datum as it can while keeping every row and column that the handed datum holds at zero (Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap)): it rounds to nearest, softens the off-diagonal entries where positivity needs it, and places what completeness leaves only on elements whose support holds it. Where no such placement exists, a last resort mixes the class with a multiple of the identity (Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv)); that may give an element a row the handed datum denies it, so the class is flagged. No claim relates a snapped class to the handed datum. A run that forms the number its key length is formed from on the reference companion (Corollary 12.1), which the reductions build with no normalizer from the handed reference, or from the instance's own snapped blocks where none is handed, evaluates every class off its snap; a run that does not, such as one whose key length is formed from a number certified on the snapped declaration in the fallback of Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i), rests, on the classes the normalizer moved, on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement),

and a run that declares the strict option refuses there instead (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)). The emission theorem (Theorem 11.2) is stated about what the normalizer produces. The six conditions its last step decides are those of Requirement "Exact verification before emission" (foundations/req:exact-emission); when a producer re-establishes its lattice at an emission, and on what it may snap at all, is Requirement "Snapping by the declarer only" (foundations/req:who-snaps). Everything the normalizer records is Requirement 17.3, and its refusals are Requirement 17.2.

Definition 11.1 (The emission normalizer, the declared datum and the declared model)**.** Let $\mathcal{J}$ be a reduction instance (Definition 10.1) with handed datum $A \in \mathcal{D}_{\mathrm{adm}}$, formed exactly. Those numbers *are* the data sectors of the emitted target (Definition 10.1 (I2)), so they are this module's own declaration to its consumer. Where the exact arithmetic that formed them has left the declared lattice (9.1), they are re-established on it, per data sector s and per class t, by the following steps, the *emission normalizer*. The steps act on the elements $A_v^{(s)}$ with $v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}$, the elements of the class t in the sector's support; the elements outside the sector support are exactly zero, lie on the lattice, and are touched by no step. Every positive semidefiniteness below is decided exactly by (9.3), on matrices with entries in $\mathbb{F}_i$. The structural support of a matrix is the set of indices i whose row i or column i holds a non-zero entry (Definition "Snapping declared data onto a lattice" (foundations/def:snap)). The *designation order* ranks the elements of a class in the sector's support by largest exact trace and, among equal traces, by the declared order of the class. *A tie is exact and ordinary here*: the traces compared are exact rationals, and a sector symmetric under a relabelling of the class ties every element of it, so "largest trace" alone names no element; the declared order is the order in which this contract reads $\mathcal{V}$, the outcome varying fastest, and it is emitted with the artefact, so the order is a function of the declaration and of nothing else. *Rounding to nearest* a matrix at κ rounds every real and imaginary part of every entry to the nearest integer multiple of $2^{-\kappa}$, *ties to even*: a part lying exactly halfway between two multiples goes to the one whose integer numerator is even. *The tie is exact*: the rounding is carried out in the substrate of Lemma 9.3 on exact rationals, so a halfway part is an exact equality rather than a floating-point near-miss. *Ties to even is odd-symmetric*: negating a part negates both of the multiples nearest it and their numerators, which preserves evenness, so the rounding commutes with $X \mapsto X^{\dagger}$ — mirror entries have equal real parts and opposite imaginary parts before it and therefore after it — and a matrix rounded from a Hermitian one is Hermitian; a zero entry stays zero.

(1) *Keep.* A class t at a sector s every element of which satisfies (9.1) entry by entry is *kept*: $A_v'^{(s)} := A_v^{(s)}$ for every $v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}$, and no further step acts on it.

(2) *Snap for fidelity.* At every other class and sector, for $w = 0$ and then $w = 2^{-\kappa}, 2^{-\kappa+1}, \ldots, 2^{-1}$ in this order, let P_v^w be $A_v^{(s)}$ with every off-diagonal entry multiplied by $1 - w$, that is $(1 - w)A_v^{(s)} + w\,T_v$ with T_v the diagonal part of $A_v^{(s)}$. Every element of the class in the sector's support receives P_v^w rounded to nearest at κ, and the *class residue* $E^w := q_t \mathbb{1}_{d_s} - \sum_{v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}} A_v'^{(s)}$, a Hermitian matrix of lattice values, is split by its *components*: the classes of the equivalence relation on the indices i whose row i of E^w is non-zero generated by $i \sim l$ whenever $E_{il}^w \neq 0$. For a component C write E_C for the matrix that agrees with E^w on the entries whose row and column both lie in C and is zero elsewhere. Each E_C is added to the first element $A_u'^{(s)}$ in the designation order whose structural support in the handed datum, that of $A_u^{(s)}$, contains C, its *host*. The first w at which every component has a host and every element so formed is decided positive semidefinite gives the family, and w is recorded. Where no w does, the class is snapped by step (3).

(3) *The last resort.* The support-safe steps of (2) produced no snap of the class, and the class is snapped by a uniform lift, applied to the class with constant q_t and starting from the handed datum. Let n be the number of elements of the class in the sector's support, v_1 the first of them in the designation order, and, for $w \in (0, 1)$,

$$B_v^w \;:=\; (1 - w)\, A_v^{(s)} + w\, \frac{q_t}{n}\, \mathbb{1}_{d_s}, \qquad v \in \mathcal{V}_t \cap \mathcal{V}^{(s)}, \tag{11.1}$$

a family that is Hermitian, complete at q_t, with entries in $\mathbb{F}_i$, and positive definite in every element, since $q_t > 0$ at a class that is not kept (at $q_t = 0$ completeness and positive semidefiniteness make every element zero, which is on the lattice). Its designation order is that of the class, $\operatorname{tr} B_v^w$ being an increasing affine function of $\operatorname{tr} A_v^{(s)}$. For $w = 2^{-\kappa}, 2^{-\kappa+1}, \ldots, 2^{-1}$ in this order, every element other than v_1 receives B_v^w rounded to nearest at κ, and v_1 receives $q_t \mathbb{1}_{d_s}$ minus their sum; the first w at which every element so formed is decided positive semidefinite gives the family. Where no w does, every element $v \neq v_1$ receives the first matrix $A_v'^{(s)}$ of the following list that is Hermitian, satisfies (9.1) entry by entry, and satisfies

$$0 \;\preceq\; A_v'^{(s)} \;\preceq\; B_v^{1/2}\,: \tag{11.2}$$

$B_v^{1/2}$ itself; then, for $r = \kappa, \kappa - 1, \ldots, 1$ in this order, the shift $B_v^{1/2} - 2^{-r}\mathbb{1}_{d_s}$ and the scaling $(1 - 2^{-r})B_v^{1/2}$, each rounded to nearest at κ; finally the zero matrix, which always qualifies, $B_v^{1/2}$ being positive semidefinite; and v_1 receives $q_t \mathbb{1}_{d_s}$ minus their sum. The weight w, or the fall-back, is recorded, and the class is flagged as one whose support

is not preserved: an element may then hold a non-zero entry in a row or column in which the handed element holds none, that is, outside its structural support (Definition "Snapping declared data onto a lattice" (foundations/def:snap)).

(4) *All six conditions are decided last*, after (2) or (3), each of which changes partial sums from the touched element onward: (a)–(e) of Definition 9.1 together with the lattice invariant (9.1) of every entry, read on the target family (10.2) the datum enters. They are the six Theorem 11.2 (ii) decides; a family for which one of them fails is refused.

A class at a sector that step (1) does not keep is *moved*, whichever of (2) and (3) snapped it. Write A' for the resulting datum — the *declared datum*, an element of $\mathcal{D}_\kappa$ by (1) or the step of (2) and (3) that snapped the class, each of which makes it Hermitian, on the lattice and class complete exactly, positive semidefinite by an exact decision in (2) and in (3) at a weight of its ladder, and by construction in the fall-back of (3), and by (4), which decides all six — and

$$M'^{\mathrm{ph}} \;\coloneqq\; M[A'], \qquad \text{that is} \qquad M'^{\mathrm{ph}}_v \;=\; M_v[A'] \quad (v \in \mathcal{V}), \tag{11.3}$$

for the *declared model*: the handed model with the datum replaced by A', as the instance's model map defines it. Per data sector s, those being the sectors the normalizer touches and the only ones both data have,

$$\Delta_s \;\coloneqq\; \max_{v \in \mathcal{V}} \left\| A'^{(s)}_v - A^{(s)}_v \right\|_{\max} \tag{11.4}$$

is the *emission displacement*; it is recorded per data sector, together with the kept and the moved classes, for a moved class the step that snapped it, the weight w of (2) or (3) or the fall-back of (3), the hosts of the components of (2), and the flag of a class snapped by (3).

The normalizer has five properties, each of which Theorem 11.2, or a claim that instantiates it, uses. *First, $A' = A$ and every $\Delta_s = 0$ whenever the exact construction already satisfies* (9.1): every class is then kept, the normalizer is the identity and nothing below has an antecedent. That is the ordinary case at a nominal declaration and is not the general one: every exact step of a construction multiplies, a product of dyadics adds significand widths, and a lattice established at the instance is closed under none of them; that is the case in which a producer re-establishes its lattice at an emission. *Second, the normalizer reads data sectors and nothing else.* No object the hypotheses (H) of the instance are about — an operator without a finite presentation, a family on a part of $\mathcal{H}(B)$ the datum does not describe, a register POVM — is read, and none is repaired; so (H) holds of M'^{ph} by the stability clause of Definition 10.1 (I1), M'^{ph} is a POVM on B by (I1)

at A', and every statement the instance makes about its model under (H) holds of M'^{ph}. For an instance whose datum is a family of preserved blocks below a cutoff, this is the statement that the declared model agrees with the handed one above the cutoff and satisfies every above-cutoff fact the handed one does. *Third, what the normalizer produces is a datum, and the family the emitted identity is about is the model the instance builds from it.* Two objects, two symbols, and every statement below names which it is on: A' is the declared datum, a finite list of sector blocks carried entry by entry in the emitted target; M'^{ph} is the declared model on B, the object an adjoint landing on B can equal (Theorem 11.2 (iii)). Where the model map is the identity on a finite B the two coincide; where the model has parts the datum does not describe — an above-cutoff family, further branches — they do not, and the instance's model map says how A' sits inside M'^{ph}. For a branching instance (Theorem 10.4) the normalizer acts branch by branch, so the declared model is the branches' declared models composed. *Fourth, this is a snap of what this module declares and never of what it was handed*, which is the condition under which a producer may snap at all. *Fifth, no step revives a zero row or column, save the last resort (3)*: for every data sector s and every $v \in \mathcal{V}$ whose class at s is not snapped by (3) the structural support of $A_v'^{(s)}$ is contained in that of $A_v^{(s)}$. In (2) the family $\{T_v\}$ of diagonal parts is complete at q_t, the diagonal of $\sum_v A_v^{(s)} = q_t \mathbb{1}_{d_s}$ being q_t in every place, and T_v is diagonal, with non-negative entries, non-zero only at indices of the structural support of $A_v^{(s)}$, so P^w is a mixture with a complete diagonal family that adds only on coordinates of each element's own structural support; P_v^w has the structural support of $A_v^{(s)}$, and a rounding to nearest keeps a zero entry zero. Each E_C has its non-zero rows and columns in C, which the host's structural support contains, and the entries of E^w between two components are zero by their definition, so $E^w = \sum_C E_C$ and every non-zero entry of the residue goes to a host. In (2) the residue need not be positive semidefinite, and the family is taken only where every element is decided positive semidefinite. No element is mixed with a multiple of the identity outside (3). In (3) the elements of the class in the sector's support may gain rows and columns, which is what its flag discloses; the elements outside the support stay zero there too. And (3) always yields a family whose elements are positive semidefinite: at a weight of its ladder by decision, and in the fall-back because every element of $B^{1/2}$ is positive definite, the rounded elements satisfy (11.2), and $A_{v_1}'^{(s)} = B_{v_1}^{1/2} + \sum_{v \neq v_1} \left(B_v^{1/2} - A_v'^{(s)}\right) \succeq B_{v_1}^{1/2} \succ 0$ by the completeness of $B^{1/2}$.

Uses Def. 4.4, Def. 9.1, Lem. 9.3, Def. 10.1, foundations/def:exact-reference, foundations/def:snap · *Used by* Cor. 10.3, Thm. 10.4, Cor. 10.5, Lem. 10.6, Thm. 11.2, Cor. 11.3, Thm. 11.5, Cor. 11.6, Rem. 11.7, Rem. 11.8, Rem. 11.9, Cor. 12.1, Req. 17.2, Req. 17.3, Req. 17.4, Prop. 20.5, Prop. 21.2, Cor. 21.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.normalizer:normalize

The emitted target is computed exactly, decided exactly admissible before emission, and squashes the declared model exactly; nearness to the handed model bounds nothing. On blocks the record names as resting on A-SNAP, the run's conclusions about the device rest on that premise.

This is the theorem every emission of the module rests on: what is emitted, the displacement record aside, is computed exactly, is decided before it is emitted, and is about the declared model of Definition 11.1. The identity instance applies it through Corollary 11.6, and each submodule's instance through its own emitted corollary. Part (v) says what relates the declared model to the handed one: nothing proved, and in particular not the size of the move, since a family near the constructed one need not have a key length near its own. A run that forms the number its key length is formed from on the reference companion (Corollary 12.1) needs no such relation: it reads the emitted target only to find points, and every data sector is evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)). Elsewhere, where the normalizer moved a class, a run on that emission rests on the premise that Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) (ii) states for a producer's snap of its own emission: every conclusion about the device that the run draws from the emitted target holds with the declared datum in place of the handed one on the sectors concerned. Since the emitted target squashes the declared model exactly (part (iii)), that premise holds in particular where the device measures the declared model there. A moved block of the declaring instance's own snap of the measurement the handed datum is formed from is in the same position; there the premise is clause (i) of the same assumption. Part (vi) states what a run on the emission then rests on. It is conditional: it asserts nothing of a run whose record names no such block, so that the premise is engaged exactly where the record names a block as resting on it. A run whose key length is formed from a number on the reference companion (Corollary 12.1) names no data sector of this emission among them, since the companion carries none of the normalizer's data, and of the declaring instance's snap only the blocks on which the companion carries the snapped block; the bound then reads the companion, and this theorem describes the target the search ran on. Where the key length is formed instead from a number certified on the snapped declaration, because the companion's bound counts but no point of it can be certified (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i)), the run reads the emitted target there, and names the data sectors with a moved class among the blocks resting on the premise. The six conditions of part (ii) are those of Requirement "Exact verification before emission" (foundations/req:exact-emission), and that no status is emitted for an approxi-

mate relation is Requirement 14.4.

Theorem 11.2 (Exact emission of a reduction instance). Let the declared data pass every refusal rule of this module's interface, so that every declared block and scalar it reads is exactly admissible on the declared lattice (Definition 9.1). Let $\mathcal{J}$ be a reduction instance (Definition 10.1) with handed datum A, and let A' be the declared datum the normalizer of Definition 11.1 produces from it (where the normalizer refuses, nothing is emitted). Let $\{F_v\} \coloneqq F[A']$, $\Lambda \coloneqq \Lambda[A']$ and $\mathfrak{L} \coloneqq \mathfrak{L}[A']$ be the target, the channel and the entry list of the instance at A', and $M'^{\mathrm{ph}} = M[A']$ the declared model (11.3). Then:

(i) *(everything on the list below is computed exactly, and the displacement Δ_s is the one emitted number that is not on it)* every number on the following list is obtained from the numbers the instance was handed by finitely many operations in $\mathbb{F}_i$, floors and comparisons: the data-sector values $A_v'^{(s)}$; the scalar values $\phi_s(v)$; the sector projectors; and every emitted constraint operator and constant of Corollary 11.3. *That list is everything this module emits with exactly one exception*: the emission displacement Δ_s of (11.4), an entrywise maximum *modulus* and hence the square root of an exact rational, in general irrational, which is emitted only as a record of what the emission normalizer did and which no statement of this contract reads. *No floating-point comparison, no eigenvalue routine, no enclosure and no interval occurs anywhere in that list*; the intermediate numbers an instance forms on the way to its datum and its scalar values are listed by the corollary that instantiates this theorem for it;

(ii) *(the emitted family is exactly admissible, and is decided so before it is emitted)* the family $\{F_v\}$, presented as the sector-block list (10.2), satisfies conditions (a)–(e) of Definition 9.1 together with (9.1) of every entry — six conditions — and each of the six is *decided*, exactly, on the numbers about to be emitted: (a) by comparison of declared labels against the instance's sector map; (b) and (d) by entrywise equality and (c) by Lemma 9.3, per sector block; (e) by Lemma 9.5, whose hypothesis (9.5) is a declared input condition; and (9.1) entry by entry. *A family for which any of the six cannot be established is not emitted. And the same numbers discharge what the consuming declaration asks of a party's side*: the tuple $\big(Q, \mathbb{S}, \{F_v\}_{v\in\mathcal{V}}\big)$ *together with the party partition* of Definition 10.1 (I2) — the default party partition, every factor Bob's, unless the instance declares another party's label, in which case every factor is that party's — satisfies the well-formedness conditions the consuming declaration places on *that party's* side (Definition "Admissible declaration" (meat-core/def:admissible-declaration) (i) and (iii), with the party labels per tensor factor demanded through

its (viii)) —

$$F_v = F_v^\dagger, \qquad F_v \succeq 0, \qquad \sum_{v \in \mathcal{V}} F_v = \mathbb{1}_Q, \tag{11.5}$$

with every F_v block diagonal for $\mathbb{S}$ and $\mathcal{H}(Q) = \bigoplus_{s \in \mathbb{S}} \mathcal{H}(Q_s)$ finite-dimensional, every factor of every $\mathcal{H}(Q_s)$ the instance's declared party's and Bob's by default. *Those are conditions on a party partition and not on a defect*, and they are a *different* condition from the conditions of Definition "Exact admissibility of a declared operator family" (foundations/def:exact-admissibility) that (a)–(e) discharge, which say nothing about who owns which factor;

(iii) *(the defining identity is about the numbers emitted, and it is an identity on B)* for every $v \in \mathcal{V}$,

$$\Lambda^\dagger(F_v) \;=\; M'^{\mathrm{ph}}_v \qquad \text{exactly}, \tag{11.6}$$

both sides operators on B, which is where $\Lambda^\dagger$ lands, Λ being a channel $\mathcal{T}(B) \to \mathcal{T}(Q)$. *Only the data sectors carry anything the normalizer produced*: M'^{ph} is the handed model with the datum replaced by A' as the instance's model map defines it, and everything the hypotheses (H) are about is the declared one, unchanged. Moreover every claim of the instance holds verbatim of the declared model — the relation (10.3) with its transported set and pull-backs, every certificate of (10.4), and the class-marginal identity (10.5) where the instance discharges it — and Theorem 3.11 and Theorem 7.4 hold of the one-link receiver segment (Definition 7.1) whose source family is (B, M'^{ph}), whose target is (Q, F) and whose channel witness is Λ, with status = exact;

(iv) *(the emitted bounds, and the direction each is rounded in)* every scalar value that was rounded to reach the lattice was rounded *down*, as the lower bound of the quantity it certifies, and every complementary value was *recomputed* from the rounded ones and never rounded on its own (Definition 10.1 (I2)); and a caller-declared constant of modulus at most one off the lattice is rounded in the direction that enlarges the feasible set for its declared relation (Corollary 11.3 (iii)), an equality constant off the lattice and any constant of modulus above one being refused. *Every other number this module emits is exact, the displacement record Δ_s of (i) aside*; these are all the rounding directions there are among the numbers a statement of this contract reads, and the direction of each is recorded;

(v) *(what relates the declared model to the handed one)* where some $\Delta_s > 0$ the emitted numbers are a declaration of a family near the constructed one, and nearness is not what relates them: *no bound relates a key length computed from M'^{ph} alone to one computed from*

the handed model $M[A]$, or from the supplier's own output, and this contract has no theorem converting an operator perturbation of the target into a security penalty, nor any other relation between A' and A on a moved class: the consuming contract's impossibility (Proposition "Instability of the rate under measurement perturbations" (meat-core/prop:optol-nobound)) is the whole of what is proved there; what a run rests on there instead is (vi);

(vi) *(what a run on the emission rests on where it reads a moved block, and only there)* let $B = B_{\mathrm{snap}} \sqcup B_{\mathrm{em}}$ be the blocks that the record of the run names, for this artefact, as resting on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), kept as two disjoint parts, since they are blocks of two different snaps and their labels may coincide: B_{em} holds the data sectors of the emission so named, and B_{snap}, where A is formed from measurement blocks of a snap (Definition "Snapping declared data onto a lattice" (foundations/def:snap)) that the declaring party made of the measurement it states, the blocks of that snap so named; the declaring party is the instance that states the measurement of the device, or a reference of it, from which A is formed. Where B is empty, this part asserts nothing, and no part of this theorem rests on that assumption. Where B is not empty, let that assumption hold. Then on every block in B_{snap}, the block of the measurement the device performs is the snapped block, with the entries recorded for that snap, from which A is formed, in place of the block of the model the declaring party states; and every conclusion about the device that the claims of the run draw from the emitted target, under their hypothesis that its data sectors carry the handed datum A, holds when they carry the declared datum A' on every data sector in B_{em}, which by (iii) is the datum whose model M'^{ph} the emitted target squashes exactly. The run carries no bound relating its key length to the declaring party's model on the blocks in B_{snap}, nor to $M[A]$ on the data sectors in B_{em}: no claim relates that model to the snapped block on a block in B_{snap}, nor A to A' on a data sector in B_{em}.

Uses Thm. 3.11, Def. 4.4, Cor. 4.7, Def. 7.1, Def. 7.2, Thm. 7.4, Def. 9.1, Def. 9.2, Lem. 9.3, Lem. 9.5, Def. 10.1, Def. 11.1, foundations/def:snap · *Assumes* A-SNAP · *Used by* Rem. 2.5, Cor. 11.3, Thm. 11.5, Cor. 11.6, Rem. 11.7, Rem. 11.8, Req. 17.2, Req. 17.3, Cor. 21.3, Cor. 21.4, Cor. 21.6, protocol-compiler/req:link-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.emit:reduce · *Oracles* registry_reductions.py

Proof. Idea. Parts (i) and (ii) follow what the instance and the normalizer compute and what the six conditions ask of it; part (iii) is the stability clause of the instance applied at the declared datum; part (iv) is the rounding rule

of the instance's scalar values and of a caller's constants; part (vi) applies the two clauses of the snapped-measurement assumption to the blocks the record names.

Part (i). Every entry of a declared block and every declared scalar is a dyadic rational (Definition 9.1 (a),(b)), hence an element of $\mathbb{F}_i$ resp. $\mathbb{F}$. The handed datum is formed from them by finitely many operations in $\mathbb{F}_i$ (Definition 10.1 (I1)); the scalar values by finitely many operations in $\mathbb{F}$ and downward roundings $\lfloor\cdot\rfloor_\kappa$, each a floor and a comparison in $\mathbb{F}$ (I2); the sector projectors are the indicator values 0 and 1; each emitted constraint operator of Corollary 11.3 is a real combination (10.4) of emitted blocks and sector projectors with lattice coefficients, and each constant is declared or rounded by a floor or a ceiling in $\mathbb{F}$; and each of the normalizer's steps (1)–(4) is a rounding to nearest, a comparison, an addition, a subtraction, a product with a rational number, a positive semidefiniteness decision, or a test of which entries are zero, and its mixing weights are the dyadic numbers 2^{-l}. Field operations, comparisons and floors in $\mathbb{F}_i$ are exact by the definition of the substrate (Definition 9.2), and the only non-arithmetical predicate that occurs — positive semidefiniteness, the form every operator inequality of this contract takes, the Löwner order being the only order it uses — is decided by sign comparisons in $\mathbb{F}$ (Lemma 9.3 (i), (9.3)); the other conditions of (ii) above are equalities, compared entry by entry in $\mathbb{F}_i$, and comparisons of declared labels. *No bound in the operator norm is decided here or anywhere on the emission path*: Lemma 9.3 (ii) is correct as it stands and is carried without a client. *The one emitted number the list omits is* Δ_s, and it is omitted because (11.4) takes an entrywise maximum *modulus* of a complex difference, which is a square root of an exact rational and in general leaves $\mathbb{F}$. It is emitted only as a record (Requirement 17.3), so the statement above is a statement about the enumerated list and about everything this module emits except that one number; no step of this proof, and no statement of this contract, is proved from its value (Remark 11.8).

Part (ii). Each of the six is decidable on a finite list of elements of $\mathbb{F}_i$: (a) is a comparison of declared labels; (b) is an entrywise equality; (c) is (9.3); (d) is an entrywise equality between two finite sums; (9.1) is, per entry, integrality of $2^\kappa x$ together with $|x| \le 1$, two comparisons in $\mathbb{F}$; and (e) is Lemma 9.5, whose hypothesis is the declared input condition (9.5) and whose conclusion covers every entry position, the accumulation being entrywise. *That (b), (c) and (d) hold of the emitted family* is, on a data sector, Definition 11.1, whose step (1) keeps exactly admissible classes, whose steps (2) and (3) establish Hermiticity, (9.1) and class completeness for a moved class and establish or decide its positivity, and whose step (4) decides all six on the result — completeness over $\mathcal{V}$ following from class completeness and $\sum_t q_t = 1$ (Definition 10.1, first consequence); and, on a scalar sector, the declared values $\phi_s(v) \in [0,1]$ on the lattice with $\sum_v \phi_s(v) = 1$ exactly (I2), which give a real, positive semidefinite and complete block outright.

So the decision of step (4) is a defensive exact check: on declared data that pass the refusal rules, as the theorem assumes, every path of the normalizer produces a family the six conditions hold of, and no path reaches the refusal it guards. The refusal of a class that can occur is the strict option's refusal of a moved class (Requirement 17.2). *The consuming declaration's own conditions follow from what has just been established*: (b), (c) and (d) are (11.5) — the families are POVMs and every F_v is sector-block-diagonal — which is Definition "Admissible declaration" (meat-core/def:admissible-declaration) (iii), block diagonality for $\mathbb{S}$ being by construction; and $\dim \mathcal{H}(Q) = \sum_{s\in\mathbb{S}} d_s$ is a finite sum of declared dimensions (Definition 10.1 (I2), its fifth consequence), so $\mathcal{H}(Q)$ is finite-dimensional and the alphabet $\mathcal{V}$ finite, which is the part of its (i) this tuple is about. *And the default party partition assigns every factor of every $\mathcal{H}(Q_s)$ to Bob, which is the assignment the physical model already makes*: the target describes what reaches Bob's receiver, the instance's system B being his. Where the instance declares another party's label (Definition 10.1 (I2)) the same sentence holds of that party, whose system B then is; what is emitted is the declared label either way, no step above reads its value, and an emission of several measuring parties carries one label per factor (Theorem 11.5 (i)). *The party labels themselves are neither (i) nor (iii)*: a declaration names a party partition $\mathfrak{P}$ (Definition "Protocol declaration" (protocol-compiler/def:declaration)) and Definition "Admissible declaration" (meat-core/def:admissible-declaration) (viii) is what demands it; what is emitted here is that partition's value on the target's sectors. Nothing in this paragraph is measured and nothing is concluded from a returned number.

Part (iii). By Definition 11.1 (1)–(4), $A' \in \mathcal{D}_\kappa \subseteq \mathcal{D}_{\mathrm{adm}}$: it is class complete at q_t by (1) or by the step of (2) and (3) that snapped the class, the last step that changes an element on every path, and Hermitian, positive semidefinite and on the lattice per data sector by the decision of (4). By the second property of that definition the normalizer read no object the hypotheses (H) are about, so (H) holds of $M[A']$ by the stability clause of Definition 10.1 (I1), and $M[A'] = M'^{\mathrm{ph}}$ is a POVM on B by (I1). Parts (I3) and (I4) of the instance are established for *every* admissible datum under (H); at A' they read $(M'^{\mathrm{ph}}; O[A']) \preceq_\Lambda (F; \hat{O}[A'])$, whose first clause is (11.6), together with every certificate of $\mathfrak{L}$ and, where the flag says so, (10.5) at A'. That is what "holds verbatim of the declared model" asserts, and nothing was re-proved: the instance at A' is the same construction with a different admissible datum, which is what the stability clause was stated for. The sector pinching in front of $\Lambda_0[A']$ changes none of it (Corollary 4.7 (i), the second consequence of Definition 10.1). Theorem 3.11 needs only the relation just established; the one-link segment with source family (B, M'^{ph}), target (Q, F), channel witness Λ, the transported pair $(\hat{O}[A'], O[A'])$, the reachable-set statement of that channel and the entry list $\mathfrak{L}$ is a receiver-stage link of Definition 7.1 and a chain of Definition 7.2 with an empty source segment, and Theorem 7.4 (iv) gives it $\mathsf{status} = \mathsf{exact}$ because (11.6) carries no error

term. *Every remaining ingredient is common to the handed model and the declared one*: B, the alphabet with its classes and weights, the sector map, the supports, the scalar values, the constructor, and every object of (H), which the normalizer never reads.

Part (iv). The scalar values are (I2) of Definition 10.1, which requires the downward direction and the recomputed complement; the constants are Corollary 11.3 (iii). *That a complement is recomputed rather than rounded is what makes a scalar sector's completeness an identity of lattice values*: the rounded values and their complement are formed from the same finitely many rounded quantities, so $\sum_v \phi_s(v) = 1$ holds exactly, and Lemma 9.5 makes the accumulation that checks it exact as well. Rounding the two independently would leave a residue of up to $2^{-\kappa}$ on the sector and would fail (d) — which is the case that Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) closes by requiring the complementary quantity to be recomputed.

Part (v). It asserts the absence of a bound and of a relation and needs no proof; the consuming contract rules the same inference out from its own side (Proposition "Instability of the rate under measurement perturbations" (meat-core/prop:optol-nobound)).

Part (vi). Where B is empty there is nothing to prove, and no step of the proofs of (i)–(v) reads Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement). Otherwise each of B_{snap} and B_{em} is, for its own snap, the set that the assumption calls B: the blocks of that snap the record of the run names as resting on it. The assumption is applied to the two snaps separately, and a label in both parts names two different blocks.

Blocks in B_{snap}. By the statement of (vi), the declaring party is the instance that states the measurement A is formed from, so the snap whose blocks form B_{snap} is the declaring instance's snap of the measurement of the device, or of a reference of it, that it states. Its D is therefore data an instance states as the measurement of a device, or a reference of it, and clause (i) of the assumption applies: on each block in B_{snap} the block of the measurement the device performs is the snapped family with the entries recorded for the snap, which are the entries A is formed from on that block, and a statement of another assumption of the run that the declaring party's model describes the device there is not asserted.

Data sectors in B_{em}. A data sector of the emission is a block of the normalizer's snap of the handed datum A to the declared datum A' (Definition 11.1), a producer's snap of what it emits whose exact construction is A, so clause (ii) applies to it with A' as the snap: every conclusion about the device that the claims of the run draw from the emitted family, under their hypothesis that it is A, holds when it is $A'^{(s)}$ on every data sector $s \in B_{\mathrm{em}}$. The data sectors of the emitted target carry the values $A'^{(s)}_v$ (part (i)), and

the target squashes $M'^{\mathrm{ph}} = M[A']$ exactly (part (iii)).

What the run does not carry. The argument is that of part (v): a nearness of the families relates no key lengths (Proposition "Instability of the rate under measurement perturbations" (meat-core/prop:optol-nobound)), and no claim of these contracts relates a moved block of a snap to the data it was made from. A block in B_{snap} is a moved block of the declaring instance's snap, by the assumption's definition of B, and a data sector in B_{em} holds a moved class of the normalizer's; so the run carries no relation on these blocks. The sentence is confined to them: it says nothing of a relation to $M[A]$ on a block in B_{snap}, where A is formed from the snapped block and part (iii) stands as it is established for the data sectors, and nothing of a block of either snap that the record does not name. □

The emitted list is empty unless the request carries entries; then it holds the instance's entries, the class-marginal entry where discharged and further entries whose certificate a claim decides. Certificates hold of the emitted numbers, and off-lattice inequality constants are rounded outward.

A request either carries entries or carries none, and there is no third case; whether the emitted constraint list is empty, and with it whether the attack-set licence is present, depends on that alone. The flag-state submodule's encoded and licensed routes are that submodule's instance of the two cases, named here for the reader; this module declares no route key of its own. The entries below are untagged; where an entry carries the party-local tag instead is Requirement 16.8, and Remark 11.4 explains why a one-factor emission carries none. Remark 3.14 identifies the expectation form of the class-marginal identity, read in part (ii), as its sound squashed form.

Corollary 11.3 (The emitted constraint entries)**.** Assume the hypotheses of Theorem 11.2 for a reduction instance $\mathcal{J}$ with declared datum A', target $\{F_v\}$, channel Λ and declared model M'^{ph}. A request *carries entries* when it selects the case in which the instance's own entries are emitted: in the flag-state submodule that case is the licensed route; in a core-only artefact it is a caller's request for an entry — for a default carried operator as an entry, or for the class-marginal entry. *What is emitted is decided by the request*: *where the request carries no entry the emitted constraint list is empty* — whatever entries the instance's own list $\mathfrak{L}[A']$ contains — and the artefact carries no attack-set licence; *where the request carries entries the emitted list is the instance's own entries, the class-marginal entry on request and any further requested entry that (iii) admits*, so that the list is non-empty and the licence of Theorem 6.7 is present. In *both* cases the certificate $(\Pi_s, \succeq, 0)$ of every sector projector of every instance is carried *per entry in the transported set*, with the certificate case of Corollary 1.7 that makes each usable, and not as a constraint entry. Then *(i) and the identity* (11.8) *of (ii) of every request, and what each of (ii) and (iii) emits of a request that carries entries*:

(i) *(the default carried operators)* every operator of the instance's *de-*

fault carried set — every sector projector Π_s, $s \in \mathbb{S}$, which Definition 10.1 (I3) puts in the transported set of every instance, together with the sums of sector projectors that instance transports besides (for a target split at a cutoff, the projector onto the blocks below it and the projector onto the flag blocks; for a branching instance, the branch projector Π_j) — is emitted at $c = 0$ with $\bowtie = \succeq$: its entries are 0 and 1, hence on the lattice at every $\kappa \geq 1$, and its pull-back

$$\Lambda^\dagger(\Pi_s) \;\succeq\; 0 \tag{11.7}$$

is exactly positive for every instance, as therefore is the pull-back of every sum of them. *It is emitted with the transported set*, per entry and whatever the request carries, as the certificate case of Corollary 1.7 (ii) that entry records, and it becomes an entry of $\mathfrak{L}$ only where a caller requests it *as an entry* — a request that makes the emitted list non-empty and so brings the licence with it;

(ii) *(the class-marginal entry)* For $\hat{O} = \sum_{v \in \mathcal{V}_t} F_v$ at a declared $t \in \mathcal{T}_{\text{set}}$ — the expectation form of the physical class-marginal identity (4.3) —

$$\Lambda^\dagger\big(\hat{O}\big) \;=\; q_t \, \mathbb{1}_B \qquad \text{exactly} \tag{11.8}$$

holds *whenever* the instance discharges the class-marginal identity (10.5), and is established by this contract *only* there — *a fact about the instance and the channel, which this part asserts of every request. Where the request carries entries*, so that the emitted list is non-empty and the licence is present, the entry is then emitted, on request, as the single equality entry $(\hat{O}, =, q_t)$ at the declared q_t; where the instance does not discharge it the request is refused, and the entry is never emitted on the strength of a branch that does not establish it;

(iii) *(any further entry — the general rule) under the same hypothesis as (ii)'s emission sentence — the request carries entries*, let $\hat{O} = \sum_{v \in \mathcal{V}} \sum_{s \in \mathbb{S}} \alpha_{v,s} \Pi_s F_v \Pi_s + Z$ with declared real lattice coefficients $\alpha_{v,s}$ and Z a fixed operator of the declared sector decomposition (a sector projector or a combination of them whose coefficients are declared lattice values, as Definition 10.1 (I4) requires), the form (10.4), with a relation $\bowtie$ and a declared real constant c. *Where a declared constant of modulus at most one does not satisfy* (9.1) *it is first rounded onto the lattice in the direction that enlarges the feasible set for* $\bowtie$ — down for $\succeq$, up for $\preceq$ — and *an equality entry whose constant is off the lattice is refused rather than split into a pair*, an equality having no loosening direction; *a declared constant of modulus above one is refused under every relation.* Suppose the certificate

$$\Lambda^\dagger(\hat{O}) \;\bowtie\; c\,\mathbb{1}_B \tag{11.9}$$

is established for the emitted numbers, at the constant c so obtained, in one of two ways: for an entry of the instance's own list $\mathfrak{L}[A']$, by (I4) of the instance; for a requested entry, by a *decided certificate* — a condition on the emitted numbers and the request's declared constants, decided exactly from them, that a claim stated for the instance proves sufficient for (11.9) under the instance's hypotheses (H) — that holds there. Then the entry is emitted at that c. *No other requested entry is emitted*: a requested operator of this form for which no claim stated for the instance gives a decided certificate, or whose decided certificate fails at the emitted numbers, is refused, and no certificate is taken from the request on trust.

Every entry that (i)–(iii) emit is *untagged*, $(\hat{O}, \bowtie, c)$, and is read in the form of the declared source mode (Definition 6.5).

Each certificate above is discharged for the emitted numbers themselves, those numbers being the family every part is read on (Theorem 11.2 (iii)), so no constant is shifted to pay for a displacement between the family a certificate is asserted of and the numbers its operator is assembled from. The rounding rule of (iii) applies only to a constant a caller declares, never to one this module or an instance computes. *Together, (i)–(iii) cover every entry this module can emit*: the default carried operators of every instance, the class-marginal entry a caller may request, and every entry of the form (11.9) — which includes every entry an instance carries in its own list, such as a flag-sector entry of a weight-preserving squasher, and the branch entries a branching instance inherits (Theorem 10.4, Corollary 10.5). *A requested operator outside that form, or of that form without a decided certificate that holds, is not emitted at some other constant; it is refused.* So the three parts are exhaustive over what is *emitted*, which is what the artefact carries. *Where the request carries no entry there is nothing for (ii)–(iii) to be exhaustive about*: $\mathfrak{L} = \varnothing$ there, whatever entries the instance's own list contains, and the certificate of every default carried operator travels with the transported set alone.

Uses Cor. 1.7, Def. 4.4, Def. 6.5, Thm. 6.7, Def. 9.1, Def. 10.1, Def. 11.1, Thm. 11.2 · *Assumes* A-CONST · *Used by* Rem. 11.4, Cor. 11.6, Rem. 11.9, Cor. 12.1, Req. 15.2, Req. 16.3, Req. 16.4, Req. 16.5, Req. 16.6, Req. 16.7, Req. 17.4, Cor. 21.3, Cor. 21.4, protocol-compiler/req:link-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.constraints:certify_carried_operators

Proof. (i) The sector projectors are determined by the declared dimensions and order alone (Definition 10.1 (I2)); each is in the transported set by (I3), and its pull-back is positive semidefinite because $\Lambda^\dagger$ is a positive map — the third consequence of Definition 10.1 — which is (6.9) with $c = 0$ and $\bowtie = \succeq$, the certificate case of Corollary 1.7 (ii). A sum of distinct sector projectors is again a projector of the declared decomposition, with entries 0 and 1 and with the sum of their pull-backs as its own, positive by additivity; those

sums are the remaining operators of the default carried set. *Where each is emitted is the transported set*, which (I3) is about and which the artefact carries in either case with the certificate case per entry; nothing in this part makes it an entry of $\mathfrak{L}$, and where the request carries no entry $\mathfrak{L}$ is empty. A caller who asks for one of them *as an entry* asks for an entry, and gets it at this constant — the list being non-empty then, and the licence present. *Nothing in this part uses which of the two cases holds*, so it is asserted of every request.

(ii) By Theorem 11.2 (iii) and linearity, $\Lambda^{\dagger}(\hat{O}) = \sum_{v \in \mathcal{V}_t} M'^{\mathrm{ph}}_{v}$, the class sum of the declared model (the fourth consequence of Definition 10.1). Where the instance discharges (10.5), it does so for every admissible datum under (H), hence at A' by Theorem 11.2 (iii), so the class sum is $q_t \mathbb{1}_B$ and (11.8) is (6.9) with $\bowtie = =$ at $c = q_t$; the constant is declared before the run (Assumption 13.7). That derivation reads the instance and the channel alone: *nothing in it uses which of the two cases holds*, so (11.8) is asserted of every request, exactly as (i) is. What the request decides is whether the entry is *emitted* — where it carries entries the list is non-empty and the licence present, and the entry joins the list on request. That is the whole of the implication, and *no converse is asserted*: where the instance does not discharge (10.5), (11.8) may still hold of the device and nothing in this contract establishes it — an uncharacterised branch contributes some operator between 0 and its branch weight, not q_t times it (Corollary 10.3) — and emitting the entry on an undischarged identity would impose $\mathrm{tr}\left[\hat{O}\sigma\right] = q_t$ on reachable states for which it may be false, the security break of Remark 3.14; so the request is refused, which is the safe direction and the only one this corollary licenses.

(iii) The entry is emitted at the constant its certificate is established at, which is the hypothesis; that the constant is declared and not observed is Assumption 13.7. An instance's own entries carry their certificates for every admissible datum (Definition 10.1 (I4)), hence at A'. A requested entry's decided certificate holds at the emitted numbers by hypothesis, and the claim that gives it proves (11.9) from it under (H), which holds of the instance at A' by the stability clause of Definition 10.1, A' being admissible (Theorem 11.2 (iii)); so (11.9) holds at the emitted numbers, and nothing about it is taken from the request. A requested operator without a decided certificate that holds has nothing that establishes (11.9) for it, and emitting it would impose on reachable states a condition that may be false of them, the security break of Remark 3.14; refusing it is the safe direction, nothing being emitted. *The rounding rule*: let c' be a declared constant off the lattice with $|c'| \leq 1$ and c its rounding in the stated direction, so $c \leq c'$ when $\bowtie$ is $\succeq$ and $c \geq c'$ when $\bowtie$ is $\preceq$; since -1 and 1 are lattice values, c lies in $[-1, 1]$ and satisfies (9.1). The certificate is established at c, the constant emitted; and wherever it holds at c' it holds at c, since $\Lambda^{\dagger}(\hat{O}) \succeq c' \mathbb{1}_B \succeq c \mathbb{1}_B$ in the first

case and $\preceq$ in the second, so the rounding loses no entry whose certificate holds at the declared constant, while $\mathfrak{A}_{\mathfrak{L}}$ only grows — which is the safe direction, a smaller admissible set being the one in which Theorem 6.7's inclusion can fail. *For an equality entry neither direction survives*, which is why such an entry is refused rather than rounded. A constant of modulus above one is refused under every relation: on its enlarging side there need be no value satisfying (9.1) at all ($\preceq$ at $c' > 1$, $\succeq$ at $c' < -1$), and refusing is the safe direction, nothing being emitted. □

The licence covers a party-local entry only through a certificate on its own factor's channel; a joint one can hold while the operator inequality fails. On a one-factor emission the untagged entry is implied by the party-local reading, and rate is lost only in untrusted-source mode at d_A > 1.

The entries of Corollary 11.3 are untagged; the party-local tag exists for a measured system that is a declared product of party-scoped factors (Requirement 16.8). This remark says what licenses the tagged reading and what a one-factor emission gives up by emitting the untagged one.

Remark 11.4 (Party-local tags and the one-factor emission). An entry scoped to a factor f of a declared product $Q = \bigotimes_f Q_f$ is read as the operator inequality (6.5) on the other systems of ω, with $\hat{O}$ an operator on Q_f alone. *What licenses that reading is a certificate discharged on the factor's own channel* Λ_f, which is the hypothesis of Theorem 6.7 (v); a certificate on the joint channel does not license it, and does not almost license it, Lemma 6.6 (iv) exhibiting a channel that satisfies the scalar bound and violates the operator one.

On a one-factor emission the untagged entry is safe, whatever the case: it is *implied* by the party-local reading of the same $(\hat{O}, \bowtie, c)$ — in untrusted-source mode it is that reading's trace (Lemma 6.6 (iii)), and in trusted-source mode it is the same operator inequality under the marginal condition $\omega_A = \tau_A$ — and the licence covers it on the very same certificate, by Theorem 6.7 (i)–(iv). *What it gives up does depend on the case, and Definition 6.5 states all three*: in trusted-source mode under the consumer's marginal condition $\omega_A = \tau_A$, and in untrusted-source mode at $d_A = 1$, the untagged entry *is* the party-local reading and nothing is given up; in untrusted-source mode with $d_A > 1$ the party-local reading is an operator inequality on A against the marginal that is actually there, of which the untagged entry is only the trace, so what is emitted is strictly weaker — a rate loss and never a soundness loss. Emitting the stronger form there would need a tag naming a scope that no declared product provides; widening it is an amendment of Definition 6.5 and of the consuming round model. Theorem 11.5 (vi) says the same of the one-factor emission as a whole.

Uses Def. 6.5, Lem. 6.6, Thm. 6.7, Cor. 11.3 · *Used by* Req. 16.8 · newly derived · accepted

Several measuring parties are emitted as one exactly admissible artefact per party with a combination rule, never a product matrix. The product target is a POVM squashing the product of the declared models, and every entry keeps its constant, scoped to its factor.

Where several parties measure, each party's measurement is its own reduction instance and is emitted as one; the theorem says what the product of the emitted factors is, without forming a product datum. At one factor its parts (i), (ii), (iv) and (v) are Theorem 11.2 again and its part (iii) is Theorem 6.7 (v) (part (vi)). A class-marginal identity is emitted in expectation form, as in Remark 3.14, and what a factor's emission records is Requirement 17.3.

Theorem 11.5 (The emission of several measuring parties)**.** Let $\mathbb{F}$ be a finite ordered set of *factors*, each f with a party label, and for each f let $\mathcal{J}^f$ be a reduction instance (Definition 10.1) on a system B_f over $\big(\mathcal{V}^f, \{\mathcal{V}_t^f\}_{t\in\mathcal{T}_{\mathrm{set}}^f}, \{q_t^f\}\big)$ to which Theorem 11.2 applies, with emitted datum A'_f, emitted target $F^f \coloneqq F^f[A'_f]$ on $Q_f = \bigoplus_{s\in\mathbb{S}^f} Q_{f,s}$, channel $\Lambda_f \coloneqq \Lambda_f[A'_f]$, declared model $M'^{\mathrm{ph},f} = M^f[A'_f]$ and entry list $\mathfrak{L}^f$. Put $B \coloneqq \bigotimes_f B_f$, $Q \coloneqq \bigotimes_f Q_f$, $\mathcal{V} \coloneqq \prod_f \mathcal{V}^f$, $\Lambda \coloneqq \bigotimes_f \Lambda_f$, all products in the order of $\mathbb{F}$, and for $v = (v_f)_f \in \mathcal{V}$

$$F_v \coloneqq \bigotimes_f F^f_{v_f}, \qquad M'^{\mathrm{ph}}_v \coloneqq \bigotimes_f M'^{\mathrm{ph},f}_{v_f}. \tag{11.10}$$

Suppose that *what is emitted* is the ordered list of the factors' artefacts, each with its party label, and the rule (11.10), and that no entry of any F_v is emitted, declared or rounded. Then:

(i) *(a measurement, block diagonal for the product sector map)* $\{F_v\}_{v\in\mathcal{V}}$ is a POVM on Q: every F_v is Hermitian and positive semidefinite and $\sum_{v\in\mathcal{V}} F_v = \mathbb{1}_Q$, as identities of the exact rational numbers the factors emitted. With the *product sector map* — index set $\prod_f \mathbb{S}^f$ in the product order, the last factor fastest, identifiers the factors' identifiers joined in that order, dimensions the products, projectors $\Pi_s = \bigotimes_f \Pi^f_{s_f}$ — every F_v is block diagonal, its block at $s = (s_f)_f$ being the Kronecker product $\bigotimes_f F^{f,(s_f)}_{v_f}$ of the factors' blocks. The party partition assigns every tensor factor of $\mathcal{H}(Q_{f,s_f})$ to the party of f.

(ii) *(the squashing relation and the declared model)* $\Lambda^\dagger(F_v) = M'^{\mathrm{ph}}_v$ for every v, exactly, so $M'^{\mathrm{ph}} \preceq_\Lambda F$; the declared joint model is the *product* of the factors' declared models. Λ is of the form the artefact records for one factor: with $\mathcal{P}_{\mathbb{S}}$ the pinching for the product projectors, $\mathcal{P}_{\mathbb{S}} \circ \Lambda = \Lambda$, and the squashed analysis may be restricted to states block diagonal for the product sector map without excluding a reachable one. The transported set contains every product projector

Π_s, every lifted projector $\mathbb{1} \otimes \Pi^f_{s_f} \otimes \mathbb{1}$ and every lifted transported operator $\mathbb{1} \otimes \hat{O} \otimes \mathbb{1}$, $\hat{O}$ transported by $\mathcal{J}^f$, with pull-back the lift of $\hat{O}$'s pull-back.

(iii) *(entries keep their constants)* for every factor f and every entry $(\hat{O}_k, \bowtie_k, c_k) \in \mathfrak{L}^f$, the certificate $\Lambda^\dagger_f(\hat{O}_k) \bowtie_k c_k \mathbb{1}_{B_f}$ which $\mathcal{J}^f$ discharges at A'_f is a certificate on that factor's channel in the sense of Theorem 6.7 (v). So the joint list $\mathfrak{L} := \bigsqcup_f \{(\hat{O}_k, \bowtie_k, c_k; f)\}$, each entry scoped to its factor and carrying the same operator, the same relation and the same constant, is licensed with its entries read party-locally (Definition 6.5), or as scalars on the lifted operators $\mathbb{1} \otimes \hat{O}_k \otimes \mathbb{1}$, or any mixture; every squashed attack sequence against M'^{ph} is admissible for it and reproduces the protocol's output state; and the lifted operators are block diagonal for the product sector map with entries that are the factor's emitted entries and zeros and ones. No constant is re-derived and none is rounded.

(iv) *(class-marginal identities)* with classes $\mathcal{V}_t = \prod_f \mathcal{V}^f_{t_f}$ and weights $q_t = \prod_f q^f_{t_f}$, $\sum_{v \in \mathcal{V}_t} M'^{\mathrm{ph}}_v = q_t \mathbb{1}_B$ holds for every t if every factor's class-marginal identity is discharged. Where it is discharged for the factors in a subset $\mathbb{G} \subseteq \mathbb{F}$ only, the identity holds for the classes that are full in every factor outside $\mathbb{G}$: $\sum_{v:\, v_f \in \mathcal{V}^f_{t_f}\ (f \in \mathbb{G})} M'^{\mathrm{ph}}_v = \prod_{f \in \mathbb{G}} q^f_{t_f} \mathbb{1}_B$. What is emitted for it is the expectation-form entry on the pulled-back class sum and never an operator identity on Q. The weight q_t is a product of declared numbers and is formed exactly by whoever reads it; it is not emitted as a number of this module, and it could not be: each $q^f_{t_f}$ lies on its factor's declared lattice (Definition 9.1 (b), Definition 4.4 (a)), and a product of two lattice numbers lies in general on a finer lattice only and need not be a binary64 number, exactly as for the blocks of (11.10).

(v) *(what was rounded)* the numbers behind (11.10) are the factors' emitted numbers and nothing else. Every displacement, kept or moved class and rounding direction is a factor's and is recorded with that factor's emission, and Theorem 11.2 (v) applies to each factor as it stands: no bound relates a key length computed from M'^{ph} alone to one computed from the product of the handed models. The product introduces no further displacement.

(vi) *(one factor)* for $|\mathbb{F}| = 1$ parts (i), (ii), (iv) and (v) are the corresponding statements of Theorem 11.2 for that instance, and what is emitted is that instance's artefact, every entry untagged. Part (iii) at one factor is Theorem 6.7 (v) at one factor, which says *more* than (i)–(iv) of that theorem in untrusted-source mode with $d_A > 1$ and the

same otherwise (Definition 6.5); the untagged entry that is emitted is implied by it (Lemma 6.6 (iii)), so the one-factor emission asserts nothing beyond Theorem 6.7 (i)–(iv).

These statements are about the factors' instance links and their emitted families.

Uses Lem. 1.2, Prop. 4.6, Cor. 4.7, Prop. 4.8, Def. 6.5, Lem. 6.6, Thm. 6.7, Def. 10.1, Def. 11.1, Thm. 11.2 · *Used by* Req. 16.8, Req. 17.6, Cor. 21.6, protocol-compiler/req:link-row-provenance · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.factors:reduce_factors

Proof. (i) By Theorem 11.2 (ii) each $F^f_{v_f}$ is Hermitian and positive semidefinite with $\sum_{v_f} F^f_{v_f} = \mathbb{1}_{Q_f}$, exactly. A tensor product of Hermitian operators is Hermitian and of positive semidefinite operators positive semidefinite, and $\sum_{v\in\mathcal{V}} \bigotimes_f F^f_{v_f} = \bigotimes_f \sum_{v_f} F^f_{v_f} = \mathbb{1}_Q$ by multilinearity. These are identities between polynomials in the emitted entries, which are dyadic rationals. Each $F^f_{v_f} = \bigoplus_{s_f} F^{f,(s_f)}_{v_f}$ is block diagonal for $\{\Pi^f_{s_f}\}$, so $\bigotimes_f F^f_{v_f} = \bigoplus_s \bigotimes_f F^{f,(s_f)}_{v_f}$ along $\mathcal{H}(Q) = \bigoplus_s \bigotimes_f \mathcal{H}(Q_{f,s_f})$, the distributive law for tensor products over direct sums; the stated order of the index set is a convention for listing the summands. The party partition is each factor's own (Definition 10.1 (I2)) with its party's label in place of the default one.

(ii) $\Lambda^\dagger(\bigotimes_f F^f_{v_f}) = \bigotimes_f \Lambda^\dagger_f(F^f_{v_f}) = \bigotimes_f M'^{\mathrm{ph},f}_{v_f}$ by Lemma 1.2 (ii) and Theorem 11.2 (iii) for each factor; this is Proposition 4.8 iterated. Each $\Lambda_f = \mathcal{P}_f \circ \Lambda_{0,f}$ with $\mathcal{P}_f$ the pinching for $\{\Pi^f_{s_f}\}$ (Definition 10.1 (I3)), and $\bigotimes_f \mathcal{P}_f$ is the pinching $\mathcal{P}_\mathbb{S}$ of Proposition 4.6 for the product projectors, because $\big(\bigotimes_f \mathcal{P}_f\big)(\xi) = \sum_s \Pi_s\, \xi\, \Pi_s$ by multilinearity; a pinching is idempotent, so $\mathcal{P}_\mathbb{S} \circ \Lambda = \Lambda$. Every F_v, every Π_s and every lifted operator named is block diagonal for the product projectors by (i), so Corollary 4.7 (iii) applies to Λ. For a transported $\hat{O}$ of factor f with pull-back O, $\Lambda^\dagger(\mathbb{1}\otimes\hat{O}\otimes\mathbb{1}) = \mathbb{1}\otimes\Lambda^\dagger_f(\hat{O})\otimes\mathbb{1} = \mathbb{1}\otimes O\otimes\mathbb{1}$ by the same lemma and unitality of the other factors' adjoints; Π_s is a product of lifted projectors and its pull-back is $\bigotimes_f \Lambda^\dagger_f(\Pi^f_{s_f})$.

(iii) The certificate of an entry of $\mathfrak{L}^f$ is discharged by $\mathcal{J}^f$ at every admissible datum, in particular at A'_f (Definition 10.1 (I4), Theorem 11.2 (iii)), and it is a statement about Λ_f and $\hat{O}_k$ alone. That is the hypothesis of Theorem 6.7 (v), whose conclusion is the admissibility in the party-local set, in the scalar set for the lifted operators, and for every mixture, together with the equality of output states and the security implication, the relation $M'^{\mathrm{ph}} \preceq_\Lambda F$ being (ii). The lifted operator is a Kronecker product of $\hat{O}_k$ with identity matrices, so its entries are entries of $\hat{O}_k$, zeros and ones, and it is block diagonal because $\hat{O}_k$ is for $\{\Pi^f_{s_f}\}$.

(iv) $\sum_{v\in\mathcal{V}_t} M'^{\mathrm{ph}}_v = \bigotimes_f \sum_{v_f\in\mathcal{V}^f_{t_f}} M'^{\mathrm{ph},f}_{v_f}$. Where factor f discharges its identity the f-th factor of this product is $q^f_{t_f}\mathbb{1}_{B_f}$ (Theorem 11.2 (iii)); sum-

ming over all t_f for $f \notin \mathbb{G}$ replaces those factors by $\mathbb{1}_{B_f}$, by completeness of $M'^{\mathrm{ph},f}$. The remark on the emitted form is Definition 10.1, fourth consequence, applied to the pulled-back class sum.

(v) (11.10) is a function of the factors' emitted numbers, and by hypothesis nothing else is emitted; so there is no number to which a further displacement could attach.

(vi) With one factor, (11.10) is $F_v = F_v^f$, the product sector map is the factor's and $\Lambda = \Lambda_f$, so (i), (ii), (iv) and (v) are Theorem 11.2's. For (iii): by Definition 6.5 the party-local reading at one factor is the entry of the source mode in trusted-source mode and at $d_A = 1$, and in untrusted-source mode with $d_A > 1$ it is an operator inequality on A whose trace is the scalar entry, Lemma 6.6 (iii); the emitted entry is the untagged one in every case, which Theorem 6.7 (i)–(iv) license on the same certificate. □

For the identity instance the normalizer moves nothing, so the emitted family is the handed datum itself, squashed by the sector pinching with no error term on an exact link. A requested entry's certificate is the operator inequality itself, decided exactly on the emitted numbers.

The identity instance is the one a protocol with an already-finite receiver would use; this corollary is Theorem 11.2 read at it.

Corollary 11.6 (The emitted family of the identity instance)**.** Let the declared data pass every refusal rule of this module's interface, let $\mathcal{J}_{\mathrm{id}}$ be the identity instance of Corollary 10.2 with handed datum A, and let A' be the datum the normalizer of Definition 11.1 declares. Then Theorem 11.2 holds of $\mathcal{J}_{\mathrm{id}}$; the normalizer moves nothing, $A' = A$ and $\Delta_s = 0$ for every data sector s; and, for every $v \in \mathcal{V}$,

$$\Lambda^\dagger(F_v) \;=\; \mathcal{P}_{\mathbb{S}}(F_v) \;=\; F_v \;=\; M'^{\mathrm{ph}}_v \;=\; \bigoplus_{s \in \mathbb{S}_0} A'^{(s)}_v \qquad \text{exactly, on } B = Q, \tag{11.11}$$

so the emitted family *is* the declared datum, which is the handed datum, exactly admissible and decided so before emission; the emitted constraint list is *empty where the request carries no entry* — this instance's own list being empty in any case (Corollary 10.2 (I4)) — and where the request carries entries it carries, on request, the class-marginal entries $\big(\sum_{v \in \mathcal{V}_t} F_v, =, q_t\big)$, which the instance discharges, and a further requested entry $(\hat{O}, \bowtie, c)$ of the form (10.4), which Corollary 11.3 (iii) emits exactly where $\hat{O} \bowtie c\,\mathbb{1}_B$ holds at the constant c it emits, this instance carrying none of its own: $\Lambda^\dagger(\hat{O}) = \hat{O}$, so the certificate (11.9) of such an entry is that operator inequality on the emitted numbers, decided exactly by finitely many field operations and sign comparisons, a decided certificate in the sense of Corollary 11.3 (iii) for every operator of that form; its default carried operators — here the sector projectors and nothing else, the transported set being $\{\Pi_s\}_{s \in \mathbb{S}_0}$ — are emitted with the transported set at $c = 0$ with $\succeq$ in either case (Corollary 11.3 (i));

no scalar value is emitted and no bound of this module's own is rounded; the chain is the identity link with status = exact. These statements are about the instance's link and its emitted family. The instance is selectable only once a declared input carries an already-finite POVM.

Uses Prop. 4.6, Thm. 7.4, Def. 9.1, Lem. 9.3, Def. 10.1, Cor. 10.2, Def. 11.1, Thm. 11.2, Cor. 11.3 · newly derived · accepted

Proof. Corollary 10.2 makes $\mathcal{J}_{\mathrm{id}}$ a reduction instance, so Theorem 11.2 applies to it with the data passing every refusal rule. Its handed datum is the declared family, every block of which is decided exactly admissible on the declared lattice on entry, so every entry satisfies (9.1), and the first of the properties of Definition 11.1 gives $A' = A$ and every $\Delta_s = 0$. Part (iii) of that theorem at this instance reads $\Lambda^\dagger(F_v) = M_v[A'] = \bigoplus_s A_v'^{(s)}$; and $\Lambda = \mathcal{P}_{\mathbb{S}}$ is self-adjoint with $\mathcal{P}_{\mathbb{S}}(F_v) = F_v$ for the block-diagonal F_v (Proposition 4.6), which gives the two middle equalities of (11.11). That the emitted family is exactly admissible and decided so is part (ii); that it is the declared datum is (I2) of the instance, there being no scalar sector. The default carried operators and the entries are Corollary 11.3 (i) and, the identity instance discharging the class-marginal identity, (ii). For a further requested entry, $\hat{O}$ is a real combination of the blocks $\Pi_s F_v \Pi_s$ and of sector projectors, hence block diagonal for $\mathbb{S}_0$ and Hermitian, so $\Lambda^\dagger(\hat{O}) = \mathcal{P}_{\mathbb{S}}(\hat{O}) = \hat{O}$ (Proposition 4.6); its entries are formed from emitted lattice entries and declared lattice coefficients by finitely many field operations, so they lie in $\mathbb{F}_i$; and the certificate (11.9) is then $\hat{O} - c\,\mathbb{1}_B \succeq 0$, $c\,\mathbb{1}_B - \hat{O} \succeq 0$ or an entrywise equality, the first two decided by (9.3) of Lemma 9.3 and the last by comparing entries. The instance's (H) being empty, that condition is a decided certificate for every operator of the form (10.4), and Corollary 11.3 (iii) emits the entry exactly where it holds. Parts (ii) and (iii) are statements about a request that carries entries; where the request carries none the emitted list is empty by that corollary's rule, whatever the instance's own list contains — and this instance's is empty in any case (Corollary 10.2 (I4)) — so the artefact then carries no entry and no licence, which is what the last sentence of that corollary says of its chain. With no scalar sector the rounding clause of Theorem 11.2 (iv) has nothing of this module's to round, and only a caller's constant can be. The chain and its status are the last sentence of Corollary 10.2 with Theorem 7.4 (iv), now at the declared datum. The last sentence is Example 7.5. □

Where the normalizer moves the datum, the claims are about the declared model, and nothing proved relates it to the handed one. A run can evaluate off the snap on the reference companion; elsewhere a disclosed premise relates them.

Remark 11.7 (What the reduction's claim is about after an emission snap). Where the exact construction stays on the declared lattice — every $\Delta_s = 0$ — there is nothing to say: the emitted target is the constructed one, the defining identity is about the handed model itself, and what the artefact asserts is what the instance's existence theorem proves (Definition 10.1 (I3)). *Where some $\Delta_s > 0$ the emitted numbers are a declaration and not the constructed ones*, and what the artefact asserts is then this. There is a datum A' — exhibited rather than posited, and carried entry by entry in the emitted target's data sectors — which is exactly admissible, class complete at the declared weights, and differs from the handed datum by at most Δ_s entrywise per data sector. *It is the datum of a whole declared model $M'^{\mathrm{ph}} = M[A']$* (11.3): the handed model with the datum replaced as the instance's model map defines it, every object the hypotheses (H) are about — an above-cutoff family, a register POVM, a further branch — being the declared one, which the normalizer never reads. *The emitted channel Λ squashes that model, exactly and with no error term* — $\Lambda^\dagger(F_v) = M'^{\mathrm{ph}}_v$ (11.6), an identity on B and not on the data sectors alone; every hypothesis (H) holds of it by the stability clause; every certificate of the instance is discharged for it; and every claim of this contract that reads a declared number reads it. *The physical content of the artefact alone is therefore a statement about M'^{ph}, of which the normalizer moved the data sectors and nothing else*, and A' stands one modelling step further from the supplier's own output than the instance's declaration already stood. Both steps are recorded, the first in the instance's record of its snap and the second in this module's emission record (Requirement 17.3).

Nothing proved relates M'^{ph} to the handed model, and distance is not such a relation. The size Δ_s of the move bounds nothing that matters to a key rate: the consuming contract shows that two exactly admissible families within any declared distance can have rate bounds far apart (Proposition "Instability of the rate under measurement perturbations" (meat-core/prop:optol-nobound)), so no bound relates a key length computed from M'^{ph} alone to one computed from the handed model, and none is asserted. The snap keeps every row and column that the handed datum holds at zero (Definition 11.1, fifth property), save on a class snapped by the last resort, which the record flags, and it may move support along other directions. A run therefore does not rest on the declared model where it can avoid it: where it forms the number its key length is formed from on the reference companion (Corollary 12.1), which the module builds with no normalizer from the handed reference, or from the instance's own snapped blocks where none is handed, every data sector is evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)), and the emitted target serves only the search. Elsewhere, and so also where the key length is formed from a number certified on the snapped declaration in place of the companion's (Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i)), the run rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) (ii), the premise for a producer's snap of its own emission: what the run concludes about the device from the emitted target holds with A' in place of A on the sectors concerned. The artefact being an exact

statement about M'^{ph}, the premise holds in particular where the device measures M'^{ph} there. The record names the moved sectors (Requirement 17.3).

Two things this does not license. It does not license repairing what this module was handed: the normalizer runs on the datum an instance computed, and on nothing else (Definition 11.1, fourth property). And it does not license emitting status = exact about a family the identity does not hold of: the value is truthful precisely because the identity is re-established for M'^{ph}, whose data sectors are A', rather than asserted of a construction the artefact does not carry.

Uses Def. 10.1, Def. 11.1, Thm. 11.2 · *Used by* Rem. 11.8 · newly derived · accepted

The squashing, transfer, calculus and core instance results read no declared number; a submodule's instance reads declared data decided exactly, the licence exactly Hermitian operators and declared constants. Where the normalizer acts, every statement is read on the declared model.

Remark 11.8 (The claims that read a declared number). *The squashing relation and its soundness, the factorization and transfer theorems, the calculus of the preorder, the branching rule, the flags-enough corollary, register-conditioned squashing and the obstruction read no declared number* and assert nothing about the emitted artefact: each is a statement about an arbitrary squashing relation, an arbitrary instrument, an arbitrary register-conditioned mixture, or an explicit counterexample. *The instance theorems of the core* — the identity, flag, branching and register instances (Corollaries 10.2, 10.3 and 10.5, Theorem 10.4) — are statements about arbitrary admissible data and arbitrary branch instances and read no declared number either. *A submodule's instance* builds its datum out of declared blocks and declared classical data; its hypotheses are conditions on a POVM family and on declared scalars and matrices, and this module's refusal rules decide every one of those conditions *exactly*, so it applies to the declared data verbatim. *The attack-set licence* (Theorem 6.7) needs its emitted constraint operators exactly Hermitian and its constants declared: the first holds by construction and is decided by the refusal rule that says so, the second by Assumption 13.7. *The chain theorem* (Theorem 7.4) asserts status = exact of the artefact, which is true of the numbers emitted (Theorem 11.2 (iii)). *The emission theorem* (Theorem 11.2) is the claim about all of the above, stated once over the reduction instance (Definition 10.1).

One conditioning survives and it is the only one: where the emission normalizer acts, each of those statements is read on the declared model of Definition 11.1 — on B, the declared model $M'^{\mathrm{ph}} = M[A']$; in the data sectors, the declared datum A' — and not on the handed datum, and Remark 11.7 is what that means. *Every residual is recorded*: the displacement and the rounding directions are (Requirement 17.3), and the consumer's key length is Theorem 11.2 (v) — not established, and no refusal rule can establish it.

Uses Def. 10.1, Def. 11.1, Thm. 11.2, Rem. 11.7 · *Assumes* A-CONST · newly derived · accepted

Data without a finite presentation, the objects of an instance's hypotheses, and combinatorial data offer no numerical predicate: the first stay hypotheses, the second are settled by inspection. The normalizer never touches the first, so the hypotheses hold of the declared model.

Remark 11.9 (The boundary of the exact decisions). Every operator predicate this module checks is decided exactly. *Two kinds of declared datum carry no such predicate.*

First: data with no declared finite presentation. The hypotheses (H) of a reduction instance (Definition 10.1 (I1)) are about exactly these: an operator family on a part of $\mathcal{H}(B)$ that no finite list of matrices describes — $\mathcal{H}(B)$ being separable and possibly infinite-dimensional (Definition 1.1) — such as the blocks of a family above a photon-number cutoff; a POVM on a register whose dimension is nowhere declared or bounded; the POVM of an uncharacterised branch, which the instance that files it deliberately declines to constrain (Corollary 10.3). *None of these has a finite matrix presentation anywhere in this contract, so none offers anything to decide*: there is no list of entries for Lemma 9.3 to run on. Their conditions — that a family is a POVM, that an above-cutoff block satisfies a certified inequality, that a register family is complete — are therefore *hypotheses of the instance*, carried in the ledger and decided by this module's refusal rules only to the extent that the declaration presents something to decide. Which objects these are for a given instance is that instance's to say; of the core's leaf instances only the flag instance has one, its branch POVM, and the branching and register instances add the hypothesis that their instrument is a generalized instrument, and for the register form that the register family is a POVM (Theorem 10.4, Corollary 10.5).

Second: data that is combinatorial rather than numerical. A coarse-graining map with its totality, surjectivity and fibre conditions; the alphabets $\mathcal{Y}$, $\mathcal{T}_{\mathrm{set}}$, $\mathcal{V}$ and their declared orders; the sector identifiers, their dimensions, their supports and their order; a branch index set; a cutoff; and the lattice exponent κ itself. *None of these is a number a condition could be decided about*, and a predicate on them is settled by finite inspection — which is the same reason the exact island (Definition "The exact island" (foundations/def:exact-island)) is drawn where it is.

Two consequences. First, the objects of the first kind are also the ones the emission normalizer never touches: it acts on the data sectors and on nothing else (Definition 11.1, second property), so none of them enters any Δ_s and none is repaired — which is what makes every hypothesis (H) hold of the declared model by the stability clause, and with it Corollary 11.3 (ii)'s identity available where the instance discharges it. Second, *extending an exact decision to such an object would first require declaring a finite representation for it*, and for an uncharacterised branch that means constraining the very model the instance leaves free; that is a widening of the instance's model rather than a change of number model. The declared classical data an instance's route reads — a stochastic matrix and its descended form, the class weights, the certified weights — *is* finitely presented, so it is decided here, exactly.

Uses Def. 1.1, Lem. 9.3, Def. 10.1, Def. 11.1, Cor. 11.3 · newly derived · accepted

Every entry this module emits holds on every reachable squashed state, as a conclusion of Theorem 7.4, Corollary 11.3 and Theorem 6.7 and not as an assumption. Inside the engine's security chain the whole list is covered by the stronger A-ATK through the engine's range lemma.

Remark 11.10 (Soundness of the emitted constraint list). Every entry of the constraint list $\mathfrak{L}$ this module emits is sound for the reduction chain that produced the declaration, in the sense of Definition 3.10: every reachable squashed state satisfies it. For the entries this module emits this is a conclusion and not an assumption: the entries are the ones the chain theorem, the emission corollary and the licence theorem put into $\mathfrak{L}$ (Theorem 7.4, Corollary 11.3, Theorem 6.7), under their hypotheses. *Inside* the consuming engine's security chain, soundness of the whole list, entries a declaration brings itself included, is implied by the stronger A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) through the engine's range lemma (Lemma "Constraint transport and range inclusion" (meat-core/lem:range) (i)), which is declared there. Of an entry this module did not emit, nothing here asserts soundness outside that chain.

newly derived · accepted

12 The reference companion

Where the declaring instance holds a reference of the measurement it snapped, the same instance run on the reference in place of the snap is again a reduction instance with the emitted alphabet, sectors and scalar values: its target is a POVM that the reference model squashes to exactly, with its own certified entries, known through enclosures formed from the reference's (Corollary 12.1). This needs only that the instance's datum, formed from any family with the properties of a reference, is admissible, which holds for the identity instance, passes through branching, and holds for the flat instance. An entry built from the reference is emitted in binary64 by rounding its operator and moving its constant outward by a bound on the rounding, which keeps a certificate and enlarges the set the entry cuts out (Lemma 12.3). Where the instance forms its datum as a non-negative combination of the blocks, as the identity and the flat instance do, factor enclosures of the reference's blocks pass to the companion's datum and target, columns side by side and cores scaled (Corollary 12.2).

Imports foundations/def:evaluated-off-snap, foundations/def:exact-reference, foundations/def:factor-enclosure, foundations/def:snap, receiver/def:complex-enclosure, receiver/lem:enclosure-arithmetic

The same instance, run on the reference in place of the snap, or on its own snapped blocks where none is handed, is a reduction instance of the emitted shape that squashes that model exactly. Its entries are enclosed from the reference, requested ones where their certificates hold there.

An emission is computed on the snapped measurement a declaring instance states, and the normalizer may move the instance's datum once more (The-

orem 11.2). Where the declaring instance holds a reference of the measurement and hands it beside its snap, the same construction can be run on the reference instead: the instance's datum formed from the reference blocks, with no normalizer. Where it hands none, the same construction run on the snapped blocks themselves gives the producer's exact construction, the handed datum, which the normalizer has not touched: the snap of the emission is then bypassed though the instance's own snap is not. Nothing has to be proved again for that, because every part of a reduction instance is established at every admissible datum; the one thing to supply is that the datum formed from the reference is admissible, which is hypothesis (H-REF) below, and the substrate in which the construction is read, since reference entries are in general irrational. What results is no declaration a consumer solves in its own right (Requirement 17.4); its entries, which are formed from the reference, are emitted in binary64 by Lemma 12.3. Part (iv) discharges (H-REF) for the identity instance and passes it through branching; the flat instance discharges it in Corollary 21.5. Part (vi) carries to the companion the entries a caller requests beside the instance's own: their operators are formed at the companion's datum by the same formula, and their certificates are re-established there, since a certificate decided on the emitted numbers says nothing about the reference; for the operator of (19.13) the flat instance does so in Corollary 21.5.

Corollary 12.1 (The reference companion of a reduction instance). Let $\mathcal{J}$ be a reduction instance (Definition 10.1) whose handed datum A is formed, by the rule of its part (I1), from a family $K = \{K_j^{(b)}\}$ of measurement blocks, over blocks b of a sector map and outcomes j of an alphabet (one such family per setting where the blocks are given per setting), that a declaring instance states as a snap (Definition "Snapping declared data onto a lattice" (foundations/def:snap)) of the measurement of a device. Let $G = \{G_j^{(b)}\}$ be a family over the same blocks, alphabet and dimensions, every block of which is Hermitian and positive semidefinite and satisfies exactly every completeness identity that the declaration of K states of K on that block. In the use this corollary is for, $G^{(b)}$ is the block of the handed reference (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)) where the declaring instance hands one, and $K^{(b)}$ elsewhere, so that $G = K$ where it hands none. Suppose

(H-REF) $\mathcal{J}$ is a reduction instance over the substrate $\mathbb{F} = \mathbb{R}$ (Definition 9.2), and the rule of its part (I1) forms, from every family with the properties just listed for G, a datum in $\mathcal{D}_{\mathrm{adm}}$.

Let A^{ref} be the datum that rule forms from G in place of K, and call the instance at A^{ref} — the same B, (H), sectors, supports, scalar values, constructor and entry rule, with A^{ref} in place of the handed datum — the *reference companion* of $\mathcal{J}$ at G. Then:

(i) *(an admissible target over the emitted shape)* $A^{\mathrm{ref}} \in \mathcal{D}_{\mathrm{adm}}$, the reference companion is a reduction instance, and its target $\{F_v[A^{\mathrm{ref}}]\}_{v\in\mathcal{V}}$ is a POVM on Q, sector-block-diagonal by construction. Its alphabet with classes and weights, its sector map (the sector list, its order, each sector's dimension and kind), its scalar values and its party partition are those of the target the instance has at every admissible datum, the declared datum the normalizer produces included, and it differs from such a target on the data sectors alone, where it carries A^{ref}. Where $G = K$, $A^{\mathrm{ref}} = A$: the reference companion is $\mathcal{J}$ at its handed datum;

(ii) *(an exact squash of the companion's model)* $\big(M[A^{\mathrm{ref}}]; O[A^{\mathrm{ref}}]\big) \preceq_{\Lambda[A^{\mathrm{ref}}]} \big(F[A^{\mathrm{ref}}]; \hat{O}[A^{\mathrm{ref}}]\big)$, so that $\Lambda[A^{\mathrm{ref}}]^{\dagger}(F_v[A^{\mathrm{ref}}]) = M_v[A^{\mathrm{ref}}]$ exactly for every v; every certificate of the entry list $\mathfrak{L}[A^{\mathrm{ref}}]$ is discharged, and the class-marginal identity (10.5) holds at A^{ref} where the instance's flag says it is discharged; and Theorem 3.11 and Theorem 7.4 hold of the one-link receiver segment (Definition 7.1) whose source family is $(B, M[A^{\mathrm{ref}}])$, whose target is $(Q, F[A^{\mathrm{ref}}])$ and whose channel witness is $\Lambda[A^{\mathrm{ref}}]$, with $\mathsf{status} = \mathsf{exact}$;

(iii) *(the companion is known through enclosures)* let every block of G be read with an entrywise enclosure (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)), with radius zero on every block where $G^{(b)} = K^{(b)}$. Where the rule of (I1) forms each entry of the datum as a finite expression in entries of the blocks and in numbers the instance was handed, and every reciprocal node of that expression receives an enclosure by part (iv) of Lemma "Arithmetic on complex enclosures" (receiver/lem:enclosure-arithmetic), as it does where the expression has none, the rules of that lemma applied to that expression give a complex enclosure of every element of A^{ref} whose centre is formed by the same expression from the centres and whose radius is zero at every entry whose inputs are all enclosed with radius zero; part (v) of that lemma, applied element by element, makes it an entrywise enclosure of A^{ref}, whose centre is the Hermitian part of that centre and whose radius is zero at an entry (i, l) where the radii at (i, l) and (l, i) both are. Every entry operator $\hat{O}_k[A^{\mathrm{ref}}]$ of (10.4), a combination of the blocks $\Pi_s F_v[A^{\mathrm{ref}}]\Pi_s$ with real coefficients $\alpha_{k,v,s}$ plus a combination Z_k of sector projectors, is then enclosed entrywise by the same combination of the centres, with radius $\sum_v |\alpha_{k,v,s}|\, R^{(s)}_{v,il}$ at the entry (i, l) of each data sector s, where (C, R) is any entrywise enclosure of the datum and $R^{(s)}_{v,il}$ is the radius of its element v at that entry, and radius zero on every scalar sector;

(iv) *(where (H-REF) holds)* (H-REF) holds for the identity instance of

Corollary 10.2; and it holds for the branching instance of Theorem 10.4, and for the register instance of Corollary 10.5, whenever it holds for every branch instance whose datum is formed from blocks of K, the data of the other branches being formed from no block of K;

(v) *(what the companion rests on, and only there)* no part of (i)–(iv) rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement). Let B be the blocks on which $G^{(b)} = K^{(b)}$ that the record of the run names as resting on that assumption. Where B is empty this part asserts nothing. Where it is not, let the assumption hold: then on every block in B the block of the measurement the device performs is the snapped block $K^{(b)}$, with the entries recorded for the snap, in place of the block of the declaring instance's model, so that the companion is formed there from the block the device then measures; and no claim relates a key length computed from the companion to that model on the blocks in B;

(vi) *(entries requested beside the instance's own)* let $(\hat{O}, \bowtie, c)$ be an entry emitted at a request that carries entries (Corollary 11.3), written in the form (10.4) with coefficients $\alpha_{v,s}$ and a combination Z of sector projectors, $\hat{O} = \sum_{v\in\mathcal{V}} \sum_{s\in\mathbb{S}} \alpha_{v,s} \Pi_s F_v[A'] \Pi_s + Z$ at the declared datum A', and call $(\hat{O}^{\mathrm{ref}}, \bowtie, c)$, with

$$\hat{O}^{\mathrm{ref}} \;:=\; \sum_{v\in\mathcal{V}} \sum_{s\in\mathbb{S}} \alpha_{v,s}\, \Pi_s F_v[A^{\mathrm{ref}}] \Pi_s \;+\; Z, \tag{12.1}$$

its *companion form* at that writing. Part (iii) encloses $\hat{O}^{\mathrm{ref}}$. Where $\hat{O}$ is a sum of sector projectors (Corollary 11.3 (i)), written with every $\alpha_{v,s} = 0$, $\hat{O}^{\mathrm{ref}} = \hat{O}$ and $\Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}) \succeq 0$. Where it is the class-marginal entry $\big(\sum_{v\in\mathcal{V}_t} F_v, =, q_t\big)$ of Corollary 11.3 (ii), written with $\alpha_{v,s} = 1$ for $v \in \mathcal{V}_t$, 0 otherwise, and $Z = 0$, $\hat{O}^{\mathrm{ref}} = \hat{O}$, and where the instance discharges (10.5), $\Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}^{\mathrm{ref}}) = q_t \mathbb{1}_B$ exactly. For any other entry this part asserts no certificate at A^{ref}. In every case, where $\Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}^{\mathrm{ref}}) \bowtie c\,\mathbb{1}_B$ holds, $\big(M[A^{\mathrm{ref}}]; O[A^{\mathrm{ref}}], \Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}^{\mathrm{ref}})\big) \preceq_{\Lambda[A^{\mathrm{ref}}]} \big(F[A^{\mathrm{ref}}]; \hat{O}[A^{\mathrm{ref}}], \hat{O}^{\mathrm{ref}}\big)$, and the companion form is an entry carrying the spectral certificate (6.9) at its constant.

Uses Def. 1.3, Thm. 3.11, Cor. 4.7, Thm. 6.7, Def. 7.1, Def. 7.2, Thm. 7.4, Def. 9.2, Def. 10.1, Cor. 10.2, Thm. 10.4, Cor. 10.5, Def. 11.1, Cor. 11.3, foundations/def:evaluated-off-snap, foundations/def:exact-reference, foundations/def:snap, receiver/def:complex-enclosure, receiver/lem:enclosure-arithmetic · *Assumes* A-SNAP · *Used by* Cor. 12.2, Req. 17.4, Cor. 21.5, Cor. 21.6, meat-core/cor:reference-companion-factors, meat-core/prop:reference-companion, meat-core/req:companion-admission, protocol-compiler/req:reference-companion-compiled · newly derived · accepted

Proof. Idea. The stability clause of Definition 10.1 was stated so that the instance at any admissible datum is again an instance; (H-REF) supplies an admissible datum formed from G, and every conclusion is a part of the instance read at that datum.

(i) The family G has the properties (H-REF) lists, so $A^{\mathrm{ref}} \in \mathcal{D}_{\mathrm{adm}}$, the set of admissible data over the substrate $\mathbb{F} = \mathbb{R}$, which Definition 9.2 admits: it leaves the field open between $\mathbb{Q}$ and $\mathbb{R}$, and entries of a reference may be irrational. The definition's closing sentence says that the instance at an admissible datum is a reduction instance whenever $\mathcal{J}$ is, by the stability clause: (H) reads no entry of the datum, so it holds of $M[A^{\mathrm{ref}}]$ as it holds of the handed model, and (I2)–(I4) are established at every admissible datum. By the first consequence of Definition 10.1, $\{F_v[A^{\mathrm{ref}}]\}$ is a POVM on Q, its data sectors carrying A^{ref} and its scalar sectors the scalar values; it is assigned one sector block at a time by (10.2), so it is sector-block-diagonal by construction. The alphabet, the classes and weights, the sector list with its order, dimensions and kinds, the scalar values and the declared party label are data of the instance that the instance at a datum keeps by definition; only the datum changes, and the datum occupies the data sectors alone. Where $G = K$, the rule of (I1) is applied to the family it formed the handed datum from, so $A^{\mathrm{ref}} = A$.

(ii) The relation is (I3) of the reference companion, which (i) shows to be a reduction instance; its first clause is $\Lambda[A^{\mathrm{ref}}]^{\dagger}(F_v[A^{\mathrm{ref}}]) = M_v[A^{\mathrm{ref}}]$ by Definition 1.3. Every certificate of $\mathfrak{L}[A^{\mathrm{ref}}]$ and, where flagged, the class-marginal identity are (I4) at A^{ref}, discharged there by the stability clause. Theorem 3.11 needs only the relation. The one-link segment with source family $(B, M[A^{\mathrm{ref}}])$, target $(Q, F[A^{\mathrm{ref}}])$, channel witness $\Lambda[A^{\mathrm{ref}}]$, the transported pair $(\hat{O}[A^{\mathrm{ref}}], O[A^{\mathrm{ref}}])$, the reachable-set statement of that channel and the entry list $\mathfrak{L}[A^{\mathrm{ref}}]$ is a receiver-stage link of Definition 7.1 and a chain of Definition 7.2 with an empty source segment, and Theorem 7.4 (iv) gives it $\mathsf{status} = \mathsf{exact}$, the relation carrying no error term. The sector pinching in front of $\Lambda_0[A^{\mathrm{ref}}]$ changes none of this (Corollary 4.7 (i)).

(iii) Each entry of A^{ref} is the value of the expression the rule of (I1) uses, at the entries of G and at the handed numbers. Those handed numbers that are not entries of a block are the same as for A and are known exactly; an exactly known number is enclosed by itself with radius zero (Definition "Complex enclosures, and enclosures of square roots and angles" (receiver/def:complex-enclosure)), and an entrywise enclosure of a block is a complex enclosure of each of its entries. The rules (i)–(iii) of Lemma "Arithmetic on complex enclosures" (receiver/lem:enclosure-arithmetic) enclose sums, products, conjugates and, where the condition of its part (iii) is met, reciprocals of enclosed numbers, with centres formed by the same operation on the centres, and keep an enclosure exact where the inputs are; by its part (iv), applied along the expression under the hypothesis that every reciprocal node receives an enclosure, they give a complex enclosure of every entry of

A^{ref}, with the centre and the radii stated. Every element of A^{ref} is a Hermitian matrix, the datum being admissible (i), so part (v) of that lemma turns the complex enclosure of each element into an entrywise enclosure in the sense of Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference), with Hermitian centre $\frac{1}{2}(X_c + X_c^\dagger)$ and symmetric radii $\frac{1}{2}(r + r^{\mathsf{T}})$, exact at an entry (i,l) where $r_{il} = r_{li} = 0$. For an entry operator, $\hat{O}_k[A^{\mathrm{ref}}]$ restricted to a data sector s is $\sum_v \alpha_{k,v,s} A_v^{\mathrm{ref},(s)}$ plus the coefficient of Π_s in Z_k times $\mathbb{1}_{d_s}$, exactly known; at each entry the triangle inequality gives $\left|\sum_v \alpha_{k,v,s}(A_v^{\mathrm{ref},(s)} - C_v^{(s)})_{il}\right| \le \sum_v |\alpha_{k,v,s}|\, R_{v,il}^{(s)}$, with (C,R) any entrywise enclosure of the datum. On a scalar sector the operator is a combination of the scalar values, which are the instance's and exact.

(iv) For the identity instance the rule of (I1) is the identity: the datum is the declared family read as a sector-block list, and it is admissible exactly when each block is Hermitian and positive semidefinite and the class sums are q_t times the identity, the completeness identities its declaration states. A family G with the properties listed has them, so the datum it forms is admissible. That the identity instance is an instance over $\mathbb{R}$ is the proof of Corollary 10.2, which reads of the substrate its field operations and its order and of a datum only admissibility: its channel, the sector pinching, does not depend on the datum. For the branching and the register instances, $\mathcal{D}_{\mathrm{adm}} = \prod_j \mathcal{D}_{\mathrm{adm}}^{(j)}$, a composite datum being admissible exactly when every branch datum is (Step 1 of the proof of Theorem 10.4, which Corollary 10.5 applies); replacing K by G changes only the branch data formed from blocks of K, each admissible by (H-REF) for its branch, and leaves the others as they are. Both results establish the composite instance from its branch instances by arguments that read of the substrate its field operations and its order, so the composite is an instance over $\mathbb{R}$ when its branches are.

(v) No step of (i)–(iv) reads the assumption. Where B is not empty, each block in B is a block of the declaring instance's snap of the measurement of a device, the data D of that snap being what the instance states as that measurement or a reference of it, so clause (i) of Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) applies to it: the block of the measurement the device performs is the snapped family there, with the entries recorded for the snap, in place of the model. On such a block $G^{(b)} = K^{(b)}$, so the companion is formed there from that snapped block. A nearness of the snapped block and the model relates no key lengths, and no claim relates the two on a block in B, which is a moved block of the snap, as the assumption's B is defined; so no claim relates the companion's key length to the model there.

(vi) The companion form is an operator of the form (10.4) at the datum A^{ref}, which (iii) encloses. Where every $\alpha_{v,s}$ is zero, (12.1) reads no datum, so $\hat{O}^{\mathrm{ref}} = Z = \hat{O}$; Z is a sum of sector projectors, each in the

transported set of the companion by (I3) at A^{ref}, and $\Lambda[A^{\mathrm{ref}}]^{\dagger}$ is a positive map (the third consequence of Definition 10.1), so $\Lambda[A^{\mathrm{ref}}]^{\dagger}(Z) \succeq 0$, as Corollary 11.3 (i) argues at A'. For the class-marginal entry, $F_v[A]$ is sector-block-diagonal at every admissible datum (i), so $\sum_s \Pi_s F_v[A] \Pi_s = F_v[A]$ and $\hat{O}^{\mathrm{ref}} = \sum_{v \in \mathcal{V}_t} F_v[A^{\mathrm{ref}}]$. On a scalar sector s the block of $\sum_{v \in \mathcal{V}_t} F_v$ is formed from the scalar values, which are the same at both data (i). On a data sector s it is $\sum_{v \in \mathcal{V}_t} A_v^{(s)}$ at the datum it is read at, the elements outside the sector support being zero; an admissible datum is class complete, so this sum is $q_t \mathbb{1}_{d_s}$ at A^{ref}, admissible by (i), and at A', admissible as an element of $\mathcal{D}_\kappa$ (Definition 11.1). The two operators therefore agree block by block, and $\hat{O}^{\mathrm{ref}} = \hat{O}$. Where the instance discharges (10.5), (ii) gives that identity at A^{ref} and $\Lambda[A^{\mathrm{ref}}]^{\dagger}(F_v[A^{\mathrm{ref}}]) = M_v[A^{\mathrm{ref}}]$, so by linearity $\Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}^{\mathrm{ref}}) = \sum_{v \in \mathcal{V}_t} M_v[A^{\mathrm{ref}}] = q_t \mathbb{1}_B$, which is (6.9) with $\bowtie$ the equality at $c = q_t$. For any other entry the certificate was established at A', for the numbers emitted there (Corollary 11.3 (iii)), and a certificate of an operator formed from one datum is no statement about the operator formed from another; so none is asserted. A writing of $\hat{O}$ in the form (10.4) need not be unique, and two writings may give different companion forms; each is an entry wherever its own certificate holds, which is all this part claims of it. Finally, where $\Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}^{\mathrm{ref}}) \bowtie c\,\mathbb{1}_B$ holds, (ii) is a squashing relation at A^{ref}, and adjoining the pair $\big(\Lambda[A^{\mathrm{ref}}]^{\dagger}(\hat{O}^{\mathrm{ref}}), \hat{O}^{\mathrm{ref}}\big)$ to its carried operators keeps (1.2), whose clause for that pair holds by the choice of its physical operator (Definition 1.3); the certificate is then (6.9) for the adjoined carried operator. □

Where an instance forms its datum as a non-negative rational combination of measurement blocks, factor enclosures of the reference's blocks give one of each element of the companion's datum, columns side by side and cores scaled, and so of the target's block on that data sector.

Corollary 12.1 (iii) carries the entrywise enclosures of the handed reference to the companion's datum through the expression the instance's rule of (I1) is. A factor enclosure passes through a narrower class of rules. A sum, with non-negative weights, of members of factor enclosures is a member of the enclosure whose columns are theirs side by side and whose core is theirs scaled by the weights, and no arithmetic on the columns is needed for it. The corollary names that class of rules and carries the factor enclosures through it. The identity instance is in it, and so is the flat instance (Corollary 21.5). A rule outside it, such as one that maps a block by a conjugation before it combines blocks, carries no factor enclosure here, and the elements it forms are known through their entrywise enclosures alone (Requirement 17.4).

Corollary 12.2 (Factor enclosures carried to the reference companion)**.** Let $\mathcal{J}$, K, G and A^{ref} be as in Corollary 12.1, the blocks $K_j^{(b)}$ indexed by a block b and an outcome j, together with the setting where the blocks are given per setting. Call the rule of part (I1) of $\mathcal{J}$ (Definition 10.1) a *non-negative*

combination when for every data sector s and every $v \in \mathcal{V}$ there are non-negative rationals $\lambda_{b,j}^{(s,v)}$ such that, for every family K' with the properties that Corollary 12.1 lists for G, the datum A' the rule forms from K' satisfies

$$A_v'^{(s)} \;=\; \sum_{b,j} \lambda_{b,j}^{(s,v)}\, K_j'^{(b)}, \tag{12.2}$$

every block $K_j'^{(b)}$ with $\lambda_{b,j}^{(s,v)} > 0$ being of dimension d_s. Let the rule be a non-negative combination, and let some blocks $G_j^{(b)}$ carry factor enclosures $\left(L_j^{(b)}, \varrho_j^{(b)}, W_j^{(b)}\right)$ (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)).

(i) Let s be a data sector and $v \in \mathcal{V}$ such that every block $G_j^{(b)}$ with $\lambda_{b,j}^{(s,v)} > 0$ carries one, and list the pairs (b, j) with $\lambda_{b,j}^{(s,v)} > 0$ in a fixed order. The triple whose matrix box has as its centre and its radii those of the boxes $\left(L_j^{(b)}, \varrho_j^{(b)}\right)$ side by side in that order, and whose core is the diagonal matrix of the $\lambda_{b,j}^{(s,v)} W_j^{(b)}$ in that order, is a factor enclosure of $A_v^{\mathrm{ref},(s)}$, its number of columns the sum of theirs; where every $\lambda_{b,j}^{(s,v)}$ is zero, it has no column.

(ii) The block of the companion's target $F_v[A^{\mathrm{ref}}]$ on the data sector s is $A_v^{\mathrm{ref},(s)}$, so the triple of (i) is a factor enclosure of that block.

(iii) The rule of (I1) of the identity instance of Corollary 10.2 is a non-negative combination: each element of its datum is the block of the family it is read from at the same sector and outcome, with the coefficient 1, every other coefficient being 0.

Uses Def. 10.1, Cor. 10.2, Cor. 12.1, foundations/def:factor-enclosure · *Used by* Req. 17.4, Cor. 21.5, meat-core/cor:reference-companion-factors · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.companion:combined_factor

Proof. (i). By Corollary 12.1, A^{ref} is the datum the rule of (I1) forms from G, a family with the properties that corollary lists, so (12.2) with $K' = G$ gives $A_v^{\mathrm{ref},(s)} = \sum_{(b,j)} \lambda_{b,j}^{(s,v)} G_j^{(b)}$, the sum over the listed pairs, the others having coefficient 0. Each listed block carries its factor enclosure, so there is a matrix $L_{j,*}^{(b)}$ that the box $\left(L_j^{(b)}, \varrho_j^{(b)}\right)$ contains with $G_j^{(b)} = L_{j,*}^{(b)} W_j^{(b)} L_{j,*}^{(b)\dagger}$ (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)); every such matrix has d_s rows, its block being of dimension d_s. Let L_* be the matrix with the $L_{j,*}^{(b)}$ side by side in the listed order and W the block-diagonal matrix of the $\lambda_{b,j}^{(s,v)} W_j^{(b)}$ in the same order. Each column of the box of the triple is a column of one of the boxes it is made of, with its centre and radii, so the box contains L_*; its centre has entries in $\mathbb{Q}(i)$ and its radii

are non-negative rationals, as those of the boxes it is made of; and W is diagonal, each $W_j^{(b)}$ being diagonal, with positive rational diagonal entries, each a product of a positive rational $\lambda_{b,j}^{(s,v)}$ and a positive rational entry of $W_j^{(b)}$. Multiplying by blocks,

$$L_* W L_*^\dagger = \sum_{(b,j)} L_{j,*}^{(b)} \big(\lambda_{b,j}^{(s,v)} W_j^{(b)}\big) L_{j,*}^{(b)\dagger} = \sum_{(b,j)} \lambda_{b,j}^{(s,v)}\, G_j^{(b)} = A_v^{\mathrm{ref},(s)},$$

which is what a factor enclosure of $A_v^{\mathrm{ref},(s)}$ asks (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)). The number of columns of L_* is the sum of those of the $L_{j,*}^{(b)}$. Where every coefficient is zero, no pair is listed, L_* has no column, and $A_v^{\mathrm{ref},(s)} = 0$, the one member of a triple with no column.

(ii). By Corollary 12.1 (i) the reference companion is a reduction instance at the datum A^{ref}, and its target is (10.2) of Definition 10.1 at that datum, whose block on a data sector s is $A_v^{\mathrm{ref},(s)}$.

(iii). The rule of (I1) of the identity instance is the identity: its datum is the declared family read as a sector-block list, $A_v^{(s)}$ being the block of M_v on the sector s (Corollary 10.2), and the same rule reads any family K' over the same sectors so. That is (12.2) with $\lambda_{s,v}^{(s,v)} = 1$ and every other coefficient 0, and the block with the coefficient 1 is of dimension d_s. □

Rounding a certified entry's operator and moving its constant outward by a bound on the rounding's norm gives an entry that is again certified and that every attack and every state satisfying the exact entry satisfies. An equality becomes two inequalities.

A constraint entry built in exact arithmetic from data off the binary64 grid has an operator that binary64 cannot carry. Rounding the operator moves the set the entry cuts out, in either direction. Moving the constant outward by a bound on the size of the rounding makes the rounded entry weaker than the exact one, so that whatever satisfies the exact entry satisfies the rounded one, and the rounded entry still carries a certificate of its own. The bound is read off the entries, since a Hermitian matrix has operator norm at most its largest absolute row sum.

Lemma 12.3 (A rounded constraint entry with its constant relaxed)**.** Let $\Lambda\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ be a channel with $(M;O) \preceq_\Lambda (F;\hat{O})$ (Definition 1.3), and let $(\hat{O}_k, \bowtie_k, c_k)$ be an entry with $\hat{O}_k$ a carried operator, $\bowtie_k \in \{\preceq, =, \succeq\}$, $c_k \in \mathbb{R}$ and the certificate $\Lambda^\dagger(\hat{O}_k) \bowtie_k c_k \mathbb{1}_B$ of (6.9). Let $\hat{O}'_k$ be a Hermitian operator on Q, let $E_k := \hat{O}'_k - \hat{O}_k$, and let $\rho_{il} \geq 0$ be numbers with $|(E_k)_{il}| \leq \rho_{il}$ for every entry in a fixed orthonormal basis of $\mathcal{H}(Q)$, and $\delta_k \geq \max_i \sum_l \rho_{il}$. Define the *relaxed entries* of $(\hat{O}_k, \bowtie_k, c_k)$ as $(\hat{O}'_k, \succeq, c'_k)$ with $c'_k \leq c_k - \delta_k$ where $\bowtie_k$ is $\succeq$; $(\hat{O}'_k, \preceq, c''_k)$ with $c''_k \geq c_k + \delta_k$ where $\bowtie_k$ is $\preceq$; and both of these where $\bowtie_k$ is $=$. Then:

(i) $-\delta_k \mathbb{1}_Q \preceq E_k \preceq \delta_k \mathbb{1}_Q$;

(ii) *(certificate)* $(M; O, \Lambda^\dagger(\hat{O}'_k)) \preceq_\Lambda (F; \hat{O}, \hat{O}'_k)$, and every relaxed entry carries the spectral certificate (6.9) at its own constant: $\Lambda^\dagger(\hat{O}'_k) \succeq c'_k \mathbb{1}_B$, respectively $\Lambda^\dagger(\hat{O}'_k) \preceq c''_k \mathbb{1}_B$;

(iii) *(attack sets)* for all finite-dimensional E', E'', every $\mathcal{A}$ in the set (6.2), (6.3) or, for an entry tagged party-local with the factor f, (6.4), of the list holding the exact entry lies in the set of the list in which the exact entry is replaced by its relaxed entries, the other entries unchanged;

(iv) *(states)* for every finite-dimensional system X and every positive semidefinite ω on $\mathcal{H}(X) \otimes \mathcal{H}(Q)$, $\operatorname{tr}_Q[(\mathbb{1}_X \otimes \hat{O}_k)\,\omega] \succeq c_k \operatorname{tr}_Q \omega$ implies $\operatorname{tr}_Q[(\mathbb{1}_X \otimes \hat{O}'_k)\,\omega] \succeq c'_k \operatorname{tr}_Q \omega$, and the same with $\preceq$, c''_k in place of $\succeq$, c'_k; with X trivial these are the scalar readings $\operatorname{tr}\left[\hat{O}_k \omega\right] \geq c_k \operatorname{tr} \omega$ and the like.

Uses Lem. 1.2, Def. 1.3, Def. 1.8, Def. 6.3, Thm. 6.7 · *Used by* Req. 17.4, meat-core/prop:reference-companion, protocol-compiler/req:reference-companion-compiled · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.companion:relaxed_entries

Proof. (i) E_k is Hermitian, as a difference of Hermitian operators, so its operator norm is the largest modulus of an eigenvalue. Let $E_k x = \lambda x$ with $x \neq 0$ and let i be an index with $|x_i| = \max_l |x_l|$. Then $|\lambda||x_i| = |\sum_l (E_k)_{il} x_l| \leq \sum_l \rho_{il} |x_l| \leq \big(\sum_l \rho_{il}\big)|x_i|$, so $|\lambda| \leq \delta_k$, and the spectral theorem gives $-\delta_k \mathbb{1} \preceq E_k \preceq \delta_k \mathbb{1}$.

(ii) $\Lambda^\dagger$ is linear, positive and unital (Lemma 1.2), so (i) gives $-\delta_k \mathbb{1}_B \preceq \Lambda^\dagger(E_k) \preceq \delta_k \mathbb{1}_B$. Where $\bowtie_k$ is $\succeq$ or $=$, $\Lambda^\dagger(\hat{O}_k) \succeq c_k \mathbb{1}_B$, hence $\Lambda^\dagger(\hat{O}'_k) = \Lambda^\dagger(\hat{O}_k) + \Lambda^\dagger(E_k) \succeq (c_k - \delta_k)\mathbb{1}_B \succeq c'_k \mathbb{1}_B$; where it is $\preceq$ or $=$, symmetrically $\Lambda^\dagger(\hat{O}'_k) \preceq (c_k + \delta_k)\mathbb{1}_B \preceq c''_k \mathbb{1}_B$. The relation with $\hat{O}'_k$ carried is (1.2) with the one further carried operator $\hat{O}'_k$ and its pull-back $\Lambda^\dagger(\hat{O}'_k)$, which holds by the definition of that pull-back; the conditions on F, M, $\hat{O}$ and O are the given relation. Each relaxed entry's certificate is the operator inequality just shown, a case of (6.9).

(iii) The adjoint of a channel, and of $\mathrm{id}_{R'} \otimes \mathcal{A}$, is unital and completely positive (Lemma 1.2). For (6.2), $-\delta_k \mathbb{1} \preceq E_k \otimes \mathbb{1}_{E''} \preceq \delta_k \mathbb{1}$ by (i), so $\mathcal{A}^\dagger(\hat{O}'_k \otimes \mathbb{1}_{E''}) = \mathcal{A}^\dagger(\hat{O}_k \otimes \mathbb{1}_{E''}) + \mathcal{A}^\dagger(E_k \otimes \mathbb{1}_{E''})$ is $\succeq (c_k - \delta_k)\mathbb{1}_{E'} \succeq c'_k \mathbb{1}_{E'}$ where $\mathcal{A}^\dagger(\hat{O}_k \otimes \mathbb{1}_{E''}) \succeq c_k \mathbb{1}_{E'}$, and $\preceq (c_k + \delta_k)\mathbb{1}_{E'} \preceq c''_k \mathbb{1}_{E'}$ where $\mathcal{A}^\dagger(\hat{O}_k \otimes \mathbb{1}_{E''}) \preceq c_k \mathbb{1}_{E'}$; an equality gives both. The same with $\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E''}$ proves it for (6.3). For (6.4), with $X \succeq 0$ on $R' R\, Q_{f^c}$,

$$X \otimes (\hat{O}'_k - c'_k \mathbb{1}) \otimes \mathbb{1}_{E''} = X \otimes (\hat{O}_k - c_k \mathbb{1}) \otimes \mathbb{1}_{E''} + X \otimes \big(E_k + (c_k - c'_k)\mathbb{1}\big) \otimes \mathbb{1}_{E''},$$

and $E_k + (c_k - c'_k)\mathbb{1} \succeq E_k + \delta_k \mathbb{1} \succeq 0$ by (i), so the last term is a tensor product of positive semidefinite operators and its image under the completely positive

adjoint is positive semidefinite; the first term's image is $\succeq 0$ where the exact entry holds. For $\preceq$ the second bracket is $E_k - (c''_k - c_k)\mathbb{1} \preceq E_k - \delta_k\mathbb{1} \preceq 0$, and the images are $\preceq 0$. An equality entry gives both. The other entries of the list are unchanged, so their conditions are the same in both sets.

(iv) Put $P \coloneqq E_k + \delta_k \mathbb{1}_Q \succeq 0$ by (i). For $\omega \succeq 0$, $\operatorname{tr}_Q[(\mathbb{1}_X \otimes P)\omega] = \operatorname{tr}_Q[(\mathbb{1}_X \otimes \sqrt{P})\,\omega\,(\mathbb{1}_X \otimes \sqrt{P})] \succeq 0$, by cyclicity of the partial trace for an operator acting on Q alone and because a partial trace of a positive semidefinite operator is positive semidefinite. Hence $\operatorname{tr}_Q[(\mathbb{1}_X \otimes \hat{O}'_k)\omega] \succeq \operatorname{tr}_Q[(\mathbb{1}_X \otimes \hat{O}_k)\omega] - \delta_k \operatorname{tr}_Q \omega \succeq (c_k - \delta_k)\operatorname{tr}_Q \omega \succeq c'_k \operatorname{tr}_Q \omega$, the last step because $\operatorname{tr}_Q \omega \succeq 0$ and $c_k - \delta_k \geq c'_k$. The case $\preceq$ uses $\delta_k\mathbb{1} - E_k \succeq 0$ in the same way. With X trivial the operators are numbers and the inequalities are the scalar ones. □

13 Assumptions

These assumptions are the premises the reductions rest on and do not prove: a measuring party's device entering only through one POVM per round, the same physical measurement in every round with no memory across rounds, a supplier's artefact being its module's own output at an accepted version and true of the device, a measuring party's private setting being its own declared randomness, output-state test statistics and declared constants. The dimensions of the systems are the setting's, and the adversary's registers are finite-dimensional as the round model's are; no assumption is made about the adversary's memory, which the consuming engine reaches at any dimension by a limit (Corollary "Security of the protocol and its net key" (meat-core/cor:composed-security)). No assumption about the distance of the declared numbers from a true operator family is made, because their exact admissibility is decided on entry; what remains a hypothesis is that the supplier's facts hold of the family declared (Assumption 13.3, Remark 13.4), which for an exact, rigorously evaluated receiver artefact the run built itself holds under the receiver's own ledger (Requirement 18.7).

Imports foundations/def:round-homogeneity

In every round each measuring party's device acts as one POVM: it emits a classical outcome, retains no quantum system, and everything after it acts on classical data. It is the hypothesis about the devices behind the output-state equality of the physical and the squashed protocol.

The assumption is what lets a round's processing factor through the measuring parties' measurement channel: once each device is one POVM whose outcome is all that leaves it, the rest of the round acts on classical data. Where several parties measure, it is asserted of each of their devices, and where the declaration declares several measured factors of the measured system, of the device at each measured factor, two factors of one party included;

where one party measures (Bob, in a prepare-and-measure declaration), it is asserted of that party's device. The round's order is not part of it; classical detector memory across rounds is excluded by the round-homogeneity assumption.

Assumption 13.1 [A-MEAS] (Each measuring party's device as one POVM per round)**.** In every round, each measuring party's interaction with the quantum system it receives satisfies: (i) it is described by a single POVM on that system; (ii) it emits the classical outcome and nothing else, and no post-measurement quantum system is retained or reused; (iii) all downstream processing (announcements, sifting, acceptance, key map, error correction, privacy amplification) acts on classical data; (iv) the round's measurement channel is the product of one measurement channel per factor, each acting on its own factor alone, for each of the following factorizations of the measured system B_j: where the declaration declares several measured factors (Definition "Protocol specification" (protocol-compiler/def:spec)), the factorization $B_j = \bigotimes_f B_{f,j}$ with one factor per measured factor, whichever parties carry their labels and two factors of one party included; and where several parties measure, the factorization into the parties' parts of Definition 2.1.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Each measuring device, at every measuring station of either party, turns the incoming quantum signal into a classical result in every round and keeps nothing quantum afterwards: no post-measurement system is stored, re-used or passed on. Everything after detection (announcements, sifting, acceptance, the key map, error correction, privacy amplification) uses classical data only. Where several measuring stations are declared, of different parties or of one, or where several parties measure inside one declared station, each device measures independently of the others, so that the round's measurement is the product of theirs: no shared randomness or shared ancilla couples them. A receiver with a quantum memory that keeps the measured light, or passes it on, violates this.

Uses Def. 2.1 · *Category* receiver · *Affects* secrecy · *Assumed by* Lem. 2.4, Thm. 3.1, Rem. 3.7, Rem. 3.8, protocol-compiler/req:link-row-provenance · *Machine key* `measurement_only_receiver` · newly derived · accepted

One physical POVM describes the measurement of every round, and each round's measurement acts as it describes whatever happened in other rounds: no detector memory and no drift. Nothing is assumed about the attacks being independent across rounds.

The measurement this assumption holds fixed is the physical one, the POVM by which the devices of the measuring parties measure the measured system of a round; where several parties measure, the assumption fixes each party's factor across the n rounds. Rounds are not averaged, and per-round variation is not modelled. The declared model and the squashing channel are objects of the analysis, not of the device, and are not part of this assumption; neither is the round's *order*, every message of the round after every

measurement of the round.

Assumption 13.2 [A-ROUND-SQ] (Round homogeneity of the physical measurement)**.** Let M be the physical POVM: the POVM on the measured system of a round that the analysis takes as the measurement the devices of the measuring parties perform on it. The family $\{M\}$ is round-homogeneous (Definition "Round homogeneity of a per-round family" (foundations/def:round-homogeneity)) in both clauses of that definition: (i) the same M describes the measurement of every round; (ii) a round's measurement acts as M describes, whatever the inputs and outputs of the other rounds, with no detector memory across rounds.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Each measuring device, at every measuring station of either party, performs the same measurement in every round: the same optics, detectors and dark-count behaviour, settings drawn with the same probabilities, and, where you declared detector-response modes, the same response in each mode. What a device does in one round does not depend on the inputs or outputs of other rounds. Drift of efficiency, alignment or dark counts violates this. Only parts of a run fixed before it starts, each processed as its own execution and composed under A-SEPARATE, can be analysed separately; a split chosen after seeing the data is not covered. Detector memory (afterpulsing, a dead time, any dependence on earlier rounds) violates it too, and nothing here repairs that. Attacks may still correlate the rounds. The engine's finite model and squashing map are its own construction, not part of this premise.

Uses Def. 2.1,
foundations/def:round-homogeneity · *Category* receiver · *Affects* secrecy · *Refines* A-RHOM · *Assumed by* Thm. 3.1, Rem. 3.6, Rem. 3.7 · *Machine key* `round_homogeneity_squashing` · newly derived · accepted

A supplier artefact read by a reduction instance is that module's own output at the accepted version it declares, and the device facts its claims state hold of the numbers exactly as declared. It is the premise behind a reduction on supplier numbers, save an exact, rigorous artefact the run built.

For the receiver supplier the numbers in question are the photon-number blocks, the high-photon weights with their quantifier domain and the dark-count channel. Which of a supplier's claims are consumed, and so which of its assumptions the imported facts rest on, is stated where the supplier's artefact is consumed; the assumption says that those facts hold of the numbers declared. Where the run built a receiver artefact itself, from a declaration with a model admissible exactly and with every weight evaluated rigorously, both clauses hold without it, by the receiver's own theorems under its own assumptions, and the engine omits its row; at a weight evaluated in floating point, or at a declaration admitted only within the receiver's tolerances, clause (ii) remains a premise.

Assumption 13.3 [A-RECV] (Supplier artefact whose claims hold of the device)**.** Every declared block, weight, matrix and scalar that a reduction instance of this module reads from a supplier's artefact satisfies: (i) the artefact is the output of the supplying module, at the module version it

declares, which is an *accepted* version of the supplying module, and on the inputs it declares, and the validation results it carries are that module's own; (ii) the facts the supplier's claims state hold of those numbers as declared, under that module's own assumption ledger.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: The receiver data your reduction uses (blocks, high-photon weights with their quantifier domain, dark-count channel) come from a receiver artefact produced by an accepted version of the receiver module, the version the artefact names, on the inputs it names, with its own validation results attached, and the device facts that module's claims state are true of those numbers exactly as declared, under its own assumptions. Where this run built it from a declaration describing a valid receiver exactly, every weight evaluated rigorously, both halves follow and the report omits this premise (Requirement 18.7). For another artefact this run built, acceptance is the status the report gives the receiver's claims and you rely on the second half, floating-point weights included; a receiver file from outside the run needs your own justification for both halves.

Category supplied artefact · *Affects* secrecy · *Assumed by* Rem. 6.8, Prop. 20.4, Rem. 20.7, Rem. 20.10, Rem. 20.11, Rem. 20.12, Prop. 21.2, Cor. 21.3, Cor. 21.4, Cor. 21.5, Cor. 21.6 · *Machine key* `receiver_artifact_accepted` · newly derived · accepted

A-RECV is stated of the exactly admissible declaration, not of an idealisation of it, and of one declared receiver. No claim relates a key length from the declaration to one from the supplier's own floating-point output; an exact, rigorous receiver artefact the run built has its row omitted.

Remark 13.4 (What the supplier assumption is stated of). Assumption 13.3 is stated of the declaration rather than of an idealisation of it: what this module is handed is exactly admissible (Definition 9.1) and is decided so on entry, and what the assumption adds is that the device facts the supplier states are true of it. Where the declaration and the supplier's own floating-point output differ, the difference is a modelling act taken at the instance and recorded there, with that party's declared tolerance as its ceiling (Requirement "One declared tolerance per instance" (foundations/req:declared-tolerance)); no claim of this contract relates a key length computed from the declaration to one computed from that output, exactly as Theorem 11.2 (v) says of this module's own emission. Every one of the imported facts is inherited here unverified, and a single device family, uncertainty set or characterisation protocol appears nowhere: this contract reduces *one* declared receiver. Which of the supplier's claims are imported is stated where its artefact is consumed (for the receiver, Remark 20.12). For a receiver artefact the run built itself, from a declaration with a model admissible exactly and with every weight evaluated rigorously, Requirement 18.7 omits the row of Assumption 13.3: both its clauses hold there by the receiver's theorems under the receiver's own ledger, with no premise about arithmetic. Assumption 13.3 is the assumption on which the quantitative content of every other reduction instance built on a supplier's numbers rests (it is part of the hypotheses (H) of such an instance, Definition 10.1 (I1)), and it is the reason Theorem 6.7 is stated with hypothesis (6.9) rather than with a claim about optics.

newly derived · accepted

A station's private per-round setting is drawn from its party's own randomness with the declared class weights, independently of the adversary and the round's quantum input, never inferred from an outcome. It lets the setting enter a joint outcome under one common squashing channel.

The setting of the optical instance is the probe setting. Drawn this way, the setting enters as part of a joint outcome, the weight inside the element, and one squashing channel serves every value of it; a setting the adversary could read or steer would instead have to be handed to her.

Assumption 13.5 [A-PROBE] (The station's private setting drawn from its own declared randomness)**.** The private per-round setting $t \in \mathcal{T}_{\text{set}}$ of a measuring party's device satisfies, in every round: (i) it is drawn from that party's own randomness with the declared class weights q_t; (ii) it is drawn independently of the adversary's actions and of the round's quantum input; (iii) it is not inferred from any outcome.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: The station's private setting in each round (for the optical instance, how its probe is switched) is chosen by a random number generator of the station's own party with exactly the declared probabilities. It is independent of anything the attacker does and of the light arriving in that round, and it is never derived from a detection result. An attacker who can predict or influence the setting violates this.

Category local randomness · *Affects* secrecy · *Assumed by* Prop. 21.2, Cor. 21.3, Cor. 21.5, Cor. 21.6 · *Machine key* `probe_setting_private` · newly derived · accepted

Every per-round public choice of known bias at the measurement the emitted family describes, its private setting included where it has one, is a component of the emitted outcome and enters only through its statistics. No derivation then assumes a state-independent test fraction.

For the optical instance the private per-round setting is the probe setting. A choice that is a component of the outcome is analysed through the statistics of the emitted family like every other component, so no fraction of rounds of one kind is fixed independently of the state.

Assumption 13.6 [A-TSM] (Output-state test statistics)**.** Every per-round public choice of known bias in the physical protocol that is applied at the measurement the emitted family $\{F_v\}$ describes, a private per-round setting of that measurement and any round-type designation the consumer adds by relabelling its outcomes included, (i) is a component of the emitted outcome v, and (ii) enters the analysis only through the statistics of $\{F_v\}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every per-round random choice with a known bias made at this station (for example its probe setting, where it has one, or whether a round is a test round, where that is added by relabelling its outcomes) is recorded as part of that round's outcome, and it enters the analysis only through the observed statistics. No derivation of this contract then assumes a state-independent test fraction.

Category protocol structure · *Affects* secrecy · *Assumed by* Rem. 4.5, Prop. 21.2, Cor. 21.3, Cor. 21.5, Cor. 21.6, meat-core/rem:not-licensed, statistics/binomial-box/rem:guard-activation, statistics/binomial-box/rem:sound-indep · *Machine key* `test_statistics_model` · newly derived · accepted

Every constant of an emitted constraint entry is fixed before the run by the device fact that certifies it and is no function of an observed statistic. Observed data reach the analysis only through the acceptance set, which belongs to another module.

The only sanctioned carrier of observed data in the analysis is the acceptance set, which belongs to a different module and appears nowhere in this contract.

Assumption 13.7 [A-CONST] (Declared constants of the constraint list)**.** Every constant c_k in an emitted constraint entry (i) is fixed before the protocol run by the device fact that certifies it, and (ii) is not a function of any statistic observed during the run.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Applies only where the reduction emits constraints on the attacker; a run that emits none does not rest on it. Every number used as a constant in such a constraint (for example a bound on a detector's multi-photon response) is fixed before the run from the device characterisation that justifies it. It is never adjusted using data observed in the run; observed data enter only through the acceptance test.

Category supplied artefact · *Affects* secrecy · *Assumed by* Rem. 3.14, Thm. 6.7, Rem. 6.8, Cor. 11.3, Rem. 11.8, Rem. 19.9 · *Machine key* `declared_constants` · newly derived · accepted

Part II
Engine requirements

What an implementation of this module must do, record or refuse; each requirement names the result that motivates it.

14 Requirements: channel witnesses, post-processing and relations

These requirements fix what the artefact carries: an explicit channel witness for every emitted link, the artefact's witness being the squashing channel followed by the sector pinching, and the carried tuple every emitted constraint is built from. They also fix what the engine refuses: a relation with an error term, a noise or coarse-graining map after a squashing link, a declared relation that fails a necessary condition, and a link of a reserved kind. Only the round-homogeneous model is emitted, and branch blocks are emitted as the branches supply them, a padded variant only where a submodule proves it and declares it in its interface.

Every link the engine emits carries an explicit channel witness, and a request naming a reduction that comes with none is refused. Without a squashing channel the transfer of Theorem 3.11 is not available (Remark 1.9).

A reduction that exhibits no squashing channel cannot be fed into Theorem 3.11 (Remark 1.9). Where the witness sits in the artefact and which rule refuses a reduction without one are the outputs table and the refusal rules of the interface (`spec/modules/reductions/interface.yaml`).

REQUIREMENT 14.1 (A channel witness for every emitted link). The engine emits every link with an explicit channel witness and refuses a request that names a reduction coming with none.

Motivated by Rem. 1.9, Thm. 3.11 · accepted

The artefact's channel witness is the squashing channel followed by the sector pinching. The block-diagonality restriction and every other emitted constraint are then sound for one channel, which a block-diagonal optimisation of the emitted target needs.

Part (iii) of Corollary 4.7 is what makes the emitted target usable by a block-diagonal optimisation, and it holds for the pinched channel; Remark 3.15 requires every constraint imposed together to be sound for one and the same channel.

REQUIREMENT 14.2 (The pinched squashing channel as the emitted witness). The engine emits $\mathcal{P}_{\mathbb{S}} \circ \Lambda$, the squashing channel followed by the

pinching onto the sectors of Corollary 4.7, as the artefact's channel witness (output `squashing_channel`), so that the block-diagonality restriction of that corollary's part (iii) and every other emitted constraint are sound for *one* channel.

Motivated by Rem. 3.15, Cor. 4.7 · accepted

Beside each link's channel witness the engine emits the carried tuple of the link's squashing relation, and it builds every emitted constraint from an operator in that tuple. Such a constraint belongs to the relation, not to one realizing channel.

A constraint built from one of the carried operators has the same pullback for every channel realizing the squashing relation, while one read off a particular channel is tied to it (Remark 3.15).

REQUIREMENT 14.3 (Emission of the carried tuple, and constraints built from it). Beside the channel witness of each link it emits, the engine emits the carried tuple of the link's squashing relation (Definition 1.3; `transported_operators`), and it builds every constraint it emits from an operator in that tuple.

Motivated by Def. 1.3, Rem. 3.15 · accepted

A declared relation carrying an error term is refused, and no link is emitted with the status reserved for approximate squashing. No theorem of this contract converts an error term into a security penalty (Remark 2.5).

Theorem 3.11 rests on an equality of output states and has no penalty slot, so an error term in a declared relation has nowhere to enter (Remark 2.5). The refusal of a chain containing a link that is not exact is Requirement 15.4.

REQUIREMENT 14.4 (Refusal of approximate squashing relations). The engine refuses a declared squashing relation that carries an error term, with the reason `approximate_squashing_unsupported`, and emits no link with the status `certified_outer`, which is reserved for approximate squashing.

Motivated by Rem. 2.5, Thm. 3.11, Req. 15.4 · accepted

The engine emits one declared model and one squashing channel for every round, built from the one physical POVM that A-ROUND-SQ holds fixed. A round-dependent squasher, for which Remark 3.6 proves the factorization, is not emitted.

The factorization holds for round-dependent squashers as well (Remark 3.6), but no declared input describes more than one round.

REQUIREMENT 14.5 (Emission of the round-homogeneous case only). The engine emits one declared model and one squashing channel for every round, both built from the one physical POVM that Assumption 13.2 holds fixed (the interface convention `round_homogeneity`). It emits no round-dependent squasher.

Motivated by Rem. 3.6, Ass. 13.2 · accepted

A noise or coarse-graining map on the physical POVM is applied before the first squashing link, and a request to apply one after a link is refused. The refusal reaches these two kinds of map and no other, so a relabelling of both sides after a link stays available.

A noise or coarse-graining map changes the physical family a squasher is built from, and a squasher built for one physical family is not a squasher to the same target for the family the map produces from it (Remark 4.5).

Requirement 14.6 (Noise and coarse-graining before the first squashing link). The engine applies a noise or coarse-graining map on the physical POVM before the first squashing link, and refuses a request to apply one after a squashing link (input `post_processing_order_declaration`, refusal `post_processing_order_violated`). The refusal reaches exactly those two kinds of map: a stochastic relabelling applied to both sides after a link, depending on the private component of the outcome only, is not refused by it (Remark 4.5, (4.4)).

Motivated by Rem. 4.5 · accepted

Every relation a caller declares is tested against both necessary conditions of Lemma 4.9, decided exactly on the declared numbers, and refused where one fails. No eigenvalue is computed and no threshold is used.

Lemma 4.9 rules out an impossible relation by inspection, and Remark 4.10 gives the exact form in which each of its conditions is decided on declared numbers.

Requirement 14.7 (Exact check of a declared relation). The engine tests every declared relation (input `declared_relations`) against both necessary conditions of Lemma 4.9 and refuses one that fails either, with `declared_relation_impossible`. With X and X' as in that lemma, it decides (i) in operator form on the rational enclosures $a \le b$ the declaration exhibits: the pair $a\mathbb{1}_Q \preceq X \preceq b\mathbb{1}_Q$ exactly by (9.3) and, where it holds, the same pair for X'; and (ii) as the exact rank over $\mathbb{F}_i$ of the matrix whose columns are the vectorized declared elements, by elimination in the field, together with its linear relations: an exact basis of the solutions (a, c) of $\sum_v a_v F_v = c\,\mathbb{1}_Q$, obtained by elimination in the field, each basis solution tested on the declared physical elements as $\sum_v a_v M_v = c\,\mathbb{1}_B$. No eigenvalue is computed and no threshold is used.

Motivated by Lem. 4.9, Rem. 4.10, Lem. 9.3 · accepted

The engine emits no link of the reserved label-space kind and refuses a request that declares one. The family-dominating kind is the domination submodule's, and no requirement emits it.

Proposition 5.9 merges branches whose targets squash to one common target; with the device label read as a branch register, it reduces a family of devices to a single squashed model (Remark 5.10). The receiver-stage link kind (Definition 7.1) for a general device-label space is reserved. The kind `family-dominating`, whose first instance is the domination link of Definition 23.1, belongs to the domination submodule, and no requirement emits it.

Requirement 14.8 (Refusal of the reserved receiver-stage link kind). The engine emits no receiver-stage link of the reserved kind `label-space`, and refuses a request that declares a link of that kind with `link_kind_deferred`.

Motivated by Prop. 5.9, Rem. 5.10, Def. 7.1, Def. 23.1 · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_link_kind

The engine does not pad the branch blocks of a composite target: a reduction instance emits the blocks its branches supply. A padded variant a submodule proves from Corollary 5.11 is emitted only where that submodule's interface declares it.

Corollary 5.11 allows the branch blocks of a composite target to be padded to a common size; this requirement says when the engine does so.

Requirement 14.9 (Branch blocks emitted as supplied). The engine does not pad the branch blocks of a composite target: a reduction instance emits the blocks its branches supply, and a padded variant a submodule proves from Corollary 5.11 is not emitted unless that submodule's interface declares it.

Motivated by Cor. 5.11 · accepted

15 Requirements: chains, constraint slots and the licence

Every artefact carries a constraint list and a licence slot, even when both are empty, and its licence record names the licensing theorem, the certificate of each entry, the claim that discharges it and, where that claim reads a supplier's fact, the supplier artefact. The engine builds no entry whose constant depends on data observed during the run, refuses a chain containing a link that is not exact, and records an identity link where nothing was reduced.

Every artefact carries a constraint list and a licence slot, even when both are empty. A record without them would exclude device knowledge that no classical target can carry.

REQUIREMENT 15.1 (Constraint and licence slots on every artefact). The engine emits every artefact with a constraint list `constraint_list` and a licence slot `attack_set_license`, even when both are empty: a record without them excludes a class of device knowledge that no classical target can carry (Proposition 6.1, Remark 6.2 (a)), and the licence slot is where the licence of Theorem 6.7 is carried.

Motivated by Prop. 6.1, Rem. 6.2, Thm. 6.7 · accepted

The licence record names the licensing theorem, the certificate per entry, the claim that discharges it and, where that claim reads a supplier's fact, the supplier artefact by version; it is emitted only when every entry has all of these. It also names the proof technique it was issued for.

Where the consumer's mechanical check degenerates, its gate rests on this record alone (Remark 6.9); so the record names a proved theorem, its hypothesis and the claim that discharges it, never a bare attestation. Every emitted certificate is a conclusion of a claim (Corollary 11.3); an instance with no supplier, such as the identity instance, has no supplier artefact to name.

REQUIREMENT 15.2 (Content and presence of the licence record). The engine emits the licence record `attack_set_license` naming the theorem that licenses the constraint list (Theorem 6.7); the hypothesis it was discharged on, which is the certificate (6.9) of each entry; and, per entry, what discharges that hypothesis: the claim that establishes the certificate under Corollary 11.3 — the instance's own claim for an entry of its list, that corollary for a default carried operator and for the class-marginal entry, the claim giving the decided certificate for a requested entry — and, where that claim reads a fact a supplier's claims state, the supplier's artefact under Assumption 13.3, named by its version. It emits the record only when every entry has its certificate and what discharges it. The artefact names the proof technique the licence was issued for, since under another technique the licence must be re-established (Remark 6.8 (e)).

Motivated by Thm. 6.7, Rem. 6.8, Rem. 6.9, Cor. 11.3, Ass. 13.3 · accepted · *Code* certiqs_key.reductions.constraints:issue_licence

A constraint entry whose constant is a function of a statistic observed during the run is refused, never built. Such an entry would be unsound regardless of provenance, the licence resting on constants declared before the run.

REQUIREMENT 15.3 (Refusal of data-dependent constants). The engine builds no constraint entry whose constant c_k is a function of a statistic observed during the run; a request for one is refused with `data_depende`

`nt_constant`. Such an entry would be unsound regardless of provenance (Remark 6.8 (c)), the constants being declared before the run (Assumption 13.7).

Motivated by Rem. 6.8, Ass. 13.7 · accepted

A chain containing a link that is not exact is refused rather than given a composite status. No theorem of this contract converts an error term into a security penalty.

Requirement 15.4 (Refusal of a chain with a link that is not exact). The engine assigns a chain no composite status other than the one of Theorem 7.4 (iv): a chain containing a link that is not exact is refused, with `approximate_squashing_unsupported`, rather than given a composite status, because no theorem of this contract converts an error term into a security penalty. The refusal is a rule of the engine and not a converse this contract proves.

Motivated by Thm. 7.4 · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_approximate

Where nothing is reduced, the artefact carries the identity link rather than an empty chain. An empty chain could not say whether a reduction was performed and found trivial or never considered.

Requirement 15.5 (Identity link recorded for an already-finite receiver). The engine records the reduction of an already-finite receiver as the identity link of Example 7.5, never as an empty chain: an empty chain would leave the artefact unable to say whether a reduction was performed and found trivial or never considered, and the composition order the chain carries as data (Definition 7.2) has to be carried even when there is nothing to order.

Motivated by Def. 7.2, Thm. 7.4, Ex. 7.5 · accepted

16 Requirements: constraint entries

The engine emits only certified constraint entries, each with its recorded certificate case, as the request decides, and carries the certificates of the transported set beside them. The class-marginal identity is emitted only in expectation form and only where the instance discharges it; its operator form, and any requested operator whose certificate no claim decides on the emitted numbers, are refused. A caller's inequality constant off the lattice is rounded outward and an equality constant off it is refused; a party-local entry carries its tag, and every entry of an emitted constraint operator satisfies the declared lattice condition, its modulus bound included.

Every constraint entry the engine emits carries exactly one certificate, named by its case of Corollary 1.7, and is of the certified kind of Corollary 3.13, never an assumption on the input. The artefact records each entry's certificate case.

A certified family is sound for every family of physical inputs, while an assumed one is sound only for the family it was assumed on (Corollary 3.13); an emitted entry therefore never rests on an assumption about the input (Remark 3.14 gives the standard way a constraint fails to be sound).

Requirement 16.1 (Emission of certified constraint entries only). Every constraint entry the engine emits carries exactly one certificate, named by its case of Corollary 1.7 (identity, positivity or spectral bounds), and is of the certified kind, case (i) of Corollary 3.13; the engine emits no entry of the assumed kind, case (ii). Each emitted entry's certificate case is recorded with the transported set, as Requirement 16.4 states.

Motivated by Cor. 1.7, Cor. 3.13, Rem. 3.14, Req. 16.4 · accepted

The engine refuses a request to impose the class-marginal identity on the target as an operator identity, a form Remark 3.14 shows unsound. Its sound squashed form is the expectation identity.

Where the physical class-marginal identity holds, its squashed counterpart holds in expectation on every reachable state but not, in general, as an operator identity on the target, and imposing the operator form where the target does not satisfy it would exclude reachable states (Remark 3.14).

Requirement 16.2 (Refusal of the operator form of the class-marginal identity). The engine refuses, with the reason `unsound_constraint_requested`, a request to impose on the squashed states the operator form $\sum_{v \in \mathcal{V}_t} F_v = q_t \, \mathbb{1}_Q$ of the class-marginal identity (Remark 3.14).

Motivated by Rem. 3.14 · accepted

Where the request carries no entry the engine emits an empty constraint list and no licence; where it carries entries it emits the instance's entries and the requested ones its claims admit, with the licence. A licence is present exactly when the list is non-empty.

A non-empty list without a licence would carry no key-length claim, and a licence without a list would attest to nothing; which case holds is the request's, as Corollary 11.3 states.

Requirement 16.3 (The constraint list and the licence decided by the request). Where the request carries no entry (Corollary 11.3), the engine emits an empty `constraint_list`, whatever entries the instance's own list contains, and no `attack_set_license`. Where the request carries entries, it emits in `constraint_list` the instance's own entries, the class-marginal entry on request and any further requested entry that Corollary 11.3 (iii) admits (Requirement 16.7), and it emits the licence of Theorem 6.7 in `attack_set_license`. The licence is present if and only if the list is non-empty

(`licence_presence`): a non-empty list without a licence is refused with `licence_missing`, and a licence without a list with `licence_without_constraints`.

Motivated by Thm. 6.7, Cor. 11.3, Req. 16.7 · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_licence_presence

Whatever the request carries, the engine emits the transported set with the certificate case of each operator. The positivity certificate of every sector projector travels there, and becomes a constraint entry only where a caller requests it as one.

Every instance transports its sector projectors, and their pull-backs are positive for every instance (Corollary 11.3 (i)); that certificate is a property of the channel and does not wait for a request.

Requirement 16.4 (Certificates carried with the transported set). Whatever the request carries, the engine emits in `transported_operators`, per operator of the transported set, the certificate case of Corollary 1.7 that makes it usable. In particular the certificate $(\Pi_s, \succeq, 0)$ of every sector projector of every instance travels there and not as a constraint entry; such an operator becomes an entry of `constraint_list` only where a caller requests it as an entry (Corollary 11.3 (i)).

Motivated by Cor. 1.7, Cor. 11.3 · implemented (effective: accepted) · *Code* certiqs_key.reductions.channels:TransportedOperator

The engine emits the class-marginal entry only in expectation form, on request, and only where the instance discharges the class-marginal identity; a request for it anywhere else is refused. The entry is never emitted on the strength of a branch that does not establish it.

Emitting the entry where the identity is not discharged would impose it on reachable states for which it may be false, which is the security break of Remark 3.14; its operator form is refused outright (Requirement 16.2).

Requirement 16.5 (The class-marginal entry only where the instance discharges it). The engine emits the class-marginal entry $\left(\sum_{v\in\mathcal{V}_t} F_v, =, q_t\right)$ at a declared t, in the expectation form of Corollary 11.3 (ii), only on request and only where the instance discharges the class-marginal identity (10.5). A request for it of an instance that does not discharge the identity is refused with `constraint_without_certificate`, and the entry is never emitted on the strength of a branch that does not establish it.

Motivated by Rem. 3.14, Def. 10.1, Cor. 11.3, Req. 16.2 · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_constraint_certificate

A caller's inequality constant off the declared lattice is rounded onto it in the direction that enlarges the feasible set, and the direction is recorded. An equality constant off the lattice, or any constant of modulus above one, is refused rather than rounded.

Rounding toward the enlarging side keeps the certificate true at the rounded constant, while the other side could exclude a reachable state (Corollary 11.3 (iii)); an equality has no enlarging side.

Requirement 16.6 (Outward rounding of a caller's constant off the lattice). Where the declared constant of a requested entry does not satisfy (9.1), the engine first rounds it onto the lattice in the direction that enlarges the feasible set for the entry's relation, down for $\succeq$ and up for $\preceq$, and records the direction taken per entry in `emission_record` (Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction)). An equality entry whose constant is off the lattice is refused with `constraint_without_certificate` rather than split into a pair, and a declared constant of modulus above one is refused with `constraint_without_certificate` under every relation, before any rounding.

Motivated by Def. 9.1, Cor. 11.3, foundations/req:rounding-direction · implemented (effective: accepted) · *Code* certiqs_key.reductions.constraints:rounded_constant

A requested operator is emitted only where a claim gives a condition on the emitted numbers implying its certificate and the engine decides that condition exactly. Every other requested operator is refused, and none is emitted at some other constant or on the caller's word.

Parts (i)–(iii) of Corollary 11.3 cover every entry the module can emit. A requested operator they do not cover has no certificate to be emitted on, and a certificate the caller declares is not one: nothing in this contract decides it, and no assumption of the ledger names it as a premise, while an entry on a false certificate excludes reachable states, the security break of Remark 3.14.

Requirement 16.7 (Refusal of a requested operator without a decided certificate). The engine emits an operator that a request carries as an entry only where Corollary 11.3 covers it: as a default carried operator (part (i)), as the class-marginal entry of an instance that discharges the class-marginal identity (part (ii)), or, under part (iii), where a claim stated for the instance gives a decided certificate for the operator's form and the engine decides that condition exactly on the emitted numbers and the request's declared constants, finding that it holds. It refuses, with `constraint_without_certificate`, every other requested operator: one outside the form of Corollary 11.3 (i)–(iii), one of that form for which no claim stated for the instance gives a decided certificate, and one whose decided certificate fails at the emitted numbers. It takes no certificate (6.9) from the request on trust, and it never emits a requested operator at some other constant.

Motivated by Rem. 3.14, Thm. 6.7, Cor. 11.3 · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_constraint_certificate

Every emitted entry carries its reading as a tag: untagged in the form of the source mode, or party-local with a factor scope only on a declared product with a certificate on that factor's channel. A party-local request on a one-factor emission gets the untagged entry.

The tag makes a consumer's gate a type check that compares no numbers, and a party-local reading is covered by the licence only through a certificate on its own factor's channel (Remark 11.4).

REQUIREMENT 16.8 (The party-local tag and its scope). The engine emits every constraint entry with the reading it is to be read in. An entry is emitted untagged, as $(\hat{O}, \bowtie, c)$, and is read in the form of the declared source mode (Definition 6.5); or, only where the measured system is a declared product $Q = \bigotimes_f Q_f$ of party-scoped factors and the entry's certificate is discharged on that factor's own channel Λ_f (Theorem 6.7 (v)), it is emitted with the tag `party_local` and a scope, as $(\hat{O}, \bowtie, c; f)$, with $\hat{O}$ an operator on Q_f alone carried as a sector-block list against that factor's sector map. In this module the tag is emitted only with the entries of several measuring parties (Theorem 11.5 (iii)). A request for the party-local form on a one-factor emission is answered with the untagged entry, at the same operator, relation and constant, and is not refused.

Motivated by Def. 6.5, Thm. 6.7, Rem. 11.4, Thm. 11.5 · implemented (effective: accepted) · *Code* certiqs_key.reductions.constraints:party_local_entry

Where the licence is present, every entry of every emitted constraint operator is decided to satisfy the lattice condition before the licence is issued, and the request is refused otherwise. The modulus bound binds these entries, not the coefficients that form them.

A certified entry is a combination of transported operators whose coefficients need only be integer multiples of $2^{-\kappa}$ (Definition 10.1 (I4)). Condition (9.1), the modulus bound $|x| \leq 1$ included, binds the entries of the operator the combination produces, and those are emitted numbers.

REQUIREMENT 16.9 (Lattice condition on the entries of emitted constraint operators). Where the licence is present, the engine decides, on the emitted numbers and before the licence is issued, that every entry of every emitted constraint operator satisfies (9.1), and refuses the request otherwise, with the reason `constraint_without_certificate` (the lattice half of the rule `licensed_hermiticity`; Corollary 11.3 (iii)).

Motivated by Def. 9.1, Def. 10.1, Cor. 11.3 · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_licensed_hermiticity

17 Requirements: exact admission and emission

The engine refuses a declaration that is not exactly representable in the field of its arithmetic, and an emitted family it cannot decide exactly admissible.

A class the normalizer moves, the flagged last resort included, is disclosed rather than refused, unless the run declares the strict option and would rest on it. It records every displacement, kept or moved class and rounding direction of an emission, emits the reference companion beside the artefact where a reference is handed, with no normalizer and its entries relaxed, emits each reduction instance's party partition with its sector map, and emits several measuring parties' artefacts with their combination rule rather than a product.

Imports foundations/def:evaluated-off-snap, foundations/def:exact-island, foundations/def:factor-enclosure, foundations/lem:recentred-enclosure, foundations/req:snap-reference

A declaration whose entries are not exactly representable in the field chosen for the arithmetic is refused, never rounded into it. A declaration on the declared lattice is never refused on this ground.

Declared operator data cross every module boundary exactly admissible (Requirement "Exact admissibility at every module boundary" (foundations/req:boundary-admission)), and a consumer that finds one of its admission predicates violated refuses rather than repairs (Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)). For this module one such predicate is that the declared entries lie in the field that carries its arithmetic.

Requirement 17.1 (Refusal of a declaration outside the substrate). The engine refuses a declaration whose entries are not exactly representable in the field $\mathbb{F}$ chosen for the arithmetic (Definition 9.2), with the reason `declaration_not_exactly_admissible`, rather than rounding the entries into that field. No declaration whose entries satisfy (9.1) is refused on this ground.

Motivated by Def. 9.1, Def. 9.2, foundations/req:boundary-admission, foundations/req:consumer-decides-exactly · accepted

The engine emits a family only after deciding the six emission conditions on the numbers about to be emitted, and otherwise refuses, repairing nothing it was handed. A moved class, the flagged last resort included, is refused only under the strict option, where it would rest on the snap.

The decision on the six conditions is taken on what comes out of the normalizer (Definition 11.1 (4), Theorem 11.2 (ii)). A class the normalizer moves is not refused by default: it is snapped without reviving a zero row or column and disclosed (Requirement 17.3), as Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii) asks of any snap of a measurement family. A class whose residue no single element can take without reviving a zero row or column has no support-safe snap in the normalizer; it is snapped by the last resort, which always yields a positive semidefinite family and is flagged in the record (Requirement "Support-safe

snaps of a measurement family" (foundations/req:support-safe-snap) (iv)), so it is not refused by default either. The usual cause of a last resort is a class whose elements are rank-deficient on the sector, which a model with the device's loss and dark counts can remove where the conditions the receiver module states for it hold (Proposition "Loss and dark counts make every dark-composed element definite" (receiver/detector-imperfections/prop:noise-floor)). One refusal remains for a moved class: a run that declares the strict option refuses wherever the class would rest on the snapped-measurement premise, which is certain at emission where no reference companion that bypasses the normalizer is formed (Requirement 17.4).

REQUIREMENT 17.2 (Refusal of an emitted family that is not exactly admissible). The engine emits a family only if the six conditions of Theorem 11.2 (ii) — (a)–(e) of Definition 9.1 together with (9.1) of every entry — hold of the numbers about to be emitted, decided exactly after the emission normalizer where that acted; otherwise it refuses with the reason `emitted_family_not_admissible`, naming the predicate that failed, and repairs nothing it was handed. Under the same reason, where the run declares the strict option `strict_snaps` (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)) and no reference companion of Requirement 17.4 is formed beside the artefact, it refuses every moved class of Definition 11.1, one snapped by the last resort (3) included, naming the data sector, the class, the cause and the fix; where a companion is formed, whether the class rests is decided by the run after its search, and the run refuses there under the same option. Where the run does not declare the option, no class is refused for being moved. The cause of a moved class that would rest is that no reference companion bypasses the normalizer, which, by Requirement 17.4, is where no claim of this module establishes (H-REF) for the instance; for a class snapped by (3) it also names that a component of its residue has no host. The fix named is declared numbers on which the exact construction stays on the lattice, so that the class is kept; for a class snapped by (3) also a model that includes the device's loss and dark counts, so that the elements of the class are positive definite on the sector. A moved class is mixed only with the family of its diagonal parts in (2), or with a multiple of the identity in the last resort (3).

Motivated by Def. 9.1, Def. 11.1, Thm. 11.2, Req. 17.3, Req. 17.4, foundations/req:snap-reference, foundations/req:support-safe-snap, receiver/detector-imperfections/prop:noise-floor · implemented (effective: accepted) · *Code* certiqs_key.reductions.validation:rule_emitted_family

Every emission records the lattice exponent and the strict option, each displacement, the kept and moved classes with the step that snapped each, and every rounding direction. It names each moved sector, which rests on the snapped-measurement premise unless evaluated off its snap.

Where the normalizer moved the datum, nothing proved relates the declared model to the handed model, and the size of the move is not such a relation (Theorem 11.2 (v), Remark 11.7); the record carries the move so that it is visible to the consumer. A run that forms the number its key length is formed from on the reference companion of Requirement 17.4, which carries no data of this normalizer, evaluates off its snap every data sector that holds a moved class (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)); elsewhere a moved sector rests on the premise Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), and the record says so in the form Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii) asks, so that a report on the key rate can flag it, as the record of a snap names every block that rests on the premise (Requirement "Snapping by the declarer only" (foundations/req:who-snaps)). A class snapped by the last resort carries its flag, since its elements may treat as possible an outcome the handed datum makes impossible (Requirement "Support-safe snaps of a measurement family" (foundations/req:support-safe-snap) (iv)).

Requirement 17.3 (The record of an emission). The engine emits with every artefact, in `emission_record`: the declared lattice exponent κ; whether the run declared the strict option of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv); the reference kind of every snap the normalizer makes, *exact*, every radius zero, the reference being the handed datum itself (Definition 11.1), so that the enclosure setting, recorded as Requirement "Snapping by the declarer only" (foundations/req:who-snaps) asks of the record of every snap, does not bear on it; per data sector, the emission displacement Δ_s of (11.4); per data sector and class, whether the class was kept (Definition 11.1 (1)) or moved, and for a moved class the step (2) or (3) of Definition 11.1 that snapped it, the weight w taken or the fall-back of (3), the hosts of the components of the residue under (2), and for (3) the flag `support_not_preserved`; and the direction of every bound it rounds, each scalar value rounded down and each complementary value recomputed (Theorem 11.2 (iv)). A data sector *rests on* Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) where some class of it is moved, save where it is evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)): in a run whose key length is formed from a number on the reference companion of Requirement 17.4, its first-order bound or a number certified on

it, which carries no data of this emission's normalizer, every data sector that holds a moved class is, a kept sector being named as kept and not counted among them, and none rests on the assumption. Where the key length is formed instead from a number certified on the snapped declaration in the fallback of Requirement "The certified evaluation of the first-order bound" (meat-core/frank-wolfe/req:certified-evaluation) (i), none is, and every data sector that holds a moved class rests on the assumption, with the cause `companion_not_certified` (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(b)). The record names every moved sector, with the flag of any class of it snapped by (3), and carries Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) among the assumptions the artefact may rest on exactly where there is one. This record is emitted with the artefact, before the run knows whether the number its key length is formed from is formed on the reference companion; what it names is therefore provisional, and the record of the run decides which sectors rest on the assumption, each with its cause and group (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii); Theorem 11.2 (vi)).

Motivated by Def. 11.1, Thm. 11.2, Rem. 11.7, Req. 17.4, foundations/def:evaluated-off-snap, foundations/req:snap-reference, foundations/req:support-safe-snap, foundations/req:who-snaps, meat-core/frank-wolfe/req:certified-evaluation · *Assumes* A-SNAP · implemented (effective: accepted) · *Code* certiqs_key.reductions.artifact:EmissionRecord

Where a reference is handed or the normalizer moved a class, the engine runs the same construction exactly on the reference, or on the snapped blocks, with no normalizer, and emits the companion's enclosures, factor ones where carried, and its entries, requested ones where certified there.

The companion is what a run evaluates its first-order bound on (Requirement "The first-order bound evaluated on the reference companion" (meat-core/frank-wolfe/req:bracket-on-reference)). It is the same chain run on other data, so it is built where the artefact is built, by the same code path, and carries the same shape by construction. It keeps the reference's exact data where the reference has them, as Gaussian-rational centres and rational radii, and is rounded to binary64 only by whoever evaluates it; its entries, which a solver's oracle reads as constraints, are the one part emitted in binary64 here, and Lemma 12.3 says how they are rounded without enlarging what they assert. Where the instance is one whose (H-REF) a claim establishes, Corollary 12.1 (iv) or Corollary 21.5, the companion is emitted. It is emitted also where the declaring instance hands no reference but the normalizer moved a class: the companion is then the instance at its handed datum, the module's exact construction, which no step of the normalizer has touched, so that the classes the normalizer moved are evaluated off their snap

while the instance's own blocks stay snapped. An entry a caller requested beside the instance's own is carried to the companion in its companion form where its certificate is established at the companion's datum, and is left out where it is not: leaving an entry out removes one constraint and so only enlarges the companion's feasible set, and no bound formed on the companion is raised by it. Where the reference also carries factor enclosures and the instance's rule combines blocks with non-negative weights, the companion carries them to its datum and target (Corollary 12.2), unaltered but for the weights.

Requirement 17.4 (Emitting the reference companion beside the artefact). Where a claim of this module establishes (H-REF) of Corollary 12.1 for the reduction instance the artefact is emitted from, and either the declaring instance hands the reference beside its snap of the measurement it states (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap) (i)) on at least one block or the emission normalizer moves at least one class (Definition 11.1), the engine emits beside the artefact, in `reference_companion`, the reference companion of that instance at the family G that carries the handed reference on every handed block and the snapped block the instance declares on every other block, so that, where no block is handed, G is the instance's snapped family and the companion's datum is the handed datum (Corollary 12.1 (i)):

(i) *The datum.* Every enclosure of a handed block is first recentred by Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure), so that every entry whose enclosure contains 0 has centre 0. The engine then forms, in exact arithmetic on the exact island (Definition "The exact island" (foundations/def:exact-island)), the enclosure of the companion's datum that Corollary 12.1 (iii) gives, from these enclosures and from the snapped blocks with radius zero, by the construction it applies to the snapped family: the same rule of (I1), the same weight route, scalar values, sectors and supports. It then recentres that enclosure again, element by element and block by block, by Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure), after making its centre Hermitian and its radii symmetric by part (iv) of that lemma where they are not, every block of the datum being Hermitian (Corollary 12.1 (i)); where the rule of (I1) combines entries of several blocks, a sum that is exactly zero may so have a centre that is not, and this second recentring sets it to zero wherever its enclosure contains zero. The entries of (iii) are formed from the datum's enclosure so recentred. It applies no normalizer (Definition 11.1), decides no lattice condition, rounds nothing, and decides nothing of the companion's admissibility, which is Corollary 12.1 (i) on the claims the handed reference names (Requirement "Snaps of a measurement family: kept, or moved and disclosed"

(foundations/req:snap-reference) (i)). It emits the centres, entries of $\mathbb{Q}(i)$, and the radii, rationals, entry by entry, so that an entry the reference knows exactly is emitted exactly, with radius zero.

(ii) *The shape.* The companion is emitted with the sector map, the alphabet with its classes and weights, the scalar values and the party partition of the artefact's target, which are equal by Corollary 12.1 (i), and with the artefact's channel witness and transported set read at the companion's datum.

(iii) *The entries.* On a route whose request carries entries (Corollary 11.3), the companion's entry list holds the instance's own entries at the companion's datum and, for every entry requested beside them that the artefact carries, its companion form (Corollary 12.1 (vi)) at the writing by which the engine recognised it, where its certificate at the companion's datum is established: for a sum of sector projectors and for the class-marginal entry, by Corollary 12.1 (vi), the operator being the artefact's, enclosed with radius zero; for another entry, where a claim of this module states for the instance a condition on an enclosure of the companion's datum under which its certificate holds, by deciding that condition exactly on the datum's enclosure recentred as (i) states (for the operator (19.13) on the flat instance's licensed route, (21.10) of Corollary 21.5). A requested entry whose certificate at the companion's datum is not so established is not formed on the companion and has no entry there in its place. For each entry $(\hat{O}_k, \bowtie_k, c_k)$ of that list, with the enclosure (C_k, R_k) of its operator that Corollary 12.1 (iii) gives, or radius zero where the operator is the artefact's: the operator $\hat{O}'_k$ whose every real and imaginary part is the binary64 number nearest to that of C_k, ties to even, which is Hermitian and has the blocks of $\hat{O}_k$; the rationals $\rho_{il} \coloneqq |\operatorname{Re}(\hat{O}'_k - C_k)_{il}| + |\operatorname{Im}(\hat{O}'_k - C_k)_{il}| + (R_k)_{il}$ and $\delta_k \coloneqq \max_i \sum_l \rho_{il}$, computed exactly; and the relaxed entries of Lemma 12.3 at δ_k, with c'_k the largest binary64 number at most $c_k - \delta_k$ and c''_k the smallest at least $c_k + \delta_k$. Where $\delta_k = 0$ the entry is $\hat{O}_k$ itself and is emitted as it is, an equality included. Each emitted entry carries the certificate case of the entry it relaxes, its δ_k, and the mark that it is relaxed, its certificate being Lemma 12.3 (ii); its operator is added to the companion's transported set with its pull-back. The lattice condition of Requirement 16.9 binds the artefact's own entries and not these.

(iv) *No normalizer.* No step of the emission normalizer (Definition 11.1) acts on the companion's datum: the companion carries the reference, not a snap of it.

(v) *The record.* The companion carries, per block of G, whether it is the

handed reference or the snapped block, the kind of every handed block (*exact*, every radius zero, or *enclosed*, with the largest radius of the block after recentring), the entries recentred in the handed blocks and in the datum, the largest radius of the datum's enclosure after its recentring, each δ_k, every requested entry formed on the companion with what established its certificate there, and every requested entry not formed with the reason; and it is marked evaluation-only: it has no run of its own, no consumer admitting it as the declaration of a run, and what a consumer decides or computes of it beside the declaration it accompanies is the consumer's (Requirement "Admission of the reference companion beside its declaration" (meat-core/req:companion-admission)). The artefact itself, its target, normalizer, emission record (Requirement 17.3) and entries, is emitted as it is without a companion.

(vi) *No companion.* Where no block is handed and the normalizer moves no class, or no claim of this module establishes (H-REF) for the instance, or a reciprocal node of the rule of (I1) receives no enclosure, so that Corollary 12.1 (iii) gives none, no companion is emitted, and the record says which of the three holds.

(vii) *Factor enclosures.* Where the handed reference carries factor enclosures (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) on some of its blocks $G_j^{(b)}$, b a handed block, as the receiver's does (Requirement "The emitted blocks are dark-composed and carry their reference" (receiver/detector-imperfections/req:dark-composed-reference)), and a claim of this module establishes that the rule of (I1) of the instance is a non-negative combination (Corollary 12.2, part (iii) for the identity instance and Corollary 21.5 for the flat instance), the engine forms, for every element of every data sector of the companion's datum every combining block of which carries one, the factor enclosure of Corollary 12.2 (i): the boxes of the combining blocks side by side in a fixed order, their centres and radii unaltered, and their cores multiplied by the coefficients in exact rational arithmetic. It emits each beside the entrywise enclosure of the datum of (i), which it does not alter, as the factor enclosure of the target's block on that data sector (part (ii) of that corollary). It forms none on any other element: none where a combining block carries none, and none for an instance no claim of this module shows to have such a rule, in particular none through a rule that maps the blocks of a data sector by a conjugation before it combines them, whose factor enclosure this module does not form. The record of (v) carries, per element of the datum, whether a factor enclosure is carried, with its number of columns, or why none is, and it names the claim that shows the rule of (I1) to be a non-negative combination, Corollary 12.2 (iii) or Corollary 21.5.

Motivated by Def. 10.1, Def. 11.1, Cor. 11.3, Cor. 12.1, Cor. 12.2, Lem. 12.3, Req. 16.9, Req. 17.3, Rem. 19.9, Cor. 21.5, foundations/def:evaluated-off-snap, foundations/def:exact-island, foundations/def:factor-enclosure, foundations/lem:recentred-enclosure, foundations/req:snap-reference, meat-core/frank-wolfe/req:bracket-on-reference, meat-core/req:companion-admission, receiver/detector-imperfections/req:dark-composed-reference · *Used by* meat-core/req:companion-admission · implemented (effective: accepted) · *Code* certiqs_key.reductions.companion:reference_companion

Each reduction instance's party partition is emitted with its sector map: the default one, every factor Bob's, unless a party label is declared. Where several parties measure, each factor's artefact carries that factor's own label.

The consuming declaration places its well-formedness conditions on a party's side (Theorem 11.2 (ii)), so the party partition of Definition 10.1 (I2) is part of what a consumer reads with the target.

REQUIREMENT 17.5 (Party partition emitted with the sector map). The engine emits the party partition of each reduction instance with its sector map (`target_descriptor`): the default party partition where the instance declares no party label, and otherwise the declared label (`party_label`) on every factor of every $\mathcal{H}(Q_s)$. Where several parties measure, each factor's artefact carries that factor's own label with its own sector map (Theorem 11.5 (i)).

Motivated by Def. 10.1, Thm. 11.2, Thm. 11.5 · implemented (effective: accepted) · *Code* certiqs_key.reductions.sectors:party_partition_statement

Where several parties measure, the engine emits one artefact per factor, each with its party label, in the declared order, together with the combination rule. It emits, declares and rounds no entry of any product element.

A Kronecker product of two families on the declared lattice lies in general only on a finer lattice, and a normalised product would declare a non-product measurement for separated parties; the product is therefore emitted as a rule, and Theorem 11.5 states what that rule gives.

REQUIREMENT 17.6 (Emission of several measuring parties as factors). Where several measuring parties measure, the engine emits in `factor_emission` the ordered list of the factors' artefacts, each with its party label, in the declared order of the factors, and the combination rule (11.10); it emits, declares and rounds no entry of any product element F_v (Theorem 11.5).

Motivated by Thm. 11.5 · implemented (effective: accepted) · *Code* certiqs_key.reductions.factors:reduce_factors

18 Requirements: the assumption ledger

The emitted assumption ledger carries the rows of the emission's assumptions under their presence rules and marks those that are never verified. It

refuses a supplier artefact of a version the caller has not declared accepted, passes the supplier's ledger through entire, refuses one with a repeated or missing key, and relies only on supplier presence rules that omit no premise an imported fact rests on. It omits the row of the supplier assumption only where the run attests that it built every receiver artefact read and each is exact with rigorous weights (Requirement 18.7). The engine declares the output-state model of the test statistics.

Imports receiver/cor:unconditioned-family, receiver/def:setting-family, receiver/def:slot, receiver/thm:click-povm, receiver/thm:coarse-graining, receiver/thm:enclosed-blocks, receiver/thm:enclosed-weight, receiver/thm:representation-independence

Every artefact carries the assumption ledger: a row per assumption of this contract or taken from another module, under its presence rule. The rows cover every such key of the producing claims' closure but those that hold vacuously and A-RECV for an exact, rigorous receiver artefact the run attests.

A consumer inherits from this module the assumptions its artefact names, and it can match an assumption against its own ledger only by machine key.

Requirement 18.1 (Rows of the emitted assumption ledger). The engine emits with every artefact the assumption ledger `assumption_ledger`: one row, carrying its machine key, for each assumption of this contract and for each assumption the interface lists as taken from another module, each under its declared presence rule. The row of A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) is present only where the emitted chain's source segment is non-empty, and there under the presence rule the interface declares for it, the row of Assumption 13.7 only where the emitted constraint list is non-empty, the row of Assumption 13.5 only where the declared setting set $\mathcal{T}_{\text{set}}$ has at least two values, the row of Assumption 13.6 only where $\mathcal{T}_{\text{set}}$ has at least two values or a public choice is applied at the measurement the emitted family describes (and also where the emission is not told whether one is), a submodule's own assumption where that submodule's instance is emitted and any narrower presence rule the submodule declares for it holds, the row of Assumption 13.3 on every emission save where Requirement 18.7 omits it, and every other row on every emission. These rows include every key of the core's own assumptions, and every key the interface lists as taken from another module, that the closure of the claims producing the artefact contains, save the row of Assumption 13.7 where the emitted constraint list is empty, a condition on the constants of emitted entries holding vacuously of an empty list, the row of Assumption 13.5 where $\mathcal{T}_{\text{set}} = \{*\}$, whose clauses hold of the one setting drawn with weight one, and the row of Assumption 13.6 where in addition no public choice is applied at that measurement, whose clauses then concern no choice, and the row of Assumption 13.3 where Requirement 18.7 omits it, whose clauses that requirement shows to hold there (the supplier's keys ride the row of Assumption 13.3 instead, under the supplier's own presence rules,

Requirement 18.3); a row may be present that no such closure contains.

Motivated by Ass. 13.3, Ass. 13.5, Ass. 13.6, Ass. 13.7, Req. 18.3, Req. 18.7, transmitter/ass:src · implemented · *Code* certiqs_key.reductions.emit:emitted_assumption_rows

The ledger marks as never verifiable the row of every assumption the engine never verifies, as its verification field says, whichever module states it. Validation decides the declared data the assumptions are read with, and no code path decides more of a never-verified one.

An assumption that is a statement about the physical devices, the source, the adversary or the run, rather than about declared data, has a clause that no code path can decide; what validation decides beside that clause does not change its mark. Which assumptions the engine never verifies is recorded in each one's verification field (`spec/claim-format.md#32-whether-the-engine-verifies-an-assump`) which the contract prints beside the statement. A row that this module takes from another module carries the mark its owner gives it, as the row of the transmitter's retention assumption (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) does.

Requirement 18.2 (Never-verified rows and validation of declared data). The engine marks as never verifiable (`never_verified`) the row, in the assumption ledger, of every assumption that the engine never verifies, as that assumption's verification field says, whether this contract, one of its submodules or another module states it; no code path decides what any of them says beyond its clauses on the declared data. The engine decides by validation the conditions on the declared data that the assumptions of this contract are read with, such as the declared dimensions of Definition 1.1 and the declared class weights of Definition 4.4.

Motivated by Def. 1.1, Def. 4.4, transmitter/ass:src · implemented · *Code* certiqs_key.reductions.artifact:assumption_ledger

The row of A-RECV carries every machine key of the supplied ledger, as supplied and in the supplier's order, beside the artefact's version. The engine keeps no list of the supplier's keys and adds no presence rule of its own.

The facts a reduction instance imports from a supplier rest on the supplier's assumptions (Assumption 13.3); a consumer inherits them only if the artefact names them by the supplier's own machine keys.

Requirement 18.3 (Pass-through of the supplier's assumption ledger). On the row of Assumption 13.3 in the assumption ledger `assumption_ledger`, the engine carries, as machine keys and beside the artefact's module version `module_version`, every key the supplied ledger carries, as supplied and in the supplier's order; a key beyond those the supplier's presence rules require is carried rather than refused. The engine holds no list of the supplier's keys and adds no presence rule of its own: what it reads from the supplying module, as it is installed where the check runs, is that module's machine

keys and the set of keys its rules make conditional, and it evaluates the condition itself on the artefact's own data.

Motivated by Ass. 13.3 · accepted

A supplied ledger that repeats a machine key is refused, and so is an artefact whose ledger lacks a key the supplier's own presence rules make present. A shortened list is never passed on in silence.

The pass-through of Requirement 18.3 carries the supplied keys as they stand, so a repeated key would be carried twice and a missing one would silently shorten what a consumer inherits.

REQUIREMENT 18.4 (Refusal of a supplier ledger with a repeated or missing key). The engine refuses a supplied ledger that carries one machine key more than once, and it refuses an artefact whose ledger lacks a key the supplier's own presence rules make present, rather than passing it on as a silently shortened list; both refusals carry the reason `receiver_spec_not_accepted`.

Motivated by Req. 18.3 · accepted

A supplier artefact of a version the caller does not declare accepted, or without its supplier's validation results, is refused. Outside the run, that the set names accepted versions and the artefact is genuine stays A-RECV; in the run, acceptance is the claims' effective status.

The version an artefact declares and the presence of its supplier's validation results are declared data, decided on entry. The check is against the caller's declared set of accepted versions (`accepted_receiver_versions`); that every version in that set is an accepted version of the supplying module and that the artefact is what it declares are Assumption 13.3 (i) for an artefact from outside the run, and that its facts hold of the device is Assumption 13.3 (ii) wherever the ledger carries its row. For every receiver artefact the run built itself, whether or not Requirement 18.7 omits that row, acceptance is no premise but the effective status of the receiver's claims in the run's closure, which the record of the run reports.

REQUIREMENT 18.5 (Refusal of a supplier artefact of an unaccepted version). The engine refuses a supplier artefact whose declared module version `module_version` is not in the caller's declared set of accepted versions `accepted_receiver_versions`, and one that does not carry the supplying module's validation results; both refusals carry the reason `receiver_spec_not_accepted`, the reason the ledger refusals of Requirement 18.4 carry as well.

Motivated by Ass. 13.3, Req. 18.4, Req. 18.7 · implemented · *Code* certiqs_key.reductions.validation:rule_receiver_spec

The supplier's presence rules, read as installed, may omit a key of the imported facts' closure only where the facts imported through it hold without it, and a change of the rules must meet this too. The pass-through then never drops a premise an imported fact rests on.

The row of Assumption 13.3 carries the supplier's ledger under the supplier's own presence rules (Requirement 18.3), and those rules are read from the supplying module as it is installed where the check runs, so a change of them takes effect here without an edit.

REQUIREMENT 18.6 (Supplier presence rules that omit no imported premise). The supplier's presence rules that the engine reads omit a key of the closure of the facts a reduction instance imports from that supplier only where the facts imported through it hold of the declaration without it; a change of those rules, which takes effect in the engine's check without an edit of this module, must itself meet this requirement.

Motivated by Ass. 13.3, Req. 18.3 · accepted

The ledger omits A-RECV only where every supplier artefact read is a receiver artefact this run built, with a model admissible exactly and every weight evaluated rigorously. Both clauses then hold by the receiver's theorems under its own ledger, with no premise on arithmetic.

Assumption 13.3 names what a reduction instance cannot see in a supplier artefact: that it is its module's output at an accepted version, on the inputs it declares, and that the facts the supplier's claims state hold of its numbers. Version, validation results and ledger are data that an artefact written by hand can carry equally; whether the run built the artefact is known to the part of the engine that ran the supplier, as for the chain of Requirement "Omission of the link-soundness row only on a proved chain" (protocol-compiler/req:link-row-provenance). Where the run built it, clause (i) holds by construction: the artefact is the installed receiver's output on the declaration the run handed it, with that receiver's own validation results, at the version it states. For every receiver artefact the run built, whether or not the row is omitted, acceptance is no premise: it is the effective status of the receiver's claims in the run's closure, which the run's record reports, as for every module the run executes; the acceptance part of clause (i) is a premise only for an artefact from outside the run. Clause (ii) concerns the facts the instance imports, those the consuming claims cite (Remark 20.12; Proposition 20.4 (iv)), and condition (iv) below makes each a theorem of the receiver with no premise on arithmetic. A declaration with a model admissible exactly (Requirement "Deciding the exact route and enclosing the port Grams" (receiver/req:exact-route-decision)) has the port Grams of a receiver instrument, which determine its click-pattern measurement (Theorem "Representation independence of the click-pattern POVM" (receiver/thm:representation-independence)), so the theorems

stated of a receiver instrument apply to it under the receiver's ledger, Assumption "Trusted and exactly characterised receiver" (receiver/ass:trust) saying that it is the physical receiver: the block diagonality in the photon number of the elements of its setting family and of their coarse-graining (Definition "Setting family" (receiver/def:setting-family), Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining), Theorem "Click-pattern POVM" (receiver/thm:click-povm), Corollary "The unconditioned receiver is a setting family" (receiver/cor:unconditioned-family)); the dark-count channel, the declared matrix or the product form, of which (D1) and (D2) hold by construction and which this module decides exactly on entry (Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm), Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts)); at a normalized station the efficiency normalization (Proposition "A receiver as uniform loss followed by its normalized station" (receiver/detector-imperfections/prop:efficiency-normalization)); and the strengthening slot (Definition "The strengthening slot" (receiver/def:slot)). Its weights carry the rigorous label (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)), so each per-setting weight is the largest binary64 number at or below an enclosed weight (Requirement "Rigorous evaluation of the multiclick weight" (receiver/req:rigorous-weight) (iv)), and Theorem "The weight on enclosed bound data bounds the multiclick element" (receiver/thm:enclosed-weight) (iii) gives its guarantee on every photon-number block above the declared cutoff at its own setting, the quantifier domain being the declared cutoff and settings; at a setting family a receiver submodule constructs, that submodule states the same of its rigorously evaluated weight (Theorem "Exact statements, floating-point and rigorous evaluation" (receiver/thm:exact-and-evaluated), lead-in). No computed eigenvalue enters such a weight, so Assumption "Declared slack on the computed block eigenvalues" (receiver/ass:eig) is invoked at no setting, and the non-negativity and the sum of the weights at a setting are decided on entry (the slot row of `spec/modules/reductions/wpfss/interface.yaml`). The photon-number blocks below the cutoff are no imported fact: they reach the instance as its declaration (Definition "Declared data and the parties to a module boundary" (foundations/def:declared-data) (ii)), beside rational enclosures that contain the model's blocks (Theorem "Rigorous enclosures of the photon-number blocks" (receiver/thm:enclosed-blocks)), against which the declaring party snaps them, a moved block resting on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) (Remark 13.4). Where a weight is evaluated in floating point, the receiver asserts no bound between the emitted number and the weight its theorems speak of (Theorem "Exact statements, floating-point and rigorous evaluation" (receiver/thm:exact-and-evaluated) (ii)), and where the declaration is admitted only within the receiver tolerances

no receiver instrument has its declared numbers; there clause (ii) stays a premise.

Requirement 18.7 (Omission of the supplier row for a receiver artefact the run built). The engine omits the row of Assumption 13.3 from the assumption ledger `assumption_ledger` of an emission only where the reduction instance read at least one supplier artefact and, for every supplier artefact it read, all of the following hold: (i) the supplying module is the receiver, and the artefact is one that the part of the engine that ran this module obtained in the same run by running the receiver module, as installed where the run runs, on the receiver declaration the run holds, and handed to this module as it was emitted; (ii) the module version `module_version` the artefact declares is the one that installed module states for the artefacts it emits, and the artefact passes Requirements 18.5 and 18.4; (iii) the record of the run carries that artefact's own assumption ledger; (iv) the artefact records that its declaration has a model admissible exactly (Requirement "Deciding the exact route and enclosing the port Grams" (receiver/req:exact-route-decision) (iv)), and it carries the evaluation label `rigorous` (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)), every per-setting weight having been formed by the rigorous evaluation of Requirement "Rigorous evaluation of the multiclick weight" (receiver/req:rigorous-weight). There clause (i) of Assumption 13.3 holds by construction, the acceptance of the version being the effective status of the receiver's claims in the run's closure, which the record of the run reports, and clause (ii) holds of the facts the instance imports, each a theorem of the receiver about the receiver instrument the declaration is, holding of the numbers emitted under the ledger (iii) carries and resting on no premise about arithmetic. That (i) and (iii) hold is established by the part of the engine that ran the receiver module and docked its artefact into the instance, and is passed to this module beside the artefact (`receiver_provenance`); the engine never reads it from the artefact. The engine decides (ii) and (iv) on the artefact. Where it omits the row, it records the omission with the artefact's version (`receiver_row_omitted_on_provenance`), and the supplier's keys, which ride that row where it is emitted (Requirements 18.3 and 22.9), reach the record of the run from the artefact's own ledger by (iii). Where `receiver_provenance` is not passed, or a condition fails for any artefact read, the row is emitted as Requirement 18.1 says.

Reading: Where this run built the receiver artefact itself from your declared receiver, your declaration describes a valid receiver exactly, and every high-photon weight was computed rigorously, the report does not list the premise that a supplied receiver file is genuine and true of your device (A-RECV). What that premise says then follows from the receiver's theorems under the receiver's own assumptions, which the report lists, and whether those theorems are accepted is shown by their status in the report. Wherever a weight was evaluated in floating point, the receiver was admitted only within its tolerances, or the receiver file came from outside the run, the report keeps A-RECV.

Motivated by Ass. 13.3, Rem. 13.4, Req. 18.1, Req. 18.3, Req. 18.4, Req. 18.5, Prop. 20.4, Rem. 20.12, Req. 22.9, foundations/ass:snapped-measurement, foundations/def:declared-data, protocol-compiler/req:link-row-provenance, receiver/ass:eig, receiver/ass:trust, receiver/cor:unconditioned-family, receiver/def:setting-family, receiver/def:slot, receiver/detector-imperfections/def:darkadm, receiver/detector-imperfections/prop:efficiency-normalization, receiver/detector-imperfections/thm:dark-counts, receiver/req:evaluation-label, receiver/req:exact-route-decision, receiver/req:rigorous-weight, receiver/thm:click-povm, receiver/thm:coarse-graining, receiver/thm:enclosed-blocks, receiver/thm:enclosed-weight, receiver/thm:exact-and-evaluated, receiver/thm:representation-independence · implemented (effective: accepted) · *Code* certiqs_key.reductions.emit:receiver_row_omitted_on_provenance

The engine declares the output-state model of the test statistics and emits no state-independent test fraction. Every public choice of known bias enters through the statistics of the emitted family, as A-TSM states.

No derivation of this contract assumes a state-independent test fraction: every per-round public choice of known bias enters only through the statistics of the emitted family (Assumption 13.6).

REQUIREMENT 18.8 (Output-state model of the test statistics). The engine emits no state-independent test fraction, and it declares the test-statistics model `test_statistics_model` to be `output_state`.

Motivated by Ass. 13.6 · accepted

Part III
Flag-state and weight-preserving squashers

This submodule consumes a receiver artefact and turns it into a finite reduction instance. It keeps the photon-number blocks below a cutoff and replaces everything above by flags: the flag-state target, which puts each outcome on its own flag, or the weight-preserving target, which puts each outcome's certified high-photon weight on every flag and one minus the total weight on its own. Its first main result is that, whenever the certified weights total less than one, the physical measurement squashes to both targets, the two squashers compose exactly, and a weight the weight-preserving target encodes holds on every squashed state (Proposition 19.4). The weight route first applies the dark-count channel on the emitted alphabet, which requires it to descend through the coarse-graining, then forms the joint family of the measuring party's private setting and routes the receiver's per-setting multiclick weights into certified weights rounded down onto the lattice. The resulting flat instance, on the encoded or the licensed route, is a reduction instance, and the second main result is its emission: the emitted family is exactly admissible and pulls back exactly to the declared model on the channel system, with its weights rounded down and still certified (Corollary 21.3). Where the emission moves a block below the cutoff, that statement is about the declared model, which nothing proved relates to the unrounded family; a run evaluates off the snap on the reference companion of the flat instance, or the moved class is disclosed. Where the receiver is compiled at its normalized station, the instance is formed on that station and precomposed with the factored loss, which changes the channel witness and nothing else emitted (Corollary 21.6). The receiver's facts enter by named claims under A-RECV, and the receiver assumptions those claims rest on enter the assumption closures with them.

19 Flag-state targets and their chain

Above a cutoff the receiver is replaced by flags: the flag-state target keeps the blocks below the cutoff and puts each outcome on its own flag, and the weight-preserving target puts each outcome's certified high-photon weight on every flag and one minus the total weight on its own. The main result, Proposition 19.4, is that the physical measurement squashes to both targets whenever the certified weights total less than one, that the chain from the physical measurement through the weight-preserving target to the flag-state target composes exactly, and that a weight the weight-preserving target

encodes holds on every squashed state, whereas under the flag-state target it holds only on reachable states. An artefact therefore takes one of two routes, and a lowered certified weight stays certified and, while the total weight is below one, gives a cleaner target (Proposition 19.7).

Certified weights are per-outcome lower bounds on a POVM's blocks above a cutoff. Both targets keep the blocks below the cutoff; the flag-state target puts each outcome on its own flag, the weight-preserving target puts its certified weight on every flag and one minus the total weight on its own.

Above a photon-number cutoff the receiver is replaced by one classical flag per outcome, and the only device knowledge kept there is a lower bound on each outcome's block. The weight-preserving target keeps these bounds in its flags, so that the corresponding inequality holds on every squashed state (Proposition 19.4 (iii)); both targets are known from the literature (Remark 19.9).

Definition 19.1 (Certified weights and the flag-state and weight-preserving targets)**.** Throughout this definition and the claims of this submodule that use it, $\mathcal{H}(B) = \mathcal{H}(B_{\le N}) \oplus \mathcal{H}(B_{>N})$ is an orthogonal two-block decomposition with projections $\Pi_{\le N}$ and $\Pi_{>N}$ — a photon-number cutoff in the optical instance, where $\mathcal{H}(B_{\le N}) = \bigoplus_{m \le N} \mathcal{H}(C_m)$ is finite-dimensional — and every M_v is block diagonal for it, $M_v = M_v^{(\le N)} \oplus M_v^{(>N)}$. $\mathcal{H}(\mathrm{Flg}) = \mathbb{C}^{|\mathcal{V}|}$ carries the orthonormal basis $\{|v\rangle\}_{v \in \mathcal{V}}$, and $\mathcal{H}(Q) = \mathcal{H}(B_{\le N}) \oplus \mathcal{H}(\mathrm{Flg})$ with block projections $\Pi_{\le N}$ (the same symbol on either space) and Π_{Flg}.

A family $(p_v)_{v \in \mathcal{V}}$ of *certified weights* for M at the cutoff consists of non-negative numbers p_v — on the emission path, integer multiples of $2^{-\kappa}$, the $p_{(y,t)}$ of (20.4) — with

$$M_v^{(>N)} \succeq p_v \, \mathbb{1}_{B_{>N}} \qquad \text{for every } v \in \mathcal{V}, \tag{19.1}$$

and we write $p \coloneqq \sum_v p_v$ for their *total weight*. Define

$$F_v^{\mathrm{FSS}} \coloneqq M_v^{(\le N)} \oplus |v\rangle\langle v|_{\mathrm{Flg}}\,, \tag{19.2}$$

$$F_v^{\mathrm{WP}} \coloneqq M_v^{(\le N)} \oplus \big(p_v \, \mathbb{1}_{\mathrm{Flg}} + (1-p)\, |v\rangle\langle v|_{\mathrm{Flg}}\big). \tag{19.3}$$

F^{FSS} is the *flag-state target* and F^{WP} the *weight-preserving target*, the multi-bound weight-preserving flag-state-squasher target, with one certified weight per outcome. The *single-bound case* is the one in which only one p_v is non-zero, and F^{FSS} is the case $p_v \equiv 0$.

Uses Def. 20.3 · *Used by* Lem. 19.2, Rem. 19.3, Prop. 19.4, Def. 19.5, Rem. 19.6, Prop. 19.7, Rem. 19.8, Rem. 19.9, Prop. 20.4, Prop. 20.5, Rem. 20.7, Cor. 21.1, Prop. 21.2, Cor. 21.3, Cor. 21.4, Cor. 21.5 · newly derived · accepted

Certified weights are non-negative, and their total is at most one whenever the space above the cutoff is non-trivial. On a trivial space above the cutoff the certified-weight condition is vacuous and bounds the total not at all.

The hypothesis on the space above the cutoff is needed: where that block is trivial, (19.1) is vacuous and constrains p not at all.

Lemma 19.2 (Bound on the total certified weight). For any certified weights, $p \geq 0$; and $p \leq 1$ whenever $\mathcal{H}(B_{>N}) \neq \{0\}$.

Uses Def. 19.1 · *Used by* Rem. 19.3, Req. 22.2 · newly derived · accepted

Proof. Summing (19.1) over v and using block completeness $\sum_v M_v^{(>N)} = \mathbb{1}_{B_{>N}}$ gives $\mathbb{1}_{B_{>N}} \succeq p\,\mathbb{1}_{B_{>N}}$, hence $p \leq 1$ whenever $\mathcal{H}(B_{>N}) \neq \{0\}$; and $p \geq 0$ termwise. □

At total weight one the certified weights are exact and the relation still holds, through a map that keeps the block below the cutoff and replaces the block above it by a fixed flag state. Refusing the case is therefore conservative rather than forced by an obstruction.

Lemma 19.2 settles two of the failure conditions of a squashing request (`spec/architecture.md#54-failure-policy`) to the extent it proves and no further: $p_v < 0$ cannot be produced by a valid weight family, and neither can $\sum_v p_v > 1$ *whenever* $\mathcal{H}(B_{>N}) \neq \{0\}$, so on that hypothesis they are impossibilities rather than error paths. Where the block above the cutoff is trivial the second is possible, the certified-weight condition being vacuous there, and the construction of Proposition 19.4 divides by $1 - p$; a total weight of one or more is therefore refused (Requirement 22.2). The remark shows that the boundary case $p = 1$ is no obstruction.

Remark 19.3 (The boundary case of total weight one). At $p = 1$ the certified weights are forced to be exact: summing (19.1) gives $\sum_v \left(M_v^{(>N)} - p_v \mathbb{1}\right) = 0$ with every summand positive semidefinite, so every summand vanishes and $M_v^{(>N)} = p_v \mathbb{1}$ for each v. The relation then still holds, with the target $F_v = M_v^{(\leq N)} \oplus p_v \mathbb{1}_{\mathrm{Flg}}$ — which is (19.3) at $1 - p = 0$ — realised by $\Lambda(\rho) = \Pi_{\leq N}\rho\Pi_{\leq N} \oplus \mathrm{tr}[\Pi_{>N}\rho\Pi_{>N}]\,\mu$ with $\mu = \sum_v p_v \,|v\rangle\langle v|$ a state: that map is a channel, its adjoint sends $X_{\leq N} \oplus X_{\mathrm{Flg}}$ to $X_{\leq N} \oplus \mathrm{tr}[\mu X_{\mathrm{Flg}}]\mathbb{1}_{B_{>N}}$, and evaluating at F_v returns $M_v^{(\leq N)} \oplus p_v \mathbb{1} = M_v$. So $p = 1$ is not an obstruction. Refusing it instead of constructing that map (Requirement 22.2) costs nothing that a declaration reaching $p = 1$ could not also obtain by lowering one weight by the smallest step the declared lattice allows, $2^{-\kappa}$.

Uses Def. 19.1, Lem. 19.2 · newly derived · accepted

With total certified weight below one, the physical POVM squashes to both targets, and the squashers compose exactly through a flag-noise channel. A weight the weight-preserving target encodes holds on every squashed state; under the flag-state target it holds only on reachable ones.

Both targets are reached by measuring the part above the cutoff and preparing a flag, the weight-preserving one after the certified weights have been taken out of the measured POVM. In the weight-preserving target the weights are then an operator inequality on the target itself (part (iii)), which is the difference between the two routes (Remark 19.6).

Proposition 19.4 (Flag-state squashers: existence, chain and encoded weights). Let (p_v) be certified weights with $p < 1$ and define the channels

$$\Lambda^{\mathrm{WP}}(\rho) \coloneqq \Pi_{\le N}\rho\,\Pi_{\le N} \;\oplus\; \sum_v \mathrm{tr}\big[M_v^{\mathrm{WP}}\,\Pi_{>N}\rho\,\Pi_{>N}\big]\;|v\rangle\langle v|_{\mathrm{Flg}}\,, \qquad M_v^{\mathrm{WP}} \coloneqq \frac{M_v^{(>N)} - p_v\,\mathbb{1}_{B_{>N}}}{1-p}, \tag{19.4}$$

$$\Lambda^{\mathrm{FSS}}(\rho) \coloneqq \Pi_{\le N}\rho\,\Pi_{\le N} \;\oplus\; \sum_v \mathrm{tr}\big[M_v^{(>N)}\,\Pi_{>N}\rho\,\Pi_{>N}\big]\;|v\rangle\langle v|_{\mathrm{Flg}}\,, \tag{19.5}$$

$$\widetilde{\Lambda}(\sigma) \coloneqq (1-p)\,\sigma + p\,\Pi_{\le N}\sigma\,\Pi_{\le N} + \mathrm{tr}\big[\Pi_{\mathrm{Flg}}\sigma\big]\,\mu, \qquad \mu \coloneqq \sum_v p_v\,|v\rangle\langle v|_{\mathrm{Flg}}\,. \tag{19.6}$$

Then:

(i) *(existence, with the cutoff projectors transported)*

$$\big(M;\Pi_{\le N},\Pi_{>N}\big) \preceq_{\Lambda^{\mathrm{WP}}} \big(F^{\mathrm{WP}};\Pi_{\le N},\Pi_{\mathrm{Flg}}\big), \qquad \big(M;\Pi_{\le N},\Pi_{>N}\big) \preceq_{\Lambda^{\mathrm{FSS}}} \big(F^{\mathrm{FSS}};\Pi_{\le N},\Pi_{\mathrm{Flg}}\big); \tag{19.7}$$

(ii) *(the chain)* $F^{\mathrm{WP}} \preceq_{\widetilde{\Lambda}} F^{\mathrm{FSS}}$, and the channels compose exactly, $\Lambda^{\mathrm{FSS}} = \widetilde{\Lambda}\circ\Lambda^{\mathrm{WP}}$ on all of $\mathcal{T}(B)$, so that

$$M \;\preceq_{\Lambda^{\mathrm{WP}}}\; F^{\mathrm{WP}} \;\preceq_{\widetilde{\Lambda}}\; F^{\mathrm{FSS}} \;\preceq_{\Lambda_{\mathrm{mp}}}\; P^{\mathrm{Flg}}; \tag{19.8}$$

(iii) *(the encoded constraint)* $F_v^{\mathrm{WP}} \succeq p_v\,\Pi_{\mathrm{Flg}}$ as operators on Q, so $\mathrm{tr}\big[F_v^{\mathrm{WP}}\sigma\big] \ge p_v\,\mathrm{tr}[\Pi_{\mathrm{Flg}}\sigma]$ holds on *every* squashed state, reachable or not. Under F^{FSS} the same inequality is only a pull-back constraint: it holds on every reachable state, since $\Lambda^{\mathrm{FSS}\dagger}\big(F_v^{\mathrm{FSS}} - p_v\Pi_{\mathrm{Flg}}\big) = M_v^{(\le N)} \oplus \big(M_v^{(>N)} - p_v\mathbb{1}_{B_{>N}}\big) \succeq 0$, and it *fails* at $\sigma = |v'\rangle\langle v'|_{\mathrm{Flg}}$ for any $v' \ne v$ with $p_v > 0$;

(iv) *(no reverse relation in general)* $F^{\mathrm{FSS}} \preceq F^{\mathrm{WP}}$ can fail: at $\mathcal{H}(B_{\le N}) = \mathcal{H}(B_{>N}) = \mathbb{C}$, $\mathcal{V} = \{1,2\}$, $M_1 = \frac{1}{2}\oplus 1$, $M_2 = \frac{1}{2}\oplus 0$, $p_1 = \frac{1}{2}$, $p_2 = 0$, one has $\|F_2^{\mathrm{FSS}}\| = 1 > \frac{1}{2} = \|F_2^{\mathrm{WP}}\|$, which Lemma 4.9 (i) forbids.

Uses Lem. 1.2, Cor. 1.7, Lem. 4.9, Lem. 5.1, Lem. 5.3, Cor. 5.8, Def. 19.1 · *Used by* Rem. 19.6, Prop. 19.7, Rem. 19.8, Rem. 19.9, Prop. 20.5, Cor. 21.1, Prop. 21.2, Cor. 21.3, Cor. 21.4, Cor. 21.5, Req. 22.2 · newly derived · accepted · *Oracles* registry_reductions.py

Proof. (i) $\{M_v^{\mathrm{WP}}\}$ *is a POVM on* $B_{>N}$: $M_v^{\mathrm{WP}} \succeq 0$ by (19.1), and $\sum_v M_v^{\mathrm{WP}} = \big(\mathbb{1}_{B_{>N}} - p\,\mathbb{1}_{B_{>N}}\big)/(1-p) = \mathbb{1}_{B_{>N}}$ by block completeness.

Λ^{WP} *is a channel.* It is the map (5.2) of Lemma 5.3 for the pinching instrument $\mathcal{I}_1(\rho) = \Pi_{\le N}\rho\Pi_{\le N}$, $\mathcal{I}_2(\rho) = \Pi_{>N}\rho\Pi_{>N}$ (Corollary 5.8 (ii)), the identity channel on branch 1, and the measure-and-prepare channel Λ_{mp} of the POVM M^{WP} (Lemma 5.1) on branch 2; part (ii) of that lemma gives complete positivity and trace preservation. The POVM that Λ^{WP} squashes is computed directly below rather than read off Lemma 5.3 (iii): for these branch data that part names the coarse-grained target $M_v^{(\le N)} \oplus |v\rangle\langle v|$, which is the flag-state target of the *modified* POVM M^{WP}, whereas F^{WP} is a different target on the same space, and the weights p_v are what the difference records.

The relation. The output of Λ^{WP} is block diagonal, so only the diagonal blocks of an observable contribute to (1.1), and for $X = X_{\le N} \oplus X_{\mathrm{Flg}}$,

$$\Lambda^{\mathrm{WP}\dagger}\big(X_{\le N} \oplus X_{\mathrm{Flg}}\big) = X_{\le N} \;\oplus\; \sum_v \langle v|\, X_{\mathrm{Flg}}\, |v\rangle \;\; M_v^{\mathrm{WP}}. \tag{19.9}$$

Applying (19.9) to F_v^{WP}, whose flag block is $p_v\mathbb{1}_{\mathrm{Flg}} + (1-p)\,|v\rangle\langle v|$ with diagonal entries $\langle v'|\cdot|v'\rangle = p_v + (1-p)\delta_{vv'}$,

$$\begin{aligned}\Lambda^{\mathrm{WP}\dagger}\big(F_v^{\mathrm{WP}}\big) &= M_v^{(\le N)} \;\oplus\; \Big(p_v \sum_{v'} M_{v'}^{\mathrm{WP}} + (1-p)\, M_v^{\mathrm{WP}}\Big) \\ &= M_v^{(\le N)} \;\oplus\; \big(p_v\,\mathbb{1}_{B_{>N}} + M_v^{(>N)} - p_v\,\mathbb{1}_{B_{>N}}\big) \;=\; M_v.\end{aligned}$$

For the transported projectors, $X = \Pi_{\le N}$ means $X_{\le N} = \mathbb{1}$, $X_{\mathrm{Flg}} = 0$, and (19.9) returns $\Pi_{\le N}$; $X = \Pi_{\mathrm{Flg}}$ means $X_{\le N} = 0$, $X_{\mathrm{Flg}} = \mathbb{1}_{\mathrm{Flg}}$, and it returns $0 \oplus \sum_v M_v^{\mathrm{WP}} = \Pi_{>N}$. The statement for Λ^{FSS} is the case $p_v \equiv 0$ of the same computation, with $M_v^{\mathrm{WP}} = M_v^{(>N)}$; note that F^{FSS} is a POVM because $\sum_v F_v^{\mathrm{FSS}} = \mathbb{1}_Q$, and so is F^{WP} because $\sum_v F_v^{\mathrm{WP}} = \mathbb{1}_{B_{\le N}} \oplus \big(p\,\mathbb{1}_{\mathrm{Flg}} + (1-p)\mathbb{1}_{\mathrm{Flg}}\big) = \mathbb{1}_Q$.

(ii) $\widetilde{\Lambda}$ is a channel: it is the convex combination, with weights $1-p$ and p, of the identity and of the channel that pinches onto the two blocks and replaces the flag block by the fixed state μ/p (for $p = 0$, $\widetilde{\Lambda} = \mathrm{id}$); trace preservation is $\operatorname{tr}\widetilde{\Lambda}(\sigma) = (1-p)\operatorname{tr}\sigma + p\,\operatorname{tr}[\Pi_{\le N}\sigma] + \operatorname{tr}[\Pi_{\mathrm{Flg}}\sigma]\,\operatorname{tr}\mu = (1-p)\operatorname{tr}\sigma + p\big(\operatorname{tr}[\Pi_{\le N}\sigma] + \operatorname{tr}[\Pi_{\mathrm{Flg}}\sigma]\big) = \operatorname{tr}\sigma$, using $\operatorname{tr}\mu = p$ and $\Pi_{\le N} + \Pi_{\mathrm{Flg}} = \mathbb{1}_Q$. Its adjoint is $\widetilde{\Lambda}^\dagger(X) = (1-p)X + p\,\Pi_{\le N}X\Pi_{\le N} + \operatorname{tr}[\mu X]\,\Pi_{\mathrm{Flg}}$, so on block observables

$$\widetilde{\Lambda}^\dagger\big(X_{\le N} \oplus X_{\mathrm{Flg}}\big) = X_{\le N} \;\oplus\; \big((1-p)\,X_{\mathrm{Flg}} + \operatorname{tr}[\mu X_{\mathrm{Flg}}]\,\mathbb{1}_{\mathrm{Flg}}\big), \tag{19.10}$$

whence $\widetilde{\Lambda}^{\dagger}(F_v^{\mathrm{FSS}}) = M_v^{(\le N)} \oplus \big((1-p)\,|v\rangle\langle v| + p_v\,\mathbb{1}_{\mathrm{Flg}}\big) = F_v^{\mathrm{WP}}$, which is the relation. For the composition, $\Lambda^{\mathrm{WP}}(\rho)$ is block diagonal, so

$$\begin{aligned}\widetilde{\Lambda}\big(\Lambda^{\mathrm{WP}}(\rho)\big) &= \Pi_{\le N}\rho\Pi_{\le N} \;\oplus\; \sum_v \Big[(1-p)\,\mathrm{tr}\left[M_v^{\mathrm{WP}}\Pi_{>N}\rho\Pi_{>N}\right] + p_v\;\mathrm{tr}\left[\Pi_{>N}\rho\Pi_{>N}\right]\Big]\,|v\rangle\langle v| \\ &= \Pi_{\le N}\rho\Pi_{\le N} \;\oplus\; \sum_v \mathrm{tr}\Big[\big((1-p)M_v^{\mathrm{WP}} + p_v\,\mathbb{1}_{B_{>N}}\big)\,\Pi_{>N}\rho\Pi_{>N}\Big]\,|v\rangle\langle v| \;=\; \Lambda^{\mathrm{FSS}}(\rho),\end{aligned}$$

since $(1-p)M_v^{\mathrm{WP}} + p_v\mathbb{1} = M_v^{(>N)}$; this holds for every ρ, including states with coherences across the cutoff, which both sides discard. The last relation in (19.8) is Lemma 5.1.

(iii) $F_v^{\mathrm{WP}} - p_v\Pi_{\mathrm{Flg}} = M_v^{(\le N)} \oplus (1-p)\,|v\rangle\langle v| \succeq 0$. The pull-back identity is (19.9) at $p_v \equiv 0$ applied to $F_v^{\mathrm{FSS}} - p_v\Pi_{\mathrm{Flg}}$, whose flag block has diagonal entries $\delta_{vv'} - p_v$, giving $M_v^{(\le N)} \oplus \big(M_v^{(>N)} - p_v\mathbb{1}\big)$, which is $\succeq 0$ by (19.1); so Corollary 1.7 (ii) applies on every reachable state. For the failure, $\mathrm{tr}\left[(F_v^{\mathrm{FSS}} - p_v\Pi_{\mathrm{Flg}})\,|v'\rangle\langle v'|\right] = 0 - p_v < 0$ when $v' \neq v$ and $p_v > 0$.

(iv) With the stated data, (19.1) holds ($1 \succeq \frac{1}{2}$ and $0 \succeq 0$) and $p = \frac{1}{2} < 1$. Then $F_2^{\mathrm{FSS}} = \frac{1}{2} \oplus |2\rangle\langle 2|$ has norm 1, while $F_2^{\mathrm{WP}} = \frac{1}{2} \oplus \big(0 \cdot \mathbb{1}_{\mathrm{Flg}} + \frac{1}{2}\,|2\rangle\langle 2|\big)$ has norm $\frac{1}{2}$. If $F^{\mathrm{FSS}} \preceq F^{\mathrm{WP}}$ held, Lemma 4.9 (i) applied with F^{FSS} in the physical role would give $\|F_2^{\mathrm{FSS}}\| \le \|F_2^{\mathrm{WP}}\|$, i.e. $1 \le \frac{1}{2}$. □

The encoded route uses the weight-preserving target and emits no constraint; the licensed route uses the flag-state target and emits the licensed entry.

An artefact carries the device knowledge above the cutoff either inside its target or as a constraint entry beside it; the two choices are named here.

Definition 19.5 (The encoded and the licensed route)**.** The *encoded* route uses the weight-preserving target F^{WP} and emits $\mathfrak{L} = \varnothing$; the *licensed* route uses the flag-state target F^{FSS} and emits the entry of Theorem 6.7.

Uses Def. 19.1 · *Used by* Rem. 19.6, Prop. 21.2 · newly derived · accepted

Both routes are sound, and the encoded one, which needs no attestation, is preferable where it is available. Pairing the licensed entry with the encoded route's channel is unsound, so an artefact takes one route.

Part (iii) of Proposition 19.4 is the whole difference between the two targets. Under F^{WP} the device fact is an operator inequality *on the target*, true of every state the adversary could possibly prepare, so it needs no constraint, no licence and no attestation. Under F^{FSS} the same fact is true only on the image, so it must travel as a declared constraint entry, and something must certify that the adversary cannot leave the image, which is exactly Theorem 6.7. Part (ii) of the proposition says which of the two hands the adversary more: by Corollary 4.2 a cleaner target never helps, and F^{FSS} is

cleaner than F^{WP}, in general strictly so: part (iv) gives data on which the reverse relation fails.

Remark 19.6 (Soundness of the two routes and one route per artefact). Both routes of Definition 19.5 are sound, and an artefact takes exactly one of them (Requirement 22.1). The encoded route is preferable wherever it is available, because it requires no attestation a program cannot check. It is *not* always available: Proposition 6.1 exhibits a device fact, a bound on a *union* of outcomes that must not be merged, that no target classical on the whole of Q can carry and that F^{WP}, one weight per outcome, does not carry, and for which the licensed route is the only route; this is why every artefact carries a constraint slot and a licence slot (Requirement 15.1).

Why one route per artefact. The two routes run on different channels, Λ^{WP} and Λ^{FSS}, and by Remark 3.15 (c) a constraint sound for one channel need not be sound for another. Here it is not: pairing the licensed entry with the encoded route's channel gives, by (19.9),

$$\Lambda^{\mathrm{WP}\dagger}\big(F_o^{\mathrm{FSS}} - p_o\Pi_{\mathrm{Flg}}\big) = M_o^{(\le N)} \;\oplus\; \Big(\tfrac{1}{1-p}\big(M_o^{(>N)} - p_o\mathbb{1}\big) - p_o\mathbb{1}\Big), \tag{19.11}$$

whose flag block is *not* implied positive by (19.1): at the tight end $M_o^{(>N)} = p_o\mathbb{1}$ it equals $-p_o\mathbb{1} \prec 0$ for $p_o > 0$. So that pairing would impose a constraint some reachable state violates, voiding Theorem 3.11, which is exactly the failure Remark 3.14 describes. Since an artefact's payload is single-valued (Requirement 22.1) and (7.1) binds each link's constraint list to *that link's* channel witness, the remedy is structural: one route per artefact, and two requests for a caller who wants both bundles, with no schema in which the unsound pairing can be expressed.

Uses Thm. 3.11, Rem. 3.14, Rem. 3.15, Cor. 4.2, Def. 19.1, Prop. 19.4, Def. 19.5 · *Used by* Req. 22.1 · newly derived · accepted

Certified weights stay certified when lowered but not below zero, and while the total weight is below one the weight-preserving target of the larger weights squashes to that of the smaller ones. By Corollary 4.2 the smaller weights give the cleaner target, which hands the adversary more.

Nothing in Definition 19.1 or Proposition 19.4 requires p_v to be the largest admissible value. The proposition says what lowering the weights does: they stay valid, and the target becomes cleaner, a statement about the pre-order that is proved rather than asserted.

Proposition 19.7 (Monotonicity of the weight-preserving target in its weights)**.** Let $(p_v)_{v\in\mathcal{V}}$ be certified weights for M with total weight p, let $p'_v \in [0, p_v]$ for every $v \in \mathcal{V}$, and write $p' := \sum_v p'_v$. Write $F^{\mathrm{WP}}[p]$ and $F^{\mathrm{WP}}[p']$ for the weight-preserving targets (19.3) built from (p_v) and from (p'_v).

(i) (p'_v) are certified weights for M, and $p' \le p$; in particular, where $p < 1$ the construction of Proposition 19.4 runs unchanged with (p'_v).

(ii) If $p < 1$, put

$$\Psi(\sigma) := \lambda\,\sigma + (1-\lambda)\,\Pi_{\le N}\sigma\Pi_{\le N} + \mathrm{tr}\big[\Pi_{\mathrm{Flg}}\sigma\big]\,\nu, \qquad \lambda := \frac{1-p}{1-p'}, \qquad \nu := \sum_v \frac{p_v - p'_v}{1-p'}\,|v\rangle\langle v|_{\mathrm{Flg}}\,. \tag{19.12}$$

Then Ψ is a channel and $F^{\mathrm{WP}}[p] \preceq_\Psi F^{\mathrm{WP}}[p']$, so by Corollary 4.2 the smaller-weight target is the cleaner one.

Uses Cor. 4.2, Def. 19.1, Prop. 19.4 · *Used by* Rem. 19.8, Prop. 20.4, Rem. 20.10, Req. 22.3 · newly derived · accepted

Proof. (i) By (19.1), $M_v^{(>N)} \succeq p_v \mathbb{1}_{B_{>N}} \succeq p'_v \mathbb{1}_{B_{>N}}$ with $p'_v \ge 0$, for every v, and summing $p'_v \le p_v$ gives $p' \le p$; so where $p < 1$ also $p' < 1$, and Proposition 19.4 applies to (p'_v).

(ii) Ψ *is a channel.* $\lambda \in [0,1]$ and $\nu \succeq 0$ because $p' \le p < 1$ and $p'_v \le p_v$, so Ψ is CP; and with $\mathrm{tr}\,\nu = (p-p')/(1-p')$ and $\Pi_{\le N} + \Pi_{\mathrm{Flg}} = \mathbb{1}_Q$, $\mathrm{tr}\,\Psi(\sigma) = \mathrm{tr}[\Pi_{\le N}\sigma] + \big(\lambda + \mathrm{tr}\,\nu\big)\,\mathrm{tr}[\Pi_{\mathrm{Flg}}\sigma] = \mathrm{tr}\,\sigma$, since $\lambda + \mathrm{tr}\,\nu = \big((1-p) + (p-p')\big)/(1-p') = 1$.

The adjoint. On block observables $\Psi^\dagger(X_{\le N} \oplus X_{\mathrm{Flg}}) = X_{\le N} \oplus \big(\lambda X_{\mathrm{Flg}} + \mathrm{tr}[\nu X_{\mathrm{Flg}}]\mathbb{1}_{\mathrm{Flg}}\big)$, and evaluating at $X_{\mathrm{Flg}} = p'_v\mathbb{1}_{\mathrm{Flg}} + (1-p')\,|v\rangle\langle v|$ gives flag block

$$\Big[\tfrac{(1-p)p'_v}{1-p'} + \tfrac{p'_v(p-p')}{1-p'} + (p_v - p'_v)\Big]\mathbb{1}_{\mathrm{Flg}} + (1-p)\,|v\rangle\langle v| \;=\; p_v\,\mathbb{1}_{\mathrm{Flg}} + (1-p)\,|v\rangle\langle v|\,,$$

because the first two brackets sum to p'_v; with the unchanged $\le N$ block this is $F^{\mathrm{WP}}[p]_v$. The conclusion is Corollary 4.2 applied to $F^{\mathrm{WP}}[p] \preceq_\Psi F^{\mathrm{WP}}[p']$.

Taking $p' \equiv 0$ recovers the flag-noise channel $\widetilde{\Lambda}$ of (19.6), as it must. □

Every smaller non-negative weight is also certified, but it gives a cleaner target, which hands the adversary more (Proposition 19.7). The largest weight the artefact certifies is therefore the one used, and a smaller one stays valid.

Remark 19.8 (Why the largest certified weight is used). Nothing in Definition 19.1 or Proposition 19.4 requires p_v to be the largest admissible value: every $p'_v \in [0, p_v]$ is again a certified weight, and the construction runs unchanged with $p' = \sum_v p'_v \le p$ (Proposition 19.7 (i)). A smaller weight is not free, however: it gives a *cleaner* target, which by Corollary 4.2 hands the adversary more (Proposition 19.7 (ii)). The largest weight the artefact certifies is therefore the best choice, and any smaller one is valid but weaker (Requirement 22.3).

Uses Cor. 4.2, Def. 19.1, Prop. 19.4, Prop. 19.7 · *Used by* Req. 22.3 · newly derived · accepted

The flag-state target is the flag-state squasher of Zhang et al., and the single-bound weight-preserving target that of Nahar et al. Their usual scalar flag-weight constraint is a carried-operator entry stronger than the default licensed one, and only its operator form is ever emitted.

Remark 19.9 (Relation to the published flag-state squashers). F^{FSS} is the flag-state squasher of [10, Thm. 1, Eq. (8)], which states exactly (19.2): $\widetilde{M}_y = M_{y,n\le k} \oplus |y\rangle\langle y|$ for a POVM block diagonal at a photon-number cutoff. The adjoint form is stated there in part: [8, Eqs. (149)–(150)] gives, for that map, the defining identity $\Lambda^\dagger(F_i) = M_i$ and the *flag* half of (19.7), $\Lambda^\dagger(\Pi_{\mathrm{Flg}}) = \Pi_{>N}$, and states no second projector identity. The preserved half follows from it in one step and is written out here rather than attributed: Λ is trace preserving, so $\Lambda^\dagger$ is unital, and $\Pi_{\le N} + \Pi_{\mathrm{Flg}} = \mathbb{1}_Q$ with $\Pi_{\le N} + \Pi_{>N} = \mathbb{1}_B$ give $\Lambda^\dagger(\Pi_{\le N}) = \mathbb{1}_B - \Lambda^\dagger(\Pi_{\mathrm{Flg}}) = \mathbb{1}_B - \Pi_{>N} = \Pi_{\le N}$. This contract proves both halves at Proposition 19.4 (i) and the citation is corroboration, which is why the narrower attribution costs nothing. The single-bound case of the weight-preserving target F^{WP} is the weight-preserving flag-state squasher of [6, Lemma 7, Eqs. (28)–(29)], whose target places f_N on the diagonal of the flag block of one distinguished element and $(1 - f_N)$ on the others; the multi-bound form (19.3) is the symmetric statement with one weight per outcome.

The scalar flag-weight constraint these targets are usually paired with is [6, Eq. (27)], and the comparison with what this module emits runs in the direction opposite to the naive one. Writing $a \coloneqq \lambda_{\min}\big(M_e^{(\le N)}\big)$ and $b \coloneqq \lambda_{\min}\big(M_e^{(>N)}\big)$ for a designated event e (declared device constants, fixed before the run and never recomputed from the declaration, Assumption 13.7), that equation reads, in full, $W \le 1 - p(m \le N) \le \big(p(e) - a\big)/(b - a)$ for the flag weight $W = \mathrm{tr}[\Pi_{\mathrm{Flg}}\sigma]$ and the observed $p(e) = \mathrm{tr}\big[F_e^{\mathrm{FSS}}\sigma\big]$; the right-hand member is the operative one and the only one compared with below. It is a member of the family Corollary 3.13 generates, namely the carried operator $\hat{O}$, which carries the certificate

$$\begin{aligned} \hat{O} &= F_e^{\mathrm{FSS}} - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}, \\ \Lambda^{\mathrm{FSS}\dagger}\big(\hat{O}\big) &= M_e - a\,\Pi_{\le N} - b\,\Pi_{>N} = \big(M_e^{(\le N)} - a\mathbb{1}\big) \oplus \big(M_e^{(>N)} - b\mathbb{1}\big) \;\succeq\; 0 \end{aligned} \tag{19.13}$$

by the definitions of a and b and the transported projectors of (19.7). The emitted constant is 0, with no shift (Corollary 11.3 (iii)). Corollary 1.7 (ii) gives $p(e) - a(1 - W) - bW \ge 0$, which rearranges to Eq. (27) when $b > a$ (the published division is by $b - a$; at $b \le a$ it reverses or is undefined, and only the operator form above is used here). It is *not* the member Proposition 19.4 (iii) supplies and not the one Corollary 21.1 emits: those use $a = 0$, where the same corollary gives only $p(e) - bW \ge 0$.

Which of the two is stronger is settled at the level of the *constraint sets*, not by comparing the two caps on W: since $\hat{O} = \hat{O}\big|_{a=0} - a\,\Pi_{\le N}$ with $a \ge 0$ and $\Pi_{\le N} \succeq 0$, one has $\mathrm{tr}\Big[\hat{O}|_{a=0}\,\sigma\Big] \ge \mathrm{tr}\Big[\hat{O}\,\sigma\Big]$ for every $\sigma \succeq 0$, so $\{\sigma : \mathrm{tr}\Big[\hat{O}\sigma\Big] \ge 0\} \subseteq \{\sigma : \mathrm{tr}\Big[\hat{O}|_{a=0}\sigma\Big] \ge 0\}$: the $a > 0$ entry cuts a set no larger and is at least as strong a constraint, unconditionally. The implication therefore runs from Eq. (27) to the $a = 0$ form and not the other way. (The comparison of the two *caps* $\big(p(e) - a\big)/(b - a)$ and $p(e)/b$ is not the right one and does not run in a fixed direction: the former is

the smaller exactly when $p(e) < b$, because $p(e)$ is itself a function of σ rather than a fixed number, so the two inequalities cut the state set differently.)

Two consequences follow. *(a)* The default licensed entry of Corollary 21.1 is weaker than standard practice, not stronger; a declaration may request the entry (19.13) as a carried operator with declared constants a and b, which need not be the least eigenvalues above. Its certificate is then not taken from the request: it is decided on the emitted numbers, where every emitted block of e below the cutoff clears a and b is at most the certified weight p_e (Corollary 21.4), and the request is refused where either fails. Both constants must be declared on the lattice of (9.1), as every declared constant must: an a or b off the lattice puts the requested operator's entries off the lattice, and the request is refused (Requirement 16.9). Smaller lattice values of a and b keep both conditions. *(b)* The *cap* in the displayed form is never emitted: its right-hand side contains the run's observed $p(e)$, whereas Assumption 13.7 admits only constants declared before the run, and an entry with such a constant is refused (Requirement 15.3). The admissible object is the operator entry (19.13) with $\bowtie = \succeq$ and $c = 0$, of which the cap is a run-time consequence and not a declaration. Every statement of this remark is proved in the claims it names (Definition 19.1, Proposition 19.4, Corollary 21.1, Corollary 21.4 and Corollary 11.3); the citations record where a reader finds the correspondence.

Uses Cor. 1.7, Cor. 3.13, Def. 19.1, Prop. 19.4 · *Assumes* A-CONST · newly derived · accepted

20 The weight route

The receiver supplies coarse-grained blocks, a dark-count channel on raw click patterns and per-setting weights for raw outcome sets. The dark-count channel can act on the coarse blocks only if it descends through the coarse-graining (Lemma 20.1); the joint family of the private setting is then formed, and each certified weight is routed to its bounded letter, which must be absorbing, sharpened by the dark counts and rounded down onto the lattice (Proposition 20.4). Blocks supplied with the dark counts already composed form the same instance wherever no snap intervenes (Proposition 20.5). The remarks show why the constancy, absorbing and fibre conditions cannot be dropped and which receiver claims are imported.

Imports foundations/def:exact-reference, foundations/def:snap, receiver/cor:unconditioned-family, receiver/def:setting-family, receiver/def:slot, receiver/detector-imperfections/def:darkadm, receiver/detector-imperfections/def:darktol, receiver/detector-imperfections/prop:efficiency-normalization, receiver/detector-imperfections/thm:dark-counts, receiver/prop:slotvalid, receiver/thm:click-povm, receiver/thm:coarse-graining, receiver/thm:emitted-bound

R descends through a coarse-graining when each fibre sum depends on a column only through its letter; then R and the coarse-graining commute via a column-stochastic R^cg. A monotone R descends if the coarse-graining is injective off the multiclick set and maps exactly that set to one letter.

The receiver emits its blocks on the coarse alphabet $\mathcal{Y}$ but supplies its dark-count channel on the raw click patterns $2^{[K]}$. This lemma says when the channel can be moved onto the coarse alphabet through the declared coarse-graining ϖ, and gives the sufficient condition the canonical threshold-detector declaration meets; the constancy clause of that condition cannot be dropped (Remark 20.2).

Lemma 20.1 (Descent of a dark-count matrix through a coarse-graining)**.** Let R be column-stochastic on $2^{[K]}$ and $\varpi\colon 2^{[K]} \to \mathcal{Y}$ surjective. Say that R *descends through* ϖ if the quantity $P(y \mid S') \coloneqq \sum_{S \in \varpi^{-1}(y)} R_{S|S'}$ depends on S' only through $\varpi(S')$; the resulting matrix $R^{\varpi}_{y|y'} \coloneqq P(y \mid S')$ for any $S' \in \varpi^{-1}(y')$ is then well defined and column-stochastic. In that case, for every family of operators $\{X_S\}$,

$$\sum_{S \in \varpi^{-1}(y)} \sum_{S'} R_{S|S'} X_{S'} \;=\; \sum_{y' \in \mathcal{Y}} R^{\varpi}_{y|y'} \sum_{S' \in \varpi^{-1}(y')} X_{S'}, \tag{20.1}$$

so applying R and then coarse-graining agrees with coarse-graining and then applying R^{ϖ}. A sufficient condition, which the canonical threshold-detector declaration satisfies, is that R be *monotone* ($R_{S|S'} > 0 \Rightarrow S' \subseteq S$) and that, with $\mathsf{MC} \coloneqq \{S : |S| \geq 2\}$,

$$\varpi \text{ is injective on } 2^{[K]} \setminus \mathsf{MC}, \qquad \varpi \text{ is } \textit{constant} \text{ on } \mathsf{MC}, \text{ say } \varpi(\mathsf{MC}) = \{y_{\mathrm{mc}}\}, \qquad \varpi^{-1}(y_{\mathrm{mc}}) = \mathsf{MC}; \tag{20.2}$$

the case $\varpi = \mathrm{id}$ is trivial.

On declared data both the hypothesis and the conclusion are read exactly. The declared R is exactly column-stochastic, and the descent condition, that $P(y|S') = P(y|S'')$ for every y and every pair S', S'' with $\varpi(S') = \varpi(S'')$, is an identity between the dyadic rationals the declaration carries and is decided as one (Definition 9.2). The representative is declared, not chosen: $R^{\varpi}_{y|y'} \coloneqq P(y|S'_{y'})$ with $S'_{y'}$ the first element of $\varpi^{-1}(y')$ in the declared order of $2^{[K]}$, so the descended matrix is a determinate object and not one of several. Column-stochasticity of R^{ϖ} needs no second decision: the identity $\sum_y R^{\varpi}_{y|y'} = \sum_S R_{S|S'}$ of the proof regroups one finite sum, and in the exact substrate a regrouped sum *is* the sum it regroups. So (20.1) holds of the declared numbers as an exact identity.

Used by Rem. 20.2, Def. 20.3, Prop. 20.4, Prop. 20.5, Rem. 20.9, Rem. 20.11, Rem. 20.12, Prop. 21.2, Cor. 21.3, Cor. 21.5, Req. 22.4, Req. 22.7 · newly derived · accepted

Proof. Well-definedness is the stated hypothesis, and $\sum_y R^{\varpi}_{y|y'} = \sum_y P(y|S') = \sum_S R_{S|S'} = 1$ gives column-stochasticity. For (20.1), exchange the two

sums on the left and group the columns by $\varpi(S')$: $\sum_{S'} P(y|S')X_{S'} = \sum_{y'} \sum_{S' \in \varpi^{-1}(y')} R^{\varpi}_{y|y'} X_{S'}$.

For the sufficient condition (20.2): if $S' \in \mathsf{MC}$ and $R_{S|S'} > 0$ then $S \supseteq S'$ by monotonicity, so $|S| \geq 2$ and $S \in \mathsf{MC}$; hence the whole column S' is supported on MC, which by the constancy and preimage clauses is exactly $\varpi^{-1}(y_{\mathrm{mc}})$. Column-stochasticity then gives $P(y_{\mathrm{mc}}|S') = \sum_{S \in \mathsf{MC}} R_{S|S'} = 1$ and $P(y|S') = 0$ for $y \neq y_{\mathrm{mc}}$, the same values for *every* $S' \in \mathsf{MC}$. This is where constancy of ϖ on MC is needed: without it, MC would split across several letters and the individual $P(y|S')$ could differ inside MC (Remark 20.2). If $S' \notin \mathsf{MC}$ then $\varpi(S')$ determines S': by the preimage clause $\varpi(S') \neq y_{\mathrm{mc}}$, so by the constancy clause no pattern of MC has the letter $\varpi(S')$, and ϖ is injective on $2^{[K]} \setminus \mathsf{MC}$. So $P(y|S')$ depends on S' only through $\varpi(S')$ in both cases. □

With four ports and independent dark counts, one coarse-graining injective off the multiclick set that splits that set over two letters has two columns in one fibre with different fibre sums, so R^cg is undefined. The constancy clause of Lemma 20.1 cannot be left unchecked.

Remark 20.2 (Necessity of the constancy clause). The constancy clause of (20.2) cannot be dropped: without it the conclusion of Lemma 20.1 fails. Take $K = 4$, independent per-port dark counts of common probability $p \in (0,1)$ (the independent product form with the same probability at every port), and let ϖ send $\{1,2\}, \{1,2,4\}, \{1,2,3,4\} \mapsto y_a$, every other multiclick pattern to y_b, and be injective with distinct fresh letters off MC. Then ϖ is injective on $2^{[K]} \setminus \mathsf{MC}$ and $\varpi^{-1}(\{y_a, y_b\}) = \mathsf{MC}$, and R is monotone, but

$$P\big(y_a \,\big|\, \{1,2\}\big) = (1-p)^2 + p(1-p) + p^2 = 1 - p + p^2, \qquad P\big(y_a \,\big|\, \{1,2,4\}\big) = (1-p) + p = 1,$$

two columns with the same ϖ-image and different values, so R^{ϖ} is not well defined. The first count is over the supersets of $\{1,2\}$ lying in $\varpi^{-1}(y_a)$, namely $\{1,2\}$ itself, $\{1,2,4\}$ and $\{1,2,3,4\}$; the second over the supersets of $\{1,2,4\}$ there, namely itself and $\{1,2,3,4\}$. The canonical declaration maps all of MC to the single letter mc and is unaffected; what the counterexample rules out is testing the other two clauses of (20.2) in place of the descent property itself (Requirement 22.4).

Uses Lem. 20.1 · *Used by* Req. 22.4 · newly derived · accepted

The receiver's blocks pass through the descended dark-count channel, unless supplied already composed, and join the private setting. Each joint outcome at a bounded letter gets its class weight times its setting's certified weight, sharpened by the dark counts and rounded down onto the lattice.

One granularity fact shapes this definition. The supplier emits the *coarse-grained* blocks $M^{(=m)}_{y|t}$, one per emitted letter $y \in \mathcal{Y}$, together with a dark-count channel R that acts on *raw* click patterns $2^{[K]}$ and weights $p_i^{(t)}$, one per setting, certified for *raw* outcome sets $\mathcal{O}_i \subseteq 2^{[K]}$. A consumer holding only the coarse blocks cannot in general apply R to them, because $S \mapsto \varpi(S)$ is many-to-one and R need not respect it. This definition therefore requires

R to *descend* through ϖ (Lemma 20.1; decided exactly, Requirement 22.4). The case $\varpi = \mathrm{id}$, in which the emitted blocks are the raw ones, satisfies the condition trivially. Remarks 20.7 and 20.8 show that neither the absorbing condition nor the fibre condition below can be dropped (without the absorbing condition the bound (20.6) for the weights of (20.4) fails in general, in the *unsafe* direction), and Remark 20.10 shows why the weight at (y_i, t) is formed from the weight of its own setting, and why a slot carrying one weight at every setting, as every one-setting slot does, gives the numbers of the uniform route, which consumes each entry's minimum over the settings. The order in which steps 1–2 precede the first squashing link is that of Remark 4.5 and Requirement 14.6; the column-stochasticity of R^{ϖ}, the fibre condition and the absorbing condition are decided by Requirements 22.7, 22.6 and 22.5; and the downward rounding of the routed weights is that of Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction), which rounds every emitted lower bound down. A declaring party may compose the dark counts into the receiver's model before it snaps the blocks, so that the snap is made on a model whose elements are positive definite where the receiver module's conditions for that hold (loss in the receiver and a dark-count probability strictly between zero and one in every port, Proposition "Loss and dark counts make every dark-composed element definite" (receiver/detector-imperfections/prop:noise-floor)); it then supplies the blocks already composed, and possibly snapped. The dark-composed form below admits them, and Proposition 20.5 shows that it forms the same instance as composing afterwards wherever no snap intervenes.

Definition 20.3 (The weight route)**.** Let a receiver artifact supply, for each setting $t \in \mathcal{T}_{\mathrm{set}}$, preserved photon-number blocks, $m \le N$, in one of the two input forms below, for a conditional POVM $\{M_{y|t}\}_{y\in\mathcal{Y}}$ on C block diagonal at the cutoff; a dark-count matrix R on $2^{[K]}$ that descends through the declared coarse-graining ϖ with descended matrix R^{ϖ} (Lemma 20.1); class weights $\{q_t\}_{t\in\mathcal{T}_{\mathrm{set}}}$ as in Definition 4.4; and a slot $\{(\mathcal{O}_i, (p_i^{(t)})_{t\in\mathcal{T}_{\mathrm{set}}})\}_{i=1}^{r}$ of certified high-photon weights, one per entry and per setting of the same $\mathcal{T}_{\mathrm{set}}$; every declared entry and scalar lies on the declared lattice (9.1), and every one of the supplier's facts about them is held under Assumption 13.3 (Proposition 20.4 (iv)). The *weight route* is the following three steps, in this order, each with the claim that licenses it. Steps 1–2 are classical post-processing of the physical outcome and precede the first squashing link.

1. *Dark counts, on the emitted alphabet.* Replace the supplied conditional POVM by

$$\widetilde{M}_{y|t} \;:=\; \sum_{y'\in\mathcal{Y}} R^{\varpi}_{y|y'}\, M_{y'|t}, \qquad y \in \mathcal{Y},\ t \in \mathcal{T}_{\mathrm{set}}. \tag{20.3}$$

   This is post-processing by a column-stochastic matrix (Lemma 4.3 (i)), so the result is a conditional POVM at each t with the same photon-

number block structure, and Lemma 20.1 shows it equals the coarse-graining of the raw-level composition.

2. *Joint alphabet.* Form the joint family of Definition 4.4 (b), $M_{(y,t)} \coloneqq q_t \widetilde{M}_{y|t}$ on the joint alphabet $\mathcal{V} = \mathcal{Y} \times \mathcal{T}_{\text{set}}$; this is a POVM by (4.2) (Lemma 4.3 (ii); the weights are the measuring party's own, Assumption 13.5), block diagonal at the cutoff, and the class-marginal identity (4.3) holds of it block by block. Its preserved blocks $M_{(y,t)}^{(=m)}$, $m \le N$, are the *post-processed datum*.

3. *The instance.* The post-processed joint family, with the routed weights below, is the datum, the hypotheses and the target data of the *flat instance* (Proposition 21.2), and its emission is that instance's.

The two input forms. The supplied preserved blocks are declared in one of two forms. In the *bare form* they are the blocks $M_{y|t}^{(=m)}$, $m \le N$, of the conditional POVM above, and step 1 acts on them. In the *dark-composed form* they are a declared family $\{\widetilde{M}_{y|t}^{(=m)}\}_{y \in \mathcal{Y}}$, $m \le N$, per setting t, each block Hermitian, positive semidefinite and complete on its block, which the model takes below the cutoff after the dark counts: in this form the family $\widetilde{M}_{y|t}$ of step 1 is defined as these blocks below the cutoff and by (20.3) above it, step 1 is not applied to the supplied blocks, and the post-processed datum is formed from them by step 2 alone. Where no snap intervened between composing and supplying, this form is the bare form composed in advance (Proposition 20.5 (ii)); where the declaring party snapped the composed blocks, they are its snapped family, and this definition asserts of them no relation to a bare family. In either form R^{ϖ} is formed and its conditions are decided as below, (20.3) defines the family above the cutoff, $\widetilde{M}_{y|t}^{(>N)} = \sum_{y'} R^{\varpi}_{y|y'} M_{y'|t}^{(>N)}$, and the routed weights below read R^{ϖ} as they stand. In either form $\{\widetilde{M}_{y|t}\}_y$ is a POVM at each t, block diagonal at the cutoff: below it by step 1 in the bare form and by the declared properties of the supplied blocks in the dark-composed form, and above it by step 1.

Exactness of steps 1 and 2. The two steps are written on the declared numbers, and their POVM conclusions hold *exactly* of them. The declared R^{ϖ} is exactly column-stochastic with non-negative entries, decided on entry, and $\{q_t\}$ is exactly a probability distribution, as Definition 4.4 requires of class weights; and the combination is formed in the exact substrate of Lemma 9.3 (Definition 9.2), so $\{\widetilde{M}_{y|t}\}_y$ and $\{M_{(y,t)}\}$ are POVMs and every claim of this submodule reads them as such. What the two steps can nevertheless do is leave the lattice: a product of two dyadic rationals adds significand widths, so the post-processed blocks $M_{(y,t)}^{(=m)}$ need not be binary64 numbers even though every factor is. That is the one place in this submodule where exact arithmetic outruns the representation; it is where the emission

normaliser of Definition 11.1 acts, and it acts on numbers this module computed and is about to declare, never on numbers it was handed.

The routed weights. They follow the same route, one bounded letter at a time. Fix an entry i whose bounded set is exactly a fibre, $\mathcal{O}_i = \varpi^{-1}(y_i)$, and whose letter is absorbing, $R^{\varpi}_{y_i|y_i} = 1$; both are declared input validity conditions, and both are decided exactly. Write

$$\begin{aligned} \hat{p}_{(y_i,t)} &\coloneqq q_t\Big((1 - r_{\min,i})p_i^{(t)} + r_{\min,i}\Big), \qquad p_{(y_i,t)} \coloneqq \big\lfloor \hat{p}_{(y_i,t)} \big\rfloor_\kappa, \\ r_{\min,i} &\coloneqq \min_{y' \neq y_i} R^{\varpi}_{y_i|y'} \quad \big(\coloneqq 0 \text{ when that minimum is over the empty set}\big), \end{aligned} \tag{20.4}$$

with $p_{(y,t)} \coloneqq 0$ for every other (y, t); the family $(p_v)_{v\in\mathcal{V}}$ is the *routed weights*, and $p \coloneqq \sum_{v\in\mathcal{V}} p_v$. Here

$$\big\lfloor x \big\rfloor_\kappa \coloneqq 2^{-\kappa}\big\lfloor 2^\kappa x \big\rfloor \tag{20.5}$$

is the largest integer multiple of $2^{-\kappa}$ that is at most x: the *downward* rounding onto the declared lattice, computed exactly on the rational $\hat{p}_{(y_i,t)}$ and equal to it whenever it is already a lattice number. The routed weights are *certified* when the element $M_{(y_i,t)}$ of step 2 satisfies, on every channel block above the cutoff,

$$M^{(>N)}_{(y_i,t)} \succeq p_{(y_i,t)}\,\mathbb{1} \qquad \text{for every } i \text{ and every } t \in \mathcal{T}_{\text{set}}, \tag{20.6}$$

which Proposition 20.4 proves from the declared input validity conditions together with the absorbing hypothesis those conditions impose, and from nothing else.

The direction of the rounding. The emitted weight is $p_{(y_i,t)}$ and not $\hat{p}_{(y_i,t)}$, and the direction of that rounding is the whole of what makes it safe: it is the one emitted number of this submodule that a downstream inequality consumes as a *bound*, the certified-weight inequality of Definition 19.1 is preserved when a certified weight is lowered and, when one is raised, is no longer proved and can be false, and $\lfloor \cdot \rfloor_\kappa$ only lowers (Proposition 20.4; Proposition 19.7 (i)). Four features of the route are relied on, and none of them may be dropped. *(a)* $r_{\min}$ is indexed by the bounded letter: it is a property of row y_i of R^{ϖ} read across the columns $y' \neq y_i$, and one entry's value is not another's. *(b)* The absorbing condition is a hypothesis, not a consequence: without it the bound (20.6) for the weights of (20.4) is no longer proved. *(c)* A non-zero $r_{\min,i}$ can occur only at $r = 1$ (Corollary 20.6), so for $r \geq 2$ *every* entry contributes the un-improved weight $\lfloor q_t p_i^{(t)} \rfloor_\kappa$, which is sound, merely not an improvement. *(d)* $p_{(y_i,t)} \geq 0$ needs no clamp: $\hat{p}_{(y_i,t)}$ is $q_t > 0$ times $(1 - r_{\min,i})p_i^{(t)} + r_{\min,i}$, a convex combination of $p_i^{(t)} \geq 0$ and 1, since $r_{\min,i} \in [0, 1]$ as an entry of the column-stochastic R^{ϖ} or as 0; and the largest lattice number at most a non-negative rational is itself non-negative.

Uses Lem. 4.3, Def. 4.4, Def. 9.1, Def. 9.2, Lem. 9.3, Lem. 20.1 · *Used by* Def. 19.1, Prop. 20.4, Prop. 20.5, Cor. 20.6, Rem. 20.9, Rem. 20.10, Rem. 20.11, Rem. 20.12, Prop. 21.2, Cor. 21.3, Cor. 21.5, Cor. 21.6, Req. 22.8 · newly derived · accepted

At absorbing bounded letters the receiver's per-setting high-photon bound survives the dark counts, sharpened by r_min, on every block of the joint family above the cutoff. Rounding it down onto the lattice keeps it certified, so the routed weights are certified weights for the flag-state targets.

The routed weight of Definition 20.3 is certified for two reasons: the receiver's per-setting bound survives the dark counts at an absorbing letter, and the emitted weight is rounded down. Had that rounding been to nearest, or upwards, the second step would fail: an emitted weight above $\hat{p}_{(y_i,t)}$ leaves (19.1) unproved, and it can make it false, the failure Assumption 13.3's ledger row calls a security break. Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) names the same direction for every emitted lower bound. The input conditions part (iv) leaves to the declaration are decided by Requirements 22.7, 22.4, 22.6 and 22.5.

Proposition 20.4 (Certified weights routed from the receiver artifact)**.** Let $\{M_{y|t}\}_{y\in\mathcal{Y}}$ be, for each $t \in \mathcal{T}_{\mathrm{set}}$, a POVM on C block diagonal at the cutoff, and let R descend through ϖ with descended matrix R^{ϖ} (Lemma 20.1). Let $\{q_t\}$ be a probability distribution, and let $\{(\mathcal{O}_i, (p_i^{(t)})_{t\in\mathcal{T}_{\mathrm{set}}})\}_{i=1}^{r}$ be the declared slot, with $\mathcal{O}_i = \varpi^{-1}(y_i)$ for pairwise distinct letters $y_1, \ldots, y_r$, with $0 \le p_i^{(t)} \le 1$, and with the per-setting guarantee $M^{(>N)}_{y_i|t} \succeq p_i^{(t)}\,\mathbb{1}$ for every $t \in \mathcal{T}_{\mathrm{set}}$ and every i. Suppose each bounded letter is *absorbing*,

$$R^{\varpi}_{y_i|y_i} = 1 \qquad \text{for every } i = 1, \ldots, r, \tag{absorbing}$$

and put $r_{\min,i} := \min_{y'\neq y_i} R^{\varpi}_{y_i|y'}$, with the convention $r_{\min,i} := 0$ when the minimum is over the empty set (that is, when $\mathcal{Y} = \{y_i\}$), the conservative value, which reduces (20.7) to the un-improved bound. Then:

(i) The joint-alphabet POVM $M_{(y,t)} := q_t \sum_{y'} R^{\varpi}_{y|y'} M_{y'|t}$ of Definitions 4.4 and 20.3 satisfies, for every i and every $t \in \mathcal{T}_{\mathrm{set}}$,

$$M^{(>N)}_{(y_i,t)} \;\succeq\; q_t\Big(\big(1 - r_{\min,i}\big)\, p_i^{(t)} + r_{\min,i}\Big)\,\mathbb{1} \;=\; \hat{p}_{(y_i,t)}\,\mathbb{1}. \tag{20.7}$$

A slot whose weights do not depend on the setting, $p_i^{(t)} = p_i$ for every t (the t-uniform form, which every one-setting slot has), is the special case, and there (20.7) is the bound in p_i.

(ii) The *emitted* weight of (20.4) is certified, because the rounding runs downwards. Every number on the right of (20.7) lies on the declared lattice: q_t and $p_i^{(t)}$ by declaration, and $r_{\min,i}$ as an entry of R^{ϖ}, which is a sum of declared entries and lies in $[0, 1]$. So $\hat{p}_{(y_i,t)}$ is an exact

element of $\mathbb{F}$ formed from them; and $p_{(y_i,t)} = \lfloor \hat{p}_{(y_i,t)} \rfloor_\kappa \le \hat{p}_{(y_i,t)}$ by (20.5), whence

$$M^{(>N)}_{(y_i,t)} \succeq p_{(y_i,t)}\,\mathbb{1},$$

which is (20.6); so the emitted (p_v) are certified weights in the sense of Definition 19.1 and they lie on the declared lattice.

(iii) Under the sufficient condition (20.2) of Lemma 20.1 and with the single entry $\mathcal{O}_1 = \mathsf{MC}$, (absorbing) holds at $y_1 = y_{\mathrm{mc}}$, and its per-setting hypothesis is the supplier's slot guarantee $\sum_{S\in\mathsf{MC}} M^{(>N)}_{S|t} \succeq p^{(t)}_1\,\mathbb{1}$ at each t, read through $\varpi^{-1}(y_{\mathrm{mc}}) = \mathsf{MC}$.

(iv) On the emission path the hypotheses about the supplier's objects hold under Assumption 13.3: the conditional families, and their block diagonality in the photon number, block by block and so in particular at the cutoff, are at every setting those of the coarse-grained elements of the setting family on which the supplier states its weights, by item (a) of Definition "Setting family" (receiver/def:setting-family) and Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining); for a receiver without probe inputs that family is the one-setting family of Corollary "The unconditioned receiver is a setting family" (receiver/cor:unconditioned-family), and the conditional families and their block diagonality are given by the receiver's Theorem "Click-pattern POVM" (receiver/thm:click-povm) (iii) and Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining); the dark-count matrix R, and the block diagonality in the photon number of the elements with the dark counts composed, by Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm) and Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts); so every element of the joint family of (i), and of the post-processed joint family that Definition 20.3 forms in either of its forms, is block diagonal in the photon number in every block, below the cutoff as above it; and the slot, its per-setting guarantee (the first conclusion of Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound)) and $0 \le p^{(t)}_i \le 1$, $\sum_i p^{(t)}_i \le 1$ at every t by Definition "The strengthening slot" (receiver/def:slot), Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound) and Proposition "Range of the strengthening-slot weights" (receiver/prop:slotvalid). The other hypotheses are not supplier facts: that R is column-stochastic and descends through ϖ, that $\mathcal{O}_i = \varpi^{-1}(y_i)$ and (absorbing) are decided exactly on the declared data, a declaration failing one being refused, and $\{q_t\}$ are the declared class weights (Definition 4.4).

Uses Def. 4.4, Def. 9.2, Lem. 9.3, Def. 19.1, Prop. 19.7, Lem. 20.1, Def. 20.3, receiver/cor:unconditioned-family, receiver/def:setting-family, receiver/def:slot, receiver/detector-imperfections/def:darkadm, receiver/detector-imperfections/thm:dark-counts, receiver/prop:slotvalid, receiver/thm:click-povm, receiver/thm:coarse-graining, receiver/thm:emitted-bound · *Assumes* A-RECV · *Used by* Cor. 20.6, Rem. 20.7, Rem. 20.8, Rem. 20.9, Rem. 20.10, Rem. 20.11, Rem. 20.12, Prop. 21.2, Cor. 21.3, Req. 22.5, Req. 22.8 · newly derived · accepted · *Oracles* registry_reductions.py

Proof. (i) Fix i and t. Split the sum over columns and use (absorbing) at that letter:

$$\sum_{y'} R^{\varpi}_{y_i|y'} M_{y'|t} = M_{y_i|t} + \sum_{y' \neq y_i} R^{\varpi}_{y_i|y'} M_{y'|t} \succeq M_{y_i|t} + r_{\min,i} \sum_{y' \neq y_i} M_{y'|t}$$
$$= (1 - r_{\min,i})\, M_{y_i|t} + r_{\min,i}\, \mathbb{1},$$

using $M_{y'|t} \succeq 0$ and $R^{\varpi}_{y_i|y'} \geq r_{\min,i}$ in the inequality and POVM completeness in the last step. All operators are block diagonal at the cutoff, so the inequality restricts to the $> N$ block, where $M^{(>N)}_{y_i|t} \succeq p_i^{(t)}\, \mathbb{1}$ and $1 - r_{\min,i} \geq 0$, every entry of R^{ϖ} lying in $[0,1]$ by non-negativity and unit columns (Lemma 20.1). Multiplying by $q_t \geq 0$ gives (20.7). Nothing in the argument couples the entries: each i uses only its own row of R^{ϖ} and its own weight, which is why $r_{\min}$ must be indexed (Corollary 20.6 says what that costs).

(ii) $\hat{p}_{(y_i,t)}$ is formed from q_t, $p_i^{(t)}$ and $r_{\min,i}$ by finitely many field operations in $\mathbb{F}$ (Lemma 9.3, Definition 9.2), so it *is* the real value of (20.4)'s expression and (20.7) is a statement about that value. Monotonicity of $\succeq$ in a scalar multiple of the identity, with $p_{(y_i,t)} \leq \hat{p}_{(y_i,t)}$, gives (20.6); and $p_{(y_i,t)} \geq 0$ because $\hat{p}_{(y_i,t)} \geq 0$ and $\lfloor \cdot \rfloor_{\kappa}$ carries a non-negative rational to a non-negative lattice value. No rounding-error budget is tracked and none is needed: the inequality is exact, and the only inexactness anywhere near it is a rounding that weakens it (Proposition 19.7 (i)).

(iii) The proof of Lemma 20.1 under (20.2) gives $P(y_{\mathrm{mc}}|S') = 1$ for every $S' \in \mathsf{MC}$, hence $R^{\varpi}_{y_{\mathrm{mc}}|y_{\mathrm{mc}}} = 1$; and $M_{y_{\mathrm{mc}}|t} = \sum_{S \in \varpi^{-1}(y_{\mathrm{mc}})} M_{S|t} = \sum_{S \in \mathsf{MC}} M_{S|t}$ by the definition of the coarse-grained POVM.

(iv) Each receiver claim cited there states the hypothesis it is cited for, of the objects the receiver artifact supplies, and Assumption 13.3 makes the artifact's facts hold of the numbers as declared. The supplier states its weights on a setting family (Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound)), whose elements at every setting commute with the photon number by item (a) of Definition "Setting family" (receiver/def:setting-family), and a coarse-grained element is a sum of them (Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining)); for a receiver without probe inputs Corollary "The unconditioned receiver is a setting family" (receiver/cor:unconditioned-family) identifies that family as the click-pattern POVM of Theorem "Click-pattern POVM"

(receiver/thm:click-povm), block diagonal by its item (iii). The remaining hypotheses are the input decisions and the declaration named there. An element of the joint family of (i), and an element of the post-processed joint family in either form, is a combination with non-negative coefficients (q_t and the entries of R^{ϖ}, or q_t alone) of the receiver's elements, conditional or with the dark counts composed, which the claims cited give block diagonal in the photon number; and a sum of operators each block diagonal in the photon number is again block diagonal in it, block by block. □

Supplying the preserved blocks already composed with the dark counts forms the same flat instance, and so the same emission, as composing them here, wherever no snap moves a block. A snap made before or after the composition changes only the data sectors, and the composed reference is a reference.

The dark counts are a trusted classical channel, and composing them before or after the supplier's blocks are handed over is the same operation; what the order changes is only where a snap, if one is needed, is made. A declaring party snaps the composed model when it wants the noise floor of the dark counts in place before its snap. This proposition says that the flat instance does not notice the order (part (ii)) and what changes when a snap intervenes (part (iii)), the composed reference being what the receiver hands beside its snap of the composed blocks. In the dark-composed form the dark-count channel is not applied to the preserved blocks again, but it is still read: it defines the family above the cutoff, and the routed weights read its minima and its absorbing letters, as in the bare form.

Proposition 20.5 (Preserved blocks supplied with the dark counts already composed)**.** Let the data of Definition 20.3 be given in the bare form, with supplied preserved blocks $B^{(m)}_{y|t}$, $m \le N$, and put

$$\widetilde{B}^{(m)}_{y|t} \;:=\; \sum_{y' \in \mathcal{Y}} R^{\varpi}_{y|y'}\, B^{(m)}_{y'|t}. \tag{20.8}$$

(i) For every t and m, $\{\widetilde{B}^{(m)}_{y|t}\}_{y\in\mathcal{Y}}$ is a family of Hermitian positive semidefinite matrices complete on its block.

(ii) The flat instance of Proposition 21.2 formed from the blocks $\widetilde{B}$ in the dark-composed form, and the one formed from the blocks B in the bare form, have the same handed datum, $A^{(m)}_{(y,t)} = q_t \widetilde{B}^{(m)}_{y|t}$, the same hypotheses (H), model map, routed weights, scalar values, channel, transported set and entries; so they are one reduction instance, and the emission normalizer gives them the same declared datum, declared model and record.

(iii) Let $\{G^{(m)}_{y|t}\}_{y\in\mathcal{Y}}$ be a reference of the bare blocks of setting t with constant 1, read through the entrywise enclosures $\big(C^{(m)}_{y|t}, r^{(m)}_{y|t}\big)$ (Definition "Reference of a measurement family, exact or enclosed" (foundations/

def:exact-reference)), and put $\widetilde{G}^{(m)}_{y|t} \coloneqq \sum_{y'} R^{\varpi}_{y|y'} G^{(m)}_{y'|t}$. Then $\widetilde{G}$ is a reference with constant 1, read through the enclosures $\big(\sum_{y'} R^{\varpi}_{y|y'} C^{(m)}_{y'|t}, \sum_{y'} R^{\varpi}_{y|y'} r^{(m)}_{y'|t}\big)$, and it is an exact reference where G is. If B is the snapped family of a snap of G and B' that of a snap of $\widetilde{G}$ (Definition "Snapping declared data onto a lattice" (foundations/def:snap)), then the handed data $A^{\mathrm{b}}_{(y,t)} = q_t \sum_{y'} R^{\varpi}_{y|y'} B_{y'|t}$ of the bare form and $A^{\mathrm{d}}_{(y,t)} = q_t B'_{y|t}$ of the dark-composed form differ in general. The two flat instances agree in everything but their data, so their emissions differ only in their data sectors.

Uses Def. 10.1, Def. 11.1, Def. 19.1, Prop. 19.4, Lem. 20.1, Def. 20.3, Prop. 21.2, foundations/def:exact-reference, foundations/def:snap · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.wpfss.weights:post_processed_preserved_blocks

Proof. (i) Each $\widetilde{B}^{(m)}_{y|t}$ is a combination of Hermitian positive semidefinite matrices with the non-negative coefficients $R^{\varpi}_{y|y'}$, and $\sum_y \widetilde{B}^{(m)}_{y|t} = \sum_{y'} \big(\sum_y R^{\varpi}_{y|y'}\big) B^{(m)}_{y'|t} = \sum_{y'} B^{(m)}_{y'|t} = \mathbb{1}$ by the column-stochasticity of R^{ϖ} (Lemma 20.1) and the completeness of the supplied family on its block.

(ii) In the bare form the handed datum is formed by steps 1–2 of Definition 20.3, $A^{(m)}_{(y,t)} = q_t \sum_{y'} R^{\varpi}_{y|y'} B^{(m)}_{y'|t}$ (Proposition 21.2 (I1)), which is $q_t \widetilde{B}^{(m)}_{y|t}$ by (20.8); in the dark-composed form it is formed by step 2 alone from the supplied $\widetilde{B}$, which gives the same matrices, entry for entry, in exact arithmetic. Every other part of the flat instance reads no supplied block below the cutoff: the family above the cutoff is $q_t \sum_{y'} R^{\varpi}_{y|y'} M^{(>N)}_{y'|t}$ in either form (Definition 20.3); the hypotheses (H1)–(H3) are about that family and the routed weights; the routed weights read R^{ϖ}, the class weights and the declared slot; the scalar values (21.4) read the routed weights; the channels (19.4) and (19.5) read the cutoff projectors, the family above the cutoff and the weights; and the transported set and the entries are built from these and from the sector projectors (Proposition 21.2 (I3) and (I4)). So the two instances coincide. The normalizer of Definition 11.1 is a function of the handed datum, the declared order, the lattice exponent and the run option, so it produces the same declared datum and records, and the declared model, the model map at that datum, is the same.

(iii) $\widetilde{G}$ is the image of G under the same non-negative combination as in (i), so its blocks are Hermitian, positive semidefinite and complete at 1. Its centres $\sum_{y'} R^{\varpi}_{y|y'} C^{(m)}_{y'|t}$ are Hermitian with Gaussian-rational entries and its radii are non-negative rationals, the entries of R^{ϖ} being non-negative dyadic rationals; and for every entry (i, l),

$$\Big|\big(\widetilde{G}^{(m)}_{y|t}\big)_{il} - \Big(\sum_{y'} R^{\varpi}_{y|y'} C^{(m)}_{y'|t}\Big)_{il}\Big| \;\le\; \sum_{y'} R^{\varpi}_{y|y'} \Big|\big(G^{(m)}_{y'|t} - C^{(m)}_{y'|t}\big)_{il}\Big| \;\le\; \sum_{y'} R^{\varpi}_{y|y'} \big(r^{(m)}_{y'|t}\big)_{il},$$

by the triangle inequality and $R^{\varpi} \geq 0$. That is Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference), and every radius is zero where every radius of G is. That the two handed data differ in general is that rounding and composing do not commute: a snap of G followed by R^{ϖ} and a snap of $R^{\varpi}G$ are different lattice families. The last sentence is (ii)'s list of what the instance reads besides its handed datum. □

With two or more absorbing bounded letters every r_min,i vanishes, so each routed weight is the un-improved lat(q_t p_i^(t)). The dark counts sharpen a weight only when the slot has a single entry.

Corollary 20.6 (Dark-count sharpening at a single bounded letter only)**.** Under (absorbing): if $r \geq 2$ then $r_{\min,i} = 0$ for *every* i, so (20.7) reads $M^{(>N)}_{(y_i,t)} \succeq q_t p_i^{(t)} \mathbb{1}$ throughout and the emitted weight is $\lfloor q_t p_i^{(t)} \rfloor_{\kappa}$. Consequently a positive $r_{\min,i}$, the only case in which the dark counts sharpen a weight, can occur for at most one bounded letter, and only when $r = 1$.

Uses Lem. 9.3, Def. 20.3, Prop. 20.4 · *Used by* Rem. 20.9, Rem. 20.10 · newly derived · accepted

Proof. Let $r \geq 2$ and fix any i; choose $j \neq i$, which is possible. Column y_j of R^{ϖ} is a probability vector and $R^{\varpi}_{y_j|y_j} = 1$ by (absorbing), so every other entry of that column vanishes; in particular $R^{\varpi}_{y_i|y_j} = 0$. Since $y_j \neq y_i$, the minimum defining $r_{\min,i}$ ranges over a set containing y_j, so $0 \leq r_{\min,i} \leq R^{\varpi}_{y_i|y_j} = 0$. As i was arbitrary, every $r_{\min,i}$ vanishes; substituting into (20.7) gives the displayed form, which is the un-improved bound and is sound, and (20.4) then reads $\hat{p}_{(y_i,t)} = q_t p_i^{(t)}$. The last sentence follows: a positive $r_{\min,i}$ is incompatible with the existence of a second absorbing bounded letter. Every step is an identity or a comparison between declared values and is decided as such in the substrate of Lemma 9.3, so the conclusion holds exactly. □

Without the absorbing condition the routed bound is unproved and fails in general, unsafely: a single-click letter loses weight to multiclick patterns under a dark count at another port, and no other column restores it. The receiver's multiclick entry is absorbing, so its slot is admitted.

Remark 20.7 (Necessity of the absorbing hypothesis)*.* Without (absorbing), (20.7) is no longer proved, and it fails in general, in the direction that certifies a weight the device does not have. Under the supplier's independent product form (Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm)), a bounded *single-click* letter $y_i = \varpi(\{k\})$ whose fibre $\varpi^{-1}(y_i)$ is $\{\{k\}\}$, as it is when ϖ is injective off the multiclick set and maps no multiclick pattern to y_i, has $R^{\varpi}_{y_i|y_i} = R_{\{k\}|\{k\}} = \prod_{k' \neq k}(1 - p_{\mathrm{dc}|k'}) < 1$ whenever some $p_{\mathrm{dc}|k'} > 0$, because a dark count at another port moves the pattern $\{k\}$ into the multiclick set and never back. And, for any monotone dark-count channel, $r_{\min,i} = 0$, because monotonicity kills *at least one* other column: any other single-click letter $y' = \varpi(\{k''\})$, $k'' \neq k$, has $R^{\varpi}_{y_i|y'} = 0$ since $\{k''\} \not\subseteq \{k\}$, which is all a minimum needs, and $K \geq 2$ supplies such a letter. Formula (20.7)

would then assert $\widetilde{M}^{(>N)}_{y_i|t} \succeq p_i^{(t)}\mathbb{1}$, while in fact the inequality that does hold is $\widetilde{M}_{y_i|t} \succeq R^{\varpi}_{y_i|y_i} M_{y_i|t}$ with $R^{\varpi}_{y_i|y_i} < 1$, and the columns that remain do not compensate: under the independent product form the only other column feeding y_i is the no-click one, $R^{\varpi}_{y_i|\varpi(\varnothing)} = p_{\mathrm{dc}|k}\prod_{k'\neq k}(1-p_{\mathrm{dc}|k'})$, which multiplies $M_{\varpi(\varnothing)|t}$ and not $M_{y_i|t}$ and carries no high-photon weight of its own. In the sub-case $p_{\mathrm{dc}|k} = 0$ with some $p_{\mathrm{dc}|k'} > 0$ that column vanishes too and the loss is exactly the factor $R^{\varpi}_{y_i|y_i} = \prod_{k'\neq k}(1-p_{\mathrm{dc}|k'}) < 1$: for a receiver whose certified weight is attained and positive, $\lambda_{\min}\big(M^{(>N)}_{y_i|t}\big) = p_i^{(t)} > 0$, the smallest eigenvalue of $\widetilde{M}^{(>N)}_{y_i|t}$ is $R^{\varpi}_{y_i|y_i}\, p_i^{(t)} < p_i^{(t)}$, and (20.7) is false. A weight asserted above the true one makes (19.1) false, so F^{WP} is not a squashing of the real receiver and the emitted licence certifies an inclusion that does not hold, the failure Assumption 13.3's ledger row calls a security break. Condition (absorbing) is therefore a declared input validity condition at every bounded letter (Requirement 22.5), and no bound is stated for a letter that fails it. The multiclick letter of the canonical declaration is absorbing (Proposition 20.4 (iii)), which is why the slot the receiver emits, the single entry $(\mathsf{MC}, (p^{(t)}_{\mathrm{mc}})_{t\in\mathcal{T}_{\mathrm{set}}})$ of its Definition "The strengthening slot" (receiver/def:slot), written $(\mathsf{MC}, p_{\mathrm{mc}})$ at a probe-free receiver, is admitted.

Uses Def. 19.1, Prop. 20.4, receiver/def:slot · *Assumes* A-RECV · *Used by* Req. 22.5 · newly derived · accepted

A certified bound on a sum of positive operators says nothing about its summands, so a certified set split over two letters gives neither a bound, and one merged with other outcomes attaches the weight to a larger fibre. Each certified set must therefore be exactly one fibre of the coarse-graining.

Remark 20.8 (Necessity of the fibre condition). Proposition 20.4 needs the certified set to be *exactly* a fibre, $\mathcal{O} = \varpi^{-1}(y_*)$. The reason is that the certified quantity is the *sum* $\sum_{S\in\mathcal{O}} M_{S|t}$, and a bound on a sum of positive operators says nothing about the individual summands. Two failures follow, in opposite directions. If ϖ *splits* $\mathcal{O}$ across two emitted letters, neither letter inherits a bound, and the natural repair, bounding each piece, is unavailable because the supplier certified only the sum. If ϖ merges $\mathcal{O}$ with outcomes *outside* it, the bound still holds for the merged letter (adding positive operators only helps), but the coarse blocks the supplier emits no longer let this module recover $\sum_{S\in\mathcal{O}} M_{S|t}$ separately, so the emitted weight is attached to a letter whose fibre is larger than the certified set: sound, and recorded as such, but only if the containment $\mathcal{O} \subseteq \varpi^{-1}(y_*)$ is validated rather than assumed. Equality is therefore required, and decided at input (Requirement 22.6). Where the pieces carry key information and must not be merged, the fact does not become encodable by re-declaring ϖ; it becomes an event-level bound, and Proposition 6.1 shows that a target classical on the whole of Q cannot in general encode it.

Uses Prop. 20.4 · *Used by* Req. 22.6 · newly derived · accepted

Exact column-stochasticity of the declared R gives that of $R^{\widehat{}}cg$ by regrouping one finite sum, so one exact decision on R serves both; the route, the post-processing step and the one-letter corollary rely on it. Every result here is proved for an exactly column-stochastic R only.

Remark 20.9 (Column-stochasticity of R and of its descended matrix). *One matrix suffices.* The declared R is decided exactly column-stochastic, $R_{S|S'} \geq 0$ for every entry and $\sum_S R_{S|S'} = 1$ for every column (Requirement 22.7). The fibres of ϖ partition $2^{[K]}$, ϖ being a total surjection onto $\mathcal{Y}$ (Lemma 20.1). So $R^{\varpi}_{y|y'} = \sum_{S \in \varpi^{-1}(y)} R_{S|S'}$ is non-negative as a sum of non-negative entries, and $\sum_{y \in \mathcal{Y}} R^{\varpi}_{y|y'} = \sum_S R_{S|S'} = 1$, which is exactly the column-stochasticity Lemma 20.1 states of R^{ϖ}, there derived from the same hypothesis on R. In the exact substrate that inference carries: both identities regroup one finite sum in $\mathbb{F}$ (Lemma 9.3, Definition 9.2), so the decision on R settles R^{ϖ} as well and no second decision is needed. A regrouped binary64 sum need not be the sum it regroups; under exact arithmetic the question does not arise.

Where the property is used. Three results rely on it. *(i)* Proposition 20.4, whose proof restricts $(1 - r_{\min,i}) M^{(>N)}_{y_i|t} \succeq (1 - r_{\min,i}) p_i^{(t)} \mathbb{1}$ and needs $1 - r_{\min,i} \geq 0$, which is non-negativity of the entries together with unit columns. *(ii)* Lemma 4.3 (i) at step 1 of Definition 20.3: post-processing by R^{ϖ} yields a POVM because the coefficients are non-negative and each column sums to one; without both, $\{\widetilde{M}_{y|t}\}$ need not be one. *(iii)* Corollary 20.6, whose proof reads a column of R^{ϖ} as a probability vector.

What the decision closes: coverage and attribution, not soundness. A violation that moves an emitted number in the unsafe direction can also be caught by the total weight: with $\sum_t q_t = 1$ (Definition 4.4) and the weights read as the values of (20.4)'s expression, the routed total is $\sum_i \left(r_{\min,i}(1 - \bar{p}_i) + \bar{p}_i \right)$ with $\bar{p}_i := \sum_t q_t p_i^{(t)}$ and $0 \leq \bar{p}_i \leq 1$ from the slot's validity condition, which gives $0 \leq p_i^{(t)} \leq 1$ at every t (at a slot with one weight at every setting $\bar{p}_i$ is that weight), so any $r_{\min,i} \geq 1$ makes the i-th summand at least 1, and the total at least 1 as long as no other summand is negative (Requirement 22.2); a negative $r_{\min,j}$, a defect of the same class, can make one negative. A defect outside that path, one that neither lifts some $r_{\min,i}$ to 1 nor shows up in the total, would meet no other rule. That path is not relied on: the decision on the declared matrix settles the two conditions directly and exactly, and the absorbing condition fixes $R^{\varpi}_{y_i|y_i} = 1$ (Requirement 22.5), so every admitted $r_{\min,i}$ lies in $[0, 1]$. Every result of this contract is proved for an exactly column-stochastic R and asserts nothing whatever about a declaration that is not one. Beyond diagnosis, the decision is used in one place: every admitted R satisfies (D1) and (D2) of Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm) exactly, which is half of Remark 20.11.

Uses Lem. 4.3, Def. 4.4, Lem. 20.1, Def. 20.3, Prop. 20.4, Cor. 20.6 · *Used by* Req. 22.7 · newly derived · accepted

The route forms each joint outcome's weight from its own setting's certified weight; consuming each entry's minimum over the settings instead would be sound but weakly smaller everywhere. A slot with one weight at every setting, as at every probe-free receiver, gives the same numbers either way.

Remark 20.10 (Per-setting weights against the minimum over the settings). *What is consumed.* The hypothesis Proposition 20.4 runs on is the guarantee the supplier's contract *states* at each setting: one weight $p_i^{(t)}$ per slot entry and per setting $t \in \mathcal{T}_{\mathrm{set}}$, with $\sum_{S\in\mathcal{O}_i} M_{S|t}^{(>N)} \succeq p_i^{(t)}\mathbb{1}$ at that setting (Definition "The strengthening slot" (receiver/def:slot) and the first conclusion of Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound)). The weight route forms the weight of a joint outcome (y_i, t) from $p_i^{(t)}$, the weight of that outcome's own setting ((20.4)). Assumption 13.3 names the supplier's weights generically, and which claims state them is Proposition 20.4 (iv).

Consuming the minimum would also be sound, and is weakly smaller at every joint outcome. The uniform weight of an entry, $p_i = \min_t p_i^{(t)}$, satisfies the supplier's guarantee at every setting, so a route consuming it in place of $p_i^{(t)}$, the *uniform route*, would be certified too. Since $1 - r_{\min,i} \geq 0$ and $p_i \leq p_i^{(t)}$,

$$q_t\Big((1 - r_{\min,i})p_i + r_{\min,i}\Big) \;\leq\; q_t\Big((1 - r_{\min,i})p_i^{(t)} + r_{\min,i}\Big) = \hat{p}_{(y_i,t)}$$

at every (y_i, t), and $\lfloor \cdot \rfloor_\kappa$ is non-decreasing, so every routed weight of that route would be at most the one (20.4) emits. *Where the per-setting total p is below one*, Proposition 19.7 (ii) and Corollary 4.2 then make that route's target the cleaner one, which grants the adversary at least as much. At a slot with a single entry, $r = 1$, that holds whenever the uniform route's total is below one. Each of the two totals is at most $\sum_t q_t\big((1 - r_{\min,1})w_t + r_{\min,1}\big) \leq 1$, $w_t \in [0, 1]$ being the weight it reads at t, and reaches one exactly when $r_{\min,1} = 1$ or $w_t = 1$ at every t, because every q_t is positive and a lattice number, which $\lfloor \cdot \rfloor_\kappa$ leaves unchanged (Definitions 4.4 and 20.3); and $p_1 = \min_t p_1^{(t)}$ is 1 exactly when every $p_1^{(t)}$ is, so the two totals reach one together. At $r \geq 2$, where every $r_{\min,i}$ vanishes (Corollary 20.6), the per-setting total can reach one where the uniform one is below it: with two settings a, b, $q_a = q_b = \frac{1}{2}$, $p_1^{(a)} = p_2^{(b)} = 1$ and the other two weights 0, the totals are 1 and 0. The per-setting route is then refused for its total (Requirement 22.2), a conservative refusal (Remark 19.3), and there is no second target to compare it with. What the per-setting route gains at entry i is, before rounding, $\sum_t q_t(1 - r_{\min,i})\big(p_i^{(t)} - p_i\big) \geq 0$ of routed weight: nothing at a setting whose weight is the minimum, and at any other setting the difference its own floor makes, for instance at a bright probe setting, where the high-photon multiclick floor is expected to be the larger one, an expectation no claim states.

The uniform case is the special case, number for number. If $p_i^{(t)} = p_i$ for every t, which holds at every one-setting family $\mathcal{T}_{\mathrm{set}} = \{\mathrm{off}\}$, whose one weight is its uniform weight (Definition "The strengthening slot" (receiver/def:slot)), and so at every probe-free receiver, (20.4) is evaluated on the same declared numbers q_t, $r_{\min,i}$ and p_i as the uniform route evaluates, in the same exact substrate and with the same downward rounding. The routed weights, their total p, the complement $1 - p$ and everything the flat instance builds from them are then the numbers that route gives, so at a declaration without a probe the two routes emit the same numbers.

Uses Cor. 4.2, Def. 4.4, Prop. 19.7, Def. 20.3, Prop. 20.4, Cor. 20.6, receiver/def:slot, receiver/thm:emitted-bound · *Assumes* A-RECV · newly derived · accepted

This submodule composes with neither of the supplier's two dark-count inequalities and proves its own bound from R itself. Every R it admits satisfies (D1) and (D2) exactly, so the supplier's cap is inactive, its correction is slack, and its exact inequality holds as stated here.

Remark 20.11 (Composition with the supplier's dark-count bounds). *Two supplier inequalities.* The supplier is the receiver at the version the artifact names (Assumption 13.3 (i)), and it separates two inequalities about the same object. The first, part of Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts) (iv), reads $\widetilde{M}_{\mathsf{MC}}^{(>N)} \succeq \big((1-r_{\min})p_{\mathrm{mc}}+r_{\min}\big)\mathbb{1}^{(>N)}$; the second, in Proposition "Multiclick bound under a dark-count channel admitted with slack" (receiver/detector-imperfections/prop:dark-slack-bound) (iii), is the same statement with the capped coefficient $\bar{r} = \min\{r_{\min}, 1\}$ in place of $r_{\min}$ and a subtracted correction δ_{dc}, both defined there. The supplier says which is which: the second is the inequality a downstream module composes with (Requirement "The emitted dark-count channel and the scalars of its composed bound" (receiver/detector-imperfections/req:dark-channel-record)), and the first is stated for an R satisfying (D1) and (D2) exactly and asserted for no other. A consumer that used the first where the second is required would overstate by $(r_{\min}-\bar{r})(1-p_{\mathrm{mc}})+\delta_{\mathrm{dc}}$ (Remark "Cost of admitting a dark-count channel with slack" (receiver/detector-imperfections/rem:d2slack) (e)), first order in the receiver tolerances and in the *unsafe* direction. Which case holds here is therefore a question this contract must answer rather than inherit.

First: this module composes with neither inequality. It consumes the matrix R itself, forms R^{ϖ} by Lemma 20.1 and its own per-letter constants $r_{\min,i} = \min_{y' \neq y_i} R^{\varpi}_{y_i|y'}$, and proves (20.7) inline in Proposition 20.4 from POVM positivity and completeness, the absorbing hypothesis (absorbing) and the exact column-stochasticity of R^{ϖ}, and from nothing else. It reads none of $r_{\min}$, $\bar{r}$ or δ_{dc} from the artifact. Those three are the supplier's *raw*-level scalars, minima and masses over MC; this module's $r_{\min,i}$ is a different object, indexed by a *coarse* bounded letter and read across the columns $y' \neq y_i$ of R^{ϖ}, and Definitions 4.4 and 20.3 say so.

Second, and this settles the question rather than leaving it to a reading: every declaration this module admits satisfies (D1) and (D2) exactly. Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm) calls R a dark-count channel when it is (D1) column-stochastic and (D2) monotone, and both are decided on the declared matrix *exactly*: (D1) by the column-stochasticity rule (Requirement 22.7), and (D2) by the descent rule, whose monotonicity half is the implication $R_{S|S'} > 0 \Rightarrow S' \subseteq S$ (Requirement 22.4). The two rules are needed together and neither covers the other: a strictly positive entry in a (D2)-forbidden position fails monotonicity while leaving column-stochasticity untouched, and a *negative* entry in such a position does the reverse, it satisfies the monotonicity implication vacuously and fails non-negativity. Consequently, on the set of declarations this module emits for, $R_{\mathsf{MC}|S'} \leq \sum_S R_{S|S'} = 1$ for every column, so the supplier's minimum $r_{\min}$ of Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts) (iv) is at most 1 and the cap in $\bar{r}$ is inactive, $\bar{r} = r_{\min}$; and Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-

counts) (iii) holds with no defect, so the δ_{dc} that contract emits is slack in a bound rather than a defect of the datum. The first term of the overstatement is therefore nil, and the second, δ_{dc}, is slack: for such an R the first inequality holds as stated (Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts) (iv)), so using it overstates nothing. That first inequality, the case of (D1) and (D2) holding exactly, is the statement available here.

What that means route by route. On the supplier's generated dark-count route, where R is its independent product form, (D1) and (D2) are a theorem (Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts) (i)) and $\delta_{\mathrm{dc}} = 0$ on that route by its definition in Proposition "Multiclick bound under a dark-count channel admitted with slack" (receiver/detector-imperfections/prop:dark-slack-bound), so the two inequalities coincide there outright. On its declared dark-count route, where Definition "Receiver tolerances deciding dark-channel admissibility" (receiver/detector-imperfections/def:darktol) admits entries down to $-\varepsilon_4$ and column sums anywhere in $[1-\varepsilon_4, 1+\varepsilon_4]$, a matrix reaches a claim of this contract only if it satisfies (D1) and (D2) exactly, and is *refused here* otherwise, under whichever of the two rules it fails. Nothing in (20.4) is restated in $\bar{r}$ or δ_{dc}, and nothing needs to be: that formula is proved for an R satisfying (D1) and (D2) exactly, and this module admits no other. What is *not* settled here is the composition in general: what a consumer that admits the supplier's widened set must do. That question is not this contract's, and this contract does not answer it: it is a question about the supplier's own receiver tolerances $\varepsilon_2, \varepsilon_4$, which this module neither reads nor converts, and Requirement "Explicit conversion between norms" (foundations/req:norm-conversion) puts any such conversion at the instance, explicitly and on that party's own authority (Assumption 13.3).

Uses Def. 4.4, Lem. 20.1, Def. 20.3, Prop. 20.4, receiver/detector-imperfections/def:darkadm, receiver/detector-imperfections/def:darktol, receiver/detector-imperfections/thm:dark-counts · *Assumes* A-RECV · newly derived · accepted

The receiver facts imported here are the claims Proposition 20.4 (iv) cites, and the emitted ledger names every supplier assumption they rest on and may name more. A-EIG enters only at an exact window evaluated in floating point; acyclicity needs no supplier assumption.

Remark 20.12 (Supplier assumptions behind the receiver imports). The receiver claims this submodule imports are those Proposition 20.4 (iv) cites for the hypotheses on the supplier's objects, read under Assumption 13.3 or, for an artifact the run built itself from a declaration with a model admissible exactly and with every weight evaluated rigorously, under Requirement 18.7, which shows them to hold of its numbers under the supplier's own ledger. They are read at the module version the artifact names, an accepted version: for an artifact from outside the run by Assumption 13.3 (i) and Requirement 18.5, and for one the run built as the effective status of the receiver's claims in the run's closure, which the record of the run reports (Requirement 18.7), whether or not the row is omitted; there each resolves and reads as Proposition 20.4 (iv) cites it; for the per-setting weights that version must state them (Requirement 22.8). Where the receiver is compiled at its normalized station, Corollary 21.6 imports in addition Proposition "A receiver as uniform loss followed by its normalized station"

(receiver/detector-imperfections/prop:efficiency-normalization), read under the same assumption. The claims named here are dependencies of this remark too, so the supplier assumptions the imported facts rest on are the assumption closure of those dependencies.

A-EIG is invoked conditionally. The supplier's evaluation model (Definition "The evaluation model: floating point, enclosures and rigour" (receiver/def:evaluation-model)), a restriction the supplier places on itself and not an assumption, governs Theorem "Exact statements, floating-point and rigorous evaluation" (receiver/thm:exact-and-evaluated) and the evaluation of the weights of Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound), hence the weights $p_i^{(t)}$: each is evaluated in floating point or, where the supplier encloses its port Grams, at a setting where the needed enclosures are formed (the artifact records which), rigorously, and its artifact says which (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)). A-EIG supports Theorem "Exact statements, floating-point and rigorous evaluation" (receiver/thm:exact-and-evaluated) (v) and *the exact-window branch* of Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound), hence the emitted $p_{\mathrm{mc}}^{(t)}$ itself at a setting carrying a depth whose weight the supplier evaluated in floating point. The supplier's Assumption "Declared slack on the computed block eigenvalues" (receiver/ass:eig) is asserted only at a setting whose declaration carries an exact-window depth m_t^* (its Definition "Exact-window weight of a setting with a declared depth" (receiver/def:pmcex)) and whose weight is so evaluated; where no depth is declared, or the weight is the supplier's enclosed weight (Theorem "The weight on enclosed bound data bounds the multiclick element" (receiver/thm:enclosed-weight)), the emitted value does not depend on it. *This submodule can neither verify nor improve it*, and its declared eigenvalue slack $\varepsilon_{\mathrm{eig}}$, which A-EIG assumes bounds the overestimate in the supplier's floating-point evaluation of an exact block minimum eigenvalue, is declared separately and never derived from the declared tolerance (Requirement "Scope of the declared tolerance" (foundations/req:tolerance-scope) (iii)), nor from anything else this module reads. Its invocation record travels with its key (Requirement 22.9), and a caller that wants it out of the chain declares no exact window, or a receiver whose port Grams the supplier encloses and at every setting of which the weight is evaluated rigorously, as the artifact records.

The emitted ledger names every assumption of that closure, and may name more. The entry of the emitted assumption ledger for Assumption 13.3 is the supplier's own ledger, not that closure (Requirements 18.3 and 18.6). The two answer two questions: the closure says which of the supplier's assumptions the facts imported here rest on; the ledger passes on what the supplier asserts of the artifact it supplied. Where Requirement 18.7 omits that row, the supplier's ledger reaches the record of the run as the artifact's own, which the run holds. Under the supplier's presence rules every row is emitted on every artifact, so no key is conditional. The ledger may name an assumption the closure is without, which names an assumption more than the imported facts need. *No assumption that a fact imported here rests on at the declaration is missing from it*: the supplier's rules omit no key, and an artifact whose ledger lacks a key they make present is refused (Requirement 18.4).

Acyclicity needs no supplier assumption. The acyclicity of a supplier's element graph is condition (E2) of Definition "Declared element graph" (receiver/def:graph), a well-formedness condition on a supplier declaration that presents its receiver as an

element graph, its route (iii), the route on which Theorem "Well-definedness of compilation" (receiver/thm:compilation-well-defined) and Theorem "Admissibility of the catalogue entries" (receiver/component-catalogue/thm:catalogue-admissibility) produce the blocks consumed here, so a declaration whose element graph has a cycle is refused before any block exists and produces no blocks to consume. A declaration on the supplier's route (i) or (ii) declares no element graph at all. None of the claims imported here rests on a graph: each is stated of the supplier's receiver instrument tuple (Definition "Receiver instrument" (receiver/def:instrument)), the object on which its three input routes converge, or of an object more abstract still: a setting family for Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound) and for the slot claims that read it, a dark-count channel for Definition "Dark-count channel and its independent product form" (receiver/detector-imperfections/def:darkadm) and Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts). The supplier's Corollary "Receiver instrument from a well-formed graph declaration" (receiver/cor:compiled-instrument), which states that a well-formed declared element graph with a declared mode space, an input split and a port map *is* a receiver instrument whose matrix is the compiled one, is deliberately not a dependency of the supplier's emitting claims, which are true of every receiver instrument and therefore on every one of its three routes.

What is deliberately not imported: the supplier's own admission procedure for a declared R. The receiver decides (D1) and (D2) for a *declared* matrix at its own receiver tolerances ε_4 and ε_2 (Definition "Receiver tolerances deciding dark-channel admissibility" (receiver/detector-imperfections/def:darktol)) and carries what that widening costs in the capped coefficient $\bar{r}$ and the correction δ_{dc} of its composed inequality. What this submodule imports is the *definition* and not that procedure: it decides (D1) and (D2) on the declared matrix *exactly* and refuses everything else (Requirements 22.7 and 22.4), so every declaration reaching a claim of this submodule satisfies (D1) and (D2) outright and Theorem "Dark counts on a photon-number block-diagonal POVM" (receiver/detector-imperfections/thm:dark-counts) applies to it verbatim (Remark 20.11). *Its residual budgets are different declared numbers and this submodule reads none of them*: the supplier's $r_1, \dots, r_6$ and the Hermiticity half of its Hermiticity-and-positivity defect are stated in the *spectral* norm (Definition "The six residuals" (receiver/def:residuals)) and are decided against its own $\varepsilon_2, \varepsilon_4$, while this module decides each of its own conditions at exactly zero (Definition 9.1) and carries no number to compare them with. *No conversion between the two is performed here and none is needed*; an instance that sizes a supplier budget performs one explicitly, with its dimension factor written down, on its own authority (Requirement "Explicit conversion between norms" (foundations/req:norm-conversion)).

Uses Lem. 20.1, Def. 20.3, Prop. 20.4, receiver/cor:unconditioned-family, receiver/def:setting-family, receiver/def:slot, receiver/detector-imperfections/def:darkadm, receiver/detector-imperfections/prop:efficiency-normalization, receiver/detector-imperfections/thm:dark-counts, receiver/prop:slotvalid, receiver/thm:click-povm, receiver/thm:coarse-graining, receiver/thm:emitted-bound · *Assumes* A-RECV · *Used by* Req. 22.9 · newly derived · accepted

21 The flat instance and its emission

On the licensed route a certified weight becomes a zero-constant entry for the flag-state target (Corollary 21.1). The post-processed joint family with its routed weights and either target is a reduction instance (Proposition 21.2). The main result, Corollary 21.3, is the core's emission theorem applied to it: on each route the emitted family is exactly admissible and pulls back exactly to the declared model on the channel system, with its weights rounded down and still certified. A requested two-sector entry is emitted where its certificate is decided on the emitted numbers (Corollary 21.4). Run on a reference of the receiver blocks in place of their snap, the flat instance has a reference companion, whose datum combines those blocks with non-negative weights, so that their factor enclosures reach it (Corollary 21.5). Formed on a normalized station and precomposed with the pure-loss channel of the factored loss, the flat instance is a reduction instance for the receiver itself, with the same emission (Corollary 21.6).

Imports receiver/detector-imperfections/def:dark-composed, receiver/detector-imperfections/def:normalized-station, receiver/detector-imperfections/def:pure-loss, receiver/detector-imperfections/prop:efficiency-normalization, receiver/thm:click-povm, receiver/thm:coarse-graining

Under the flag-state target, the certified weight of an outcome becomes a constraint entry with constant 0 whose certificate holds, so the one-entry list is licensed. Read on the family the artefact carries, that constant is emitted unshifted.

On the licensed route a certified weight travels as a constraint entry rather than inside the target (Definition 19.5). The corollary is read on the family the artefact carries: its $c = 0$ is the constant for the family the target is built from, and on the emission path that is the declared model M' of the flat instance (Proposition 21.2, Corollary 21.3 (iii); Definition 11.1, Theorem 11.2 (iii)). So the constant emitted is 0, unshifted (Corollary 11.3 (iii)), and nothing in the statement changes; a branching instance that files the flat instance as a branch carries this entry as the core's register instance says (Corollary 10.5 (ii)). The corollary is about the flag-state squasher alone, which no further link follows on any route the engine emits.

Corollary 21.1 (Licensed flag-sector entry of a certified weight)**.** In the situation of Proposition 19.4 with the flag-state target F^{FSS}, fix an outcome $o \in \mathcal{V}$ with certified weight p_o and put

$$\hat{O} := F_o^{\mathrm{FSS}} - p_o\,\Pi_{\mathrm{Flg}}, \qquad \bowtie\, = \succeq, \qquad c = 0. \tag{21.1}$$

Then (6.9) holds, so Theorem 6.7 applies and the single-entry list $\mathfrak{L} = \{(\hat{O}, \succeq, 0)\}$ is licensed with the admissible set

$$\mathfrak{A}_{\mathfrak{L}}(E', QE'') = \Big\{\mathcal{A} :\ \mathcal{A}^{\dagger}\big(F_o^{\mathrm{FSS}} \otimes \mathbb{1}_{E''}\big) \ \succeq\ p_o\,\mathcal{A}^{\dagger}\big(\Pi_{\mathrm{Flg}} \otimes \mathbb{1}_{E''}\big)\Big\}. \tag{21.2}$$

Uses Thm. 6.7, Def. 19.1, Prop. 19.4 · *Used by* Prop. 21.2, Cor. 21.3, Cor. 21.5 · newly derived · accepted

Proof. By Proposition 19.4 (i), $\Lambda^{\mathrm{FSS}\dagger}(F_o^{\mathrm{FSS}}) = M_o$ and $\Lambda^{\mathrm{FSS}\dagger}(\Pi_{\mathrm{Flg}}) = \Pi_{>N}$, so by linearity $\Lambda^{\mathrm{FSS}\dagger}(\hat{O}) = M_o - p_o\Pi_{>N} = M_o^{(\le N)} \oplus \big(M_o^{(>N)} - p_o\mathbb{1}\big) \succeq 0$, the first block being positive because M_o is and the second by (19.1). That is (6.9) with $c = 0$ and $\bowtie = \succeq$. The displayed form of the set is $\mathcal{A}^\dagger\big((F_o^{\mathrm{FSS}} - p_o\Pi_{\mathrm{Flg}}) \otimes \mathbb{1}_{E''}\big) \succeq 0$ rearranged by linearity of $\mathcal{A}^\dagger$. □

The post-processed receiver family with its routed weights defines a reduction instance, with the weight-preserving target on the encoded route and the flag-state target on the licensed route. Its declared model agrees with the handed one above the cutoff and keeps every certified weight.

The flat instance is the reduction instance of an optical receiver with no branch register: the joint family the weight route forms, with its blocks below the cutoff as the finite datum, and the blocks above the cutoff with the routed weights entering only through hypotheses that read no entry of the datum.

Proposition 21.2 (The flat instance of the post-processed receiver)**.** Let the data of Definition 20.3 pass every refusal rule of this module's interface; let $\{M_v\}_{v\in\mathcal{V}}$ be the post-processed joint family of its step 2 on C, with $\mathcal{H}(C) = \bigoplus_{m\ge 0}\mathcal{H}(C_m)$, $\mathcal{H}(C_{\le N})$ finite-dimensional, every M_v block diagonal at the cutoff N; let (p_v) be its routed weights with $p = \sum_v p_v < 1$; and let a route, encoded or licensed, be declared (Definition 19.5). Then the *flat instance* $\mathcal{J}_{\mathrm{flat}}$, given by

(I1) $B = C$; the data sectors (m), $0 \le m \le N$, in that order, with $d_m = \dim\mathcal{H}(C_m)$ and the default supports $\mathcal{V}^{(m)} = \mathcal{V}$; the handed datum $A_v^{(m)} := M_v^{(=m)}$, formed from the declared blocks, R^{ϖ} and $\{q_t\}$ by steps 1–2 of Definition 20.3 in the bare form, and from the declared blocks and $\{q_t\}$ by step 2 alone in the dark-composed form; the hypotheses (H): *(H1)* for each $t \in \mathcal{T}_{\mathrm{set}}$ the above-cutoff blocks $\{M_{y|t}^{(>N)}\}_{y\in\mathcal{Y}}$ of the supplied conditional POVM form a POVM on $C_{>N}$ (the supplier's fact, Assumption 13.3, Proposition 20.4 (iv)); *(H2)* the routed weights are certified, (20.6) (Proposition 20.4); *(H3)* $p < 1$, decided on entry; and the model map

$$M_v[A'] := \bigoplus_{m\le N} A_v'^{(m)} \oplus M_v^{(>N)}, \qquad M_{(y,t)}^{(>N)} = q_t \sum_{y'\in\mathcal{Y}} R^{\varpi}_{y|y'} M_{y'|t}^{(>N)}; \tag{21.3}$$

(I2) the scalar sectors (f), $f \in \mathcal{V}$, in the order of $\mathcal{V}$, each of dimension 1, with

$$\phi_f(v) := p_v + (1-p)\,\delta_{vf} \quad \text{(encoded route)}, \qquad \phi_f(v) := \delta_{vf} \quad \text{(licensed route)}, \tag{21.4}$$

so that $\mathcal{H}(Q) = \mathcal{H}(C_{\le N}) \oplus \mathcal{H}(\mathrm{Flg})$ and the target (10.2) is $F_v[A'] = F_v^{\mathrm{WP}}$ resp. F_v^{FSS} of Definition 19.1 for the model $M[A']$;

(I3) $\Lambda = \mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{WP}}$ resp. $\mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{FSS}}$ of Proposition 19.4, independent of the datum, with the transported set $\{\Pi_{\le N}, \Pi_{\mathrm{Flg}}\} \cup \{\Pi_{(m)}\}_{m \le N} \cup \{\Pi_{(f)}\}_{f \in \mathcal{V}}$ and pull-backs $\Pi_{\le N}$, $\Pi_{>N}$, the photon-number projectors P_m^C, and M_f^{WP} resp. $M_f^{(>N)}$ read as operators on C supported in $C_{>N}$;

(I4) an empty entry list on the encoded route, and on the licensed route the flag-sector entries $\big(F_o^{\mathrm{FSS}} - p_o \Pi_{\mathrm{Flg}}, \succeq, 0\big)$ of Corollary 21.1 for every bounded letter $o = (y_i, t)$; the class-marginal identity discharged,

is a reduction instance over the joint alphabet of Definition 20.3. Its handed model $M[A]$ is the post-processed joint family $\{M_v\}$ itself where every M_v is also block diagonal for the decomposition $\mathcal{H}(C_{\le N}) = \bigoplus_{m \le N} \mathcal{H}(C_m)$, the photon-number blocks the datum reads, and otherwise the family's pinching onto those blocks below the cutoff; on the emission path the first case holds, the receiver's elements being block diagonal in every photon number (Proposition 20.4 (iv)), so that there the handed model is the post-processed joint family itself. Its declared model, at the declared datum A' the normalizer of Definition 11.1 produces from A, is

$$M'_v \;:=\; M_v[A'] \;=\; \bigoplus_{m \le N} A_v'^{(m)} \;\oplus\; M_v^{(>N)}, \tag{21.5}$$

a POVM on C block diagonal at the cutoff, agreeing with the handed one above it and satisfying every certified-weight inequality (20.6) the handed one does.

Uses Def. 1.1, Lem. 4.3, Def. 4.4, Cor. 4.7, Def. 9.1, Def. 10.1, Def. 11.1, Def. 19.1, Prop. 19.4, Def. 19.5, Lem. 20.1, Def. 20.3, Prop. 20.4, Cor. 21.1 · *Assumes* A-RECV, A-PROBE, A-TSM · *Used by* Prop. 20.5, Cor. 21.3, Cor. 21.4, Cor. 21.5, Cor. 21.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.wpfss.construction:flat_instance

Proof. (I1) *Formation and admissibility of the handed datum.* For $v = (y, t)$, $A_v^{(m)} = M_{(y,t)}^{(=m)} = q_t \sum_{y'} R^{\varpi}_{y|y'} M_{y'|t}^{(=m)}$ by (20.3) and (4.2) read block by block — finitely many products and sums of declared entries in $\mathbb{F}_i$. It is Hermitian as a real combination of Hermitian blocks; positive semidefinite as a combination of positive semidefinite blocks with non-negative coefficients ($R^{\varpi} \ge 0$ entrywise, $q_t > 0$); and class complete: $\sum_{y \in \mathcal{Y}} A_{(y,t)}^{(m)} = q_t \sum_{y'} \big(\sum_y R^{\varpi}_{y|y'}\big) M_{y'|t}^{(=m)} = q_t \sum_{y'} M_{y'|t}^{(=m)} = q_t \mathbb{1}_{C_m}$ by column-stochasticity of R^{ϖ} (Lemma 20.1) and completeness of the conditional POVM at t read on block m. In the dark-composed form $A_{(y,t)}^{(m)} = q_t \widetilde{M}_{y|t}^{(=m)}$ with the supplied blocks, a product of declared entries with q_t; it is Hermitian and positive

semidefinite because the supplied blocks are, and $\sum_y A^{(m)}_{(y,t)} = q_t \mathbb{1}_{C_m}$ because the supplied family is complete on its block (Definition 20.3). The default supports meet every class. So $A \in \mathcal{D}_{\mathrm{adm}}$ in either form. *Stability.* (H1) is about the blocks $m > N$, which are not in the datum; (H2) is an inequality between above-cutoff blocks and scalars — the proof of Proposition 20.4 restricts to the $> N$ block and reads no preserved block; (H3) reads the weights. None reads an entry of the datum. *The model map.* For every $A' \in \mathcal{D}_{\mathrm{adm}}$, $M_v[A']$ is Hermitian and positive semidefinite block by block; it is complete below the cutoff by class completeness and $\sum_t q_t = 1$ (the first consequence of Definition 10.1), and above it by (H1) and column-stochasticity: $\sum_{(y,t)} M^{(>N)}_{(y,t)} = \sum_t q_t \sum_{y'} \big(\sum_y R^{\varpi}_{y|y'}\big) M^{(>N)}_{y'|t} = \sum_t q_t \mathbb{1}_{C_{>N}} = \mathbb{1}_{C_{>N}}$. It is block diagonal at the cutoff by construction, and $\mathcal{H}(C)$ is separable (Definition 1.1). So $M[A']$ is a POVM on C under (H), for every admissible datum. At the handed datum, $M_v[A] = \bigoplus_{m \le N} M_v^{(=m)} \oplus M_v^{(>N)}$ keeps the diagonal blocks of M_v for the blocks $m \le N$ and $> N$ and drops its off-diagonal blocks, of which only those between two blocks $m \ne m' \le N$ can be nonzero, M_v being block diagonal at the cutoff. So $M_v[A]$ is the pinching of M_v onto those blocks, and it is M_v exactly where those off-diagonal blocks vanish. On the emission path they do: every M_v is block diagonal in every photon number by Proposition 20.4 (iv), read under Assumption 13.3.

(I2) On the encoded route every p_v is a lattice value rounded *down* from $\hat{p}_v$ ((20.4), (20.5)), a certified lower bound by (H2), and $1-p$ is recomputed from the rounded p_v and is a lattice value as their complement; $\phi_f(v) \ge 0$ because $p_v \ge 0$ and $p < 1$, and $\phi_f(v) \le p_v + (1-p) \le p + (1-p) = 1$; and $\sum_v \phi_f(v) = p + (1-p) = 1$ exactly. On the licensed route $\phi_f(v) = \delta_{vf} \in \{0,1\}$ with $\sum_v \delta_{vf} = 1$. Reading (10.2) at these sectors, $F_v[A'] = \bigoplus_{m \le N} A'^{(m)}_v \oplus \bigoplus_f \phi_f(v) = M[A']^{(\le N)}_v \oplus \big(p_v \mathbb{1}_{\mathrm{Flg}} + (1-p)\,|v\rangle\langle v|_{\mathrm{Flg}}\big)$, the operator $p_v \mathbb{1}_{\mathrm{Flg}} + (1-p)\,|v\rangle\langle v|$ being diagonal in the flag basis with entries $\phi_f(v)$ — that is F^{WP}_v of (19.3) for the model $M[A']$; on the licensed route the flag block is $|v\rangle\langle v|$ and the target is F^{FSS}_v of (19.2).

(I3) Fix $A' \in \mathcal{D}_{\mathrm{adm}}$ under (H). By (I1), $M[A']$ is a POVM on C block diagonal at the cutoff; (p_v) are certified weights for it in the sense of Definition 19.1, because (19.1) concerns the above-cutoff blocks only, which are the handed ones, and at a bounded letter it is (H2) while at every other v it reads $M_v^{(>N)} \succeq 0$; and $p < 1$ by (H3). So Proposition 19.4 (i) applies to $M[A']$: $(M[A']; \Pi_{\le N}, \Pi_{>N}) \preceq_{\Lambda^{\mathrm{WP}}} (F^{\mathrm{WP}}; \Pi_{\le N}, \Pi_{\mathrm{Flg}})$, resp. with Λ^{FSS} and F^{FSS}. The channels (19.4) and (19.5) read $\Pi_{\le N}$, $\Pi_{>N}$, the above-cutoff blocks and the weights, and no entry of the datum, so they are independent of A'. The sector projectors: with $\Pi_{(m)} = P^C_m \oplus 0$ and $\Pi_{(f)} = 0 \oplus |f\rangle\langle f|$, the adjoint (19.9) gives $\Lambda^{\mathrm{WP}\dagger}(\Pi_{(m)}) = P^C_m$ and $\Lambda^{\mathrm{WP}\dagger}(\Pi_{(f)}) = \sum_v \langle v|\,|f\rangle\langle f|\,|v\rangle\, M_v^{\mathrm{WP}} = M_f^{\mathrm{WP}}$, and at $p_v \equiv 0$ the same computation gives $M_f^{(>N)}$ for Λ^{FSS}. Every transported operator is block diagonal for $\mathbb{S}$, $\Pi_{\le N}$ and Π_{Flg} being sums of

sector projectors, so Corollary 4.7 (i) gives the relation (10.3) for $\mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{WP}}$ resp. $\mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{FSS}}$ with the same pull-backs.

(I4) On the encoded route the list is empty; the certificate every instance carries for a sector projector is not an entry of it but travels with the transported set (the third consequence of Definition 10.1, Corollary 11.3 (i)). On the licensed route, for a bounded letter o, $\hat{O} = F_o^{\mathrm{FSS}} - p_o \Pi_{\mathrm{Flg}} = \sum_{s \in \mathbb{S}} \Pi_s F_o \Pi_s - p_o \sum_f \Pi_{(f)}$ has the form (10.4) with $\alpha_{o,s} = 1$ for every s, every other $\alpha = 0$, $Z = -p_o \Pi_{\mathrm{Flg}}$ and the lattice coefficient p_o; it is a combination of transported operators; and its certificate is the computation of Corollary 21.1, $\Lambda^{\mathrm{FSS}\dagger}(\hat{O}) = \bigoplus_{m \le N} A_o'^{(m)} \oplus \big(M_o^{(>N)} - p_o \mathbb{1}\big) \succeq 0$, the first summand positive by admissibility of A' and the second by (H2) — for every admissible datum. The class-marginal identity: $\sum_{v \in \mathcal{V}_t} M_v[A'] = \bigoplus_{m \le N} q_t \mathbb{1}_{C_m} \oplus \sum_y M_{(y,t)}^{(>N)}$ by class completeness, and $\sum_y M_{(y,t)}^{(>N)} = q_t \sum_{y'} \big(\sum_y R^{\varpi}_{y|y'}\big) M_{y'|t}^{(>N)} = q_t \mathbb{1}_{C_{>N}}$ by column-stochasticity and (H1) — which is Lemma 4.3 (ii) read block by block; so it is discharged for every admissible datum.

The declared model is (11.3) at the model map (21.3), which is (21.5); it agrees with the handed model above the cutoff because the normaliser touches data sectors only, and (20.6) is a statement about those blocks alone. □

The flat instance's emitted family is exactly admissible and pulls back exactly to the declared model on the channel system, its weights rounded down yet certified. Where the normalizer moved a block below the cutoff, nearness to the unrounded family bounds nothing.

The core's emission theorem, specialised to the flat instance: it adds the numbers of the weight route to the list of exact ones, names the rounding directions the emission record carries (Requirement 17.3), and says, route by route, which constraint entries the artefact carries. The six conditions of part (ii) are those of Requirement "Exact verification before emission" (foundations/req:exact-emission), and a family that fails one of them is refused by Requirement 17.2.

Corollary 21.3 (Exact emission of the flat instance). Let the declared data pass every refusal rule of this module's interface, let $\mathcal{J}_{\mathrm{flat}}$ be the flat instance of Proposition 21.2 at the declared route with handed datum A, and let A' be the datum the normaliser of Definition 11.1 declares. Then Theorem 11.2 holds of $\mathcal{J}_{\mathrm{flat}}$, and in particular:

(i) every number on the list of Theorem 11.2 (i) and, in addition, the descended matrix R^{ϖ}, the constants $r_{\min,i}$, the exact weights $\hat{p}_v$ of (20.4), the total p and its complement $1 - p$ are formed from the declared entries and scalars by finitely many operations in $\mathbb{F}_i$, floors and comparisons; the emission displacement Δ_m per preserved block is the one emitted number that is not;

(ii) the emitted family, F^{WP} resp. F^{FSS} of the declared model (21.5),

satisfies conditions (a)–(e) of Definition 9.1 together with (9.1) of every entry and is decided so before it is emitted;

(iii) for every $v \in \mathcal{V}$,

$$\Lambda^{\dagger}(F_v) \;=\; M'_v \;=\; \bigoplus_{m \le N} A'^{(m)}_v \;\oplus\; M^{(>N)}_v \qquad \text{exactly, on } C, \quad (21.6)$$

both sides operators on C itself: the physical system of this instance is the channel system and the identity is on it, not on an ampliation. M' is a POVM on C block diagonal at the cutoff for which (p_v) are certified weights in the sense of Definition 19.1, and Proposition 19.4, Proposition 20.4 and Corollary 21.1 hold verbatim of it;

(iv) the per-outcome weights are $p_v = \lfloor \hat{p}_v \rfloor_\kappa$, rounded *down*, and (20.6) holds for M' at those values (Proposition 20.4); the complement $1-p$ is *recomputed* from the rounded weights and is never rounded on its own; a caller's constant off the lattice is rounded toward the enlarging side (Corollary 11.3 (iii)); and there are no other rounding directions;

(v) where some $\Delta_m > 0$ no bound relates a key length computed from M' alone to one computed from the pre-snap post-processed family, or from the receiver's own output, and nothing else relates them (Theorem 11.2 (v)).

The emitted constraint list is *empty on the encoded route*; on the licensed route it carries the flag-sector entries $(F_o^{\mathrm{FSS}} - p_o \Pi_{\mathrm{Flg}}, \succeq, 0)$ of Corollary 21.1 at $c = 0$, unshifted, and, on request, the class-marginal entries $\big(\sum_{v \in \mathcal{V}_t} F_v, =, q_t\big)$, which the flat instance discharges, a default carried operator as an entry (Corollary 11.3 (i)), and any further requested entry that Corollary 11.3 (iii) admits, which for this instance is the two-sector entry $(F_e^{\mathrm{FSS}} - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}, \succeq, 0)$ where its decided certificate of Corollary 21.4 holds, all under the licence of that route (Corollary 11.3). The instance's default carried operators, the sector projectors and the two projectors $\Pi_{\le N}$ and Π_{Flg} it transports besides, *are emitted with the transported set at $c = 0$ with $\succeq$ on either route* (Corollary 11.3 (i), Proposition 21.2 (I3)). These statements are about the instance's link and its emitted family.

Uses Def. 9.1, Def. 9.2, Lem. 9.3, Def. 10.1, Def. 11.1, Thm. 11.2, Cor. 11.3, Def. 19.1, Prop. 19.4, Lem. 20.1, Def. 20.3, Prop. 20.4, Cor. 21.1, Prop. 21.2, Cor. 21.4 · *Assumes* A-RECV, A-PROBE, A-TSM · *Used by* Cor. 21.6, Req. 22.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.wpfss.api:reduce_flat

Proof. Proposition 21.2 makes $\mathcal{J}_{\mathrm{flat}}$ a reduction instance, so Theorem 11.2 applies to it with the data passing every refusal rule.

(i) The theorem's list, and in addition: every entry of R^{ϖ} is a finite sum of declared entries of R (Lemma 20.1); $r_{\min,i}$ is a minimum over finitely

many of them, decided by comparisons; $\hat{p}_{(y_i,t)}$ is two products and two sums of lattice values; p is a finite sum and $1-p$ a difference; and $\lfloor\cdot\rfloor_\kappa$ is a floor and a comparison in $\mathbb{F}$, all exact in the substrate (Lemma 9.3, Definition 9.2). The displacement (11.4) at the data sector (m) is Δ_m.

(ii) is the theorem's part (ii), the emitted family being (10.2) at the flat instance, which Proposition 21.2 (I2) identifies with F^{WP} resp. F^{FSS} of the declared model.

(iii) is the theorem's part (iii) with the model map (21.3): $M'^{\mathrm{ph}}_v = M_v[A'] = M'_v$ of (21.5), an operator on $B = C$. That (p_v) are certified weights for M' is (H2) at A', available by the stability clause because (20.6) is a statement about the above-cutoff blocks, which the normaliser does not touch; that M' is a POVM block diagonal at the cutoff is (I1) at A'; and $p < 1$ is (H3). Those are exactly the hypotheses Proposition 19.4 and Corollary 21.1 place on their data, so each applies to M' word for word, and Proposition 20.4 is a statement about the above-cutoff blocks and the declared scalars, which are common to the handed and the declared model.

(iv) The weights are Proposition 20.4 through Definition 20.3; the complement and the constant rule are the theorem's part (iv), the scalar values (21.4) being the only rounded numbers of this instance's own.

(v) is the theorem's part (v).

The constraint list. The default carried operators, with the transported set and, where a caller requests one as an entry, in the list, and the entries are Corollary 11.3 (i), (ii), the flat instance discharging the class-marginal identity, and (iii) for the instance's own list, which on the licensed route is the flag-sector entry of Corollary 21.1 at its constant 0; on the encoded route that corollary emits nothing, the instance's own list being empty there (Proposition 21.2 (I4)). A further requested entry is emitted by (iii) only on a decided certificate that a claim stated for the flat instance gives; the one such claim is Corollary 21.4, whose part (iii) is that decided certificate for the two-sector entry at the declared datum. □

On the licensed route the requested two-sector entry keeps its certificate wherever every block of its outcome below the cutoff clears its first constant and its second constant is at most the outcome's certified weight. Both conditions are decided exactly on the emitted numbers.

Remark 19.9 compares the scalar flag-weight constraint of the literature with the operator entry (19.13), whose two constants are the least eigenvalues of the two halves of an element. A request declares the constants instead, and the entry is emitted only where its certificate is decided on the emitted numbers. The above-cutoff half has no finite presentation, so it is decided through the certified weight; the below-cutoff half is decided on the emitted blocks themselves.

Corollary 21.4 (The requested two-sector entry, decided on the emitted numbers)**.** Let $\mathcal{J}_{\mathrm{flat}}$ be the flat instance of Proposition 21.2 on the licensed

route, with channel Λ and routed weights (p_v), and let A'' be an admissible datum of it over any substrate of Definition 9.2 over which it is a reduction instance, $\mathbb{F} = \mathbb{R}$ included. Let $e \in \mathcal{V}$, let a and b be declared real constants, and put

$$\hat{O}[A''] \;\coloneqq\; F_e^{\mathrm{FSS}}[A''] - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}, \tag{21.7}$$

written in the form (10.4) with coefficient 1 for e on every sector, every other coefficient 0, and $Z = -a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$. Then:

(i) its pull-back is

$$\Lambda^{\dagger}\big(\hat{O}[A'']\big) \;=\; \bigoplus_{m \le N} \big(A_e^{\prime\prime(m)} - a\,\mathbb{1}_{d_m}\big) \;\oplus\; \big(M_e^{(>N)} - b\,\mathbb{1}\big); \tag{21.8}$$

(ii) if

$$A_e^{\prime\prime(m)} - a\,\mathbb{1}_{d_m} \;\succeq\; 0 \quad (m \le N) \qquad \text{and} \qquad b \;\le\; p_e, \tag{21.9}$$

then $\Lambda^{\dagger}(\hat{O}[A'']) \succeq 0$, which is the certificate (6.9) of the entry $(\hat{O}[A''], \succeq, 0)$;

(iii) the first condition of (21.9) is necessary for that certificate. At the declared datum A' of Theorem 11.2, whose blocks and routed weights are emitted lattice values, and for constants a and b that are lattice values, both conditions are decided by finitely many field operations and sign comparisons on these numbers, and (21.9) is a decided certificate for the entry $(\hat{O}[A'], \succeq, 0)$ in the sense of Corollary 11.3 (iii).

Reading: A constraint you request in the published two-constant form is added to the model only after the engine has checked exactly, on the numbers it emits, that the receiver blocks below the photon-number cutoff clear the first constant and that the second constant does not exceed the certified high-photon weight the receiver supplies. Where either check fails the request is refused; the constraint is never taken on your word.

Uses Thm. 6.7, Def. 9.2, Lem. 9.3, Def. 10.1, Thm. 11.2, Cor. 11.3, Def. 19.1, Prop. 19.4, Prop. 21.2 · *Assumes* A-RECV · *Used by* Cor. 21.3, Cor. 21.5 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.constraints:_two_sector_certificate

Proof. No step of (i) and (ii) reads a property of the substrate beyond its field operations and its order, so they hold over each substrate the statement admits. (i) The instance at A'' is a reduction instance by the stability clause of Definition 10.1, and (I3) of Proposition 21.2 holds there: its channel $\Lambda = \mathcal{P}_{\mathbb{S}} \circ \Lambda^{\mathrm{FSS}}$ does not depend on the datum, $\Lambda^{\dagger}(F_e^{\mathrm{FSS}}[A'']) = M_e[A'']$, and $\Pi_{\le N}$ and Π_{Flg} are transported with the pull-backs $\Pi_{\le N}$ and $\Pi_{>N}$ (Proposition 19.4 (i)). By linearity of the adjoint and the model map (21.3),

$$\Lambda^{\dagger}\big(\hat{O}[A'']\big) \;=\; M_e[A''] - a\,\Pi_{\le N} - b\,\Pi_{>N} \;=\; \bigoplus_{m \le N} \big(A_e^{\prime\prime(m)} - a\,\mathbb{1}_{d_m}\big) \oplus \big(M_e^{(>N)} - b\,\mathbb{1}\big),$$

since $\Pi_{\le N}$ is the identity on every photon-number block $m \le N$ and zero above the cutoff, and $\Pi_{>N}$ the reverse.

(ii) The blocks below the cutoff are positive semidefinite by the first condition. The block above the cutoff is the handed instance's, the model map reading no entry of the datum there; the routed weights are certified weights by (H2) of Proposition 21.2, so (19.1) gives $M_e^{(>N)} \succeq p_e \mathbb{1} \succeq b\,\mathbb{1}$ by the second condition. A direct sum of positive semidefinite blocks is positive semidefinite, so $\Lambda^\dagger(\hat{O}[A'']) \succeq 0$, which is (6.9) with $c = 0$ and $\bowtie = \succeq$.

(iii) A direct summand of a positive semidefinite operator is positive semidefinite, so the certificate implies the first condition. At A' the blocks are lattice values, $A' \in \mathcal{D}_\kappa$ by Theorem 11.2 (ii), and so are the routed weights, a and b; lattice values are dyadic rationals. Each $A_e'^{(m)} - a\,\mathbb{1}_{d_m}$ is therefore Hermitian, A' being admissible and a real, with entries in $\mathbb{Q}(i)$, and its positivity is decided by (9.3) of Lemma 9.3 over the substrate $\mathbb{F} = \mathbb{Q}$, which Definition 9.2 admits; $b \le p_e$ is one comparison of two rationals. The condition is thus one on the emitted numbers and the declared constants, decided exactly from them, and by (ii) it is sufficient for (6.9) under the instance's hypotheses (H), which is what a decided certificate is. □

Run on any Hermitian, positive receiver blocks complete per setting, the flat instance forms, as a non-negative combination of them, an admissible datum over the reals, as the core companion asks. A requested two-sector entry keeps its certificate where the centres clear its constant by the radius.

The core states the reference companion of any reduction instance under one hypothesis, (H-REF): that the instance's rule forms an admissible datum from any family of blocks with the properties a reference has, and that the instance holds over the reals. For the flat instance both follow from the proof that it is an instance at all (Proposition 21.2), which reads of the receiver blocks only those properties and of an admissible datum only its admissibility. The companion's entries on the licensed route are therefore the flag-sector entries of Corollary 21.1 at the datum formed from the reference, and its emission is the flat emission (Corollary 21.3) with the datum formed from the reference in its data sectors and no normalizer. A caller may request besides, on the licensed route, the operator (19.13) of Remark 19.9. Its certificate on the emitted numbers says nothing about the reference, so the corollary re-establishes it at the companion: the below-cutoff half from the centres of the companion's datum, which must clear the declared constant by the radius, and the above-cutoff half by deciding the declared constant against the certified weight of that block (19.1), which no block of the receiver enters.

Corollary 21.5 (The flat instance has a reference companion)**.** Let the flat instance $\mathcal{J}_{\text{flat}}$ of Proposition 21.2 be given, at either route and in either form, with K the declared low-photon blocks from which its handed datum is formed by Definition 20.3 (the conditional POVM blocks $M_{y'|t}^{(=m)}$, $m \le N$,

of each setting t in the bare form, the supplied dark-composed blocks $\widetilde{M}^{(=m)}_{y|t}$ in the dark-composed form). Then $\mathcal{J}_{\mathrm{flat}}$ satisfies (H-REF) of Corollary 12.1: for every family G over the same blocks, settings, outcomes and dimensions whose blocks are Hermitian and positive semidefinite, with $\sum_{y'} G^{(m)}_{y'|t} = \mathbb{1}_{C_m}$ exactly for every t and $m \le N$, the datum the same rule forms from G is admissible, and $\mathcal{J}_{\mathrm{flat}}$ is a reduction instance over the substrate $\mathbb{F} = \mathbb{R}$. Its rule of (I1) is a non-negative combination in the sense of Corollary 12.2: the element (y,t) of the datum it forms from a family K' with the properties just listed for G is, on the data sector (m), $q_t \sum_{y'} R^{\varpi}_{y|y'} K'^{(m)}_{y'|t}$ in the bare form and $q_t K'^{(m)}_{y|t}$ in the dark-composed form. At the datum A^{ref} so formed, its target is $F^{\mathrm{WP}}[A^{\mathrm{ref}}]$ on the encoded route and $F^{\mathrm{FSS}}[A^{\mathrm{ref}}]$ on the licensed route, and on the licensed route its entries are $\big(F^{\mathrm{FSS}}_o[A^{\mathrm{ref}}] - p_o \Pi_{\mathrm{Flg}}, \succeq, 0\big)$ for every bounded letter o, with the routed weights p_o of the handed instance. On the licensed route, let the request carry besides an entry $(\hat{O}, \succeq, 0)$ with $\hat{O} = F^{\mathrm{FSS}}_e - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$, at an outcome $e \in \mathcal{V}$ and declared real constants a and b, written in the form (10.4) with coefficient 1 for e on every sector, every other coefficient 0, and $Z = -a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$, so that its companion form (Corollary 12.1 (vi)) is $\hat{O}^{\mathrm{ref}} = F^{\mathrm{FSS}}_e[A^{\mathrm{ref}}] - a\,\Pi_{\le N} - b\,\Pi_{\mathrm{Flg}}$. Let (C, R) be an entrywise enclosure of A^{ref} whose centres are Hermitian, and put $\rho_m \coloneqq \max_i \sum_l R^{(m)}_{e,il}$ for $m \le N$. If

$$C^{(m)}_e - (a + \rho_m)\,\mathbb{1}_{d_m} \;\succeq\; 0 \quad (m \le N) \qquad \text{and} \qquad b \;\le\; p_e, \tag{21.10}$$

then $\Lambda^{\dagger}(\hat{O}^{\mathrm{ref}}) \succeq 0$, with Λ the instance's channel, which is the same at every datum: the companion form carries the certificate of the entry it is formed from.

Uses Thm. 6.7, Def. 9.2, Def. 10.1, Cor. 12.1, Cor. 12.2, Def. 19.1, Prop. 19.4, Lem. 20.1, Def. 20.3, Cor. 21.1, Prop. 21.2, Cor. 21.4 · *Assumes* A-RECV, A-PROBE, A-TSM · *Used by* Cor. 21.6 · newly derived · accepted

Proof. Admissible datum. Part (I1) of the proof of Proposition 21.2 forms the datum as $A^{(m)}_{(y,t)} = q_t \sum_{y'} R^{\varpi}_{y|y'} M^{(=m)}_{y'|t}$ in the bare form and $q_t \widetilde{M}^{(=m)}_{y|t}$ in the dark-composed form, and shows it Hermitian, positive semidefinite and class complete using, of the blocks, only that they are Hermitian, positive semidefinite and complete on their block at each setting; of R^{ϖ}, that it is entrywise non-negative and column-stochastic (Lemma 20.1); and $q_t > 0$. Read with G in place of K, the same argument gives $A^{\mathrm{ref}} \in \mathcal{D}_{\mathrm{adm}}$: R^{ϖ}, q_t and the supports are the handed instance's and unchanged.

An instance over $\mathbb{R}$. The rest of the proof of Proposition 21.2 establishes the model map, (I2), (I3) and (I4) for every admissible datum under (H1)–(H3), and reads of a datum only its Hermiticity, positivity and class completeness: the model map is a direct sum of the datum and the above-cutoff blocks; the scalar values (21.4) are formed from the routed weights

alone; the channels of Proposition 19.4 read the above-cutoff blocks, the projectors and the weights and no entry of the datum; and the certificate of each licensed entry is the computation of Corollary 21.1, positive on the data sectors by admissibility and on the flag sectors by (H2). (H1)–(H3) are about the above-cutoff blocks and the weights, none of which is a block of K, so they hold unchanged. Every step is a field operation, a comparison in the Loewner order or an identity of operators, so the argument holds over the real substrate as over any other that Definition 9.2 admits, and the flat instance is a reduction instance over $\mathbb{R}$ (Definition 10.1).

A non-negative combination. The expressions of the first paragraph, read for any family K' with the properties listed, are the rule of (I1). Their coefficients $q_t R^{\varpi}_{y|y'}$ and q_t are non-negative rationals: $q_t > 0$ is a declared class weight, every entry of R^{ϖ} is a sum of entries of the declared R (Lemma 20.1), and every declared entry and scalar lies on the declared lattice of Definition 20.3, so is a dyadic rational; R^{ϖ} is entrywise non-negative. Neither q_t nor R^{ϖ} is formed from a block, so the coefficients are the same for every family. Every block combined into the element on the data sector (m) is a block of photon number m, of the dimension d_m of that sector. That is (12.2) of Corollary 12.2.

The target and the entries. The target at A^{ref} is (10.2) at the flat instance's sectors, which (I2) of the proof of Proposition 21.2 identifies with F^{WP} and F^{FSS} for the model $M[A^{\mathrm{ref}}]$; the entry list of its (I4) is empty on the encoded route and on the licensed route consists of the flag-sector entries of Corollary 21.1 at the datum, whose weights p_o are the routed weights and are not formed from K.

The requested entry. The companion is the instance at the admissible datum A^{ref}, and $\hat{O}^{\mathrm{ref}}$ is the operator (21.7) of Corollary 21.4 at $A'' = A^{\mathrm{ref}}$, with the instance's channel Λ, which is the same at every datum. By part (ii) of that corollary it suffices that $A_e^{\mathrm{ref},(m)} \succeq a\,\mathbb{1}_{d_m}$ for every $m \le N$ and $b \le p_e$; the second is the second condition of (21.10). Fix $m \le N$ and put $E \coloneqq A_e^{\mathrm{ref},(m)} - C_e^{(m)}$. It is Hermitian, A^{ref} being an admissible datum and the centre Hermitian by hypothesis, and $|E_{il}| \le R^{(m)}_{e,il}$ for every entry. Let $Ex = \lambda x$ with $x \neq 0$ and let i be an index with $|x_i| = \max_l |x_l|$; then $|\lambda||x_i| = |\sum_l E_{il}x_l| \le \sum_l R^{(m)}_{e,il}|x_l| \le \rho_m|x_i|$, so every eigenvalue of E is at least $-\rho_m$ and $E \succeq -\rho_m\mathbb{1}_{d_m}$. With the first condition of (21.10), $A_e^{\mathrm{ref},(m)} = C_e^{(m)} + E \succeq (a+\rho_m)\mathbb{1}_{d_m} - \rho_m\mathbb{1}_{d_m} = a\,\mathbb{1}_{d_m}$, and Corollary 21.4 (ii) gives $\Lambda^{\dagger}(\hat{O}^{\mathrm{ref}}) \succeq 0$, which is (6.9) with $c = 0$ and $\bowtie = \succeq$. □

The flat instance built on a normalized station, precomposed with the pure-loss channel of the factored loss, is a reduction instance for the receiver itself. Its emission is the normalized one with that channel in front; nothing else emitted changes.

The normalization changes the device a reduction is built on, not the reduction: the receiver is the normalized station preceded by a known loss (Proposition "A receiver as uniform loss followed by its normalized station" (receiver/detector-imperfections/prop:efficiency-normalization)), and a known channel in front of a reduction instance is absorbed by Lemma 10.6.

Corollary 21.6 (The flat instance of a normalized station)**.** Let $\mathfrak{R}$ be a receiver instrument without declared source modes, with efficiencies η_k and $\eta_{\max} > 0$, let $t \in [\eta_{\max}, 1]$, and let $\mathfrak{R}'$ be a normalized station of $\mathfrak{R}$ at the factored loss t (Definition "Normalized station and factored loss" (receiver/detector-imperfections/def:normalized-station)), both with the channel system C. Let the data of Definition 20.3 be supplied for $\mathfrak{R}'$, with $\mathcal{T}_{\text{set}} = \{\text{off}\}$ and $q_{\text{off}} = 1$, pass every refusal rule of this module's interface, and let $\mathcal{J}'_{\text{flat}}$ be the flat instance of Proposition 21.2 at the declared route. Put $\Phi := \Lambda_t^{\text{loss}}$, the pure-loss channel of transmittance t on C (Definition "Pure-loss channel on the channel modes" (receiver/detector-imperfections/def:pure-loss)), and $\mathcal{J}_{\text{flat}} := \mathcal{J}'_{\text{flat}} \circ \Phi$.

(i) *(the instance)* $\mathcal{J}_{\text{flat}}$ is a reduction instance on C with the datum, sectors, scalar values, target, transported set, entries, constants and class-marginal flag of $\mathcal{J}'_{\text{flat}}$; its channel is that of $\mathcal{J}'_{\text{flat}}$ precomposed with Φ, the pull-backs of its transported set are $\Phi^\dagger$ of those of $\mathcal{J}'_{\text{flat}}$, and its reachable states (Definition 3.10) are those of that composite channel. Where the supplied blocks are those of $\mathfrak{R}'$'s coarse-grained click-pattern POVM, in the bare form, or of its dark-composed family with the declared dark-count matrix, in the dark-composed form, its handed model is the post-processed joint family of steps 1–2 of Definition 20.3 formed from $\mathfrak{R}$'s own measurement with the same dark-count matrix.

(ii) *(the emission)* Theorem 11.2 holds of $\mathcal{J}_{\text{flat}}$: its declared datum, emitted family, routed weights, entries and constants are those Corollary 21.3 gives for $\mathcal{J}'_{\text{flat}}$, its channel is $\Lambda \circ \Phi$ with Λ that of $\mathcal{J}'_{\text{flat}}$, and for every $v \in \mathcal{V}$

$$(\Lambda \circ \Phi)^\dagger(F_v) \;=\; \Phi^\dagger(M'_v) \qquad \text{exactly, on } C, \tag{21.11}$$

with M' the declared model (21.5) of $\mathcal{J}'_{\text{flat}}$. Where moreover the normalizer kept every class, the right-hand side is the family of (i) formed from $\mathfrak{R}$'s measurement.

(iii) *(several measuring factors)* Where each factor f of Theorem 11.5 is such an instance $\mathcal{J}^f_{\text{flat}} = \mathcal{J}'^f_{\text{flat}} \circ \Lambda^{\text{loss}}_{t_f}$, each at its own factored loss t_f,

that theorem applies, with channel $\bigotimes_f \left(\Lambda_f \circ \Lambda_{t_f}^{\mathrm{loss}}\right)$ and every entry at the constant of its factor.

(iv) *(the reference companion)* $\mathcal{J}_{\mathrm{flat}}$ satisfies (H-REF) of Corollary 12.1, and its reference companion at a family G is that of $\mathcal{J}'_{\mathrm{flat}}$ at G precomposed with Φ: the same target and entries, the companion's channel precomposed with Φ, and $\Phi^\dagger$ of the companion's pull-backs.

Reading: When the receiver is analysed at its normalized station, the finite model the key rate is computed on is exactly the one computed for the normalized station. It is a valid model of your actual receiver where the receiver facts the run carries as A-RECV hold and where the largest detector efficiency may be factored out as loss in front of the receiver, which the reading of Proposition "A receiver as uniform loss followed by its normalized station" (receiver/detector-imperfections/prop:efficiency-normalization) conditions on the receiver's device assumptions and which an efficiency mismatch the adversary can exploit breaks. The factored loss is then counted as part of the channel the adversary controls. The certified high-photon weights, constraints and their constants are those of the normalized station.

Uses Def. 1.1, Def. 3.10, Lem. 10.6, Thm. 11.2, Thm. 11.5, Cor. 12.1, Def. 20.3, Prop. 21.2, Cor. 21.3, Cor. 21.5, receiver/detector-imperfections/def:dark-composed, receiver/detector-imperfections/def:normalized-station, receiver/detector-imperfections/def:pure-loss, receiver/detector-imperfections/prop:efficiency-normalization, receiver/thm:click-povm, receiver/thm:coarse-graining · *Assumes* A-RECV, A-PROBE, A-TSM · *Used by* Req. 22.10, Req. 22.11 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.reductions.wpfss.api:FlatInstanceBuilder

Proof. $\mathcal{H}(C)$ is separable (Definition 1.1) and Φ is a channel on it, so Lemma 10.6 applies to $\mathcal{J}'_{\mathrm{flat}}$, which is a reduction instance by Proposition 21.2.

(i) The first sentence is Lemma 10.6 (ii), the reachable states being by Definition 3.10 the images of the physical inputs under the instance's channel. For the second: every click-pattern element commutes with the photon number (Theorem "Click-pattern POVM" (receiver/thm:click-povm) (iii)), so do the coarse-grained elements (Theorem "Coarse-graining of the click-pattern POVM" (receiver/thm:coarse-graining)) and the dark-composed ones (Definition "The dark-composed family" (receiver/detector-imperfections/def:dark-composed)), and so does the post-processed joint family, a non-negative combination of them; it is therefore block diagonal for $\bigoplus_{m\le N} \mathcal{H}(C_m)$, and the handed model of $\mathcal{J}'_{\mathrm{flat}}$ is that family itself (Proposition 21.2). Steps 1–2 of Definition 20.3 form each element as a fixed linear combination, with coefficients from the dark-count matrix and $q_{\mathrm{off}} = 1$, of the coarse-grained click-pattern elements, which are fibre sums of the click patterns; by Proposition "A receiver as uniform loss followed by its normalized station" (receiver/detector-imperfections/prop:efficiency-normalization) (ii) the same combination formed from $\mathfrak{R}$'s measurement is $\Phi^\dagger$ of the one formed from $\mathfrak{R}'$'s. That is the model map of $\mathcal{J}_{\mathrm{flat}}$ at the handed datum (Lemma 10.6 (ii)).

(ii) $\mathcal{J}_{\mathrm{flat}}$ is a reduction instance whose handed datum, class weights and lattice exponent are those of $\mathcal{J}'_{\mathrm{flat}}$, so Theorem 11.2 applies to it, and by Lemma 10.6 (iii) its declared datum, emitted family, entries and constants are those of $\mathcal{J}'_{\mathrm{flat}}$, which Corollary 21.3 describes, the routed weights being scalar values of the target. Its emitted channel is $\Lambda \circ \Phi$ and its declared model $\Phi^{\dagger}(M')$, so (21.11) is (11.6) for $\mathcal{J}_{\mathrm{flat}}$. Where the normalizer kept every class, the declared datum is the handed one, M' is the handed model of $\mathcal{J}'_{\mathrm{flat}}$, and (i) identifies $\Phi^{\dagger}(M')$.

(iii) Theorem 11.5 asks of each factor only that it be a reduction instance to which Theorem 11.2 applies, which (ii) gives, with channel $\Lambda_f \circ \Lambda^{\mathrm{loss}}_{t_f}$ and the entries and constants of $\mathcal{J}'^{f}_{\mathrm{flat}}$.

(iv) $\mathcal{J}'_{\mathrm{flat}}$ satisfies (H-REF) (Corollary 21.5); $\mathcal{J}_{\mathrm{flat}}$ has the same rule forming its datum and is a reduction instance over the same substrate by Lemma 10.6 (ii), so it satisfies (H-REF) as well. Its reference companion at G is $\mathcal{J}_{\mathrm{flat}}$ at the datum A^{ref} formed from G, which is $\mathcal{J}'_{\mathrm{flat}}$ at A^{ref} precomposed with Φ by Lemma 10.6 (iii). □

22 Requirements

The engine emits one route per artefact, refuses certified weights that total one or more, and uses and records the largest certified weight. It decides descent, absorbing letters, fibres and column-stochasticity exactly on the declared data and refuses a declaration that fails one of them. It reads per-setting weights only from a receiver version that states them, and carries the supplier's eigenvalue-slack assumption beside the settings at which it was invoked. For a receiver compiled at a factored loss below one it records the loss in front of the link's channel witness, and refuses such a receiver as a branch of a register instance.

Every artefact is emitted on exactly one route, encoded or licensed, and a request for both is refused. A caller wanting both issues two requests, since pairing one route's entries with the other's channel is unsound (Remark 19.6).

The two routes run on different channels, and the licensed entry paired with the encoded route's channel is not sound (Remark 19.6, (19.11)); an artefact that carried both would admit that pairing.

Requirement 22.1 (One route per artefact). The engine emits every artefact on exactly one route: the request's `route` is `encoded` or `licensed`, the emitted payload is single-valued, and a request asking for both is refused with `route_not_single_valued`. A caller wanting both issues two requests and receives two artefacts.

Motivated by Rem. 19.6 · accepted

Certified weights whose total is one or more are refused, whether or not the space above the cutoff is trivial, and no squasher is constructed at total weight one. The construction of Proposition 19.4 divides by one minus the total.

On a non-trivial space above the cutoff a total weight above one is impossible (Lemma 19.2), but on a trivial one the certified-weight condition is vacuous, so the test is the only thing that bounds the total there. At total weight one the relation still holds (Remark 19.3); the refusal is the conservative choice.

Requirement 22.2 (Refusal of a total weight of one or more). The engine refuses certified weights whose total weight $p = \sum_v p_v$ is not below one, with `total_weight_not_below_one`. It decides this for every declaration, whether or not $\mathcal{H}(B_{>N})$ is trivial, and it constructs no squasher at $p = 1$, where the construction of Proposition 19.4 would divide by $1 - p = 0$.

Motivated by Lem. 19.2, Rem. 19.3, Prop. 19.4 · accepted

The engine uses the largest weight the artefact certifies for each outcome and records the weights it used. It never silently substitutes a larger one.

A smaller certified weight is valid but gives a cleaner target (Proposition 19.7, Remark 19.8), so the choice of weight is recorded rather than implied.

Requirement 22.3 (Use and record of the largest certified weight). The engine uses, for each outcome, the largest weight the artefact certifies and records the weights it used in `wpfss_data`, and it never silently substitutes a larger one.

Motivated by Prop. 19.7, Rem. 19.8 · accepted

The declared dark-count matrix is decided exactly to be monotone and to descend through the declared coarse-graining, by testing descent itself; a declaration that fails either is refused. The canonical sufficient condition serves only as a shortcut checked in full.

The descent condition is an identity between declared dyadic rationals, so it can be decided exactly. The sufficient condition (20.2) with its constancy clause left out would admit a declaration whose descended matrix is not defined (Remark 20.2), so a shortcut through that condition is checked in full. Its monotonicity half is clause (D2) of the supplier's dark-count channel, which every admitted matrix therefore satisfies exactly (Remark 20.11).

Requirement 22.4 (Exact decision of monotonicity and descent). The engine decides exactly, on the declared dark-count matrix R, that R is monotone, $R_{S|S'} > 0 \Rightarrow S' \subseteq S$, and that it descends through the declared coarse-graining ϖ in the sense of Lemma 20.1, $P(y|S') = P(y|S'')$ for every y and every pair S', S'' with $\varpi(S') = \varpi(S'')$, each an identity or a comparison between the dyadic rationals the declaration carries (Definition 9.2). It tests the descent property itself, and treats the sufficient condition (20.2) only as a shortcut that must itself be checked in full. A declaration that fails either condition is refused with the reason `dark_counts_do_not_descend`.

Motivated by Def. 9.2, Lem. 20.1, Rem. 20.2, Rem. 20.11 · accepted

Every bounded letter of the slot must be absorbing under the descended dark-count matrix, decided exactly on input. A declaration with one non-absorbing letter is refused as a whole; no entry's weight is zeroed in its place.

Without (absorbing) the routed bound (20.7) is no longer proved and fails in general, in the unsafe direction (Remark 20.7), and no bound is stated for a letter that fails it.

Requirement 22.5 (Absorbing bounded letters as an input validity condition). The engine takes (absorbing) as a declared input validity condition: it decides $R^{\varpi}_{y_i|y_i} = 1$ exactly at every bounded letter y_i of the slot, and refuses a declaration in which some bounded letter fails it, with the reason `bounded_letter_not_absorbing`. The refusal is of the whole artifact: the failing entry's weight is not set to zero and the rest kept.

Motivated by Prop. 20.4, Rem. 20.7 · accepted

Every certified set of the slot must be exactly one fibre of the declared coarse-graining; the equality is decided on input and a slot that fails it is refused. A split set leaves neither letter with a bound.

A bound on a sum of positive operators says nothing about its summands, so a certified set split over two letters leaves neither with a bound, and one merged with other outcomes attaches the weight to a larger fibre (Remark 20.8).

Requirement 22.6 (Certified sets equal to fibres of the coarse-graining). The engine requires every bounded set of the slot to be exactly a fibre of the declared coarse-graining, $\mathcal{O}_i = \varpi^{-1}(y_i)$, validates the equality at input, and refuses a slot that fails it with the reason `bound_set_not_a_fibre`.

Motivated by Rem. 20.8 · accepted

The declared dark-count matrix is decided exactly to have non-negative entries and unit column sums, and is refused otherwise. No second rule decides the descended matrix, which inherits the property exactly.

Every result of this contract is proved for an exactly column-stochastic R, and the descended matrix inherits the property by regrouping one finite sum (Remark 20.9).

Requirement 22.7 (Exact column-stochasticity of the declared dark-count matrix). The engine decides *exactly* that the declared dark-count matrix R is column-stochastic, $R_{S|S'} \geq 0$ for every entry and $\sum_S R_{S|S'} = 1$ for every column, and refuses a declaration that is not, with the reason `dark_counts_not_column_stochastic`. No second rule decides the property of the descended matrix R^{ϖ} of Lemma 20.1.

Motivated by Lem. 20.1, Rem. 20.9 · accepted

Per-setting weights are read only from an artifact whose receiver version states them as the content of a claim. An artifact of any other version is refused, a probe-free one included, and nothing it reports is read per setting.

A hypothesis of Proposition 20.4, whose part (iv) discharges the per-setting guarantee by Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound), is discharged only by a claim that states it. At a receiver version whose claims state only the minimum over the settings, the per-setting values an artifact reports beside it are the content of no claim, and the weight route reads every slot per setting (Definition 20.3), so such an artifact supplies nothing it can read.

Requirement 22.8 (Receiver versions from which per-setting weights are read). The engine reads the per-setting weights $p_i^{(t)}$ of a receiver artifact (`high_photon_bounds`) only where the artifact's module version resolves to an accepted receiver version that states them: one whose Theorem "Multiclick bounds carried by the emitted weights" (receiver/thm:emitted-bound) carries the per-setting inequality as its first conclusion and whose Definition "The strengthening slot" (receiver/def:slot) carries a weight per entry and setting. An artifact whose version does not state them, a probe-free one included, is refused with `high_photon_slot_ill_formed`: the per-setting values it reports beside the minimum are not read from it, and its minimum is not read at every setting in their place.

Motivated by Def. 20.3, Prop. 20.4, receiver/def:slot, receiver/thm:emitted-bound · accepted

Where the emitted ledger carries A-RECV, it carries the supplier's A-EIG there beside the settings at which the supplier says it was invoked. It is the one supplier record carried beside its key; every other key is carried alone.

Requirement 22.9 (Invocation record of the eigenvalue-slack assumption in the ledger). On the row of Assumption 13.3, wherever the emitted assumption ledger `assumption_ledger` carries it (Requirement 18.7), the engine records the supplier's eigenvalue-slack assumption A-EIG (Assumption "Declared slack on the computed block eigenvalues" (receiver/ass:eig)) beside the settings at which the supplier says it was invoked. Of the supplier's record behind a key, only A-EIG's is carried beside it; every other key is carried alone, and its record stays in the artifact, which the entry names by version.

Motivated by Ass. 13.3, Req. 18.7, Rem. 20.12, receiver/ass:eig · accepted

At a factored loss below one, four slots of the emitted link change: the witness and the carried pull-backs gain the pure-loss channel in front, the source family is the receiver's, the reachable set the composite's. Target, weights, entries and constants do not change.

Corollary 21.6 makes the precomposed instance, and not the flat instance of the normalized station, the reduction of the receiver. By Lemma 10.6 (ii) the two links of Definition 7.1 differ in four slots: the channel witness, the source family, the pull-backs of the carried tuple and the reachable-set statement. Every slot is part of what a link carries to be re-checked (Requirement 14.1), so the record states all four.

REQUIREMENT 22.10 (The link of a normalized station records the factored loss). Where the receiver artefact's `efficiency_record` carries a factored loss $t < 1$, the engine emits the artefact of the precomposed instance $\mathcal{J}'_{\mathrm{flat}} \circ \Lambda_t^{\mathrm{loss}}$ of Corollary 21.6, and its link records, in place of the normalized station's: (i) the channel witness as the flat instance's witness preceded by the pure-loss channel of transmittance t on the channel modes, with t recorded under `factored_loss`; (ii) the source family as the receiver's measurement; (iii) each pull-back of the carried tuple as the pull-back through that loss of the flat instance's pull-back, not as the flat instance's pull-back itself; and (iv) the reachable-set statement as the one for that composite channel (Definition 3.10). The emitted target, routed weights, transported operators, entries, constants, certificate cases and licence record are those emitted for the normalized station, unchanged. The reference companion's link is recorded the same way: its channel and pull-backs carry the same loss in front, and its target and entries are unchanged.

Motivated by Def. 3.10, Def. 7.1, Lem. 10.6, Req. 14.1, Cor. 21.3, Cor. 21.6 · implemented (effective: accepted) · *Code* certiqs_key.reductions.wpfss.construction:normalized_precomposition

An artefact that embeds the flat instance as a branch of a branching instance, the register instance included, reads only a receiver artefact compiled without a factored loss, and refuses one compiled at a factor below one.

Corollary 21.6 absorbs the factored loss for the flat instance on the channel system. A branching instance (Theorem 10.4), of which the register instance of Corollary 10.5 is one, places the flat instance beside other branches, which need not be the pull-back of anything through a loss on the channel modes, so no claim here moves the loss in front of it.

REQUIREMENT 22.11 (A normalized receiver is not read as a branch of a branching instance). An artefact that embeds the flat instance as a branch of a branching instance, the register instance included, reads only a receiver artefact whose `efficiency_record` carries the factored loss $t = 1$, and refuses one with $t < 1$ with `normalized_receiver_in_branch`.

Motivated by Thm. 10.4, Cor. 10.5, Cor. 21.6 · implemented (effective: accepted) · *Code* certiqs_key.reductions.wpfss.validation:rule_normalized_branch

Part IV
Domination links

This submodule relates two measurement families not by the size of their difference but by domination: one family dominates the other at one weight per sector. Its main result, Theorem 23.2, derives from the branching calculus that the dominating family is then a noise channel followed by the dominated one together with classical flags, the channel keeping each sector with probability one minus its weight. That relation is a typed link of its own, the domination link, placed on the target system after a squashing link, and a squashing link followed by it is an exact chain whose security transfers with the same ε (Corollary 24.1). No requirement of the engine uses these claims: a snapped measurement is not related to its model by domination, and a run evaluates its key length on the model instead where it holds one. They are kept for a declared device weight (A-DOM), and the remark on rotated projective elements says why an idealised model needs its noise floor before it dominates a lattice family at a small weight.

23 The domination link and its noise channel

A domination link runs from a finite target family to a family it dominates at one weight per sector, its flags prepared into sectors added for the purpose or into indicator sectors the target already has. The main result, Theorem 23.2, derives its channel from the generalized instrument that keeps or flags each sector and from the common-target rule. The closing remark says why a rotated projective element dominates no lattice element, so that a model needs its noise floor before domination can relate a lattice family to it at a small weight.

Imports foundations/def:domination, foundations/lem:absorbing-domination, foundations/lem:domination, foundations/lem:rational-obstruction

A domination link runs from a target family G to a family F it dominates at one weight per sector, through a noise channel that keeps each sector with probability one minus its weight and otherwise measures a residual family and prepares a flag, in added sectors or in sectors F already has.

A family G that dominates a family F at weights below one is a noise channel followed by F, together with classical flags [5, Thm. 3, Eqs. (20)–(24)]: by Definition "Domination of one operator family by another at block weights" (foundations/def:domination) each element splits as $G_v = (1-\vartheta_s)F_v + \vartheta_s R_v$ on sector s, with a residual family R that is again complete, and a channel can keep the sector with probability $1-\vartheta_s$ and otherwise measure R and prepare a flag (Theorem 23.2, which also shows that the tuple defined here is a receiver-stage link with an exact identity). The link defined here

records that channel. It is placed on the *target* side of a chain: it runs from the target family a squashing link produces at the exact datum to the target family at a datum it dominates, both on one finite target system with one sector map. The relation could equally be placed on the channel system before the squasher, from the model at the exact datum to the model at the other datum together with flags; the composite would be the same exact relation, but the intermediate descriptor would then be the infinite-dimensional channel system with flags, whereas on the target side every descriptor of the chain after the first link is the finite target system with its sector map (Corollary 24.1). The flags are prepared either into $|\mathcal{V}|$ one-dimensional sectors added to the target for the purpose, or, where the target already has one one-dimensional sector per outcome carrying that outcome's indicator, as the flag-state target does, into those sectors, and the target system does not change [5, Technical aside 1].

Definition 23.1 (The domination link)**.** Let $\mathbb{S}$ be a finite ordered set of sectors with dimensions $d_s \geq 1$, $\mathcal{H}(Q) = \bigoplus_{s\in\mathbb{S}} \mathbb{C}^{d_s}$ with sector projectors Π_s, and let $G = \{G_v\}_{v\in\mathcal{V}}$ and $F = \{F_v\}_{v\in\mathcal{V}}$ be POVMs on Q over one finite alphabet $\mathcal{V}$, both block diagonal for $\{\Pi_s\}$, $G_v = \bigoplus_s G_v^{(s)}$ and $F_v = \bigoplus_s F_v^{(s)}$. Let $\vartheta = (\vartheta_s)_{s\in\mathbb{S}}$ with $\vartheta_s \in [0,1)$ be weights at which G dominates F, every block complete at 1 (Definition "Domination of one operator family by another at block weights" (foundations/def:domination)):

$$G_v^{(s)} - (1-\vartheta_s)\, F_v^{(s)} \;\succeq\; 0 \qquad \text{for every } v \in \mathcal{V} \text{ and } s \in \mathbb{S}; \tag{23.1}$$

they are the *domination weights* of the pair. Let $R_v \coloneqq \bigoplus_s R_v^{(s)}$ with $R^{(s)}$ the residual family of Definition "Domination of one operator family by another at block weights" (foundations/def:domination) on sector s. A family F on Q has *indicator sectors* when there are pairwise distinct sectors $(f) \in \mathbb{S}$, one for each $f \in \mathcal{V}$, with $d_{(f)} = 1$ and $F_v^{((f))} = \delta_{vf}$ for every v; write $\Pi_{(f)}$ for the projector onto the sector (f), a rank-one projector.

(a) *(added flag sectors)* Let Q^+ be the system with $\mathcal{H}(Q^+) = \mathcal{H}(Q) \oplus \mathcal{H}(\mathrm{Flg}^+)$, $\mathcal{H}(\mathrm{Flg}^+) = \mathbb{C}^{|\mathcal{V}|}$ with orthonormal basis $\{|v\rangle\}_{v\in\mathcal{V}}$, whose sector set is $\mathbb{S}$ followed by $|\mathcal{V}|$ one-dimensional sectors $(f)^+$, $f \in \mathcal{V}$, in the order of $\mathcal{V}$, the sector $(f)^+$ spanned by $|f\rangle$. Put $F_v^+ \coloneqq F_v \oplus |v\rangle\langle v|_{\mathrm{Flg}^+}$, so that the added sectors are indicator sectors of F^+, and

$$\Phi_\vartheta^+(\sigma) \;\coloneqq\; \sum_{s\in\mathbb{S}} (1-\vartheta_s)\, \Pi_s \sigma \Pi_s \;\oplus\; \sum_{v\in\mathcal{V}} \Big(\sum_{s\in\mathbb{S}} \vartheta_s \operatorname{tr}\big[R_v\, \Pi_s \sigma \Pi_s\big] \Big) |v\rangle\langle v|_{\mathrm{Flg}^+} \tag{23.2}$$

for $\sigma \in \mathcal{T}(Q)$.

(b) *(shared flag sectors)* Suppose F has indicator sectors (f) and $\vartheta_{(f)} = 0$

for every $f \in \mathcal{V}$. Put

$$\Phi_\vartheta(\sigma) \;\coloneqq\; \sum_{s\in\mathbb{S}} (1-\vartheta_s)\,\Pi_s\sigma\Pi_s \;+\; \sum_{f\in\mathcal{V}} \Big(\sum_{s\in\mathbb{S}} \vartheta_s \operatorname{tr}\big[R_f\,\Pi_s\sigma\Pi_s\big] \Big) \Pi_{(f)}. \tag{23.3}$$

A *domination link* is the tuple (7.1) of a receiver-stage typed link (Definition 7.1) with stage = receiver, kind family-dominating and source family (Q, G); with target (Q^+, F^+) and channel witness Φ^+_ϑ in case (a), or target (Q, F) and channel witness Φ_ϑ in case (b); with carried operators the projectors $\Pi_s \oplus 0$, $s \in \mathbb{S}$, and the flag projector $\Pi^+_{\mathrm{Flg}} \coloneqq 0 \oplus \mathbb{1}_{\mathrm{Flg}^+}$ in case (a), and the projectors Π_s of the sectors that are not indicator sectors and the flag projector $\Pi_{\mathrm{Flg}} \coloneqq \sum_f \Pi_{(f)}$ in case (b), each with its pull-back under the channel witness; with the reachable-set statement of Definition 3.10 for that channel; with the constraint entries and licence record of the chain it ends; and with status = exact.

Uses Def. 1.3, Def. 3.10, Def. 7.1, Def. 10.1, foundations/def:domination · *Used by* Thm. 23.2, Rem. 23.3, Cor. 24.1 · newly derived · accepted

If G dominates F sector by sector, G squashes exactly to F with added flag sectors, or to F itself where F has flag sectors, through a noise channel that keeps each sector with probability one minus its weight. The adjoint returns each sector projector scaled by that probability.

The channel is the one of [5, Thm. 3, Eqs. (20)–(24)], derived here from the branching calculus of this contract: a generalized instrument decides, sector by sector, whether to keep the input (probability $1-\vartheta_s$) or to flag it (probability ϑ_s); the keep branch passes the input on unchanged, and the flag branch measures the residual family and prepares the flag of its outcome. The domination $G = (1-\vartheta)F + \vartheta R$ sector by sector is exactly the statement that the two branches together reproduce G. Remark 5.10 reads the same split as a dominated detector. Part (iii), in which the flags land in sectors the target already has, is the use of an existing flag space that [5, Technical aside 1] describes.

Theorem 23.2 (The noise channel of a domination). In the setting of Definition 23.1, let G dominate F at the domination weights ϑ and let R be the residual family.

(i) $R = \{R_v\}_{v\in\mathcal{V}}$ is a POVM on Q, block diagonal for $\{\Pi_s\}$, and $G_v^{(s)} = (1-\vartheta_s)F_v^{(s)} + \vartheta_s R_v^{(s)}$ for every v and s.

(ii) *(added flag sectors)* Φ^+_ϑ of (23.2) is a channel $\mathcal{T}(Q) \to \mathcal{T}(Q^+)$, and

$$\big(G;\ (1-\vartheta_s)\Pi_s\ (s\in\mathbb{S}),\ \textstyle\sum_s \vartheta_s\Pi_s\big) \;\preceq_{\Phi^+_\vartheta}\; \big(F^+;\ \Pi_s\oplus 0\ (s\in\mathbb{S}),\ 0\oplus\mathbb{1}_{\mathrm{Flg}^+}\big). \tag{23.4}$$

(iii) *(shared flag sectors)* If F has indicator sectors (f) with $\vartheta_{(f)} = 0$ for every f, then Φ_ϑ of (23.3) is a channel $\mathcal{T}(Q) \to \mathcal{T}(Q)$ with $G \preceq_{\Phi_\vartheta} F$,

and its adjoint takes the sector projectors and the flag projector to

$$\Phi_\vartheta^\dagger(\Pi_s) = (1-\vartheta_s)\Pi_s \quad (s \text{ not an indicator sector}),$$
$$\Phi_\vartheta^\dagger(\Pi_{(f)}) = \Pi_{(f)} + \sum_s \vartheta_s\, \Pi_s R_f \Pi_s, \qquad \Phi_\vartheta^\dagger(\Pi_{\mathrm{Flg}}) = \Pi_{\mathrm{Flg}} + \sum_s \vartheta_s \Pi_s. \tag{23.5}$$

(iv) *(output)* The outputs of both channels are block diagonal for the sectors of their target, so either channel is unchanged when the sector pinching of its target is applied after it.

(v) *(weight zero)* If $\vartheta_s = 0$ for every s, then $F = G$, and Φ_ϑ, where defined, is the sector pinching $\sigma \mapsto \sum_s \Pi_s \sigma \Pi_s$.

In particular the domination link of Definition 23.1, in either case, is a receiver-stage link of Definition 7.1: its channel witness relates its source family and carried operators to its target and carried operators exactly, as (7.1) requires with status = exact.

Uses Def. 1.3, Lem. 5.1, Def. 5.2, Lem. 5.3, Prop. 5.9, Def. 7.1, Def. 23.1, foundations/def:domination, foundations/lem:domination · *Used by* Cor. 24.1 · newly derived · accepted

Proof. Idea. The two channels are the composite channels of Lemma 5.3 and Proposition 5.9 for one generalized instrument with a keep branch and a flag branch.

(i) On every sector s, $G^{(s)}$ and $F^{(s)}$ are Hermitian, positive semidefinite and complete at 1, G and F being POVMs block diagonal for $\{\Pi_s\}$. Lemma "Elementary properties of domination" (foundations/lem:domination) (vi) gives that $R^{(s)}$ has positive semidefinite elements, is complete at 1 and satisfies the stated split; so $R_v = \bigoplus_s R_v^{(s)} \succeq 0$ and $\sum_v R_v = \bigoplus_s \mathbb{1}_{d_s} = \mathbb{1}_Q$.

The instrument. Put $\mathcal{I}_{\mathrm{k}}(\sigma) \coloneqq \sum_s (1-\vartheta_s)\Pi_s\sigma\Pi_s$ and $\mathcal{I}_{\mathrm{f}}(\sigma) \coloneqq \sum_s \vartheta_s \Pi_s \sigma \Pi_s$, both maps $\mathcal{T}(Q) \to \mathcal{T}(Q)$. Each is completely positive, a sum of maps $\sigma \mapsto c\,\Pi_s\sigma\Pi_s$ with $c \geq 0$, and $\operatorname{tr}\mathcal{I}_{\mathrm{k}}(\sigma) + \operatorname{tr}\mathcal{I}_{\mathrm{f}}(\sigma) = \sum_s \operatorname{tr}[\Pi_s\sigma\Pi_s] = \operatorname{tr}\sigma$, since $\sum_s \Pi_s = \mathbb{1}_Q$. So $\{\mathcal{I}_{\mathrm{k}}, \mathcal{I}_{\mathrm{f}}\}$ is a generalized instrument (Definition 5.2). Its adjoints are $\mathcal{I}_{\mathrm{k}}^\dagger(X) = \sum_s (1-\vartheta_s)\Pi_s X \Pi_s$ and $\mathcal{I}_{\mathrm{f}}^\dagger(X) = \sum_s \vartheta_s \Pi_s X \Pi_s$. In the keep branch take the POVM F on Q with the identity channel, so $F \preceq_{\mathrm{id}} F$; in the flag branch take the POVM R on Q with the measure-and-prepare channel $\Lambda_{\mathrm{mp}}(\tau) = \sum_v \operatorname{tr}[R_v\tau]\,|v\rangle\langle v|_{\mathrm{Flg}^+}$, so $R \preceq_{\Lambda_{\mathrm{mp}}} P^{\mathrm{Flg}}$ by Lemma 5.1. By Lemma 5.3 (i) the branch elements are $M_{v,\mathrm{k}} = \mathcal{I}_{\mathrm{k}}^\dagger(F_v) = \sum_s (1-\vartheta_s)\Pi_s F_v \Pi_s$ and $M_{v,\mathrm{f}} = \mathcal{I}_{\mathrm{f}}^\dagger(R_v) = \sum_s \vartheta_s \Pi_s R_v \Pi_s$, and since F_v and R_v are block diagonal, $M_{v,\mathrm{k}} + M_{v,\mathrm{f}} = \bigoplus_s \big((1-\vartheta_s)F_v^{(s)} + \vartheta_s R_v^{(s)}\big) = G_v$ by (i).

(ii) Lemma 5.3 (iii) gives $\{M_{v,\mathrm{k}} + M_{v,\mathrm{f}}\}_v \preceq_\Lambda \{F_v \oplus |v\rangle\langle v|\}_v$ with $\Lambda(\sigma) = \mathcal{I}_{\mathrm{k}}(\sigma) \oplus \Lambda_{\mathrm{mp}}\big(\mathcal{I}_{\mathrm{f}}(\sigma)\big)$, a channel by Lemma 5.3 (ii). The left side is G, the right side is F^+, and $\Lambda_{\mathrm{mp}}(\mathcal{I}_{\mathrm{f}}(\sigma)) = \sum_v \sum_s \vartheta_s \operatorname{tr}[R_v \Pi_s \sigma \Pi_s]\,|v\rangle\langle v|$, so $\Lambda = \Phi_\vartheta^+$

of (23.2). For the carried operators, (5.3) gives $(\Phi_\vartheta^+)^\dagger(X \oplus Y) = \mathcal{I}_{\mathrm{k}}^\dagger(X) + \mathcal{I}_{\mathrm{f}}^\dagger\big({\Lambda_{\mathrm{mp}}}^\dagger(Y)\big)$, with ${\Lambda_{\mathrm{mp}}}^\dagger(Y) = \sum_v \langle v|\, Y\, |v\rangle\, R_v$. At $X = \Pi_s$, $Y = 0$ this is $(1-\vartheta_s)\Pi_s$; at $X = 0$, $Y = \mathbb{1}_{\mathrm{Flg}^+}$ it is $\mathcal{I}_{\mathrm{f}}^\dagger(\sum_v R_v) = \mathcal{I}_{\mathrm{f}}^\dagger(\mathbb{1}_Q) = \sum_s \vartheta_s \Pi_s$, which is Lemma 5.3 (iv) for the flag branch. That is (23.4) in the notation of Definition 1.3.

(iii) Let $W\colon \mathbb{C}^{|\mathcal{V}|} \to \mathcal{H}(Q)$ be the isometry with $W\,|f\rangle$ the unit vector spanning the indicator sector (f), and $\Psi(\tau) \coloneqq W\tau W^\dagger$, a channel $\mathcal{T}(\mathrm{Flg}^+) \to \mathcal{T}(Q)$ with adjoint $\Psi^\dagger(X) = W^\dagger X W$. Because every F_v is block diagonal and equals δ_{vf} on the sector (f), $\Psi^\dagger(F_v) = \sum_f \delta_{vf}\, |f\rangle\langle f| = |v\rangle\langle v|$, so $P^{\mathrm{Flg}} \preceq_\Psi F$. With $F \preceq_{\mathrm{id}} F$ in the keep branch, F is a common target of the two branch targets F and P^{Flg}, and Proposition 5.9 gives $\{M_{v,\mathrm{k}} + M_{v,\mathrm{f}}\}_v \preceq_{\Lambda'} F$ with $\Lambda' = \mathrm{id} \circ \mathrm{id} \circ \mathcal{I}_{\mathrm{k}} + \Psi \circ \Lambda_{\mathrm{mp}} \circ \mathcal{I}_{\mathrm{f}}$, a channel by that proposition. The left side is G, and $\Psi(\Lambda_{\mathrm{mp}}(\mathcal{I}_{\mathrm{f}}(\sigma))) = \sum_f \sum_s \vartheta_s \,\mathrm{tr}[R_f \Pi_s \sigma \Pi_s]\, W\,|f\rangle\langle f|\, W^\dagger$ with $W\,|f\rangle\langle f|\,W^\dagger = \Pi_{(f)}$, so $\Lambda' = \Phi_\vartheta$ of (23.3). Its adjoint is $\Phi_\vartheta^\dagger(X) = \mathcal{I}_{\mathrm{k}}^\dagger(X) + \mathcal{I}_{\mathrm{f}}^\dagger\big(\sum_f \langle f|\, W^\dagger X W\, |f\rangle\, R_f\big)$, by linearity and the adjoints of the three maps. For a sector s that is not an indicator sector, $W^\dagger \Pi_s W = 0$, so $\Phi_\vartheta^\dagger(\Pi_s) = (1-\vartheta_s)\Pi_s$. For an indicator sector, $\vartheta_{(f)} = 0$ gives $\mathcal{I}_{\mathrm{k}}^\dagger(\Pi_{(f)}) = \Pi_{(f)}$, and $W^\dagger \Pi_{(f)} W = |f\rangle\langle f|$ gives the second term $\mathcal{I}_{\mathrm{f}}^\dagger(R_f) = \sum_s \vartheta_s \Pi_s R_f \Pi_s$. Summing over f, with $\sum_f R_f = \mathbb{1}_Q$ by (i), $\Phi_\vartheta^\dagger(\Pi_{\mathrm{Flg}}) = \Pi_{\mathrm{Flg}} + \sum_s \vartheta_s \Pi_s$.

(iv) Every term of (23.2) and (23.3) is $\Pi_s \sigma \Pi_s$ for a sector s, a multiple of $|v\rangle\langle v|_{\mathrm{Flg}^+}$, the projector onto the added sector $(v)^+$ of Q^+, or a multiple of $\Pi_{(f)}$; each is block diagonal for the sectors of the target, and the sector pinching fixes it.

(v) If every $\vartheta_s = 0$, (23.1) reads $G_v^{(s)} - F_v^{(s)} \succeq 0$ for all v and s, and both families are complete at 1 on every sector, so $F^{(s)} = G^{(s)}$ by Lemma "Elementary properties of domination" (foundations/lem:domination) (iii). The second sum of (23.3) vanishes and the first is the sector pinching.

The closing sentence. The carried operators of the link in case (a) are those of (23.4), and in case (b) the projectors of the sectors that are not indicator sectors and Π_{Flg}, whose pull-backs are (23.5); so (ii) and (iii) are the relation the channel-witness slot of (7.1) asks for, with no error term. □

A projective element on a direction with no Gaussian-rational vector dominates no non-zero lattice element, and completeness then forces a weight of one, so no domination link runs from that block. A noise floor in the model can remove it; after another channel it applies anew.

The remark is about domination only, the relation the link of Definition 23.1 carries, whose channel acts on each sector as the identity followed by a branching. A reduction that first applies another channel to the reference changes what is compared: the obstruction then applies to the image of the reference, compared with a lattice family written in the new basis, where the supports may have rational directions that they lacked. Such reductions

are planned beyond these contracts (`spec/future-scope.md`).

Remark 23.3 (Rotated projective elements and the noise floor). Let a reference family G, complete at 1 on a block, have an element $G_j = |\psi\rangle\langle\psi|$ that is a rank-one projector on a direction containing no non-zero vector of $\mathbb{Q}(i)^d$, as happens for an ideal analyser whose analysed polarisation $\cos\alpha\,|H\rangle + \sin\alpha\,|V\rangle$ has an irrational $\tan\alpha$ on its one-photon block, or for the two-photon block of a diagonal-basis analyser written in the orthonormal occupation basis. Any family F with Gaussian-rational entries that G dominates at a weight below one has $\operatorname{supp} F_j \subseteq \operatorname{supp} G_j$ by Lemma "Elementary properties of domination" (foundations/lem:domination) (ii), so $F_j = \lambda\,|\psi\rangle\langle\psi|$ with $\lambda \geq 0$, and a non-zero such F_j would have a column that is a non-zero Gaussian-rational vector on the direction ψ; hence $F_j = 0$: G_j dominates no non-zero lattice element. Completeness then forces the weight up to one: Lemma "Rationality obstruction to the domination of a lattice family" (foundations/lem:rational-obstruction) (v) gives $\vartheta \geq \lambda_{\max}(G_j)/1 = 1$, so no weight below one exists, at any lattice exponent. No domination link runs from the reference on a block that holds such an element, and no lattice family is related to it by domination there; this is the obstruction Remark "What domination controls, and why it needs a noise floor" (foundations/rem:noise-floor) explains. The remedy within domination is in the model and not in the lattice. With the device's loss and dark counts composed into the reference, every element is positive definite on every block under the conditions that the receiver module states (Proposition "Loss and dark counts make every dark-composed element definite" (receiver/detector-imperfections/prop:noise-floor): loss in the receiver and a dark-count probability strictly between zero and one in every port); loss alone or dark counts alone need not suffice. Where those conditions hold, the obstruction disappears: a lattice family whose elements but one lie between zero and their reference elements, the last taking what completeness leaves, is dominated at a weight of at most $\delta/(\lambda + \delta)$, with δ the norm of the residue that last element absorbs and λ the least eigenvalue of the reference's element at the same outcome, and not of the lattice element, which the residue raises above it (Lemma "Domination of a family that absorbs its residue in one element" (foundations/lem:absorbing-domination) (iv); Remark "What domination controls, and why it needs a noise floor" (foundations/rem:noise-floor)). After another channel the obstruction is to be read in the new basis, where it may not arise: an ideal analyser at an angle with irrational $\tan\alpha$ is the image, under one unitary applied before every outcome, of the same analyser at an angle where its elements are lattice values, and the isometry from the two-photon block into $\mathbb{C}^2 \otimes \mathbb{C}^2$ carries the diagonal-basis element above to a projector on a rational vector; neither route is a claim of this contract. The emission normalizer does not meet the obstruction in the form above: the datum it snaps has Gaussian-rational entries, so every support it meets is spanned by Gaussian-rational vectors (Lemma "Rationality obstruction to the domination of a lattice family" (foundations/lem:rational-obstruction) (ii)).

Uses Def. 23.1, foundations/def:domination, foundations/lem:absorbing-domination, foundations/lem:domination, foundations/lem:rational-obstruction · newly derived · accepted

24 Domination in a reduction chain

A squashing link at an exact datum followed by the domination link to another datum it dominates is an exact chain, so security transfers through it with the same ε and every entry is decided on the composite pull-back (Corollary 24.1). No requirement emits such a chain; it is the relation a declared device weight would use.

Imports foundations/def:domination

If an exact datum dominates another datum, the instance's squashing link at the exact datum followed by the domination link is an exact chain from the model at the exact datum to the other target with flags. Security of that target transfers to the model with the same ε.

This is Theorem 7.4 for a chain of two receiver-stage links: the squashing link of a reduction instance at an exact datum, and the domination link from its target to the target at another datum. Where the exact datum is formed from a reference by trusted classical post-processing — the dark counts and the class weights — and the domination is established after that post-processing, every map on the physical outcome acts before the first squashing link, and the domination link is a squashing link on the target system with an exact identity, not a post-processing map on an outcome; the order of Requirement 14.6 is therefore kept (Remark 4.5). Hypothesis (a) below is the existence theorem (I3) of the instance where the exact datum is admissible, with $M^\circ = M[A^\circ]$ and $\Lambda^\circ = \Lambda[A^\circ]$; it is stated on its own so that an exact datum whose entries are not in the substrate, such as a reference with square-root entries, can be used where an instance proves the relation for it directly.

Corollary 24.1 (A squashing link followed by a domination link)**.** Let $\mathcal{J}$ be a reduction instance with sector set $\mathbb{S} = \mathbb{S}_{\mathrm{dat}} \sqcup \mathbb{S}_{\mathrm{sc}}$, target family $F[\cdot]$ and model map $M[\cdot]$ (Definition 10.1), and let A° and A' be data for its data sectors such that

(a) $A' \in \mathcal{D}_{\mathrm{adm}}$; the blocks $A_v^{\circ(s)}$ are Hermitian, positive semidefinite and class complete, so that the target family $F[A^\circ]$ of (10.2) is a POVM on Q; and there are a POVM M° on B, a channel $\Lambda^\circ\colon \mathcal{T}(B) \to \mathcal{T}(Q)$ and a finite set $\hat{O}^\circ$ of operators on Q containing every sector projector, with pull-backs O°, such that $\big(M^\circ; O^\circ\big) \preceq_{\Lambda^\circ} \big(F[A^\circ]; \hat{O}^\circ\big)$;

(b) A° dominates A' at weights $\vartheta_s \in [0,1)$, $s \in \mathbb{S}_{\mathrm{dat}}$: $A_v^{\circ(s)} - (1-\vartheta_s) A_v^{\prime(s)} \succeq 0$ for every $v \in \mathcal{V}$ (Definition "Domination of one operator family by another at block weights" (foundations/def:domination)).

Put $\vartheta_s := 0$ on every scalar sector. Then:

(i) $F[A^\circ]$ dominates $F[A']$ at the domination weights $\vartheta = (\vartheta_s)_{s\in\mathbb{S}}$, as POVMs on Q block diagonal for the sectors.

(ii) Let ℓ_{dom} be the domination link (Definition 23.1) with source family $(Q, F[A^\circ])$ and target family $F[A']$, with added flag sectors (case (a) of that definition), or with shared flag sectors (case (b)) where $F[A']$ has indicator sectors among its scalar sectors; write F^{dom} and Φ for its target and channel witness. Let ℓ_1 be the receiver-stage link with source family (B, M°), target $(Q, F[A^\circ])$, channel witness Λ° and target-side carried operators the pull-backs under Φ of the carried operators of ℓ_{dom}, with $\mathsf{status} = \mathsf{exact}$. Then $(\ell_1, \ell_{\mathrm{dom}})$ is a composable and tuple-chained receiver segment, and with an empty source segment a reduction chain (Definition 7.2).

(iii) Its receiver composite $\Phi \circ \Lambda^\circ$ satisfies $M^\circ \preceq_{\Phi\circ\Lambda^\circ} F^{\mathrm{dom}}$ exactly, its composite status is exact, and Theorem 3.1 and Theorem 3.11 apply to it: the protocol whose measurement is M° and the squashed protocol with target F^{dom} have the same output state, and security of the squashed protocol over analysed inputs that contain the composite's image transfers to the protocol with M° with the same ε.

(iv) An entry $(\hat{O}, \bowtie, c)$ declared on F^{dom} is sound for the composite, and Theorem 6.7 applies to the list of such entries with the composite channel, when $(\Lambda^\circ)^\dagger\big(\Phi^\dagger(\hat{O})\big) \bowtie c\,\mathbb{1}_B$. Every projector onto a sector of the target has a positive semidefinite composite pull-back.

(v) Where $\sum_{v\in\mathcal{V}_t} M_v^\circ = q_t \mathbb{1}_B$ for every $t \in \mathcal{T}_{\mathrm{set}}$, the class-marginal identity (10.5) of M°, the composite pulls $\sum_{v\in\mathcal{V}_t} F_v^{\mathrm{dom}}$ back to $q_t\mathbb{1}_B$ for every $t \in \mathcal{T}_{\mathrm{set}}$, so the class-marginal entry of Definition 10.1 (I4) is certified for the composite.

Uses Cor. 1.7, Thm. 3.1, Thm. 3.11, Thm. 6.7, Def. 7.1, Def. 7.2, Thm. 7.4, Def. 10.1, Def. 23.1, Thm. 23.2, foundations/def:domination · newly derived · accepted

Proof. (i) Both families are POVMs on Q, $F[A^\circ]$ by (a) and $F[A']$ by the first consequence of Definition 10.1, and both are block diagonal by (10.2). On a data sector s the blocks are $A_v^{\circ(s)}$ and $A_v'^{(s)}$, and (23.1) at s is (b). On a scalar sector both blocks are $\phi_s(v)\mathbb{1}_{d_s}$, the scalar values reading no datum, so the difference at weight 0 is zero.

(ii) Composability (C1): the target of ℓ_1 is the source family of ℓ_{dom}. Tuple chaining (C2): by Theorem 23.2 (ii) and (iii) the pull-back under Φ of every carried operator of ℓ_{dom} (Definition 23.1) is a real linear combination of sector projectors of Q, namely $(1-\vartheta_s)\Pi_s$, $\sum_s \vartheta_s\Pi_s$ or $\Pi_{\mathrm{Flg}} + \sum_s \vartheta_s\Pi_s$; every sector projector is in the transported set of (I3), and by the last sentence of (I3), read for the relation of (a), such a combination is transported by Λ° to the same combination of the pull-backs of the sector projectors, by linearity of $(\Lambda^\circ)^\dagger$. So ℓ_1 carries exactly the operators ℓ_{dom} needs on its source side, and its relation, which is (a) with those operators, carries no error term. Both links are receiver-stage links of Definition 7.1: ℓ_1 by (a),

ℓ_{dom} by Definition 23.1 and Theorem 23.2. Stage order (C3) is structural, the source segment being empty.

(iii) Theorem 7.4 (i) gives the relation for the composite, with $(\Phi\circ\Lambda^{\circ})^{\dagger} = (\Lambda^{\circ})^{\dagger}\circ\Phi^{\dagger}$; both links are exact, so Theorem 7.4 (iv) gives the composite status exact; and Theorem 7.4 (ii) is the application of Theorem 3.1 and Theorem 3.11.

(iv) is Theorem 7.4 (iii) with the composite pull-back of (iii). For a sector projector Π of the target, $\Phi^{\dagger}(\Pi)\succeq 0$ because the adjoint of a channel is a positive map, and $(\Lambda^{\circ})^{\dagger}$ is positive for the same reason; so the certificate $(\Pi,\succeq,0)$ of Corollary 1.7 (ii) holds for the composite.

(v) The domination link relates $F[A^{\circ}]$ to F^{dom} exactly, $\Phi^{\dagger}(F_v^{\mathrm{dom}}) = F_v[A^{\circ}]$, so by linearity $\Phi^{\dagger}\big(\sum_{v\in\mathcal{V}_t} F_v^{\mathrm{dom}}\big) = \sum_{v\in\mathcal{V}_t} F_v[A^{\circ}]$, and by (a) $(\Lambda^{\circ})^{\dagger}$ of that is $\sum_{v\in\mathcal{V}_t} M_v^{\circ}$, which is $q_t\mathbb{1}_B$ by hypothesis. □

Interface

The declared inputs and outputs of the module and its submodules: each key with its symbol, when it is present and the claims that fix it. The domains, types and refusal rules are in the interface files themselves.

Inputs: Proof reductions by squashing (reductions)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`lattice_exponent` kappa	always; REQUIRED, no default and no fallback	Def. 9.1, Lem. 9.5
`strict_snaps` bool	read from the run configuration; off when the run does not declare it, and recorded either way	Def. 11.1, Req. 17.2, Req. 17.3, foundations/req: snap-reference, foundations/req: support-safe-snap
`detection_alphabet` Y	always	Def. 4.4
`probe_declaration` T, {q_t}	always	Def. 4.4, Ass. 13.5, Lem. 9.5
`joint_alphabet` V = Y x T with its classes {V_t} and weights {q_t}	always	Def. 4.4
`sector_declaration` S = S_dat + S_sc, with dimensions d_s, supports V^(s), and the order	always	Def. 10.1, Def. 9.1
`declared_datum` A = {A^(s)_v}	always	Def. 10.1, Def. 9.1
`reduction_instance` J = (I1)-(I4)	always	Def. 10.1
`receiver_artifact`	always for an instance that reads a supplier artifact	Ass. 13.3, Req. 18.5, Req. 18.4
`handed_reference` the reference G of the declaring instance's snapped receiver blocks, per setting and block: centres C (Gaussian rationals) and radii r (rationals)	where the declaring instance hands the reference beside its snap of the receiver blocks (foundations/def:evaluated-off-snap (i)), on every block it hands	Cor. 12.1, Req. 17.4, foundations/def: evaluated-off-snap, foundations/lem: recentred-enclosure
`accepted_receiver_versions`	always	Req. 18.5, Ass. 13.3

Key and symbol	Presence	Fixed by
`receiver_provenance`	optional; passed only by the part of the engine that ran the receiver module in the same run and docked its artifact into the instance, and absent on every other route	Req. 18.7, Ass. 13.3, receiver/req:exact-route-decision, receiver/req: evaluation-label
`link_declarations` l_1, ..., l_k	always	Def. 7.1, Def. 7.2, Rem. 8.8, Thm. 8.6
`carried_operators` hat O_1, ..., hat O_m	always; the DEFAULT is the instance's DEFAULT CARRIED SET	Def. 10.1, Cor. 11.3, Cor. 1.7, Req. 16.4, Req. 16.7
`requested_assurance`	always	foundations/req: tiers, foundations/def:tiers
`source_mode` tau_A or nomarg	always	Def. 6.5, Def. 6.3
`declared_relations` (M, F) pairs	optional; empty by default	Lem. 4.9, Rem. 4.10, Req. 14.7
`post_processing_order_declaration`	always	Rem. 4.5, Def. 4.4, Req. 14.6
`declared_orders`	always	Def. 10.1, Def. 4.4
`public_choice_applied`	OPTIONAL, one per call (per station). ABSENT MEANS NOT TOLD, and is the default.	Req. 18.1, Ass. 13.6

Outputs: Proof reductions by squashing (reductions)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`source_descriptor` B	always	Def. 10.1, Def. 1.1
`target_descriptor` Q, S, the sector map and the party partition	always	Def. 10.1, Def. 9.1, Thm. 11.2, Thm. 11.5, Req. 17.5
`target_povm` {F_v}_{v in V}	always	Def. 10.1, Thm. 11.2

Key and symbol	Presence	Fixed by
`factor_emission` the ordered factor list, with the combination rule	PRESENT EXACTLY WHEN THE CALL DECLARES MORE THAN ONE INSTANCE – one per measuring party. ABSENT FOR A ONE-FACTOR EMISSION, and nothing else in this file changes there.	Thm. 11.5, Def. 10.1, Prop. 4.8, Cor. 4.7, Req. 17.6
`outcome_correspondence` id_V	always	Def. 4.4, Rem. 4.5
`squashing_channel` Lambda = Pin_S o Lambda_0	always	Def. 10.1, Prop. 4.6, Cor. 4.7
`exact_identity` Lambdaˆdagger(F_v) = Mˆdec_v	always	Thm. 11.2, Def. 11.1
`transported_operators` (hat O_k, O_k)	always	Def. 1.3, Cor. 1.7, Cor. 3.13, Def. 10.1, Req. 16.4
`reachable_set_statement` Sˆrch	always	Def. 3.10, Cor. 3.12
`constraint_list` CList = {(hat O_k, bowtie_k, c_k)}	always	Def. 6.5, Def. 6.3, Cor. 11.3, Rem. 11.4, Req. 16.3, Req. 16.8, Req. 15.3, Thm. 6.7, Ass. 13.7
`attack_set_license`	always as a slot (req:constraint-and-licence-slots); its value is the licence record EXACTLY WHEN the constraint list is non-empty, and null otherwise	Thm. 6.7, Def. 6.3, Rem. 6.9, Cor. 11.3, Req. 15.2, Req. 16.3
`reduction_chain` Chain = (Chain_src, Chain_rec)	always	Def. 7.1, Def. 7.2, Thm. 7.4, Thm. 8.6, Rem. 8.8
`status`	always	Def. 7.1, Thm. 7.4, Thm. 11.2, Req. 15.4, Req. 14.4
`verification_report`	always	Ass. 13.3

Key and symbol	Presence	Fixed by
`assumption_ledger` A-...	always	Ass. 13.3, Rem. 11.10, Lem. 1.2, meat-core/lem: memory-lift, Rem. 20.12, Req. 18.1, Req. 18.3, Req. 18.4, Req. 18.5, Req. 18.6, Req. 18.2
`receiver_row_omitted_on_provenance`	always	Req. 18.7
`emission_record` kappa, (Delta_s)	ALWAYS PRESENT	Def. 11.1, Thm. 11.2, Rem. 11.7, Req. 17.3
`reference_companion` the instance at the datum A^ref formed from the handed reference, or at the handed datum A where no block is handed	exactly where a claim of this module establishes (H-REF) of cor:reference-companion for the instance (cor:reference-companion (iv); wpfss/cor:flat-companion), and either the declaring instance hands the reference on at least one block or the emission normalizer moves at least one class; otherwise absent, the record saying which of the two failed (req:reference-companion (vi)). Where no block is handed, the companion carries the snapped block on every block, so its datum is the handed datum A, the module's exact construction before the normalizer (cor:reference-companion (i)): the moved classes are evaluated off their snap, and the instance's moved blocks still rest on A-SNAP.	Cor. 12.1, Cor. 12.2, Lem. 12.3, Req. 17.4, Cor. 21.5
`mathematical_spec_version`	always	—

Inputs: Flag-state and weight-preserving squashers

From `wpfss/interface.yaml`.

Key and symbol	Presence	Fixed by
`photon_blocks` blk M^{=m}_{y|t}, 0 <= m <= N	always	Def. 20.3, Def. 9.1, Prop. 20.4, receiver/thm:click-povm, receiver/thm:coarse-graining
`photon_blocks_form` bare | dark_composed	always; NO DEFAULT – an artefact that does not state the form is refused	Def. 20.3, Prop. 21.2, Prop. 20.5
`high_photon_bounds` {(O_i, (p_i^(t))_{t in T})}_{i=1}^{r}	always	Def. 20.3, Prop. 20.4, receiver/thm:emitted-bound, receiver/def:slot, receiver/prop:slotvalid, Rem. 20.10, Req. 22.8
`bound_quantifier_domain` N, T	always	Def. 20.3, Prop. 20.4
`dark_count_channel` R	always	Def. 20.3, Lem. 20.1, Prop. 20.4, receiver/detector-imperfections/def:darkadm, receiver/detector-imperfections/thm:dark-counts, Rem. 20.7, Req. 22.7, Req. 22.4, Req. 22.5
`outcome_coarse_graining` cg	always	Def. 20.3, Lem. 20.1, Rem. 20.8, Req. 22.6
`bound_evaluation`	always	Rem. 20.12, Ass. 13.3, Def. 9.1, receiver/req:evaluation-label
`factored_loss` t	always	Cor. 21.6, Req. 22.10, Req. 22.11, receiver/detector-imperfections/prop:efficiency-normalization
`photon_cutoff` N	always	Def. 20.3, Prop. 21.2, Def. 19.1

Key and symbol	Presence	Fixed by
`route`	always	Def. 19.5, Rem. 19.6, Def. 19.1, Prop. 21.2, Req. 22.1

Outputs: Flag-state and weight-preserving squashers

From `wpfss/interface.yaml`.

Key and symbol	Presence	Fixed by
`wpfss_data` {p_v}, flag basis, N, p = sum_v p_v	always, for an artifact emitted from the flat instance	Def. 19.1, Def. 20.3, Lem. 19.2, Rem. 19.3, Rem. 19.8, Req. 22.3, Req. 22.2
`route`	always, for an artifact emitted from the flat instance	Def. 19.5, Rem. 19.6, Prop. 21.2, Req. 22.1

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